Meeting of the Board
10 – 13 July 2023
Songdo, Incheon, Republic of Korea
Provisional agenda item 12

GCF/B.36/02/Add.04

19 June 2023

Consideration of funding proposals – Addendum IV

Funding proposal package for FP209

Summary

This addendum contains the following seven parts:

a) A funding proposal titled "Climate Change Resilience through South Africa's Water Reuse Programme ("WRP")";
b) No-objection letter issued by the national designated authority(ies) or focal point(s);
c) Environmental and social report(s) disclosure;
d) Secretariat's assessment;
e) Independent Technical Advisory Panel's assessment;
f) Response from the accredited entity to the independent Technical Advisory Panel's assessment; and
g) Gender documentation.

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Funding Proposal

Project/Programme title:	Climate Change Resilience through South Africa's Water Reuse Programme ("WRP")
Country(ies):	South Africa ("SA")
Accredited Entity:	Development Bank of Southern Africa
Date of first submission:	[2021/12/17]
Date of current submission	[2023/04/20]
Version number	[V.7]

GREEN CLIMATE FUND

Contents

Note to Accredited Entities on the use of the funding proposal template

- Accredited Entities should provide summary information in the proposal with cross-reference to annexes such as feasibility studies, gender action plan, term sheet, etc.
- Accredited Entities should ensure that annexes provided are consistent with the details provided in the funding proposal. Updates to the funding proposal and/or annexes must be reflected in all relevant documents.
- The total number of pages for the funding proposal (excluding annexes) **should not exceed 60**. Proposals exceeding the prescribed length will not be assessed within the usual service standard time.
- The recommended font is Arial, size 11.
- Under the GCF Information Disclosure Policy, project and programme funding proposals will be disclosed on the GCF website, simultaneous with the submission to the Board, subject to the redaction of any information that may not be disclosed pursuant to the IDP. Accredited Entities are asked to fill out information on disclosure in section G.4.

Please submit the completed proposal to:

fundingproposal@gcfund.org

Please use the following name convention for the file name:

“FP-DBSA-South Africa-2022/07/06”

GLOSSARY

5CM	5 Case Model
AADD	Annual Average Daily Demand
AE	Accredited Entity
AMA	Accreditation master agreement
AML	Anti-Money Laundering
APR	Annual Project Report
AR	Assessment Report
ARA	Adaptation Results Area
ATP	Advanced Treatment Plant
BAC	Biologically Activated Carbon
BFS	Blended Finance Solution
BOOT	Build-Own-Operate-Transfer
BRCCJ	Building Resilience to Cope with Climate Change in Jordan
BUR	Biennial Update Report
CEC	Chemicals of Emerging Concern
CFC	Customer Foreign Currency
CFT	Combating the Financing of Terrorists
CHP	Combined Heat and Power
CMIP	Coupled Model Intercomparison Project
CP	Conditions Precedent
DART	Depth To Water-Level Change (D), Aquifer Type (A), Recharge (R) and Transmissivity (T)
DBSA	Development Bank of Southern Africa
DCM	Debt Capital Market
DCOG	Department of Cooperative Government
DFFE	Department of Forestry, Fisheries and Environment
DFI	Development Finance Institution
DPR	Direct Potable Reuse
DSCR	Debt Service Cover Ratio
DWS	Department of Water and Sanitation
E&SSS	Environmental and Social Safeguards Standards

EBC	Early Business Case
EI	Ecological Infrastructure
EPC	Engineering Procurement Contract
ERWAT	City of Ekurhuleni
ESG	Environmental, Social, and Governance
ESIA	Environmental and Social Impact Assessment
ESMF	Environmental and Social Management Framework
ESMP	Environmental and Social Management Plan
ESMS	Environmental and Social Management System
FAA	Funded Activity Agreement
FBC	Full Business Case
FIRR	Financial Internal Rate of Return
FNPV	Financial Net Present Value
FNWS	Faure New Water Scheme
GAC	Granular Activated Carbon
GCF	Green Climate Fund
GDP	Gross Domestic Product
GHG	Greenhouse Gas
GTAC	Government Technical Advisory Centre
GVA	Gross Value Added
HACCP	Hazard Analysis Critical Control Points
IBC	Intermediate Business Case
IBCF	Independent Blended Capital Facilitator
IDP	Integrated Development Plan
INR	Industrial Reuse
IPCC	Intergovernmental Panel on Climate Change
IPP	Indigenous Peoples Plan
IPR	Indirect Potable Reuse
IR	Irrigation Reuse
IRR	Internal Rate of Return
ISA	Infrastructure South Africa
JW	City of Johannesburg Metropolitan Municipality

LMU	Loan Management Unit
MDB	Multilateral Development Banks
MFMA	Municipal Finance Management Act
MISA	Municipal Infrastructure Support Agency
MOA	Memorandum of Agreement
MoV	Means of Verification
MRA	Mitigation Results Area
NCCAS	National Climate Change Adaptation Strategy
ND-GAIN	Notre Dame – Global Adaptation Initiative
NDC	Nationally Determined Contribution
NDP	National Development Plan
NIP	National Infrastructure Plan
NWPP	National Water Partnerships Programme
NWRS	National Water Resource Strategy
NWSMP	National Water and Sanitation Masterplan
OC	Oversight Committee
OEU	Operations Evaluation Unit
PCG	Partial Credit Guarantee
PICC	Presidential Infrastructure Co-Ordinating Commission
PIP	Programme Implementation Plan
PPP	Public Private Partnership
REIPP	Renewable Independent Power Producer Programme
RFP	Request for Proposal
SALGA	South African Local Government Association
SANS	South African National Standards
SDG	Sustainable Development Goal
SPEI	Standardised Precipitation-Evapotranspiration Index
SPV	Special Purpose Vehicle
SSP	Shared Socio-Economic Pathways
UN	United Nations
UNFCCC	United Nations Framework Convention on Climate Change
UNSCR	United Nations Security Council Resolutions

USD	United States Dollar
UV	Ultraviolet
WASH	Water, Sanitation and Hygiene
WHO	World Health Organisation
WPO	Water Partnerships Office
WRC	Water Research Commission
WRI	World Resources Institute
WRP	Water Reuse Programme
WRU	Water Reuse Unit
WSA	Water Service Authority
WTP	Water Treatment Plant
WTW	Water Treatment Works
WUA	Water Use Application
WWTW	Wastewater Treatment Works
ZAR	South African Rand

A. PROJECT/PROGRAMME SUMMARY

A.1. Project or programme	Programme	**A.2. Public or private sector**	Public
A.3. Request for Proposals (RFP)	*If the funding proposal is being submitted in response to a specific GCF Request for Proposals, indicate which RFP it is targeted for. Please note that there is a separate template for the Simplified Approval Process and REDD+.* Not applicable		

A.4. Result area(s)

*Check the applicable GCF result area(s) that the overall proposed project/programme targets below. For each checked result area(s), indicate the estimated percentage of **GCF and Co-financers' contribution** devoted to it. The total of the percentages when summed should be 100% for GCF and Co-financers' contribution respectively.*

	GCF contribution	Co-financers' contribution[1]
Mitigation total	Enter number %	Enter number %
☐ Energy generation and access	Enter number %	Enter number %
☐ Low-emission transport	Enter number %	Enter number %
☐ Buildings, cities, industries and appliances	Enter number %	Enter number %
☐ Forestry and land use	Enter number %	Enter number %
Adaptation total	Enter number %	Enter number %
☒ Most vulnerable people and communities	20 %	20 %
☒ Health and well-being, and food and water security	40 %	40 %
☒ Infrastructure and built environment	40%	40%
☐ Ecosystems and ecosystem services	Enter number %	Enter number %

A.5. Expected mitigation outcome *(Core indicator 1: GHG emissions reduced, avoided or removed / sequestered)*	None	**A.6. Expected adaptation outcome** *(Core indicator 2: direct and indirect beneficiaries reached)*	7302542	
			3424737	3877805
			5,8%	6,5%
A.7. Total financing (GCF + co-finance[2])	1.472 billion USD	**A.9. Project size**	Large (Over USD 250 million)	
A.8. Total GCF funding requested	235 million USD			

A.10. Financial instrument(s) requested for the GCF funding	*Mark all that apply and provide total amounts. The sum of all total amounts should be consistent with A.8.* ☒ Grant 35 million USD ☒ Loan 200 million USD ☐ Guarantee 0 ☐ Equity Enter number ☐ Results-based payment Enter number		
A.11. Implementation period	a) 10 years	**A.12. Total lifespan**	>20 years
A.13. Expected date of AE internal approval	Click or tap to enter a date.	**A.14. ESS category**	B
A.15. Has this FP been submitted as a CN before?	Yes ☒ No ☐	**A.16. Has Readiness or PPF support been used to prepare this FP?**	Yes ☒ No ☐
A.17. Is this FP included in the entity work programme?	Yes ☒ No ☐	**A.18. Is this FP included in the country programme?**	Yes ☒ No ☐
A.19. Complementarity and coherence	Yes ☐ No ☒		
A.20. Executing Entity information	The Executing Entity is the same as the Accredited Entity, namely the Development Bank of Southern Africa (DBSA). DBSA will manage the GCF proceeds. The DBSA will work closely with the National Water Partnerships Programme (NWPP) custodian, the South African national Department of Water and Sanitation (DWS). DWS, as national water sector leader, will ensure Country ownership of the WRP. The National Designated Authority is the national Department of Forestry, Fisheries and Environment, (DFFE). The NWPP will report to the DFFE to support that it executes its reporting obligations and commitments as National Designated Authority.		

A.21. Executive summary (max. 750 words, approximately 1.5 pages)

Climate change problem

South Africa is a water-scarce country, and the sustainable provision of water is amongst its most significant challenges. The country is located within southern Africa's 'drought belt' and, according to the World Bank, is the fifth most water-scarce country in sub-Saharan Africa[i]. Since the 1960s, South Africa has already experienced a rise in average annual temperatures country-wide, by 1.5°C. Temperature has increased more markedly across arid, inland areas of the country, with records showing that both maximum and minimum daily temperatures have risen, in all seasons. Rainfall trends display less clarity than temperature, with significant inter-annual variability. There are also considerable geographic variances in historic rainfall patterns. Since the 1960s, a marginal reduction in rainfall was experienced during the early rainfall season. Observations point to significant decreases in the number of rain days across almost all hydrological zones and indicates a tendency towards an increase in the intensity of rainfall events, coupled with prolonged dry spells. [ii]

With mean annual precipitation being only 60% of the global average and with this being spatially and temporally variable, the country has challenges in meeting water requirements with some regional disparities. This is exacerbated by major urban centres and growth nodes not being situated in alignment with water resource availability. These nodes, such as Cape Town, Port Elizabeth, and Johannesburg (amongst others) are therefore interconnected with key water resources through a network of bulk water transfers. These transfers have enabled the continued socio-economic development of the country, however, this also means that the status of water resources within these water supply systems is very dependent on water resources within other catchments in other parts of the country and the associated climate risks. In 2017-19 due to the “Day Zero” drought induced by climate change, the City of Cape Town (CCT) with a population of 5 million had insufficient water supplies to meet basic human requirements over a period of two years creating a dual shock on public health and on socio-economic growth. As a result, the Government initiated the National Water Partnerships Programme (NWPP) with a series of sub-programmes, including the proposed national Water Reuse Programme (WRP) presented here for GCF co-financing.

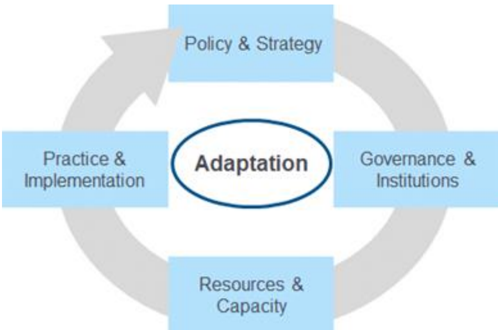
According to World Resources Institute’s (WRI’s) water risk index, at the country level South Africa faces “extremely high” baseline physical water risk[iii]. Many parts of South Africa have reached or are fast approaching the point at which all viable freshwater resources are fully utilised. According to the National Water Resource Strategy (NWRS)[iv], 98% of all surface water resources have been utilised, allowing only 2% for future growth. Due to the interconnectedness of the country’s basins through bulk water transfers, climate variability and climate change have significant impact across the country and not only in localized areas.

South Africa is ranked amongst the top half of most vulnerable countries in terms of climate change vulnerability overall, and in particular water-related climate change vulnerability.[v] It is also ranked amongst the top half of countries that have suffered the most climate change-related historic losses in the last two decades.[vi] In terms of baseline water stress, South Africa falls within the top quarter (25%) of the world’s most water-stressed nations.[vii] According to one global multi-variate index of water risk, parts of South Africa display some of the highest baseline water risk levels in the African continent and across the world.[viii] These baseline risks will be magnified and elevated in many areas by climate change. The likelihood of annual severe drought is projected to increase by 39% in the 2050s compared to 1986-2005 baseline under a high emissions future scenario (RCP8.5).[ix] Temperatures have been rising steadily in South Africa. Very high temperatures mean that parts of the country will be much drier and increased evaporation will cause an overall decrease in water availability in large parts of an already water stressed country.[x] A study by leading South African climate scientists,[xi] using CCAM (widely used by CSIRO in Australia and CSIR in South Africa as a preferred model due to its finer resolution at a local scale) and six downscalings shows that in the period 2035 to 2064, there is a high likelihood of increased conditions of drought, due to the drastic increase in maximum temperatures and very hot days.[xii] In the 2044 – 2064 period, projected change in the drought tendencies (i.e., number of cases exceeding near-normal per decade) over South Africa relative to 1986-2005 baseline period under a low mitigation scenario (RCP 8.5) will increase in frequency and duration, at a 95% statistical significance level.[xiii] But waiting until 2044 or even 2035 is not an effective adaptation option; adaptive capacity needs to be built now and strengthened progressively as the impacts of climate change further worsen.

South Africa has submitted its first adaptation communication as a component of its Nationally Determined Contribution (A-NDC) in line with the Paris Agreement. The adaptation communication provides detailed information on South Africa's planned contribution to the global adaptation goal during the NDC period, anticipated climate impacts, a description of the National Climate Change Adaptation Strategy, and details of planned adaptation actions. [xiv] The plans note the vulnerabilities that exist at municipal level and the need to strengthen water security at these levels, with water reuse being identified as one of the key interventions to address these vulnerabilities.

Proposed Intervention

Through the NWRS as well as the 2018 National Water and Sanitation Masterplan (NWSMP)[xv], the DWS has proposed a suite of interventions to address the current and future and water challenges which are projected as being a 17% deficit by the year 2030. These interventions include decreasing water demand and water losses through non-revenue water, as well as other interventions to supplement the water supply mix. These includes a national water reuse strategy, and this funding proposal recommends the commitment of USD $235 million of catalytic capital to support the establishment of a national WRP and to enable the scale-up of a pipeline water reuse projects that will support the improvement of water security and climate resilience at local and regional scales, where climate vulnerability is the driver. Broadly, GCF funding will allow for a number of key enablers to be realized, namely, by (i) establishing a programmatic approach to the development and implementation of water reuse infrastructure through support to municipalities to identify, prepare and structure bankable water reuse projects, by (ii) enhancing access to more affordable finance through using GCF funding to reduce the cost of borrowing on a project level, with these lower financing costs (and potentially longer tenors) helping in the facilitation of tariffs that are more attractive and affordable to end-customers, thereby enhancing project bankability, and by (iii) enhancing access to private capital by blending the GCF funding with private investor funding thereby enabling private funders and municipalities to lower their risk exposure on a project-by-project basis. The WRP has a blended finance structure, which enables investors (predominantly debt providers) to participate across numerous individual water reuse projects. The investment scale of the WRP and the design of the Blended Finance structure results in a New Asset Class (this is described in detail in the Annexure 3 Financial Architecture report). Increased participation from the private sector in this newly created asset class would, over time, allow for private sector support to be sustainable without continued GCF participation. However, GCF funding is critical to initially mobilise the programme and create the new asset class for future participation by the private sector. GCF funding will therefore be catalytic in the establishment of the WRP with an indicative total budget value of USD $1,472 billion.



Climate Adaptation Benefits

By establishing a programmatic response that recognizes the importance of water reuse in improving South Africa's water security and climate resilience, the WRP will provide a Government-wide response that involves institutions across the water value chain and will support in providing financial resources as well as building institutional capacity, particularly at municipal level where climate induced water insecurity impacts upon socio-economic aspects and local livelihoods. The programmatic support to ensure the efficacy of innovative water reuse projects will not only be transformational in introducing more resilient approaches at these local levels, but also the use of catalytic GCF funds will support in establishing a paradigm shift in the structuring of blended finance options and the creation of a new asset class. The indicative initial pipeline of 10 Advanced Treatment Plants would result in an additional 440Ml/day of water.

Water Reuse as a new asset class is in line with the GCF result area Adaptation Results Area 2 (ARA2): Health, well-being, food and water security in that it *reduces pressures on water supply; and supports efforts to improve the resilience of cities by improving water, sanitation, management systems and infrastructure within urban areas*. It also supports paradigm shift development pathway: "Strengthen integrated water resources management and water management" under "alternative water sources" of the upcoming GCF Water Security Sector Guide.

Additionally, due to the nature of water reuse projects this programme will also service ARA1: Most vulnerable people and communities as well as ARA 3: Infrastructure and built environment.

The proposed Programme expects to directly increase the resilience to climate change of 3,424,737 beneficiaries (based upon an initial pipeline of 10 projects) and an additional 3,877,805 indirect beneficiaries (including a potential additional 17 projects).

The following adaptation co-benefits have been identified:

- Co-benefit 1: *Environmental*. Through the implementation of the WRP there will be improvement in the quality of rivers and groundwater with there being a reduction in the discharge from wastewater treatment works, with these waters being redirected to water reuse plants. Also, with the reduced demand upon local catchments the water requirements of ecosystems may be met.
- Co-benefit 2: *Economic*. With municipalities having the mandate to drive local economic development, improvements in water security would enable further expansion of economic activities including local agriculture and industry, as well as other businesses.
- Co-benefit 3: *Gender*. In many communities women and girl children are often responsible for the collection of water. Improvements in water security can have profound impacts on the availability of supply and the improvement of water service standards.

B. PROJECT/PROGRAMME INFORMATION

B.1. Climate context (max. 1000 words, approximately 2 pages)

South Africa is ranked amongst the top half of most vulnerable countries in terms of climate change vulnerability overall, and in particular water-related climate change vulnerability.[xvi] Under the 2021 Notre Dame – Global Adaptation Initiative (ND-GAIN) Index (reflecting 2019 data), South Africa's overall ranking is 95 out of 182 countries (for its vulnerability score, it is ranked 83 out of 182 and for its readiness it is ranked 113 of 192 countries).[xvii] It is also ranked amongst the top half of countries that have suffered the most climate change-related historic losses in the last two decades, by the Germanwatch index.[xviii] In terms of baseline water stress, South Africa falls within the top quarter (25%) of the world's most water-stressed nations.[xix] According to one global multi-variate index of water risk, parts of South Africa display some of the highest baseline water risk levels in the African continent and across the world.[xx] Water supply vulnerability, the ability of supply to meet demand under current climate regimes are predicted to increase dramatically across the country under future growth scenarios.

Whilst acknowledging that there will be considerable spatial and temporal variability of water availability in South Africa (or indeed, anywhere), all the available climate science information creates a consistent narrative that water scarcity for South Africa will increase due to climate change. Demographic changes and increases in the use of water for agriculture and associated food security will compound the problem, but there can be little doubt that climate change is likely to be a significant driver.

This analysis brings together a number of different climate information sources (from the IPCC, recent peer reviewed literature, regional climate model information, and syntheses at the national level). Whilst the different sources are in broad agreement, any differences in projected climate conditions are highlighted and explained. Beginning with the synthesis of global climate models (GCMs), the IPCC AR6 Working Group 1 Summary for Policy Makers shows that at any amount of future warming South Africa will receive reduced rainfall.

Even small percentage changes from this model synthesis are likely to result in larger percentage changes for areas with dry baseline conditions (at a spatial resolution smaller than the GCMs) (Figure 1). This figure is derived from the multi-model mean of the Coupled Model Intercomparison Project Phase 6 (CMIP6). All models in the CMIP6 project are subjected to routine evaluation using tools made available by the CMIP coordination group (https://wcrp-cmip.org/cmip-phase-6-cmip6/#evaluation).

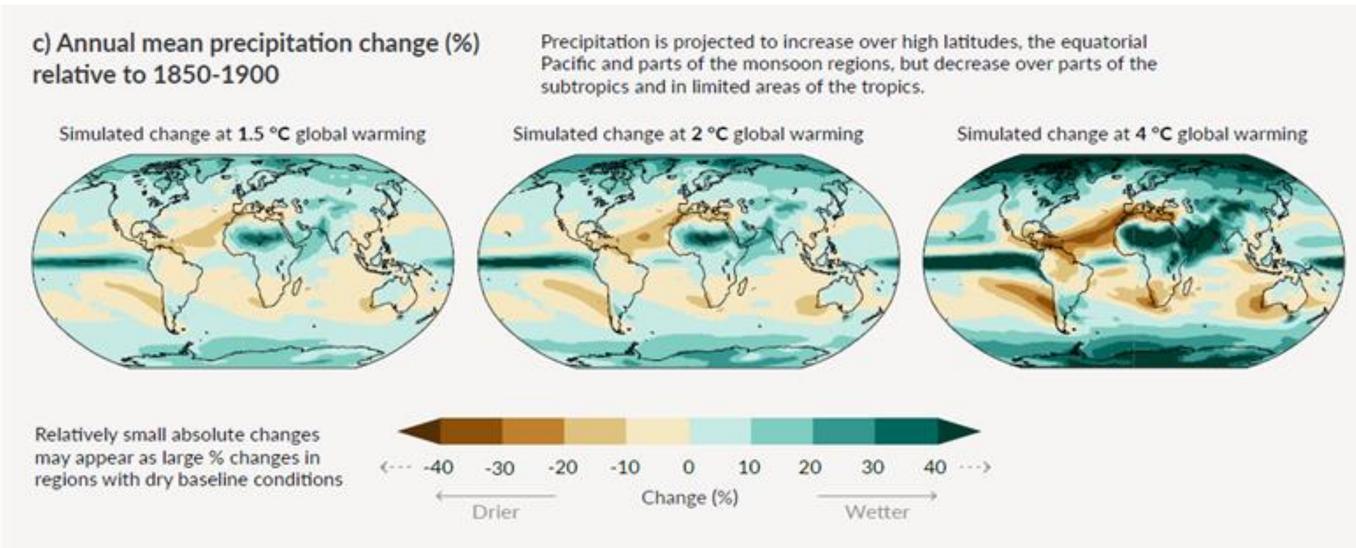


Figure 1: Annual mean precipitation changes, relative to 1850-1900, for various future degrees of warming from the CMIP6 ensemble

The model ensemble has been shown to outperform individual models and captured spatial trends across climatic zones, for Africa and the Arabian Peninsula[xxi]. In a study aimed specifically at understanding future rainfall patterns in southern Africa, Sian et al. (2021)[xxii] compared the historical data of 23 CMIP6 GCMs against rain gauge-based data to assess the individual models' performance in simulating annual precipitation variation, seasonal inter-annual variability and trends. On the basis of the best performing models, their results confirm an overall drying pattern over all sub-regions of southern Africa under SSP2-4.5 (and they note that it is imperative to properly manage water resources through sound planning).

The IPCC report concludes with high confidence (AR6 WG1 Technical Summary TS.4.3.2) that with 2°C of global warming southern Africa will experience increases in drought and aridity as early as the mid-21st century. The report also places high confidence in a projected decrease in total precipitation for southernmost Africa regions (TS.4.3.2.1). The impacts of these physical changes are summarized in the Working Group II report, "Climate Change 2022: Impacts, adaptation and vulnerability".

Since GCMs cannot simulate precipitation well over areas of fast-changing topography or high elevation, when examining future rainfall projections at a sub-national scale, finer resolution Regional Climate Models (RCMs) are used. The climate information platform (https://ssr.climateinformation.org/) developed by GCF and WMO gives access to future projections from the Coordinated Regional Climate Downscaling Experiment (CORDEX) where – for Africa – 18 RCMs are downscaled for this domain. An evaluation of the precipitation characteristics of the CORDEX-Africa models when compared with rain gauges and satellite observations was carried out by Karypidou et al. (2022)[xxiii] who concluded that the CORDEX-Africa RCMs provide results that are in closer agreement to observations than GCMs and can be considered a suitable dataset for climate impact assessments over southern Africa. The best agreement with observed indices was obtained when the ensemble average precipitation values were used, confirming previous advice that the best consensus information for climate impact studies derives from multi-model ensemble averages (Duan et al., 2019)[xxiv].

Country-level projections are now available from the sixth Coupled Model Intercomparison Project (CMIP-6), with the Shared Socio-economic Pathways (SSP) scenarios consistent with the Intergovernmental Panel on Climate Change's (IPCC's) sixth assessment report. In all future time periods, under all scenarios except the single most optimistic (and at this time unrealistic), South Africa will experience temperature rise due to climate change. It is expected that under a Business-as-Usual future, by mid-century the South African coast will warm by 1-2 ^{o}C, and the interior as much as 3 ^{o}C. Future projections of precipitation for the period 2020-2039, under all SSPs (barring the most optimistic and unrealistic), indicate there is a decrease in total annual average rainfall volume (
Figure 2.

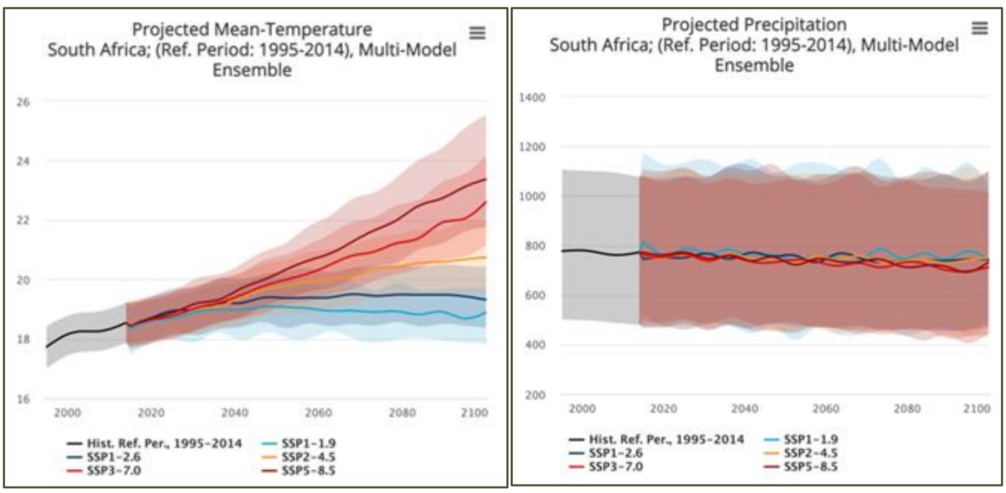


Figure 2: Comparison of historic reference mean annual temperature and rainfall and future temperature and rainfall under different SSPs, through 2100 2100 (Source: World Bank CCKP)[xxv]

Figures 3,4 and 5 below provide some extracts from the GCF-WMO climate information platform which will be one of the resources used in the DBSA WRP Project Evaluation and Screening Process. To demonstrate the future projections of rainfall related variables (and temperature), the figures below focus (arbitrarily) on Johannesburg and Cape Town, for the future period 2041-2070 (compared to 1981-2010), and for the climate change scenario RCP4.5 and these further demonstrate the expected decreases in rainfall and water discharge by mid-century.

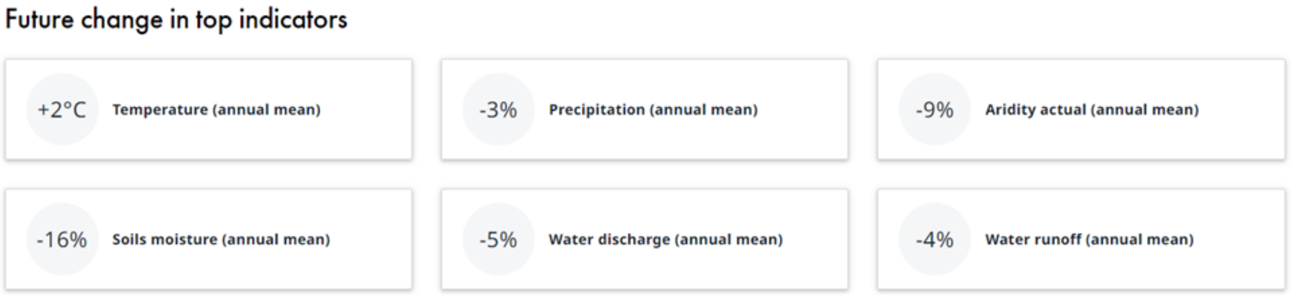


Figure 3: Future projections for Johannesburg for 2041-2070 for RCP 4.5

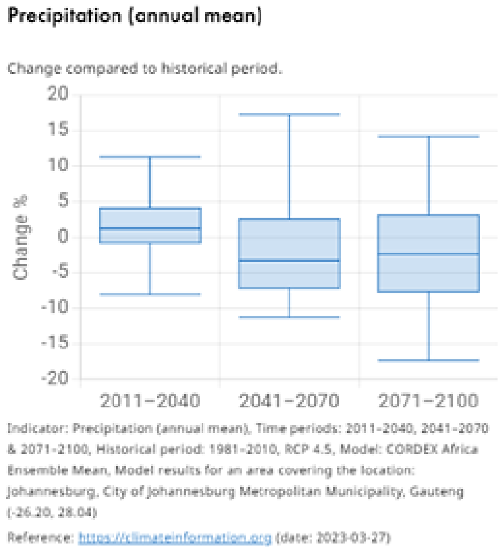


Figure 4: Spread of the 18-model ensemble for mean precipitation (2041-2070, RCP4.5) for Johannesburg

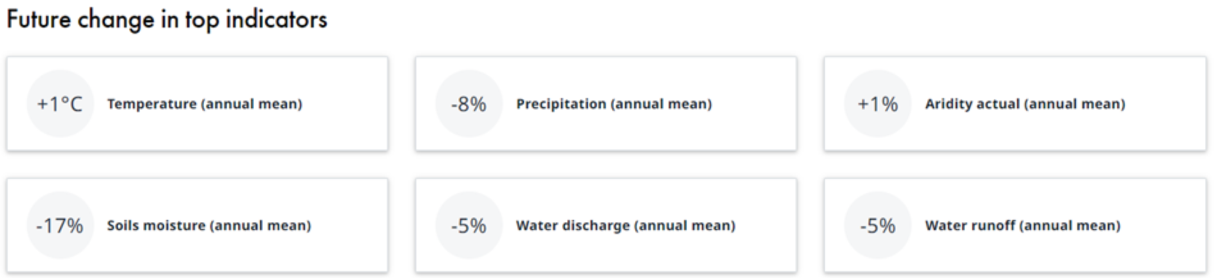


Figure 5: Future projections for Cape Town for 2041-2070 for RCP 4.5

When the WRP toolkit is used for screening, it will consider a range of scenarios with the actual geographical focus (full catchment area) of the proposed sub-projects.

Over the past four decades, South Africa has experienced more than 82 hydro-meteorological hazards (floods, storms, landslides, wildfires, droughts and extreme temperatures) that have resulted in the death of an estimated 1,692 people and affected more than 21 million people.[xxvi] The intensity and frequency of hydro-meteorological hazards have increased in South Africa the past four decades and will likely continue to be more damaging to people, property, and both natural and manmade systems in the future, due to climate change.[xxvii]

The Project Evaluation and Screening Process also draws on information from the online national “Green Book” planning tool (https://greenbook.co.za/about-the-green-book.html). The Green Book was co-funded by the South African CSIR and the International Development Research Centre (IDRC) and was developed between 2016 and 2019. The CSIR has partnered with the National Disaster Management Centre (NDMC) and co-developed this product with universities, government departments, NGOs and other peer groups. The Green Book is understood nationally to be a reliable platform for understanding climate vulnerability at both national and local (municipal) level. For each climate change hazard presented by the Green Book is a detailed technical report, including a specific analysis for surface water which gives a water supply climate risk narrative at the settlement level. Underlying all of the analyses is a suite of climate model experiments based on a 50km resolution RCM, developed by the Commonwealth Scientific and Industrial Research Organization (CSIRO) in Australia, and forced with six GCM simulations from the CMIP5 ensemble, using future climate scenarios RCP4.5 and RCP8.5. The model suite’s capability to simulate present-day southern African climate has been demonstrated and well-published (e.g. Engelbrecht et al., 2011; Engelbrecht et al., 2013; Engelbrecht et al., 2015)[xxviiixxixxxx].

The surface water analysis of the Green Book combines population data with climate model data. The vulnerability of individual towns was calculated based on the estimated current demand and supply as recorded across the country by the Department of Water and Sanitation’s (DWS) All Towns study of 2011. The future vulnerability was calculated by adjusting the water demand for each town proportional to the increase in population growth for both high and medium growth scenarios. There is a complete technical report for the water study (Cullis and Phillips, 2019)[xxxi].

Annexure 28 to the FP provides a “Climate Change Risk & Vulnerability Analysis – Water Sector and Select Municipalities (WRP Priority Cities)”, which demonstrates that the Green Book municipal vulnerability tool can provide a similar set of water-related results to the GCF-WMO climate information platform but augmented by the additional information about projections of water use at the local level. This annex presents Table 1, below, which shows water stress variables across ten municipalities. The ten locations were selected purely to demonstrate the utility of the software and are not intended to indicate the location of potential projects. The table shows that water stress is driven by significant climate change factors at eight our of ten of the locations, which is consistent with the information coming from both global and regional climate models.

Table 1. Water stress projections for 2050 under RCP8.5 from the Green Book municipal vulnerability tool.

Potential Municipalities	**2050 Projection using RCP8.5**		
	Change in Mean Annual Temp (°C)	**Change in Mean Annual Precipitation (%)**	**Change in Mean Annual Evaporation (%)**
Cape Town	**+1**	**-2.55**	**+4.18**
Nelson Mandela Bay	**+1**	**-1.88**	**+2.75**
City of Ekhurleni	**+1**	**-3.85**	**+11.35**
City of Johannesburg	**+1**	**-3.89**	**+11.15**
City of Tshwane	**+1**	**-5.52**	**+11.72**
City of eThekwini	**+1**	**+3.12**	**+3.36**
Drakenstein	**+1**	**-9.47**	**+9.22**

Mangaung	**+1**	**+8.66**	**+7,89**
Indicative Project A	**+1**	**-2.55**	**+4.18**
Indicative Project B	**+1**	**-5.52**	**+11.72**

Summarizing the multiple sources of information above, the IPCC syntheses, global and regional climate models, and the nationally funded portal for climate change and vulnerability in South Africa all describe a similar exposure and vulnerability to water stress in the future with climate change being a credible factor. Changes in temperature (both means and extremes) will be largely similar across the whole country, so there will be spatial differences but everywhere will be exposed to higher evaporation and evapotranspiration. Rainfall patterns by their nature will exhibit greater spatial variability, but the evidence is very strong for an overall tendency towards dryness for many parts of South Africa.

Thus, the country's already-stressed water resources will be put under greater pressure by climate change, with South Africa expected to become drier as the southern African region warms at nearly twice the rate of the rest of the world, becoming a climate change hotspot in terms of both hot extremes and drying.[xxxii] Noting the recent severe droughts (particularly in parts of the Western and Eastern Cape) the region has experienced and the widespread water-efficiency measures put into action to combat the extreme water shortage, the increasing risk of depleted precipitation that these results imply would indicate that such efficiency measures will become more frequently strained and relied upon.

For many South Africans, this will translate into severe water insecurity and an exacerbation of socio-economic fragility. The interconnectedness of the water supply systems means that this vulnerability will have impact at regional levels. The 2019 Sustainable Development Goals Country Report noted that 20% of the population did not have access to safely managed drinking water services (Indicator 6.1.1). Annual per capita freshwater availability in South Africa is only 786 cubic meters a year, compared to the global average of 5,732 cubic meters a year and even to the Sub-Saharan Africa average of 3,700 cubic meters a year, placing South Africa amongst the 40 countries with the lowest per capita freshwater availability on the planet.[xxxiii]

Slow-onset climate hazards like drought are expected to become far more frequent, with 'once in a 100-year drought events' likely to occur multiple times in a century. Model projections for future rainfall in southwestern South Africa (in the Western Cape region around Cape Town), for instance, indicate several multi-year droughts occurring between the present day and 2100. These multi-year drought events were responsible for "day-zero" water crisis in Cape Town- during 2018/19. Currently the city of Port Elizabeth is experiencing the same situation with severe water restrictions in place. The Green Book tool using climate model projections (multi-model ensembles) for future time-periods consistently points to a drier future for most of South Africa, across all scenarios, with relative water supply risk increasing (Figure 6).

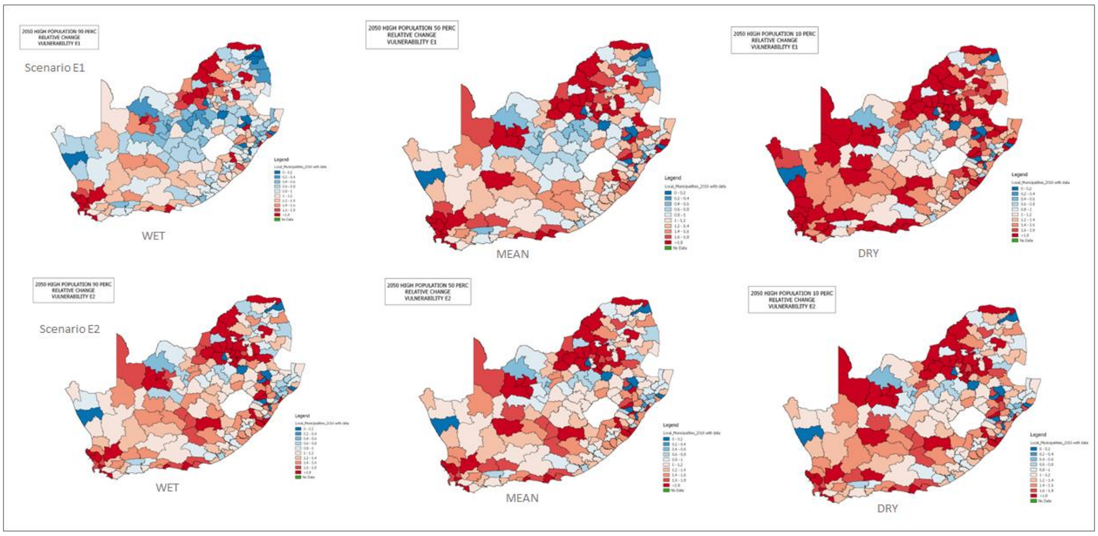


Figure 6: Relative change in water supply risk, under high growth and climate change scenarios, across municipalities when water supply is dependent on local water resources alone (E1) and with interconnected and augmented supply (E2)

Based on current demand projections (driven in part by population growth) and without effective intervention, in addition to the water conservation sub-programmes included in the NWPP, the water deficit confronting the country may range from 2.7 to 3.8 billion cubic meters by 2030, a gap of approximately 17% of available water sources.[xxxiv] At the municipal level, climate change – coupled with population growth and increased water demand – is likely to result in heightened water supply vulnerability in the majority of municipalities and particularly in the key growth nodes of the country in the Western Cape (Cape Town and linked secondary towns), Eastern Cape (Nelson Mandela Bay), Gauteng (Johannesburg, Pretoria and Ekurhuleni) and Kwa-Zulu Natal (eThekwini). While non-climate stressors such as inefficiencies in governance and suboptimal water infrastructure management (including water losses) do contribute to the challenge, it is climate change-driven variability and decrease in rainfall over several regions of the country that underpin the expected water deficit. Climate change adaptation is thus central to any, and all solutions.

Country Ownership of the Proposed Programme and Alignment with National Priorities

The Department of Water and Sanitation (DWS) has established the National Water Partnerships Programme and Water Partnerships Office (WPO) in the DBSA. The WPO is a ring-fenced entity housed in DBSA. A steering committee chaired by the DG: DWS oversees the work of the WPO. Roles of the WPO include:

- develop standardized national programmes for private sector participation in municipal water and sanitation services, to make it easier, quicker and cheaper for municipalities to enter into partnerships, without having to 'reinvent the wheel' for each partnership
- support municipalities to participate in the programmes and preparation of projects – where municipalities are lacking in the required expertise and funding to undertake bankable feasibility studies and financial structuring
- where appropriate, facilitate blended financing, including participation by DFIs.

The WPO has received seed funding from NT and DWS has requested a budget for the WPO. The WPO structure includes a head of office and a programme head for each of the standard programmes, in addition to other resources that will be added over time. Posts in the WPO are currently being established and will be filled shortly. In the interim the WPO is being run by existing DBSA officials.

The WPO has identified the following standardized programmes (other programmes, such as an Off-grid / non-sewered sanitation programme may be added in future) presented in Table 2.

Table 2: Programmes to be initiated under the NWPP

No	Programme	Description
1	Non-revenue water	Comprehensive water conservation and water demand management and cost recovery programme focusing on reducing losses, reducing over-consumption and improving cost recovery. Private sector obtains return on investment from savings through reducing losses or from increases in revenue
2	Management contract	Programme in which private sector provides overall management support to W&S function in municipality, including both engineering and non-engineering (e.g. revenue collection) aspects. Private sector returns funded from savings through efficiency improvements or from increased revenue
3	Wastewater treatment	A programme focusing on assisting municipalities to upgrade, refurbish and rehabilitate their waste water treatment facilities. Funding and implementation will be done through the PSP Model which leverages a portion of future grant funding
4	Water re-use	Programme focusing on further treatment of municipal wastewater to enable it to be resold for either potable, industrial or agricultural use. Related resource recovery includes energy generation and sludge beneficiation. Private sector obtains return on investment from sale of re-used water, generation of energy and sludge beneficiation
5	Sea water desalination	Focus on independent water production by producing potable water through seawater desalination in coastal cities. Private sector obtains return on investment from sale of desalinated water to municipalities or other customers (with the approval of the WSA)

To prepare for a hotter and drier future with dwindling freshwater resources and growing water demand, South Africa developed its National Water and Sanitation Master Plan in 2018. The Master Plan (which built on an earlier National Strategy for Water Re-use)[xxxv] identifies water re-use as one of the central strategies in its overall approach to ensuring water security and climate resilience.

The National Climate Change Adaptation Strategy (NCCAS) provides for a clustered approach that outlines a range of interventions to increase human resilience, economic resilience, environmental and ecological infrastructure resilience, and physical infrastructure resilience. It recognizes the social and technical capacity barriers, the institutional challenges and fragmentation in approach, the need for new technologies as well as the key financial constraints that inhibit adaptive response. Implementing water re-use nationwide does support the major tenets of NCCAS, which urges water management institutions to adopt more adaptive responses (action 1.1.28) and underscores the need for more water-wise water management practices in urban areas – including water re-use (action 1.1.32).[xxxvi]

South Africa's updated draft Nationally Determined Contribution (NDC)[xxxvii] reinforces this commitment further through its Goal 3: implementation of the NCCAS interventions for the period 2021-2030, including in the priority sector of water.[xxxviii] The government has communicated to the United Nations Framework Convention on Climate Change (UNFCC), and the broader global community, that water re-use is amongst its highest climate change adaptation priorities by showcasing the National Water Conservation and Water Demand Management Flagship Programme (within which the water re-use programme is situated) in South Africa's most recent Biennial Update Report (BUR).[xxxix] Water re-use is also consistent with South Africa's National Development Plan (NDP), applicable through 2030, as it aligns with the NDP's water access and water security targets (such as ensuring all South Africans have access to clean running water in their homes, ensuring that there is enough water for agriculture and industry, and developing a national water conservation programme to improve water use and efficiency).[xl]

The NCCAS was developed in 2019 and is aligned with the Paris Agreement's Article 7 and the associated aspects of the Paris Rulebook. The Strategy further outlines nine strategic objectives to which sectoral responses need to be aligned. The NCCAS, therefore, is the key domestic policy instrument to guide implementation, and has informed an updated adaptation communication as a component of its Nationally Determined Contribution (A-NDC). The A-NDC has outlined a number of relevant goals and associated actions as tabulated in Table .

Table 3: South Africa's First Adaptation Communication[xli]

Elements	**Undertaking for the period** **2021-2030**	**Assumptions / Methodologies / Context**	**Efforts**	**Adaptation investment** **(2021 - 2030)**
Element (c) of the Annex to decision 9/CMA.1 National adaptation priorities, strategies, plans, goals and actions	**Goal 3**: Implementation of NCCAS adaptation interventions for the period 2021 to 2030	The local government plays a key role in climate change response, and therefore building the capacity of the local sphere of government will be significant in achieving adaptation goals. This capacity support should be inclusive of human resources; institutionalisation of climate change response; financial resources and technological and/or technical support. The cities will play a pivotal role in leading climate change response in the country by virtue of urbanisation trends and services offered to the community.	The NDC will prioritise specific sectors and work towards the development of climate response plans to address the specific sector priorities. The priority sectors are health, water, biodiversity, agriculture and human settlements. **Water sector**: enhance water security; effectively deploy flood protection measures, and hydro-metrological monitoring systems. **Infrastructure**: Ensure the development and deployment of climate-resilient infrastructure that enhance water and energy security. Integration of climate information into infrastructure development planning.	USD 3 - 4 billion required for the implementation of the NCCAS for a period 2021 to 2030 to support South Africa.
Element (d) of the **Annex to decision** **9/CMA.1** Implementation and support needs of, and provision of adaptation support to South Africa	**Goal 4**: Access to funding for adaptation implementation through multilateral funding mechanisms	The 2009 UNFCCC report estimates that developing countries such as South Africa will require between USD $23-67 billion by 2030 to adapt to climate change. The domestic financial sector should play a pivotal role in terms of helping investors in adaptation space to satisfy funding requirements to meet the NDC goals. Direct unilateral access to advance adaptation finance by the private sector is still a major major issue. The National Treasury, South African Reserve Bank, financial sector regulators and Department of Forestry and Environment (DFFE) work collaboratively on issues of Sustainable finance initiatives in the sphere of private finance.	Development of climate change adaptation investment pipeline projects.	Adaptation needs and costs for the period 2021 – 2030 is USD 16 – USD 267 billion. Adaptation needs and costs adapted by a minimum of 4% GDP impact is USD 122 billion by 2025 and USD 37 5 billion by 2030.

From a climate change adaptation perspective, water re-use is already a proven resilience strategy in South Africa after being successfully deployed as a response measure[xlii] by the CCT in the aftermath of the 2018 “Day Zero” drought – a climate-driven event that saw a city of nearly five million run out of water (attribution studies[xliii] have established that beyond the probability linked to natural climate variability, the drought was made between three[xliv] to six[xlv] times as likely[xlvi] by climate change). During the crisis, the City commissioned a water re-use plant to provide 10 million litres a day.[xlvii] The City already sells treated wastewater for irrigation, industrial, and construction purposes, and would be well-placed to increase reuse for potable and other non-potable water use.[xlviii] Other water-stressed cities in the country must follow suit.

For the cost-effectiveness and sustainability of water re-use as a climate change adaptation measure, it is important that cities in South Africa mainstream such approaches. As a recent study noted, “prior to the drought, seawater desalination and water reuse ranked lower in the CCT’s water reconciliation strategies –

only to be accorded priority during the drought period when the City experienced severe water shortages. These options should not be considered crisis options, but rather be considered as normal options at all times. At such instances, these projects are hastily considered and implemented usually at a high premium. Such an approach is not cost efficient, resulting in end-users paying exorbitant price for the water."[xlix]

It is with this intent, i.e. to establish and scale-up water re-use infrastructure as the norm where there is climate induced water insecurity in municipalities, that South Africa has designed the proposed national water re-use programme.

It is important to consider the impacts of climate change in the context of current vulnerability as well as future resilience to climate shocks and the potential for adaptation. The Green Book risk assessment[l] (CSIR and Aurecon, 2019) has provided the most thorough assessment of national and municipal vulnerability in South Africa to date, being completed in 2019 and being updated regularly. This assessment has shown by looking at climate change impact in conjunction with water supply contexts (local supply E1 and regional supply E2), against differing population growth scenarios that there is significant vulnerability across many municipalities. By taking a median climate scenario and looking at the adjustments in vulnerability from the current context to futures with medium and high population growth, these vulnerabilities often increase between 20—40% and in some instances more. This particularly in key country economic nodes of Gauteng, Kwa-Zulu-Natal, Eastern Cape and Western Cape, where rapid urbanisation around the cities of Johannesburg/ Tshwane, eThekwini, Nelson Mandela Bay and Cape Town are taking place.

In looking at possible pathfinder projects to be undertaken by the WRP, a longer-term pipeline was shortlisted to ten indicative projects across the country. While this assessment is not exhaustive it provides a useful indication of potential pipeline for embedding the programme. Using these localities, an assessment of vulnerabilities was undertaken using the Green Book municipal vulnerability assessment tool as well as the WRI Aquaduct Risk Tool[li] for water stress by the year 2040.

The vulnerabilities at municipal level are summarised in Table . While all show increased risk and vulnerability due to the impacts of climate change a number of these do not demonstrate strong climate arguments. Thus, in preparing the project pipeline for the WRP the need for more detailed vulnerability assessments is imperative and will be undertaken as a first step prior to projects being provided project preparation support. Only projects with clear climate arguments will receive financial support from the WRP and this has been incorporated into the project selection criteria.

The Environmental and Social Management Framework for the WRP outline the importance of ensuring that all projects adequately address all related environmental and social risk. These detailed vulnerability studies must address these risks.

Table 4: Summary of municipal vulnerabilities for potential projects

PILOT MUNICIPALITIES (Showing Treated Water Output from WRP Ml/d)	Green Book Projected Vulnerability 2050 Median Climate Scenario Exposure E2		WRI Aquaduct Risk Tool
	Medium Population Growth	High Population Growth	Water Stress 2040
Eastern Cape			
Nelson Mandela Metropolitan Municipality (40 Ml)	40-60% deficit	60-80% deficit	>80% Extremely High
Free State			
Mangaung Metropolitan Municipality (25 Ml)	0-20% surplus	0 -20% deficit	20-40% Medium High
Gauteng			

City of Ekurhuleni (60 Ml)	40-60% deficit	60-80% deficit	20-40% Medium High
City of Johannesburg Metropolitan Municipality (50 Ml)	60-80% deficit	100% deficit	20-40% Medium High
City of Tshwane Metropolitan Municipality (30 Ml)	40-60% deficit	60-80% deficit	>80% Extremely High
Kwa Zulu Natal			
eThekwini Metropolitan Municipality (100 Ml)	20-40% deficit	40-60% deficit	20-40% Medium High
Western Cape			
City of Cape Town Metropolitan Municipality (40 Ml)	20 -40% deficit	40-60% deficit	>80% Extremely High
Drakenstein Local Municipality (10 Ml)	0 - 20% surplus	0 – 20% deficit	>80% Extremely High
Indicative Projects			
Indicative Project A	20 -40% deficit	40-60% deficit	>80% Extremely High
Indicative Project B	40-60% deficit	60-80% deficit	>80% Extremely High

B.2 (a). Theory of change narrative and diagram (max. 1500 words, approximately 3 pages plus diagram)

There is a growing realization that the South African water sector needs progressive and innovative approaches to enable adaptation to climate change impacts and ensure a resilient and water secure future. The provision of local water services to communities and business is the responsibility of municipal government who are required to initiate and undertake technical projects such as water reuse projects. This programme will transform the provision of these water services while building resilience to climatic variabilities and the adverse effects of climate change. In doing so it will strengthen the technical and financial capacity of these institutions as well as the broader water sector.

This is aligned to the 2018 NWSMP, the 2019 National Climate Change Adaptation Strategy and the 2021 First Nationally Determined Contribution that make clear statements regarding the importance of addressing a range of systemic issues, undertaking infrastructural interventions as well as looking to ecosystem-based approaches in a structured manner to revolutionize urban water systems and ensure sustainable adaptation to climate change related water security challenges.

Therefore,

> **If** technical and financial support is provided to municipalities in the identification, preparation and implementation of water reuse projects for climate adaptation at scale through a pipeline of projects, **Then** the local socio-economic resilience of communities to climate change will be strengthened through improved water security, **Because** transformational change in public and private sector funding of water reuse infrastructure will enable the establishment of a new asset class that will underpin the upscaling of water reuse nationally. .

The WRP will therefore directly reduce the climate vulnerability of communities through the scaling-up of water reuse projects at municipal scale. These infrastructure projects will be enabled through the introduction of programmatic support to municipalities to develop prepare and implement these projects as well as by developing and providing innovative public-private blended financing arrangements. This will not only support the national drive to attain the Sustainable Development Goals as well as aspects of gender equality and social inclusion with these projects impacting upon local economies and livelihoods. As such, the WRP expects the GCF catalytic capital to create market level impact by developing municipal capacity to realise climate adaptive projects, by establishing water reuse infrastructure as a new asset class, and through creating greater awareness and knowledge that will further encourage investment and a long-term pipeline of projects.

The WRP aligns with these opportunities and trends and is designed to take a holistic view of water reuse opportunities, supporting optimal projects on a case-by-case basis <u>without a pre-determined leaning towards specific technologies</u> with the aim of ensuring more efficient water use with water quality that is fit-for-purpose. The approach proposed by the WRP focuses on creating a financial and institutional enabling framework that will drive a fundamental shift in the approach to building climate resilient urban waste-water systems.

In so doing, the WRP will realise a number of important co-benefits that are supportive of the overall impact that the programme will have. Through the upscaling of water reuse projects the programme will result in reduced water demand and reduced discharge of wastewater into the environment. This will have environmental benefit for downstream aquatic ecosystem health through the maintenance of water quality and water quantity requirements. The improvements in terms of water security will also be supportive of local economic activities, and have the co-benefit of supportive localized development, particularly where many local economic activities are dependent upon water for their processes and are operational within municipal areas. Additionally, in some instances the water reuse projects will be specifically focused on <u>irrigation and industrial reuse where this has the most impact in addressing climate vulnerability</u>. Lastly, water insecurity in South Africa has significant gendered impacts with women and girl-children often being responsible for household level water security. The programme, while addressing water security, will also have a focus upon improved levels of capacity regarding water security and developing climate resilience which will enable improved understanding of adaptive and climate resilient practices.

The WRP will create indirect impacts by catalysing climate resilient solutions within municipal settings with a focus on innovation and new technologies. These projects will create a demonstration effect that will further encourage public and private sector engagement in future water reuse projects. Private sector investment in water related projects in South Africa have been very limited to date and as such the WRP is expected to catalyse greater levels of private sector financing to these investments. Furthermore, with WRP being linked to the National Water Partnerships Programme it is expected that the WRP will catalyse adaptation action and investment beyond the scope of the WRP. As such the establishment of the Water Partnerships Office (WPO) will be catalytic in the early stages of the National Water Partnerships Programme and will provide a range of tools and approaches that will guide other adaptation focused interventions and investments. The WRP will therefore be structured to address the following key barriers:

- **Water security initiatives are fragmented, inconsistent, and often involve duplication of effort**: Due to the multi-sectoral nature of water security, the limited understanding of climate focused resilience and the complexity of the water, energy, food nexus there are often multiple projects being undertaken, often with poor alignment. The provision of programmatic support to municipalities will assist in ensuring effective alignment and improved climate resilient outcomes and will align water reuse. interventions with other initiatives including aspects such as non-revenue water, improved regulation and so forth.
- **Inability of municipalities to technically conceptualise and prepare water reuse projects**: Many municipalities have limited technical capacity regarding climate change and the adaptive responses that are required and as such struggle to introduce innovative solutions. The establishment of a centre of excellence that provides support to these municipalities will aid in unlocking catalytic climate smart projects.

- **Limited municipal capacity to procure and structure contracts to support new business models**: Many municipalities have weak supply chain management capacities that are not able to develop new institutional and contractual arrangements. The provision of support through the WPO will provide assistance in structuring these arrangements as well as build longer term capacity to oversee the implementation of projects.
- **Limited access to funding for adequate project preparation**: Access to catalytic finance is limited and when combined with the weak capacity of staff to develop funding applications, often municipalities cannot unlock the climate adaptation projects they require. The structured provision of project preparation and market sounding support will address risk-sharing and financial structuring to prepare bankable solutions.
- **Lack of private sector engagement in the financing, development and ownership of climate resilience interventions**: The urgent and substantial infrastructure funding gap in South Africa's water sector calls for alternate financing solutions. Currently, private sector investment in climate resilient interventions are limited and this is often related to the risk profiles which are often prohibitive. In addition, there is often mistrust between public sector and private sector actors. An opportunity exists for municipalities to access a mix of different capital sources (referred to as "blended finance"), and to develop risk-sharing protocols that facilitate project bankability and enable private sector engagement. Also, the support of the WRP will provide a transaction advisory role that will facilitate the arrangements between public and private sector actors.
- **Limited understanding of climate impact on water security and interventions required**: There is still a poor understanding of the impact that climate change will have on water security at national and local scales, and the importance of developing a range of alternative supply options such as water reuse. It is well known internationally that there is a public mistrust of water reuse and undertaking a structured approach to develop awareness and exchange knowledge will be essential to the WRP's success.

The introduction of programmatic technical, financial and procurement support will assist municipalities to develop adaptive capacity and will have profound impact upon how municipalities strengthen their abilities to manage water security challenges, thereby changing the nature of the water supply discourse. This is indeed a key learning from the Cape Town "Day Zero" water crisis. This support will be provided to a pipeline of sub-projects that the WRP will develop working with municipalities as well as national government departments. Sub-projects in the pipeline will be supported according to a suite of criteria to ensure they only address climate related vulnerabilities, are well-structured and bankable. Through this support capacity will be built within municipalities, and between municipalities, with important support from partnerships between a range of key governmental institutions including the Programme owner, DWS and others including Municipal Infrastructure Support Agency (MISA), Infrastructure South Africa (ISA), and the South African Local Government Association (SALGA), amongst others. The WRP as a centre of excellence for water reuse will develop and provide a range of guidance documents, implementation protocols, standardised procurement documents and other tools that will assist municipalities to support sub-projects.

Through the roll-out of these sub-projects the country will see significant developments in the engagement of the private sector as well as the establishment of a new asset class that will support the ongoing development of innovative structuring and financing of water reuse projects.

The WRP will drive communications and capacitation initiatives that will have targeted impact at institutional and societal levels towards improved understanding of climate adaptation and the role of water reuse in this. Noting the often-negative perceptions of water reuse, improving the levels of understanding of water reuse and its benefits will be important in embedding projects and creating a supportive environment. It will be equally important for the WRP to work closely with key national government departments, as well the municipalities, to improve the understanding of the technical, financial and regulatory dimensions of these sub-projects. This will not only support the initial pipeline of sub-projects but will also facilitate the further development of the pipeline and assist municipalities to understand the various steps and complexities.

The Theory of Change for the WRP is provided in Figure .

The introduction of the WRP as a dedicated facility to invest in climate adaptation measures, integrating climate risk assessment and management tools with local water security approaches, and providing key stakeholders technical assistance and capacity building will contribute to a paradigm shift in South Africa's financial and water sector to spur the prioritization climate adaptation measures at scale. This transformative shift will ensure the avoidance of maladaptation from spontaneous adaptation efforts by financial institutions and municipalities through supporting effective coordination and integration between institutions, plans and programmes, thereby demonstrating the increased additional value from adaptation in terms of long-term water security and the advancement of the sustainable development goals.

The programme will shift the development pathway by support integrated and transformational planning (Component 1), by catalysing innovative technologies and approaches (Component 2), by mobilising finance at scale (Component 2), and by sharing knowledge of successful innovations at scale and developing climate expertise (Component 3).

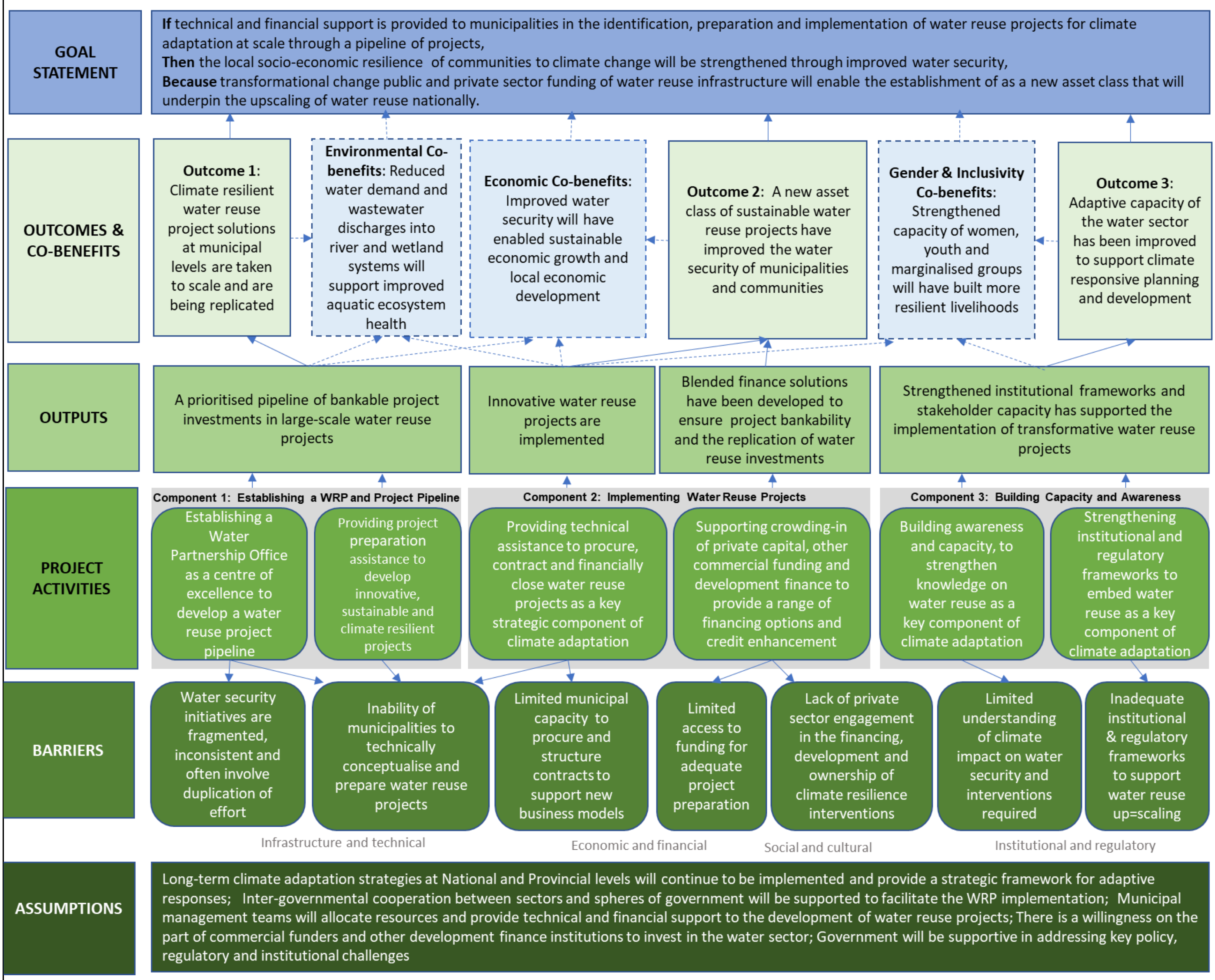


Figure 7: The WRP Theory of Change

Therefore, as part of the response to South Africa's vulnerability to climate change the WRP therefore will support a climate adaptive development pathway in five ways. **Firstly**, by reducing water demand through improved natural, technical, and institutional system efficiency introduced by water reuse sub-projects. **Secondly**, by broadening treatment options by developing technologies that lead to more resilient systems, ensuring water security, incorporating managed natural systems into urban/rural water infrastructure, as well as considering wastewater as resources through energy and nutrient recovery. **Thirdly**, by establishing an enabling environment by explicitly addressing institutional, and financial challenges related to developing water reuse sub-projects to underpin climate adaptation, to account for co-benefits, and to manage trade-offs among alternatives. **Fourthly**, by creating broader awareness with regards to importance of adopting water reuse as a key component of developing climate resilience, and, **lastly** by addressing policy, legal and regulatory frameworks that are necessary to ensure the systematic investment in water reuse as a part of climate resilient development.

B.2 (b). Outcome mapping to GCF results areas and co-benefit categorization

Outcome number	**GCF Mitigation Results Area (MRA 1-4)**				**GCF Adaptation Results Area (ARA 1-4)**			
	MRA 1 Energy generation and access	**MRA 2** Low-emission transport	**MRA 3** Building, citiies, industries, appliances	**MRA 4** Forestry and land use	**ARA 1** Most vulnerable people and communities	**ARA 2** Health, well-being, food and water security	**ARA 3** Infrastructure and built environment	**ARA 4** Ecosystems and ecosystem services
Outcome 1	☐	☐	☐	☐	☒	☒	☐	☐
Outcome 2	☐	☐	☐	☐	☒	☒	☒	☐
Outcome 3	☐	☐	☐	☐	☒	☒	☐	☐

Co-benefit number	**Co-benefit**					
	Environmental	**Social**	**Economic**	**Gender**	**Adaptation**	**Mitigation**
Co-benefit 1: Reduced water demand and impact on environmental water requirements	☒	☐	☐	☐	☐	☒
Co-benefit 2): Improved water security enables sustainable economic growth and development	☐	☒	☒	☐	☐	☐
Co-benefit 3): Strengthened capacity of women, youth	☐	☐	☐	☒	☐	☐

and marginalised groups to build more resilient livelihoods						

The following adaptation co-benefits have been identified:

- Co-benefit 1: *Environmental*. Through the implementation of the WRP there will be improvement in the water quality of rivers and groundwater with there being a reduction in the discharge from wastewater treatment works, with these waters being redirected to water reuse plants. In a number of instances WWTW do contribute to nutrient enrichment of water courses and as such these Advanced Treatment Plants would have significant impact in reducing these nutrient loads. Also, with the reduced demand upon local catchments the water requirements of aquatic ecosystems may be met with South Africa having specific environmental flow requirements in terms of the National Water Act and the supporting water use authorisations. The eligibility criteria for sub-projects under the WRP have included as secondary criteria the possibility of introducing secondary beneficiation of energy and nutrients and in so doing these sub-projects will determine their contribution to reduction in GHG emissions.
- Co-benefit 2: *Economic*. With municipalities having the mandate to drive local economic development, improvements in water security would enable the maintenance of economic activities including local agriculture and industry, as well as other businesses. For example, the impacts of droughts in Cape Town and in Nelson Mandela Bay have demonstrated the severe impact of these events on the local and regional economies, over and above the domestic supply challenges.
- Co-benefit 3: Gender. In many communities women and girl children are often responsible for the collection of water. Yet, women in South African communities are generally subjected to unequal decision-making power, lack of access to resources, gender-based discrimination and a general lack of capacity to enjoy an equal social, cultural, economic, and political status in their communities. Improvements in water security can have profound impacts on the availability of supply and the improvement of water service standards, as well as local livelihoods. The engagement of women, youth and marginalized groups in planning, investment and decision making from the initiation of sub-projects will assist in ensuring that sub-projects are sensitive to the needs of the groups, as well as building resilience along value chains and in local livelihoods. Nevertheless, women are often not engaged in processes that impact upon them. While the WRP has clear actions towards improved awareness and the building of capacity, the engagement of women in the implementation of water reuse sub-projects will provide opportunity for their needs and concerns to be addressed. Improvements in water security will also be supportive of their local economic activities that typically involve women such as local and community agriculture.

B.3. Project/programme description (max. 2500 words, approximately 5 pages)

Programme Context and Market

The WRP is a national intervention to address climate induced water insecurity in South Africa and forms part of a larger NWPP, with the establishment of the NWPP being approved in principle by the Council of the Presidential Infrastructure Co-ordinating Commission (PICC) on 10 April 2021[lii]. The PICC Council is established in terms of the Infrastructure Development Act [liii] which includes as its main objective to "provide for the facilitation and co-ordination of public infrastructure which is of significant economic or social importance" to South Africa. The NWPP is further discussed in the draft National Infrastructure Plan 2050

(NIP 2050)[liv] developed by ISA[lv] in the office of the Presidency and issued by the Minister of Public Works and Infrastructure in terms of the Infrastructure Development Act in August 2021.

The NWPP establishment was mandated in 2021 with the support of National Government and key sector partners to address a range of water sector challenges with a particular focus on innovative technical and infrastructural solutions. Therefore, GCF engagement at this early stage is catalytic in driving market change and in creating a new asset class under the WRP. The WRP will be the initial programme under the banner of the NWPP which will expand to include programmes on such aspects as non-revenue water reduction, private sector engagement in water and sanitation services, off-grid sanitation services and rural/ community water services.

The WRP is aimed at identifying, encouraging and assisting municipalities with the scaling of their water reuse sub-projects and programmes. South Africa is a water-stressed nation, and its national water resource system is continually being subjected to pressures, with a potential 17% water deficit forecast by 2030. Several interventions have been initiated by national government already to avoid this projected water deficit with a key element of these interventions being to develop an enhanced level of diversification in relation to the “mix” of water supply sources. The South African National Water and Sanitation Master Plan (2018) makes a specific note of the need to reduce water demand and increase water supply through amongst others the “re-use of effluent from wastewater treatment plants, water reclamation, as well as desalination and treated acid mine drainage”. At present, most effluent discharge and urban run-off are not reused and only 14% of water supply is from water reuse [lvi]. The opportunity to initiate a framework for the scaled development of water reuse infrastructure to strengthen water security is evident and aligned to national water sector and climate adaptation strategies.

Likewise, South Africa is becoming increasingly energy constrained with rolling blackouts over the last decade[lvii] and an increasingly important need to diversify the energy mix while addressing the country’s contributions to Greenhouse Gas emissions[lviii]. There are opportunities to realise sludge beneficiation with this being a potential source of energy and nutrients. This is discussed in the Market Study (Annex 2) and outlines potential options for this including sludge to energy, sludge to gas, composting and pelletisation. <u>These would be determined and developed on a case-by-case basis considering the existing infrastructure, sludge volumes, and biogas potential of the sludge</u>. While not the primary aims of this programme the introduction of this beneficiation as a secondary eligibility criterion will encourage these interventions.

Historically, water systems have been developed using a linear design approach, i.e., source, treat, transport, distribute, collect, treat and dispose. This approach is removed from the people it serves and results in primarily technocratic solutions and the fragmented management of the urban water cycle i.e. water management silos. Complex challenges of climate change, increased water scarcity compounded by rapid urban growth require innovative solutions that can work across institutional, sectoral, and geographic boundaries to develop efficient and robust urban water systems to improve supply resilience for future generations. In this regard and aligned to the concept of Water Sensitive Cities [lix], the South African Department of Cooperative Government has developed an Integrated Urban Development Framework that aims to support more resilient and sustainable cities and recognizes that urban resilience and sustainable development requires improved climate and risk-sensitive development planning, supported by improved governance and capacity as well as coherent systems, services and resources.[lx]

The need for water reuse as a means of augmenting and diversifying the water mix in South Africa is widely acknowledged in national policies and guidelines yet with limited uptake to date. There are a range of technical water reuse options that enable the development of bespoke solutions for local contexts (**Error! Reference source not found.**). Acceptance for water reuse from municipal wastewater treatment works (WWTW) as an alternative water supply has largely been seen in the industrial sector. However, the industrial use of water in South Africa is estimated to be only 8% of the total annual water demand and there is a gap in the expansion of the use of reused water to other sectors, particularly for the domestic and agricultural sectors.

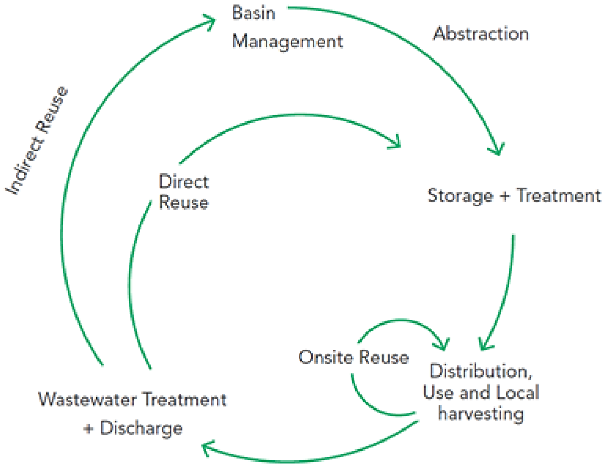


Figure 8: Water reuse opportunities across the water value chain [lxi]

Globally, there is recognition that public acceptance of water reuse, in the context of Direct Potable Reuse is a significant issue and a detailed Water Research Commission on water reuse noted the importance of developing awareness creation and knowledge building tools to address these concerns[lxii]. Through this study a survey found that only 48.5% of respondents would support direct potable use in times of extreme need such as a drought, while the support for industrial reuse was only slightly higher, at 50.3%. This demonstrates the need for significant effort to address these issues at sub-project level. Hence, the development of a communications strategy and plan for the programme (Annex 24).

South Africa has in excess of 1,000 municipal WWTW that produce and discharge approximately 2.1 km^3/year of treated effluent to receiving waters[lxiii]. There is a strong correlation between potable use and effluent discharge in cities, where approximately 60% of the annual average daily demand (AADD) is returned to WWTWs for processing and subsequent discharge as treated effluent. This creates an opportunity for reuse in growing cities, as effluent quantities generally increase in proportion to increased AADD. Importantly, water reuse will require the need for tertiary treatment and in some instances will require improved secondary treatment performance.

Various water reuse archetypes that may be considered as potential project implementation options within the WRP (see Table 5). Until recently, wastewater treatment plants have been interventions to ensure that sewage does not damage the environment, while being a generator of a side stream waste in the form of sludge. However, with a move to Circular Economy thinking and Water Sensitive Urban Design, there is a paradigm shift needed to see WWTWs as potential sources of valuable resources such as water, energy and nutrients. As such, WWTWs may be viewed as Water Resource Centres, where one is able to extract these resources and reuse them within a city environment, thus reducing the pressures placed on non-renewable and other stressed resources. Options for the WRP include:

- **Direct Potable Reuse (DPR)**: The treatment of final wastewater effluent in an advanced treatment plant (ATP) and transfer of the product water to the inlet of a bulk water treatment plant or directly into the water distribution network. These projects tend to have high capital and operational costs due to the advanced technology and extent of controls required to ensure the strict water quality standards.
- **Indirect Potable Reuse (IPR)**: The treatment of final wastewater effluent in an advanced treatment plant and the transfer of the product water to an environmental buffer (such as an aquifer or surface water reservoir) where it will blend with raw water before treatment in a bulk water treatment plant. These projects tend to have high capital and operational costs due to the advanced technology and

extent of controls required to ensure the strict water quality standards but should be lower than direct potable reuse plants as there are fewer steps in the process.

- **Industrial Reuse**: The treatment of final wastewater effluent in an advanced treatment plant to a quality suitable for industrial purposes and the transfer of the product water to a specific industry or generally to an industrial area. There is variability in the capital and operational costs of such plants depending on the water quality requirements.
- **Irrigation Reuse**: The transfer of the polishing final wastewater effluent to quality standards required by agriculture or other irrigation users. Irrigation reuse projects require a relatively minimal additional treatment before the treated water is distributed into an irrigation reuse network or a rising main and therefore, the subsequent capital investment on the associated WWTW is relatively minor.

Table 5: Characteristics of water reuse archetypes (the longer bar indicates a more favourable solution)

Reuse characteristics	Direct Potable (DPR)		Indirect Potable (IPR)		Industrial		Irrigation	
Advanced treatment plant product quality:	Highest	■■■■■	Very High	■■■■	Medium	■■■	Lowest	■■
Range of usability of water:	Highest	■■■■■	Highest	■■■■■	Medium	■■■	Lowest	■
Capital Cost:	Highest	■	High	■■	Medium	■■■■	Lowest	■■■■■
Operational Cost:	Highest	■	High	■■	Medium to High	■■■■	Lowest	■■■■■
Public Acceptance:	Lowest	■■	Medium	■■■	High	■■■■	Highest	■■■■■
Operational Expertise Required:	Highest	■	High	■■	Medium	■■■	Lowest	■■■■■

An important part of potable reuse projects is the need to conduct regular highly specialist effluent quality verification that is over and above conventional laboratory testing. These specialist tests are particularly needed for Chemicals of Emerging Concern (CECs) in potable reuse projects. The existing national potable water quality standard, South African National Standards (SANS) 241, defines the minimum water quality required for human consumption. It was however developed with surface and groundwater sources in mind and is not extensive enough to cover the specific water pollutants found in potable reuse projects. For instance, it does not have a standard for phosphate or organic Chemical Oxygen Demand and the microbiological standard is not approached from a log reduction perspective as has been adopted internationally. There is thus a need to develop a national water quality standard for potable reuse. However, in the absence of such a standard, existing reuse projects have generally adopted the California Department of Health standards for potable reuse, as these are state-of-the-art standards internationally. These set a require standard for pathogen log reduction across a potable reuse treatment train as well as maximum CEC levels in the final product water. This is all in-line with the World Health Organisation's Potable Reuse Guidelines. The approach to these will need to be developed with the South African Bureau of Standards together with outlining the monitoring protocols which are likely to range from daily to weekly.

The WHO guidelines note a complex array of criteria with focus on the CECs. This is increasingly complex and the WHO is their Potable Water Reuse Guideline[lxiv] notes that:

- Water quality targets (chemical guideline values) and microbial performance targets (log reductions of pathogens in sources waters) are the primary health-based targets for potable reuse. These targets will be underpinned by health outcome targets set by public health authorities or drinking-water regulators.
- Microbial performance targets can be identified by using system specific pathogen data in source waters or by using default pathogen concentrations.

- Default performance targets identified are 8.5-log reduction of enteric bacteria, 9.5-log reduction of enteric viruses and 8.5-log reduction of enteric protozoa. They do not represent guideline values and are not intended to encourage pathogen monitoring in drinking-water, which would be impractical and of little value for routine monitoring.
- While there tends to be considerable interest in CECs, such as pharmaceuticals and personal care products, the concentrations in drinking-water are generally low and generally do not warrant setting of new guideline values in national or regional standards.
- In specific circumstances, where a chemical with no guideline value is identified as a concern, approaches for developing screening values should be identified to support investigations into potential risks and the need for implementation of additional control measures.

It is important to note that the Municipalities, as Project Owners, will be responsible for the attainment of water quality standards. This is regulated by the DWS and the Blue Drop Assessments provide regular reviews of municipal progress in this regard.

From the market study undertaken of the 14 water reuse projects that are operational, 8 are focused on DPR and 4 on direct industrial reuse. Projects that are being initiated or planned demonstrate a similar mix of water reuse archetypes. In order to assess the potential pipeline of sub-projects that the WRP could undertake in its initial phases, the following eligibility criteria were applied:

- **Climate vulnerability:** Demonstrating a clear water supply vulnerability as a result of climate change, considering the impact of bulk water transfers and the importance of these to the local social economy, is the primary criterion for support by the WRP. In this regard, a projected climate induced water supply deficit of 20% and more would be an important prerequisite for support from the WRP.
- **Treatment Plant Design Capacity:** The size of the WWTW as it directly impacts on the size of a potential water reuse sub-project and the associated investment costs for that project. Greater than 20 Ml/day to be feasible in supporting a water reuse advanced treatment plant.
- **Compliance:** DWS publishes a periodic report containing the overall effluent compliance for each municipality based on the data submitted to the DWS by that municipality. This data provides an insight into how well the wastewater systems are managed in that municipality. Since the final effluent from a WWTW is the raw water for the ATP within a Water Resource Centre, the effluent compliance is an important indicator of the technical viability of a potential water reuse sub-project. Best practice dictates that final effluent should maintain a very high level of compliance for a reuse project to be viable. Greater than 50% compliance with physical, chemical, and microbiological standards is therefore essential.
- **WWTW Type:** There are essentially two types of WWTW technology in South Africa, namely biofiltration plants and activated sludge plants. Biofiltration is an older technology that has little to no control mechanisms for ammonia removal and they also tend to struggle to de-nitrify. The effluent compliance is thus often variable. As a result, biofiltration plants are not recommended for potable or industrial reuse projects and treatment plants with this technology were excluded from the list.
- **Reuse Archetypes:** Four different water reuse project archetype options will be considered, which include Direct Potable Reuse (DPR), Indirect Potable Reuse (IPR), Industrial Reuse (INR) and Irrigation Reuse (IR). Priority will be focused on projects that engage DPR and expect that these projects could likely include support to industrial reuse, particularly where industries are operating with municipal jurisdictions. However, the selection of the project archetype will be appropriately context specific.
- **Already Initiated Reuse Projects:** Where municipalities have already initiated these projects there will be an understanding of the technical complexities as well as greater possibilities for innovation.

This then results in a potential/indicative sub-project pipeline as provided in Table .

Table 6: Draft potential sub-project pipeline depending upon climate vulnerability assessments and sub-project preparation support to determine the most appropriate solutions

Province and Municipality	No. of projects	Potential Project Size (Ml/d)	Total Reuse Flow (Ml/d)	Already initiated Reuse projects	Potential reuse types
Eastern Cape					
Buffalo City Local Municipality	1	10	10	N	DPR, INR.
Nelson Mandela Metropolitan Municipality	1	40	40	Y	DPR, INR.
Free State					
Mangaung Local Municipality	1	25	25	Y	IPR
Gauteng					
City of Ekurhuleni (ERWAT)	3	50, 30, 33	113	Y	DPR, INR.
City of Johannesburg Metropolitan Municipality (JW)	3	100, 50, 50	200	N	IPR, INR.
City of Tshwane Metropolitan Municipality	2	80, 30	110	N	IPR, INR
Emfuleni Local Municipality	1	30	30	Y	IPR
Kwa-Zulu Natal					
eThekwini Metropolitan Municipality	3	42, 100, 20	162	Y	DPR, IPR and INR.
Msunduzi Local Municipality (Umgeni Water)	1	30	30	Y	DPR, INR.
Indicative Project A					
Limpopo					
Polokwane Local Municipality	1	10	10	Y	IPR
Mpumalanga					
Mbombela/Umjindi Local Municipality	1	10	10	N	DPR, INR.
Steve Tshwete Local Municipality	1	13	13	N	DPR, INR, IR
Northern Cape					
Indicative Project B	1	15	15	N	IPR
North-West Province					
Rustenburg Local Municipality	1	23	23	Y	
JB Marks	1	20	20	N	IPR, INR, IR
Western Cape					
Breede Valley Local Municipality	1	15	15	N	DPR, Agr.
City of Cape Town Metropolitan Municipality	2	70, 40	110	Y	DPR
Drakenstein Local Municipality	1	10	10	Y	IPR
Stellenbosch Local Municipality	1	10	10	N	IR, INR
Total	27	1031	1031	11	

If the ATP uses reverse osmosis membranes, a concentrated brine stream will be produced that can pose challenges for treatment plants located inland. As such, typically inland plants may be more suited to other reuse archetypes, such as, Indirect Potable Reuse (IPR), Industrial Reuse (INR) and Irrigation Reuse (IR), where such extensive treatment using reverse osmosis membranes may not be required. However, recent technologies do not produce a brine effluent and this will be a priority for these plants. The Goreangab water treatment plan in Windhoek is a direct potable reuse plant that has been operating since 1968 and does not produce a brine due to the choice of treatment options. However, in instances where brine disposal is required at inland sites, this will be incorporated in the design during the feasibility assessment. The costs of brine handling can be significant and have the potential to compromise the project viability. Hence, it is paramount that this is incorporated in the design from the beginning.

Example Project Summary for an Indicative Project

A municipality undertook a feasibility study looking at a potential wastewater industrial reuse project to be implemented as a Public Private Partnership (PPP). For the Municipality, site-specific future climate projections indicate the following for the 2011-2040 period under both a moderate climate change future (RCP 4.5 scenario) and an extreme climate change future (RCP 8.5 scenario) (Table 7).

Table 7: Site-specific future climate projections for the Municipality 2011-2040 under two scenarios

Climatic Variable (Indicator)	RCP 4.5	RCP 8.5
Temperature (annual mean)	+1ºC	+1ºC
Rainfall (annual mean)	-3%	-5%
Aridity (annual mean)	+2%	+1%
Soil moisture (annual mean)	-5%	-6%
Water discharge (annual mean)	-3%	-4%
Water runoff (annual mean)	-3%	-4%
Length of longest dry spell	A small increase (<5 days a year)	A small increase (<5 days a year)
Number of dry spells a year	A small decrease (<25% change)	A small decrease (<25% change)

Thus, the Municipality is projected to face rising temperatures from climate change. In addition to decreases in rainfall, runoff, and discharge, the rising heat is likely to drive an increase in aridity, leading to somewhat drier conditions (including lower soil moisture).

As a result, water deficits are projected as being:

- 20 -40% deficit under medium population growth scenarios by 2050, and
- >80% high water stress by 2040 as predicted by the WRI Aquaduct Risk Tool.

The feasibility study indicates that there is, on average, at least 82 Ml/d available for tertiary advanced treated at a proposed regional site. The transaction advisor engaged with potential used for the product water and identified a total of five potential end users who could potentially purchase up to 71.7 Ml/d on average of the product water.

A total of 7 potential technical options and configurations were identified and analysed through a comprehensive weighting and assessment system. The option of a new regional treatment (capacity 75 Mℓ/day) at a new regional site with a pump collection system was selected as the preferred option.

A preferred site for the treatment plant has been selected noting the proximity to off takers as well as a range of environmental considerations. Noting the brine to be produced by the plant the proximity to the ocean for discharge is important. The requirements for this discharge in terms of the National Environmental Management Act (Act 107 of 1998) and the Integrated Coastal Management Act (Act 24 of 2008) have been included in the feasibility study and are central to the Environmental and Social Impact Assessment and associated management plan. A total of 25 ha will be required for the regional treatment facility.

Pilot plant testing has been undertaken and indicated that the effluent can be treated to an acceptable standard by utilising filtration (presumably ultra-filtration) and reverse osmosis processes. Some additional pre-treatment steps will be necessary.

The total capital cost estimate in 2017 (including land costs) amounted to R1 736 209 178. The Unitary Reference Value was estimated at R10.82 / m^3.

The potential for biogas generation was also include in the study and it was recommended that thermophilic digestion for CHP engines be implemented as part of the project. It was however stated that the viability of the project was not dependant on the biogas electricity generation. Furthermore, composting of sludge was also included as part of the project to ensure waste beneficiation.

A collection system consisting of five pumpstation and rising mains will be required to collect and transfer the effluent to the regional treatment facility. Preliminary designs of these collection pipelines indicated a total of 31.6 km of pipelines varying from 500mm to 864mm pipelines will be required. A total installed power of 585 kW will be required for pumping.

The regional facility will produce two quality streams. One of the larger users will receive lower quality re-use water directly from the WWTW (30 Ml/d). The other users will receive a higher quality water from the tertiary treatment plant (45Ml/d). Treatment at the regional facility will take place in three treatment modules as follows:

- 50 Ml/d domestic WWTW
- 35 Ml/d Industrial treatment work
- 10 Ml/d water treatment works for higher quality water

A 50 Ml/d Biological Nutrient removal activated sludge plant will treat the raw sewage diverted from the local WWTW. The 35 Ml/d industrial treatment plant will treat industrial effluent from the pulp and paper mills using conventional water treatment, plus advanced oxidation and reverse osmosis membrane treatment. The 10 Ml/d water treatment plant will treat a portion of the treated effluent from the WWTW to an industrial standard using conventional treatment processes (Flocculation – sedimentation -filtration) plus advanced oxidation (Ozonation) and Granular Activated Carbon.

The distribution network to off-take users will be via a gravity driven network. This is made possible by the location of the regional facility. An estimated 27 km of pipelines ranging from 225mm diameter to 762mm diameter pipes will make up the network.

As part of the feasibility study, the feasibility of implementing value adding technology in the form of the potential to convert biogas to energy and the potential to convert sludge to a useable end product were investigated and found to be feasible producing sufficient power to significantly reduce the load drawn from the national grid.

Benefits to Ecosystem Services

Currently, treated effluent from the macerators and WWTWs at the municipality is discharged into nearby streams, natural water courses or directly into the ocean through deep sea outfalls. High volumes of discharge that can create significant pollutant loading into these sources can have an effect on these aquatic water bodies, on coastal systems, and on the composition of the water in these sources. A decrease in the amount of effluent released into surface water sources can go a long way to reducing adverse environmental impact. Additionally, substances and microorganisms that are present in treated effluent can have an adverse impact on stream habitats and wetlands, and on the wildlife that is dependent upon sustained water quality attributes of these environments. The diversion of water from surface water bodies can also have a harmful impact on the flora and fauna of ecosystems that exist within these water bodies. By reducing the diversion of water from surface water bodies as well as reducing the amount of effluent that is emitted into these environments through increased wastewater recycling, this can significantly lessen the deterioration of ecosystem health and hence improved wildlife subsistence in these areas.

For the purposes of the economic appraisal, it is assumed that the implementation of the project shall divert the use and demand for potable water and therefore present an economic cost avoided to the value of R739 million at an 8% social discount rate.

Project Selection Criteria

Four stage gates are applied to the project selection process with the primary screening and the secondary and tertiary criteria being applied for project preparation support (Figure 8). The primary screening and the use of the secondary criteria will be exclusionary in nature, with the tertiary criteria being used to assist project trade-offs and prioritisation. These first three stage gates are common for all sub-projects and for grant funding. A financial assessment (utilising the DBSA comprehensive investment analyses) will be undertaken as a fourth stage gate towards providing loans to support project implementation. The sub-project assessment process is illustrated in Figure 9.

The primary screening is solely based upon climate vulnerability criteria and only projects that meet these will be supported by the WRP for project preparation and funding for implementation.

Following acceptance of a sub-project for project preparation (or where a sub-project is already prepared as requires funding for implementation) applying the primary screening and secondary criteria outlined below, includes an additional financial assessment criterion that further determines a sub-project's eligibility for GCF concessional funding. Should a sub-project not meet these criteria, it will not be eligible for support by the WRP.

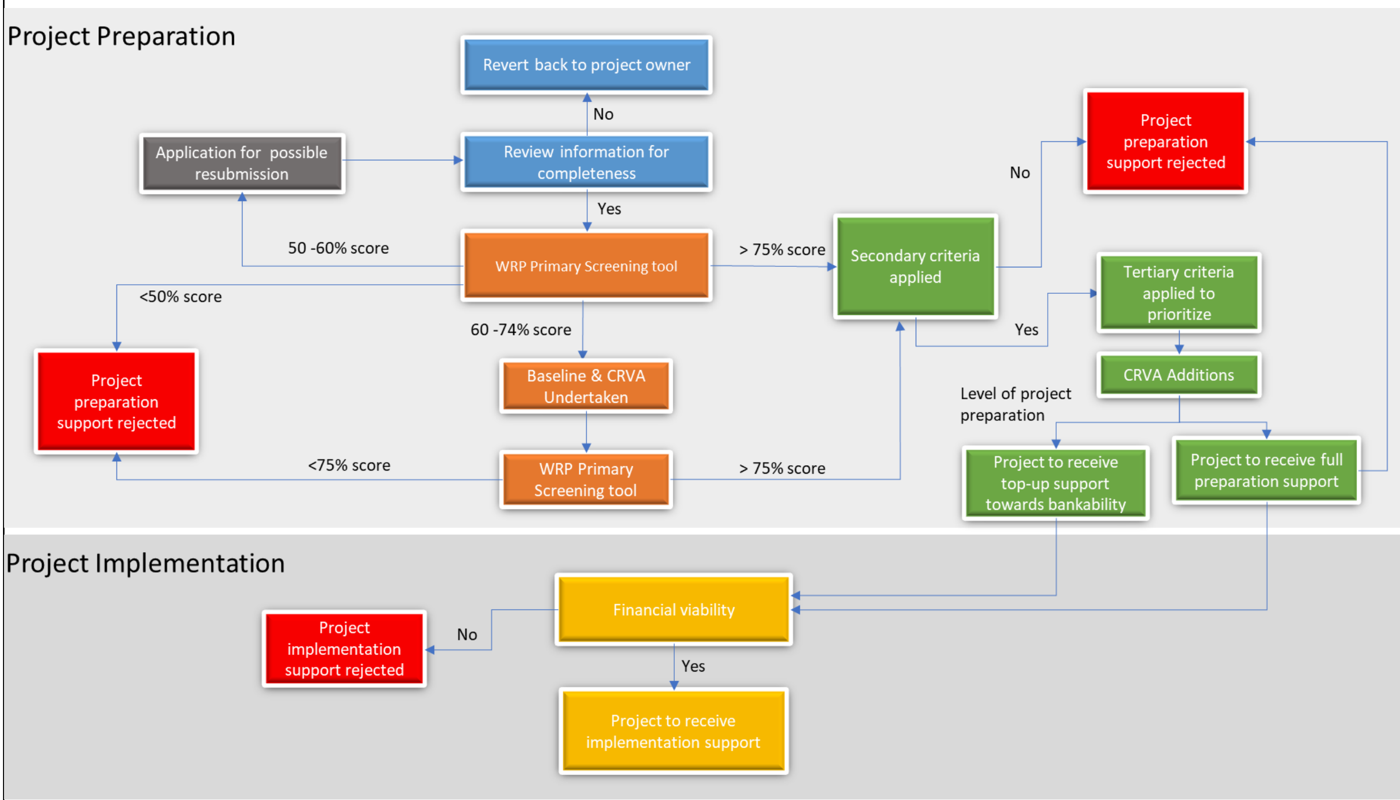


Figure 9: Sub-project assessment process

A more comprehensive eligibility rubric for project preparation selection will be applied by the WRP in finalising the future pipeline, as provided in **Error! Reference source not found.**8.

Table 8: Primary, secondary and tertiary criteria for the selection of the potential reuse sub-projects

Criteria	Value	Requirement
Primary Screening (Exclusionary)		
Water insecurity demonstrably driven by current or climate change	Scorecard includes climate change and risk vulnerability, drought exposure, resilience benefits and adaptation optimization	Clear climate vulnerability with the screening scorecard scores greater than 60%.

vulnerability, within the bounds of uncertainty	Post screening climate risk vulnerability assessment to be completed.	Only sub-projects where 70% or more of the water supply deficit is driven by climate change
Secondary Screening (Exclusionary)		
WWTW Design Capacity	Greater that 20 Ml/d	Yes, greater than 20-Ml/d
Regulatory Compliance (Chemical)	Greater than 50% regulatory compliance	Yes, greater than 50% compliance
WWTW Technology Type	Only activated sludge plants will be supported to ensure the consistency of effluent quality required.	Yes, incorporates and activated sludge plant.
Environmental Safeguards	All sub-projects shall comply with both GCF and DBSA Environmental and Social Safeguards Standards, with only Category B and C levels of impact being acceptable.	Yes, meets the safeguard requirements of GCF and DBSA and risk is at levels B and C.
Financial viability	All sub-projects that move beyond project preparation into implementation must be financially viable, and this will be required to be demonstrated in the Project-specific feasibility studies / project information memorandums to be undertaken prior to Programme funding commitment.	Yes the sub-project is financially viable
Tertiary Criteria (Prioritisation)		
Beneficiation	Priority will be provided to sub-projects that provide for mitigation including biogas, solar power, sludge management and sludge beneficiation.	
Linkage to strategic projects	Priority will be provided to water reuse sub-projects in supporting of strategic projects.	
Offtake agreements	Priority will be afforded to sub-projects where the existence of a confirmed offtake agreements	
Financial Viability (Exclusionary)		
The sub-project feasibility study will need to demonstrate affordability which will include the need to have certainty around off-take arrangements. Projects will be assessed using the financial model developed for the WRP, against a target minimum senior debt service cover ratio (DSCR) which provides an indication of a project being 'at risk' for bankability and requiring WRP concessional funding to be viable and deemed bankable. The DSCR will be fixed during the financial assessment stage of project preparation and shall be fixed prior to the sub-project being taken to implementation.		

This, and the associated primary climate change screening process has been further outlined in Annexure 23 and in the Annexure 14: Term Sheet. A primary climate change screening scorecard has been developed (Annexure 25) and this provide an assessment rubric comprised of ten climate focused criteria.

Primary Screening criteria: To ascertain suitability for support by the WRP, at the identification and selection stage, firstly a screening scorecard will be applied. This is to ensure that only sub-projects that reflect a climate change induced water insecurity are supported by the programme. This scorecard provides a consolidated assessment as to eligibility of a sub-project for WRP funding support and assesses whether the sub-project addresses *climate vulnerability*, *drought exposure*, *resilience benefits, optimisation of adaptation* as well as aspects of *mitigation*. The ten criteria and weightings of the scores in the climate change screening scorecard are outlined in Table 9.

Table 9: Climate change basis screening scorecard criteria

No.	Criterion	Assessment	Weighting
1	Level of local baseline water scarcity	Level of current baseline water supply vulnerability through assessing the ratio of total water withdrawals to available renewable surface and groundwater supplies.	1
2	Level of local water scarcity/ stress induced by climate change under optimistic future climate scenarios	Level of water scarcity risk in a world with sustainable socio-economic development (SSP1) and moderate reduction of GHG emissions (RCP2.6 /RCP4.5), a pathway which will lead to an increase of global mean surface temperature of approximately 1.5°C by the end of the 21st century.	1
3	Level of local water scarcity induced by climate change under current/ business as usual climate scenarios	Level of water scarcity risk in a world similar to current socio-economic development trends (SSP2) and intermediate GHG emission levels (RCP4.5 /RCP6.0), a pathway which will lead to an increase of global mean surface temperature of approximately 2°C by the end of the 21st century.	1
4	Level of local water scarcity induced by climate change under pessimistic future climate scenarios	Level of water scarcity in a world with unequal and unstable socio-economic development (SSP3) and high GHG emission levels (RCP6.0 /RCP8.5), a pathway which will lead to an increase of global mean surface temperature of approximately 3.5°C by the end of the 21st century.	1
5	Climatic history (rainfall)	A review of historical rainfall datasets (using trend analysis) shows potentially no increase in the average rainfall, or increases (wetting) or decreases (drying), compared to known trends in water demand. All submissions will be required to submit a rainfall trend analysis.	2
6	Climatic history (drought tendencies)	Assessing the projected change in drought tendencies (i.e. the number of cases exceeding near-normal per decade) for the period 2035–2064 relative to the 1986–2005 baseline period, under the low mitigation scenario (RCP 8.5). A negative value is indicative of an increase in drought tendencies per 10 years.	2
7	GCF-WMO Climate Information Tool ensemble assessment	Using the GCF-WMO climate information platform, using the emissions scenario 4.5 and the time period 2041-2070 are future projections of changes in temperature, rainfall, aridity and water discharge, when combined with knowledge of water demand, demonstrating that climate change is a significant factor in water scarcity. Assess through the future change top indicators as well as through the spread of the 18-model ensemble.	2
8	Vulnerability	The level to which the needs of the most vulnerable would be addre considering the levels of service in the municipality, as a % of the unse All submissions will be required to provided referenced statistics in terr populations living in poverty and those who do not receive safe and se basic water supply.	2
9	Mitigation impact (tCO2e)	The level to which the sub-project contributes to the reduction of GHG emissions based on submitted information, and that will be required from all submissions.	1
10	Mitigation impact through beneficiation	The level to which the sub-project will contribute to the mitigation of impacts through sludge beneficiation, based on submitted information required from all submissions.	1

All sub-projects to pass this screening must present a strong climate argument to justify WRP support. This screening assessment will be supported by climate specialists to ensure the necessary scientific rigour is applied.

The screening will be supported by a detailed climate vulnerability assessment that will include a baseline assessment of municipal water supply including current and projected future demands. The review will assess the baseline climate and projected future climate, determining the impact of future climate projections on the ability to meet supply requirements. Noting that increased demand is a part of the vulnerability of certain municipalities, consideration of this will be considered and the following criteria will be used to move past the primary screening:

a) **% of water supply deficit driven by climate change**: Sub-projects must be greater than or equal to 70% to quality for support.

b) **% of water supply deficit driven by increased demand:** Sub-projects must be equal to or less than 30% to qualify for support.

Secondary criteria: At the secondary level, five exclusionary criteria will be applied to ensure that the potential water reuse sub-projects selected would be able to make a strong case for inclusion in the WRP project pipeline. These criteria are listed below and must all be addressed for a project to obtain WRP support:

a) **Treatment plant design capacity:** Only sub-projects with a WWTW with a design capacity of equal to, or greater than, 20 Ml/d would be supported by the WRP.
b) **Compliance**: Only sub-projects that achieve greater than 50% compliance with discharge standards will be supported.
c) **WWTW Technology type**: Only activated sludge plants will be supported to ensure the consistency of effluent quality required.
d) **Environmental and social safeguards:** All sub-projects shall comply with both GCF and DBSA Environmental and Social Safeguards Standards, with only Category B and C levels of impact being acceptable.

Tertiary criteria: Noting that the selection to this point would provide potentially a longer list of sub-projects, tertiary criteria will be applied to assist in prioritising sub-projects. These would include the following criteria:

a) **Beneficiation**: Priority will be provided to sub-projects that provide for mitigation including biogas, solar power, sludge management and sludge beneficiation.
b) **Linkage to strategic projects:** Priority will be provided to water reuse sub-projects in supporting of strategic projects.
c) **Offtake agreements:** Priority will be afforded to sub-project where the existence of a confirmed offtake agreement is confirmed.

Financial viability: All sub-projects that move beyond project preparation into implementation must be financially viable, as will be required to be demonstrated in the Project-specific feasibility studies / project information memorandums to be undertaken prior to Programme funding commitment. The Project feasibility study will need to demonstrate affordability which will include the need to have certainty around Project off-take arrangements. Projects will be assessed using the financial model developed for the WRP, against a target minimum senior debt service cover ratio (DSCR) which provides an indication of a project being 'at risk' for bankability and requiring WRP concessional funding to be viable and deemed bankable. The DSCR will be fixed during the financial assessment stage of project preparation and shall be fixed prior to the sub-project being taken to implementation. The criteria for the selection of beneficiaries and borrowers are provided in Table 10, below.

Table 10: Beneficiary and borrower selection criteria

	Project preparation funding and capacity building funding (components 1 and 3)	**Loan funding (component 2)**
Eligible projects: public sector	Water reclamation and/or reuse sub-projects developed and implemented by public sector entities (municipalities, utilities, water boards, water user associations and departments) responsible for and mandated with the treatment of sewerage effluent and provision of water supply	
Eligible projects: private sector	Water reclamation and/or reuse sub-projects developed and implemented by private sector entities on behalf of municipalities, utilities, water boards, water user associations and departments, responsible for and mandated with the treatment of sewerage effluent and provision of water supply	

Eligible recipients/borrower: public sector	Public sector entities (including municipalities, utilities, water boards, water user associations and departments) mandated and responsible for the treatment of sewerage effluent and responsible for water supply and water reclamation, recovery and reuse, who will act as project owners and/or sponsors	(a) Public sector entities (including municipalities, utilities, water boards, water user associations and departments) mandated and responsible for the treatment of sewerage effluent and responsible for water supply and water reclamation, recovery and reuse, who will act as project owners and/or sponsors (b) Project SPVs, with a majority public sector shareholding, mandated and appointed by the public sector for the preparation, finance, implementation and operation and maintenance of water reclamation, recovery and reuse sub-projects
Eligible recipients/borrower: private sector	Private sector entities (including project SPVs) who obtained a contractual mandate for the treatment of sewerage effluent and production of water and water reclamation, recovery and reuse, from a public sector entity, and who will act as project sponsors on behalf of project owners	(a) Project SPVs mandated and appointed by the public sector for the preparation, finance, implementation and operation and maintenance of water reclamation, recovery and reuse sub-projects following a competitive bidding process (b) Private sector SPVs which has a majority private sector shareholding in such an SPV
Application of proceeds	Proceeds will be used to prepare water reuse sub-projects, including but not limited to detailed climate assessment, scoping investigations, pre-feasibility studies, bankable feasibility studies, structuring, business cases, as part of component 1. Proceeds can also be applied to capacity building as part of component 3	Loan funding will be made available to borrowers who will apply such funding for the implementation (Engineering, Procurement and Construction) and operation and maintenance of water reclamation and water reuse sub-projects
Process	Applicants will be required to apply for such funding from the WPO who will then apply the eligibility criteria to determine whether funding will be made available to applicants for project preparation and/or capacity building. Upon approval, the WPO will enter into an agreement with the recipient which will govern the relationship and the process of application of the funding. The first application of funding (irrespective of the stage of the project) will be to conduct a detailed climate assessment (primary screening criteria) on the sub-project. If the climate requirements are met, the sub-project will benefit from further project preparation and or funding support, but if the criteria is	Borrowers will apply for funding support from the WRP through the Water Partnerships Office. The DBSA will apply its due diligence processes to determine whether GCF loan funding can be applied to water reuse sub-projects as part of the blended finance solution. Should sub-projects request funding without having been supported with project preparation, such sub-project will first be required to meet the primary screening criteria.

	not met, the sub-project will not be considered further	

Financial assessment criteria

GCF funding will be utilised in a manner that ensures the GCF concessionality will reach and benefit the beneficiaries of the sub-projects. The DBSA and the IBCF will do the assessment and will have access to all project related documentation such as the PIM, financial model, etc. A physical inspection may in certain instances be performed.

Minimum Target DSCR

- The individual project-level cashflows (Expenditures and Revenues) will be assessed over the life of the sub-project to determine the Financial Net Present Value (FNPV) of each sub-project and the Financial Internal Rate of Return (FIRR) on each sub-project investment. The financial analysis will demonstrate the financial viability of the sub-project with and without GCF funding and credit enhancement, thereby justifying the need for and extent of GCF participation.
- A financial assessment (using the WRP project financial model) will be done on a sub-project by estimating the total forecast net cashflows (Expenditure and Revenues) available for debt service at the individual sub-project level and comparing each project's minimum DSCR against the Target Minimum Senior Debt DSCR for the sub-project. Sub-projects with minimum Senior Debt DSCR that fall below the minimum Target Senior Debt DSCR, indicate a potential funding gap and these sub-projects will be eligible for GCF concessional loans in order to improve their financial viability.
- This target minimum DSCR is only as an example. Sub-project specific DSCR to be considered when sub-projects are considered for funding and GCF will be appraised of these considerations for their inputs through the financial structuring process, prior to finalisation.

Minimum Concessionality

- Employ the minimum level of concessionality to address financing requirements, market entry barriers or incremental costs in line with the WRP's financial model for utilising concessional funding for sub-projects of the WRP

All water reuse sub-projects proposed for financing by the DBSA (incl utilisation of GCF funding for the WRP) are assessed for eligibility within the DBSA investment decision making structures. The DBSA applies an integrated environmental, social, economic, financial and sector assessment of each of the proposed sub-projects. Sub-projects are screened according to the DBSA Environmental and Social Safeguards Standards (E&SSS) Framework.

DBSA has a well-established set of tools and procedures to classify green operations and assess and monitor environmental and social projects, supported by internationally accredited environmental and social safeguards standards which align with the legislation in all of the jurisdictions in which the Bank operates (Figure 10).

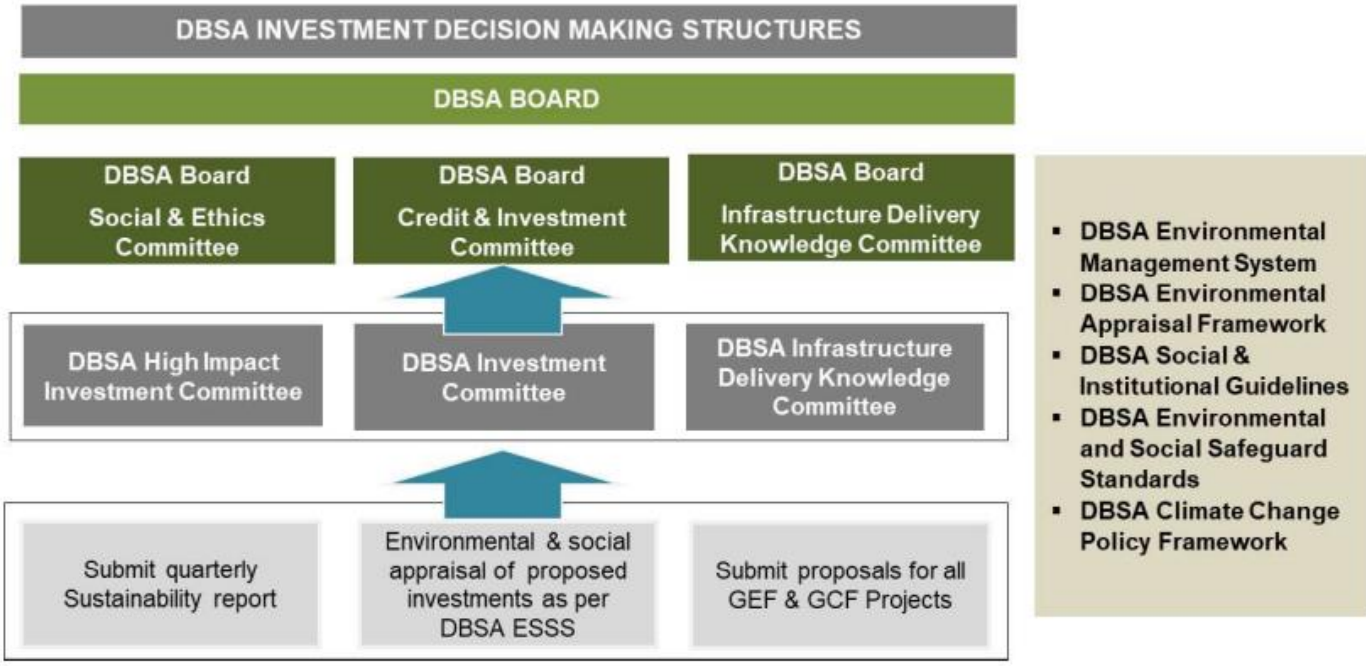


Figure 10: DBSA investment decision making structures

During project preparation, the WRP must consider various issues (Figure 11) and considerations that influence project structuring, with a number of parameters informing what is achievable and feasible in the local context. There are several technical, social and environmental considerations:

1. **Scope**: What activities is the contract to regulate, considering the activities of design, build, operate, finance which influences the risk allocation, determines roles and responsibilities, legal and contractual arrangements etc).
2. **Key objective**: Provides the framework for establishing Key Performance Indicators and motivating for the funding arrangements.
3. **Parties**: Who does the municipality intend contracting with? Private sector or public sector and the associated procurement strategy.
4. **Performance**: Determining how performance will be measured. This is especially relevant for the undertaking regarding wastewater input from the water services provider; and reclaimed water output to be provided to off-takers. All performance requirements must be identified as these need to be costed and allocated.
5. **Funding mechanism**: Outlining how activities in the contract will be paid for. Is there grant funding for the infrastructure and is there a secured revenue stream against the supply of reclaimed water. Clarifying who will use and pay for the reclaimed water (off-take agreements).
6. **Project Site**: Determining where the reuse sub-project infrastructure will be developed and operated. If it is on the municipality's land it impacts the risk of hand back, especially in the case of early termination, interface with water services provider function, environmental compliance requirements and if it can be linked to existing infrastructure operated with similar conditions (WWTW etc). Determining the capital cost of land acquisition where the sub-project infrastructure is developed. Noting that under certain contractual arrangements (i.e. public private partnerships) there is no requirement for the PPP contractors to own the land, and just need contractually secured and uninterrupted right of access.
7. **Water Quality**: Aspects of locsal water quality and its management and monitoring will be an eseential consideration noting that water quality standards for water inputs into advance treatment plants and water quality discharge standards for water from water reuse plants need to be adhered to and be supported by ongoing monitoring and testing.

8. **Offtake Agreements**: The development of offtake agreements for water reuse will be an advantage, but in some instances these will only be developed as the water reuse sub-project is scoped and prepared. Nevertheless, it is valuable to have this in place and as such is incorporated into the project eligibility criteria.

These considerations, amongst others, will be worked through during the project preparation phase and into the feasibility assessments that will cover such aspects as:

- **Section 1:** Needs analysis
- **Section 2:** Technical solution options analysis
- **Section 3:** Implementation options analysis
- **Section 4:** Project due diligence
- **Section 5:** Project Value assessment
- Summary of results: Net Present Value
- **Section 6:** Comments and representations
- **Section 7:** Treasury Views and Recommendations
- **Annexures**

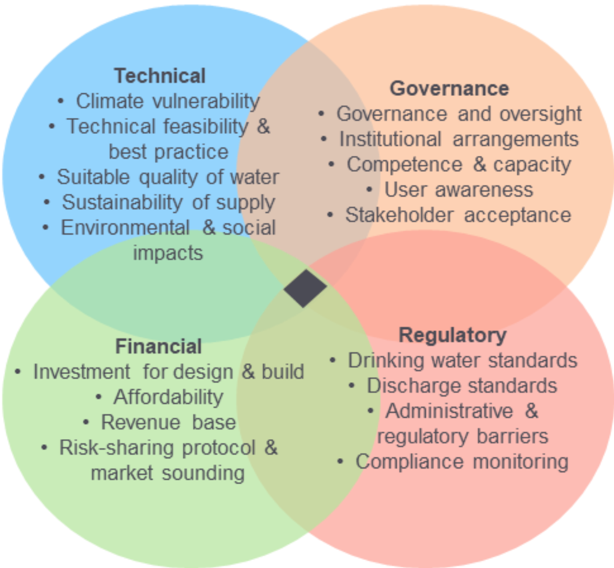


Figure 11: Key considerations for the WRP

WRP project preparation process

A methodology has been conceived for the development of the project pipeline, with the aim being that based on certain constraints and exclusionary criteria, potential water reuse sub-projects may be assessed for funding support (both preparation and implementation) through the WRP.

The project preparation process will follow the following high-level stages: the 3-stage and 5-Case (strategic, economic, commercial, financial, management) Model (5CM) process which develops over time. The 5CM comprises 3 stages, the Early Business Case (EBC), the Intermediate Business Case (IBC) and then the Full Business Case (FBC). Each one of the cases comprises a Strategic Case, Economic Case, Commercial Case, Financial Case and Management Case. These 5 cases therefore form part of each of the EBC, IBC and FBC and are progressively developed as one proceeds through the 3 stages (Table 11).

Table 11: Outline of the project preparation process

Project Preparation Stage	Outcome	PPP Process	ISA 3-Stage 5-Case Model Process
Identification & Selection	Project Owner planning and application for project preparation support. Apply WRP Primary Criteria for project selection to join water reuse unit (WRU) on NWPP	PPP registration	
Climate vulnerability assessment	Demonstrable climate vulnerability as a prerequisite for support by WRP (ie go or no-go)		
Scoping	Pre-feasibility study to motivate inclusion in WRU project pipeline. Apply secondary criteria and more technical rigour.	PPP Inception	**EBC: Strategic Outline** • Step 1: determining the strategic context Step 2: making the case for change
Feasibility	Options analysis, feasibility study & project structuring, procurement strategy and planning	PPP Feasibility	**IBC: Business Outline** • Step 3: exploring the preferred way • Step 4: Determining potential value for money Step 5: preparing for the potential Deal

Procurement	Project procurement Project financial close	PPP Procurement	**FBC** Step 6: Ascertaining affordability
Implementation	Water Reuse Contract Management	PPP Contract Management	Step 7: Planning for successful delivery
Monitoring & Evaluation	Programme and project objectives achieved		

Programmatic Support

With the intention to strengthen local climate adaptation through the preparation and implementation of water reuse sub-projects, to address climate induced water insecurity, the WRP will comprise of:

- The establishment of a WPO as a centre of excellence that will provide transversal project preparation support and will prepare a pipeline of water reuse projects.
- The identification and conceptualisation of large-scale water reuse sub-projects with tangible climatic impacts and significant ecosystem/water dependencies (impacted by climate change) and the provision of focussed project preparation support to progress municipal reuse sub-projects to bankability. The GCF grant support will be essential for project preparation and will be co-financed by funding from project sponsors, project owners (municipalities), the DBSA, the SA Government and others, which will cover the costs up to financial close. All investments falling under the WRP will require specific gender assessment/analysis/action plan as well as a specific environmental assessment/analysis/action plan and ensure a net positive social and environmental impact. Bankable feasibility studies will be prepared with project preparation resources for all projects in the pipeline to ensure that they are fully prepared, appropriately structured in terms of risk and ready for implementation.
- The introduction of a blended finance solution that will allow municipalities an alternative (and competitively priced) option to fund the implementation of such water reuse sub-projects and programmes. The proposed Blended Finance Solution (BFS) is designed to finance water reuse sub-projects in a way that (a) optimises the use of climate finance, and (b) enhances the use of private capital, thereby taking this critical infrastructure to market.
- The undertaking of capacity building and awareness creation activities that will strengthen the relative support for and uptake of water reuse sub-projects.

Water reuse sub-projects under the WRP will not be limited to only water reuse but may include all related opportunities, assessed on a project-by-project basis including i) water reclamation (including associated improvements in ecological infrastructure), ii) sludge management / beneficiation; and iii) biogas to energy generation, and iv) upgrading wastewater treatment processes, thereby realising potential mitigation co-benefits on a case-by-case basis. Such opportunities will only be identified during the project preparation of individual sub-projects that have passed the climate vulnerability stage gate.

As a consequence of implementing water reuse, there will likely be no increase in GHG emissions from the facility as the ATP will be additional to an existent WWTW. However, there will be opportunities to reduce the emissions from the current WWTW and to explore the use of biogas to reduce the energy footprint of these works. Figure 12 presents the proposed plan to reduce the additional emissions due to water reuse. The Scope 2 and direct emissions of the existing WWTW will be assessed. The increase in emissions of a consequence of water reuse will be quantified. Opportunities to mitigate the increased emissions will then be investigated. These include: optimisation of the direct emissions and energy consumption of the WWTW and the activated sludge unit in particular, ensuring optimal energy consumption of the ATP design, and assessing the potential of sludge beneficiation to reduce GHG emissions.

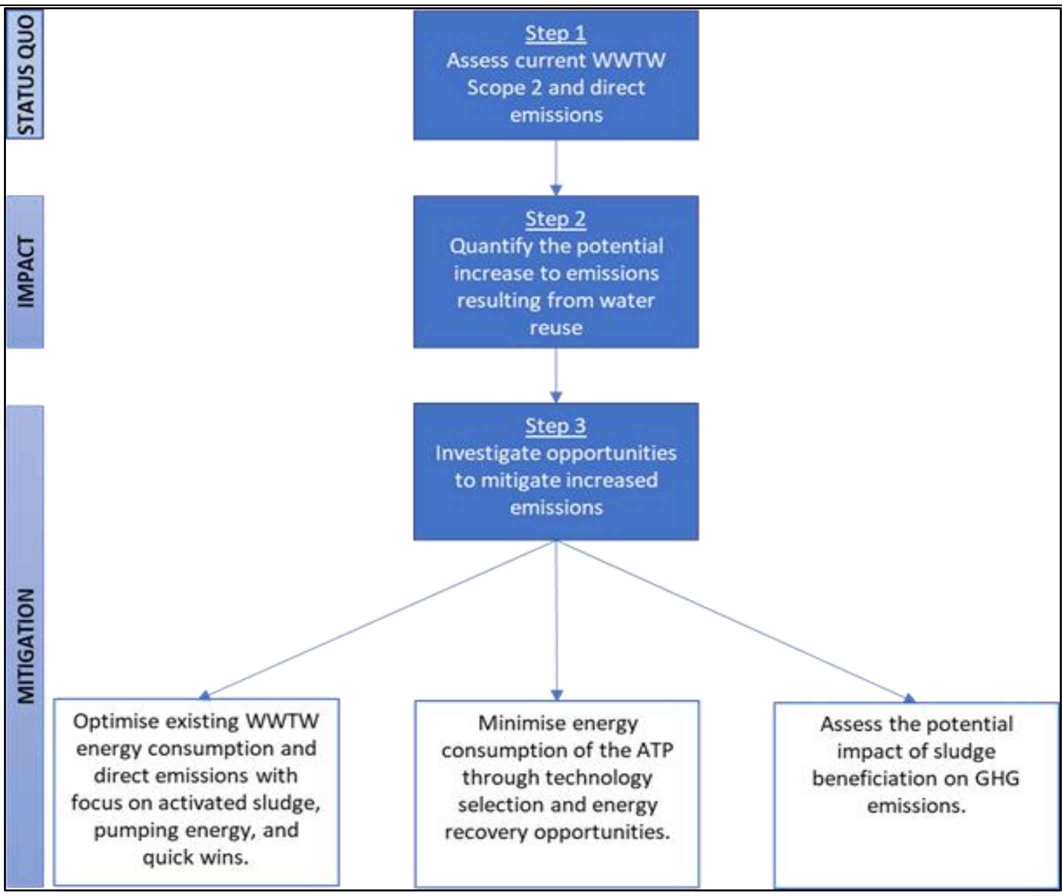


Figure 12: Plan to reduce additional GHG emissions from water reuse

Programme Components

This WRP is being developed through the implementation of 3 components. A first component comprises interrelated activities to support the establishment of the WRP, its governance arrangements and the operationally focused WPO that will develop the project pipeline and provide project preparation support. This component will result in the transformative programming and planning needed to drive market change. The second component will provide technical and financial transaction advisory services to the implementation of sub-projects through procured and contracted service providers and will support these through the deployment of blended finance financial instruments such as concessional loans, subordinated debt, guarantees and grants to de-risk private sector investments by offering new financing products, while establishing a new asset class that will support climate resilience and improved water security. Lastly, a third component of institutional capacity building, knowledge exchange and information sharing that will complement and support components 1 and 2, and importantly strengthen the adaptive capacity of institutions to ensure ongoing climate resilience.

<u>Component 1 – Establishing a WRP project pipeline</u>

The establishment and resourcing of a WRP will provide an opportunity to support municipalities as primary custodians of local climate adaptation plans to undertake innovative and targeted projects that will underpin climate resilience interventions towards improved water security. The programme will identify, prepare, finance and implement water reuse sub-projects at scale to ensure water security due to the impacts of climate change. These projects will provide much needed support to climate vulnerable local economies and communities. It is envisaged that this WRP will become the “one stop solution provider” for municipal water reuse sub-projects in the country. The proposed establishment of the WRP will include:

- the procurement and management of a panel of service providers to assist in the scoping and preparation of water reuse sub-projects;
- the standardization of contracting documents, procurement of professional service providers, contractors and operators;
- the application of best practice design and innovative climate-smart technologies including seeking ecosystem-based approaches;
- the application of lessons learnt (locally as well as internationally) and the achievement of cost optimization and savings;
- the determination of aquatic ecosystem and water reuse allocations for individual municipal areas within water catchments; and
- the establishment of a centralised communication platform (in support of Component 3).

The WRP will work closely with water sector and environmental regulators to strengthen the regulatory frameworks, specifically for abstractive water use and wastewater discharge. This approach will work together with the national certification programmes to support the ongoing uptake of water reuse as a critical intervention to address climate fragility at local levels.

Activities under this component include the following:

Sub-component 1.1: <u>Establishment and Operationalisation of the WRP</u>.

The WRP will be a pathfinder programme under the NWPP, with the NWPP aiming to provide support to the other aspects of the water value chain and will be further developed over time. The programmatic approach to the development of the WRP will therefore provide 'proof of concept' guidance in the development of later municipal water-related programmes with certain aspect of the WRP being leveraged to support the other programmes. It is also important to recognize that there is significant scope for projects under the WRP. The needs at municipal level vary technically and in scale and, with many municipalities needing support, it is necessary to have a phased and progressive approach to developing the WRP and providing project preparation support to municipalities. To achieve the objectives, the WRP will establish a national WPO that will support and facilitate a programmatic approach that will guide project identification and prioritization, will drive project preparation, will facilitate funding solutions, and monitor project implementation.

The DBSA will play several roles in the WRP. It is the accredited entity of the Green Climate Fund (GCF). It is a recognised developer of national programmes (through its Programme Development Unit) and provides project preparation support (through its Project Preparation Unit) with an established track record in both aspects (e.g. the renewable energy and student accommodation programmes) . It is an independent development financier and arranger. If an application to GCF is successful, it will sign the Funded Activity Agreement (FFA) GCF as executing entity. It will participate in the OC as (i) the accredited entity and (ii) as the managing agent of the NWPP (executing authority), being sure to keep its roles separated. It will provide all administrative services such as procurement of goods and services through DBSA supply chain management processes, procurement of panels of service providers, entering into legal contracts with service providers and employees. It will also operate a ring-fenced bank account for the WPO function and make payments on behalf of the WPO. The DBSA will receive fees for these services on a cost recovery basis – approved annually by the OC.

<u>Activity 1.1.1</u>: *Establish the WRP within the DBSA*. A Memorandum of Agreement (MOA) between the Programme owner DWS and, DBSA and SALGA have been signed. This sets out the objective of the NWPP and WPO and the respective WRP, the roles and responsibilities of the respective parties, funding arrangements, and the obligations regarding the WPO. It will include a terms of reference for the Oversight Committee (OC). The DBSA will facilitate this contracting process and once concluded will act as secretariat of the OC to facilitate contract implementation and until such time that the WPO has been formally established.

Activity 1.1.2: *Establish Oversight Committee and develop governance modalities*. The Terms of Reference for the OC will guide the establishment of this committee. Other interested stakeholders (Department of Cooperative Government (DCOG), MISA, National Treasury and DFFE) may be invited by DWS and DBSA to participate in the OCDBSA will participate in its capacity as GCF Accredited Entity (AE) and WPO managing agent. DBSA will act as management agent of the WPO for a period of five years or until such time that the WPO is institutionalized. The WPO will not have its own legal personality during this initial period and this will only be developed over time in conjunction with the NWPP.

Activity 1.1.3: *Appoint staff and operationalise the WPO and technical Water Reuse Unit*. It is envisaged in the NIP 2050 that a number of operational units will be established within the WPO, including the Water Reuse Unit (WRU) which will house the technical support for the identification, preparation and implementation oversight of water reuse sub-projects. As such the DBSA will support the development of the organisational requirements including the WPO and WRU organograms, job specifications, salaries and staff benefits, the contractual arrangements to support staff appointments, as well as the development of a phased staffing plan aligned to the programme implementation plan (PIP) (see below).

Activity 1.1.4: *Develop PIP and annual workplan and budgets*. The WPO PIP will be developed annually and submitted to the OC for approval. Once approved, the WPO will implement and report against this. This plan will set out the roles and budget of DBSA in Programme implementation, including both the NWPP and WRP development and water reuse sub-project preparation, as well as the capacity requirements and how they will be met. It will include the budgeting of the various units, differentiating between the sub-programme budgets (i.e., each sub-programme will be ring-fenced with its own costing and funding structures). Thus, the WRP implementation plan will be developed by the Water Reuse Unit and submitted to the WPO for consideration and approval. It will set out all project information and capacity and budget requirements for the WRP. The WPO will include this detail in the PIP. In so doing alignment between the NWPP and the WRP (as well as later programmes) will be ensured. An initial WRU implementation timeline is provided in Figure 13, and takes into consideration the phased approach of the development of the WRP.

A key element of the annual workplan and budgets will be the assessment of risk as well as outlining the mitigation measures that will be undertaken to manage these risks. For example, water use licensing and environmental authorisation processes should not prohibit the structuring and implementation of water reuse sub-projects and work under Component 3 will address institutional and regulatory alignment. Undertaking targeted action with the key Ministries will be important in addressing such risks.

In regard to the mitigation of identified risks, the support of lead Ministries such as the DWS, DFFE and National Treasury has been clear in the development of the WRP and they will be central in assisting to ensure that risk are mitigated. The DBSA as AE will also play a key role in supporting the WPO to address these.

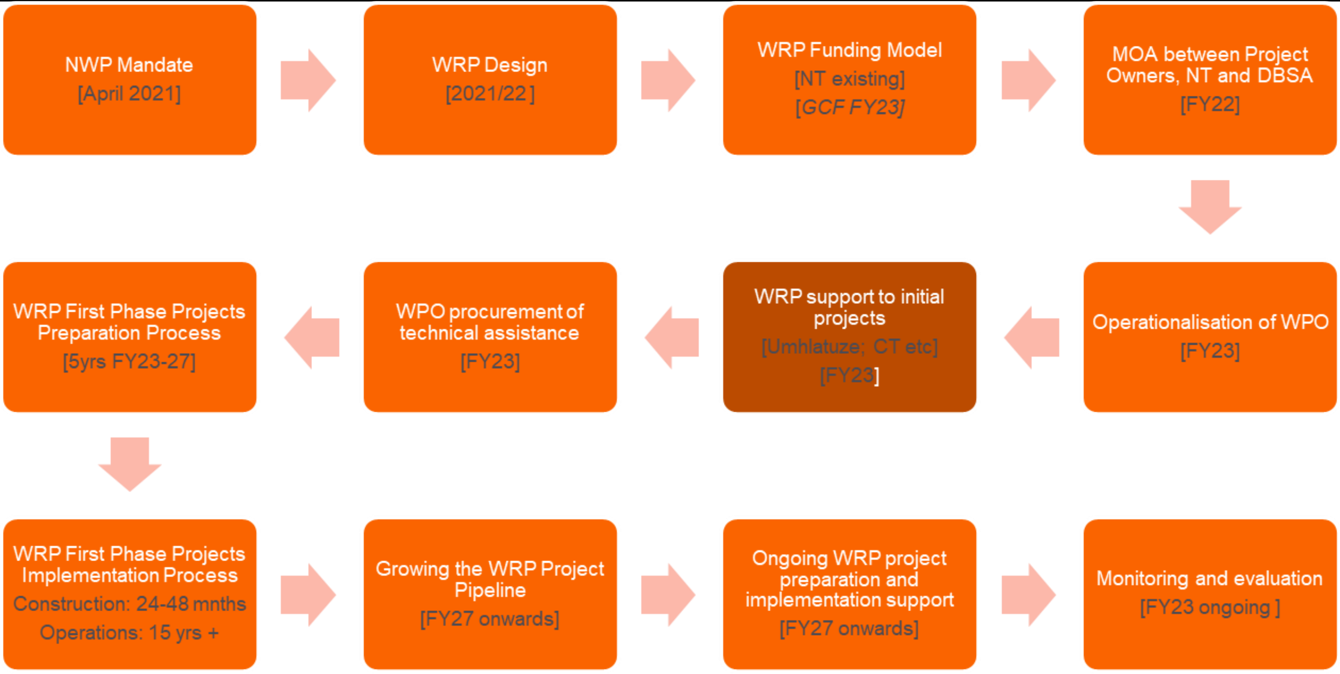


Figure 13: WRP implementation timeline

Activity 1.1.5: *Finalise operational manual and supporting tools*. It is a key objective of the WRP to drive efficiency by standarising processes and documentation. There will need to be development and where appropriate standardisation of WRP documentation and water reuse sub-project documentation regarding project preparation (scoping, feasibility and procurement) and implementation (contracting and monitoring and reporting). A key value that DBSA brings to the programme development process is its existing experience and capacity in developing and implementing programmes and a such has indicated it has a suite of standard documentation. This will also include the development of controls to be used to ensure that any materials or technology procured are used only for the purposes intended and are not diverted or misused for unauthorized, improper or illicit purposes.

Activity 1.1.6: *Establish financial arrangements and reporting regimes*. This activity will formalize the financial management arrangements and the necessary reporting that would underpin this. The program will apply the AE's financial management policies and guidelines for capital provided through loans from the DBSA itself (provided from GCF concessional capital). An Independent Blended Capital Facilitator would manage other capital in alignment with DBSA processes and these modalities will be clarified. The financial management of disbursements will be primarily managed under the DBSA Loan Management Unit (LMU). Loan Management forms part of the broader discipline of portfolio management, with the role of the LMU being to maintain financial sustainability through effective financial administration and management of the portfolio. Its purpose is to ensure the continuous monitoring of all relevant covenants throughout the lifetime of the facility. This relates to both financial and non-financial covenants.

Once the Loan or Facility has been disbursed in full, the LMU becomes the first point of contact and will be accountable for the administration of the loan; and also monitors borrower compliance and performance. Loan Management may if necessary, also consult with Legal, to ensure that the full implications of any deviation from conditions or breach of covenant are understood by the DBSA and appropriate mitigation measures are initiated.

Conditions Precedent (CPs) stipulated in the legal agreements must be fulfilled by the borrower before disbursement. If and when CPs have been fulfilled, the borrower will commence drawdown. Disbursement of funds will take place either through one drawdown or on a staged basis, depending inter alia on considerations such as the project timetable and achievement of construction milestones.

These modalities and arrangements for the WRP will then be clarified and documented as a financial management manual that will outline the systemic and procedural aspects of ensuring effective financial management. This will include clarified business processes and standard operating procedures based upon those of the DBSA as AE.

Activity 1.1.7: *Develop monitoring and evaluation and reporting regimes*. Programme level monitoring, reporting and evaluation will be undertaken in compliance with the internal systems established by the DBSA and will be documented to ensure clarity as to the requirements and alignment in process. The Monitoring and Evaluation of the sub-projects and the programme will be aligned to the DBSA Development Results Reporting Framework and relevant requirements of GCF as provided under the Accreditation master agreement (AMA) including the GCF policy on prohibited practices adopted by the Board in Decision B.22/19, as amended by the Board in Decision B.23/08. The WPO will undertake a range of activities that assess progress and support the adaptive management of sub-projects. This will be supported by the Project Owners (municipalities) that will undertake monitoring, evaluation and reporting on each water reuse sub-project as is appropriate at the various project development phases.

Sub-component 1.2 Project pipeline preparation.

The WPO will undertake a suite of actions to support sub-projects and the implementation of the Programme. This will focus upon the developing the project pipeline in the first instance, supported by the procurement of service providers to support project preparation and programme implementation. The Programme will have both strategic and operational capacity to support Project Owners to conceptualise, prepare, finance and implement water reuse sub-projects. There will be a number of standardised steps to be followed, with various stakeholders having clear roles and responsibilities. The WPO will support the WRU and the Project Owners to follow due process by establishing transversal standardised operating procedures.

Activity 1.2.1: *Design and launch an open-ended Request for Proposal (RFP)*. The WPO will have outlined the process to support the development of a robust project pipeline for the WRP. During the programme design phase an initial pipeline of potential sub-projects have been identified using agreed upon criteria. The WRP will only support sub-projects that directly address clear climate vulnerabilities. The WPO will therefore initiate a call for proposals and will enter into discussions with identified priority project owners in order to develop a short-list of initial projects for the programme. Each sub-project will undergo a detailed climate vulnerability analysis as the primary eligibility criterion for further support by the programme. Should a sub-project meet this criterion, then the WRU will thereafter provide support to municipalities to assist in the conceptualisation and prioritisation of their large-scale water reuse sub-projects, prior to the project preparation and the development of implementation-ready plans. Additionally, the OC and DWS will also liaise with various role-players in the market to select and prioritise projects for the WRP noting that the initial suite of sub-projects with be critical in providing proof of concept.

Activity 1.2.2: *Procure and manage a panel of project preparation service providers*. The provision of technical and financial skills support to assist municipalities with project preparation is key aspect of assisting these institutions to address climate vulnerabilities. The establishment of a panel of service providers to provide the WRU and the municipalities with professional services for project identification and preparation will be an important part of programme efficiency and effectiveness. The DBSA's procurement rules and guidelines will be applied to procure these service providers as well as to oversee the implementation of their support projects.

Activity 1.2.3: *Provide technical assistance for project structuring and preparation*. Technical assistance will be provided to municipalities using grant support to prepare bankable water reuse sub-projects. This will include supporting with the development of strategic cases, and full detailed business cases. In certain instances it may be of value to the sub-project to appoint service providers to support projects from end to end, however, this will need to be in alignment with South African procurement laws and regulations. In preparing these projects due diligence in terms of maladaptation will be essential and will cover maladaptation linked to infrastructure, to institutions and to behaviours. This will be incorporated into project scope and become part of the project monitoring and evaluation regime.

Effective engagement with stakeholders will be an imperative aspect of ensuring that the risk of maladaptation is duly covered during this preparatory stage.

The capacity support could also be expanded to the procurement of implementation support services if the delivery models and funding opportunities require that function. Key activities of the project preparation service providers will be to directly support municipalities to undertake their project preparation activities including scoping, feasibility and procurement. Although procured and contracted by the WPO (DBSA), the service providers will work directly with the project owners in the project preparation process. This structure will assist in the delivery of the key value proposition of the WRP, that the WRP will enable continuity of care between stages on the project preparation process. This capacity will further enable the achievement of the objective of becoming a centre of excellence.

Component 2 – Implementing water reuse sub-projects

The WRP will support climate innovation and market transformation through the implementation of a range of technical interventions that will support the implementation of water reuse sub-projects and the establishment of water reuse infrastructure as a new asset class. The WRP will establish a panel of service providers to provide municipalities (as the project owners) with professional services support to undertake their project preparation activities including scoping, feasibility and procurement. Although procured and contracted by the WRP the service providers will work directly with the project owners in the project preparation process. This structure will assist in the delivery of the key value proposition of the WRP, that the WRP will enable continuity of care between stages on the project preparation process. This capacity will further enable the achievement of the objective of becoming a centre of excellence.

The WRP project owners are primarily the municipalities as water services authorities for their areas of jurisdiction. The project owners will engage the WRP for support and engagement will be through formal written correspondence including on-boarding contractual commitments, signed by the required authority, as required at various stages and gates of the project preparation process.

To support the implementation of sub-projects, the approach will be to create an enabling blended financing environment through alternative funding solutions. These include Infrastructure finance focussing on:

- Securing purchasing agreements with industrial and agricultural users for reuse water.
- Providing cost savings and revenue flows to be found in improved water use and energy efficiency such as including biogas production at wastewater plants.

Identifying and promoting opportunities to attract financing through traditional methods (such as through municipal budgets and commercial loans) and more innovative approaches (such as municipal bonds, PPPs, project bonds and impact investing). The opportunity exists to develop an alternative funding solution to support the scaled roll out / implementation of water reuse sub-projects in SA. The key outcome associated with the development of such a funding solution includes:

- To create a new asset class by using credit enhancement to crowd in private sector funding targeted towards developing debt capital market (DCM) instruments around a specific asset class. The asset class will offer acceptable financial returns but remain in line with Environmental, Social, and Governance (ESG) impacts and help to meet the United Nation’s (UN’s) Sustainable Development Goals (SDGs) (Goal 6 – safe and clean water and sanitation; Goal 11 – more sustainable cities; Goal 12 – more sustainable consumption and production; and Goal 13 – combat climate change)

- The use of credit enhancement to support the blended finance approach through the employment of first loss / subordinated facilities, loan tenor extensions, or the employment of guarantees.

The proposed BFS is designed to finance water reuse sub-projects in a way that (a) optimises the use of climate finance, and (b) enhances the use of private capital, thereby taking this critical infrastructure to market.

It achieves this by blending numerous sources of capital and their underlying instruments, for each distinct sub-project. Blending numerous sources of capital is not straightforward, given the multiplicity of conditions associated with each source. The full BFS comprises the following four primary capital stack components: (i) Equity and/or Budgeted Grants; (ii) Private Debt Capital; (iii) programmatic Development Capital; and (iv) a Partial Credit Guarantee. As noted above, the BFS is conceptualised as an unincorporated managed fund, meaning it will facilitate the flow of capital to the WRPs underlying sub-projects, on a project-by-project basis. On the surface, blended finance is both a novel and attractive approach to financing infrastructure in South Africa; however, no truly like-for-like local examples exist to demonstrate how this might be achieved in practice. Furthermore, whilst it is being proposed that the financing of water reuse happens at a sub-project-level, there needs to be protocols in place at the programme-level to ensure a consistent and standardised approach across all projects that interact with the WRP. This will underpin the WRP as a "centre of excellence" in terms of infrastructure finance solutions.

Applying a developed and agreed upon suite of blended capital principles for the WRP, an Independent Blended Capital Facilitator (IBCF) will act as a **specialist service provider** contracted into the WPO, working alongside the Water Reuse Unit and the Project Preparation Panel to source and appropriately blend capital for water reuse infrastructure sub-projects as approved by the Water Reuse Unit and WPO. The IBCF will prepare the term sheet and project prospectus for lenders which shall include the following details, inter alia: the financial aspects of the project, its technical aspects and legal aspects. The IBCF will then solicit bids from private capital providers, evaluate these and then develop an optimum selection for each sub-project.

The ticket size will be determined on a project by project basis and the principle of least concessionality will be applied to determine the GCF contribution at sub project level.

Activities under this component include the following:

Sub-component 2.1 <u>Provision of Technical Assistance for Project Procurement</u>.

The key objective of the procurement stage is for the WRP to continue to support the Project Owner to take the project through procurement, to contracting and financial close. Support will continue to be provided by the WRP by the panel of service providers procured to support Project implementation. The Project Preparation service providers will be appointed on a project by project basis and will be required to provide the necessary technical, legal and financial support to the Project Owner including developing a detailed procurement implementation plan, drafting of all required documentation and reports, supporting regulatory processes and decision-making process. The approach will depend on the recommended way forward in the Feasibility Study undertaken during the project preparation phase.

<u>Activity 2.1.1</u>: *Provide technical assistance for detailed design and implementation of sub-projects*. Utilising the findings of the feasibility study, technical support will be provided to undertake the detailed design of the water reuse sub-project including the development of an implementation plan. The Service Provider shall investigate all applicable existing and pending statutory and regulatory requirements relevant to existing water sector services and facilities as well as any proposed means for the implementation of the sub-project. At a minimum, this will include any laws and regulations from all applicable local, provincial, national governmental and international sources. To achieve this regulatory review, the service Provider will consult with applicable local, provincial, national and international regulatory agencies to advise them on the sub-project and to solicit their input into the necessary activities.

At this stage, the Service Provider must initiate discussion between the Project Owner and the DWS to discuss the water use licensing implications and the potable water reuse standards/guidelines that will be used in this sub-project. Once the market study is complete and the off-take arrangements are known, at that stage the Service Provider must ensure that the Project Owner and the DWS agree on the licensing requirements and the associated water reuse standards/ guidelines.

Activity 2.1.2: *Facilitate procurement and contracting*. The Service Provider will draft and submit a procurement plan, which demonstrates that the Project Owner has the necessary capacity and budget to undertake the procurement of recommended option. The procurement plan must identify and schedule all activities including all the necessary Council and National Treasury approvals. This plan will reside with the DBSA's Supply Chain Management Unit who will oversee and monitor compliance.

It must include a procurement strategy and key contracting structures and alignment to the risk identified in the Value Assessment process. If an external procurement or a PPP solution is decided on and Council approves in-principle to proceed to procurement, the Service provider will be required to provide the necessary technical, legal and financial support to achieve financial closure of a Public Private Partnership in terms of the Municipal Finance Management Act (MFMA) (s120 PPP, s33 future financial commitments), the PPP Regulations and PPP Guidelines (Module 5) and the Municipal Asset Transfer Regulations. The Service Provider must prepare a complete set of procurement documents that comply with all applicable public sector procurement law, policies and guidelines, and that are in accordance with the required tendering processes of the Project Owner. This means that all procurements will be undertaken in terms of the public sector procurement laws, regulations and guidelines.

Activity 2.1.3: *Monitor, evaluate and report on project development*. Irrespective of the type of contract concluded the Project Owner must monitor performance. The Service Provider will, in close coordination with the Project Owner, draft a comprehensive Project management and monitoring plan. At a minimum, this will define all procedures required for monitoring the performance of the agreement/s concluded during the full-term of the sub-project. The management plan will be in accordance with the provisions of the agreement concluded, and follow the guidance provided in the PPP Guideline. The Project Owner will be required to report on a monthly basis to the WRU regarding sub-project progress, with the WRU in turn reporting to the WPO on a monthly basis. The WPO will report to the OC on a quarterly basis regarding progress across all sub-projects. The OC will report twice annually to key stakeholders. The DBSA, in its capacity as Accredited Entity Executing Entity will report to the GCF.

As part of the monitoring and evaluation process, controls will be used to ensure that any materials or technology procured under this sub-project are used only for the purposes intended and are not diverted or misused for unauthorized, improper or illicit purposes.

Sub-component 2.2 Development of the Blended Financing Solutions.

It is intended that the WRP will have a Blended Finance solution, which will make use of: (a) pooled programmatic capital; (b) private capital through different instruments; and (c) credit enhancement instruments from DFIs and others.

Activity 2.2.1: *Develop principles and approved protocols for developing the BFS*. A BFS concept has been designed with a sequential logic in mind, using the various equity, debt and guarantee instruments. The BFS is not intended to be a standalone institutional fund or separate incorporated legal entity. Instead, it will be an unincorporated managed fund embedded in the WPO of the WRP. The facilitation of the flow of capital will be in accordance with a Blended Finance protocol, to ensure that the objectives of the DBSA and GCF, as well as the GCF Investment Criteria, are met. In this regard, this activity will clarify and formalise these protocols that will include approaches to blending capital and how GCF funding is to be blended with private investor funding, to enable private funders and municipalities to lower their risk exposure on a project-by-project basis. Principles will cover approaches to effective blending, to efficient blending and independent blending.

Activity 2.2.2: *Project finance support through the BFS.* The BFS is designed to provide blended project finance to a portfolio of sub-projects in the WRP, on a project-by-project basis. Each sub-project will be prepared with its own financial analysis and due diligence. This will inform the type and corresponding magnitude of finance that will be allocated to the sub-project through the BFS. The types of financial instruments that could be allocated by the BFS in a blended fashion, include: equity, senior loans, subordinated loans, equity, and local government budget allocations. In addition to these project finance instruments, a partial credit guarantee instrument is used, to attract a greater portion of senior debt, and reduce the portion of subordinated debt. The deliverables of this activity are: (i) project finance analysis, and blended finance allocation report, per project; (ii) Term Sheet of instrument pricing, per project, as well as (iii) fund deployment and reporting.

Activity 2.2.3: *Establish the IBCF function and appoint specialist independent service provider*. To give effect to the abovementioned blended capital principles for the WRP, the role of an IBCF is recommended. The roles and responsibilities of the IBCF as well as the relationships with the organs and projects of the WRP and the Blended Finance Solution will be formalised as will the contracting arrangements and period.

Activity 2.2.4: *Provide financial structuring services on a project-by-project basis*. The IBCF will in terms of its mandate: advise on the setup, management, and capital stacking protocol(s) of the BFS. This includes the distinct prerequisites, conditions, and due diligences required from the WRP, to enable the sourcing of, determination of, and flow of different capital sources, across all three components of the BFS, to each WRU approved underlying sub-project, namely: a) Private Sector Capital, (b) Credit Enhancement Instruments, and (c) allocation of the Programmatic Concessional Financing. Critically, it must include the requirements of the GCF Investment Criteria and be wholly compliant with the Mandate as developed by the WRP and BFS. The IBCF mandate, and the usage of the climate finance element of the BFS is what sets the Water Reuse Programme apart from other local infrastructure technical assistance facilities, and is therefore an element that the IBCF will invest significant intellectual capacity in developing.

Activity 2.2.5: *Audit and report to measure impact of the BFS use and efficacy*. The IBCF will create an Impact & ESG Framework with a two-fold purpose: (a) to measure key impact metrics that may be attractive to a wide array of private capital providers; and (b) to help funders, financiers and investors integrate ESG factors into their portfolios. The IBCF will coordinate the continual measurement, verification and reporting of Impact Metrics, against the Impact & ESG Framework, with the support of the WRU and WPO. The IBCF will be responsible to report on all of its undertaking on a quarterly basis to the WPO, as well as provide an overview of the BFS's use, status and efficacy. In support of the WPO, the IBCF will prepare supporting documentation required for any third-party auditing required by the GCF, according to their timelines.

Component 3 – Building capacity and creating awareness

A national and regional public awareness and education processes will support the implementation of existing national, provincial and local climate adaptation strategies and plans. These knowledge exchange interventions will underpin the development of awareness regarding climate change impacts at local levels and will support the successful scaling and implementation of water reuse sub-projects in South Africa as an adaptation response. This stakeholder engagement, learning and communication strategy will also be driven and coordinated by the WRP. While the DBSA will take the lead in the rollout of this component due to its strong institutional position in the water sector, the WPO will undertake key supportive actions. It is important to note the important role of DWS in supporting these important processes, particularly as the water sector leader.

From a perspective of public perception, education and awareness:

- Climate change impacts and the severity of water security issues at local levels requiring innovative water reuse interventions to address local water developmental needs.
- Public perception, social, cultural and religious beliefs often prevent direct potable reuse options from being considered, despite being able to treat water to required standards.
- Indirect potable reuse and blending options are available that may be socially more acceptable than direct (“toilet-to-tap”) potable reuse options (perceptions).
- Current water reuse interventions are limited and often with unintended consequences due to these interventions often being unplanned, uncontrolled and unmonitored with negative environmental and health implications. Strategic, intentional, and planned water reuse sub-projects will effectively manage risks, as well as monitor and control quality for realising impact in adapting to climate change.

The WRP will develop a strategy on how to educate and communicate the details of water reuse as a means of improving the public’s perception of the approach. The key drivers affecting water reuse choices include climate change impacts and importance of water security, water quality, the cost relative to other water supply options and the social, cultural, and religious perceptions. In support of this, as well as informing Components 1 and 2, the maintenance of a ‘Community of Experts’ as an advisory forum will be supportive of developing innovation and supporting knowledge exchange. Equally, for specified sub-projects this group will be asked to provide strategic review and guidance. Lastly, it is essential to maintain strategic partnerships that will underpin and inform the approach towards climate adaptation this will assist in working towards improved institutional and regulatory frameworks that support adaptation and resilience building.

Activities under this component include the following:

Sub-component 3.1: <u>Communications and awareness creation.</u>

South Africans need to come together to secure the country’s water future. This requires everyone, from the general public to decision makers, to understand South Africa’s climate vulnerability and the impacts that this will increasingly have on the water situation. Efforts to build this understanding are important in catalysing transformational changes are essential in developing an acceptance of water reuse as a responsible approach to climate adaptation and improving water security.

<u>Activity 3.1.1</u>: *Develop, manage, and maintain the WRP brand*. Communication is a key enabler of the successful implementation of the WRP. The important role of the WRP in providing support through a ‘centre of excellence’ will need to be communicated via a communications strategy that aims to improve public knowledge of South Africa’s climate change vulnerability and water situation, the WPO and its role in supporting municipalities, and to empower municipalities to communicate clearly, consistently and transparently about water reuse sub-projects. This will include the development of a corporate identity as well as the establishment of an information hub to support information exchange will be a critical part of this process.

<u>Activity 3.1.2</u>: *Development of knowledge products and collateral*. In support if the communications strategy the WRU will develop a range of materials and knowledge products that will support in improving the levels of understanding with regard to climate change impacts and water reuse sub-projects as an adaptation response. These materials will also assist municipalities where water reuse sub-projects are to be implemented, to build awareness and understanding of their stakeholders.

<u>Activity 3.1.3</u>: *Undertake outreach and awareness creation events*. The WPO and the WRU will utilise various climate and water sector events as well as arrange specific market engagement sessions to provide information and create awareness regarding the WRP and the intervention support that is offered by the WPO.

<u>Activity 3.1.4</u>: *Facilitate and manage a community of experts forum*. The WRP will identify and invite members to participate as a Community of Experts. This forum will include water reuse sector experts to

provide technical support to the preparation, design and structuring of water re-use sub-projects. The Community of Experts will report to the WRU and will provide strategic guidance and input into the works done by the service providers on water reuse sub-projects, taking into consideration the scope of work and work plan. It will make recommendations to the WRU. The work of the Community of experts will be limited to strategic guidance and input and it will not be able to alter the scope of work, budget, timeline or design of the sub-projects or service providers. Typical participants in the WRP Community of Experts will include SALGA, Water Research Commission, Water Institute of Southern Africa, Water Boards, Trans Caledon Tunnel Authority, National Treasury (and the Government Technical Advisory Centre (GTAC)), Municipal Infrastructure Support Agent, as well as others.

The participation will be regulated by a terms of reference which will be issued by the WPO to the experts.

Activity 3.1.5: *Provide strategic review and guidance on specific sub-projects and interventions*. The WPO will facilitate and coordinate the support that the Community of Experts will provide. This will include such activities as providing strategic guidance and input into the work done by the WRU Service Providers, reviewing and making recommendations to the WRU on interim and final deliverables produced by the Service Providers, and advising the Service Provider of work done previously, or on process that may impact on or be relevant to the programme design, deliverables and intended outcomes.

Sub-component 3.2: Strengthen institutional and regulatory frameworks.

While the DWS will be engaged in the WRP oversight via the OC, there will be an ongoing need to continue regular and structured communications with key national, provincial and local level stakeholders. These engagements will not only build capacity of these key institutions but will also support in engaging with these stakeholders to address important institutional and regulatory aspects that will assist in building adaptive capacity and in entrenching water reuse as a strategically important part of climate resilience.

Activity 3.2.1: *Develop a key stakeholder engagement framework*. The development of an overarching stakeholder engagement framework is essential to guide the approach to ensuring regular interactions and discussions, the exchange of information and data, as well as the differing approaches to be used to build awareness and understanding. This will include the approach to the management of key strategic partnerships.

Activity 3.2.2: *Undertake targeted strategic outreach*. Aligned to the stakeholder engagement framework the WPO will undertake focused engagements with developers, commercial banks, DFI and other key stakeholders. These engagements may be focused on support needs for sub-projects and may be catalytic in terms of unlocking financial and technical aspects of sub-projects or in introducing innovative approaches.

Activity 3.2.3: *Establish and maintain strategic partnerships with key actors and institutions*. The maintenance of key strategic partnerships will be essential and will include such stakeholders as Government departments; utilities; business and industry associations, financiers and developers, amongst others. These partnerships will work collectively at building longer-term institutional capacity as well as addressing various regulatory issues and other barriers to the development of adaptive capacity and the roll-out of water reuse sub-projects.

B.4. Implementation arrangements (max. 1500 words, approximately 3 pages plus diagrams)

Key Stakeholders

As water sector leader, the Programme custodian is the South African DWS. It undertakes the Programme with the knowledge and support of the National Designated Authority, DEFF (see Annex 1). DWS has a national mandate to establish the NWPP and the Memorandum of Agreement for the support of the NWPP between DWS, DBSA and SALGA was signed in May 2022.

As programme custodian the DWS will contract with DBSA, in its capacity as experienced programme developer and as GCF Accredited Entity and Executing Entity for the WRP, to establish and support the implementation of the governance arrangements to give effect to the NWPP, including the WRP.

At a strategic level, an OC has been be established by DWS, SALGA and DBSA which will govern the operations of the WPO. It may include representation by invitation from other stakeholders such as National Treasury, DEFF, and representatives of local government. It will also have access to an advisory committee with sector expertise to guide the OC. Core functions of the OC include to: provide strategic direction; oversee operational management; maintain institutional linkages; and undertake periodic monitoring and evaluation.

At an operational level, the DWS, SALGA and DBSA have by agreement establish a programme management office known as the Water Partnerships Office (WPO). DBSA is the Programme Executing Entity and the WPO is established in the DBSA. All Programme contracting and procurement will be undertaken by DBSA, thereby providing shared services and systems that will assist in the operationalization of the WPO. The technical capacity for the WRP will be in a technical unit of the WPO, with access to a community of experts. It will also have access to the panel of service providers who will be procured for the WPO, by the DBSA, to assist with the identification, structuring and implementation of water reuse sub-projects. Water re-use sub-projects will be implemented by agreement with the local government structures who are empowered by law to implement water reuse sub-projects in their respective jurisdiction.

The governance structure, distinguishing strategic programme governance through the OC and operational programme management and project implementation through the WPO, is indicated diagrammatically as follows in Figure 14.

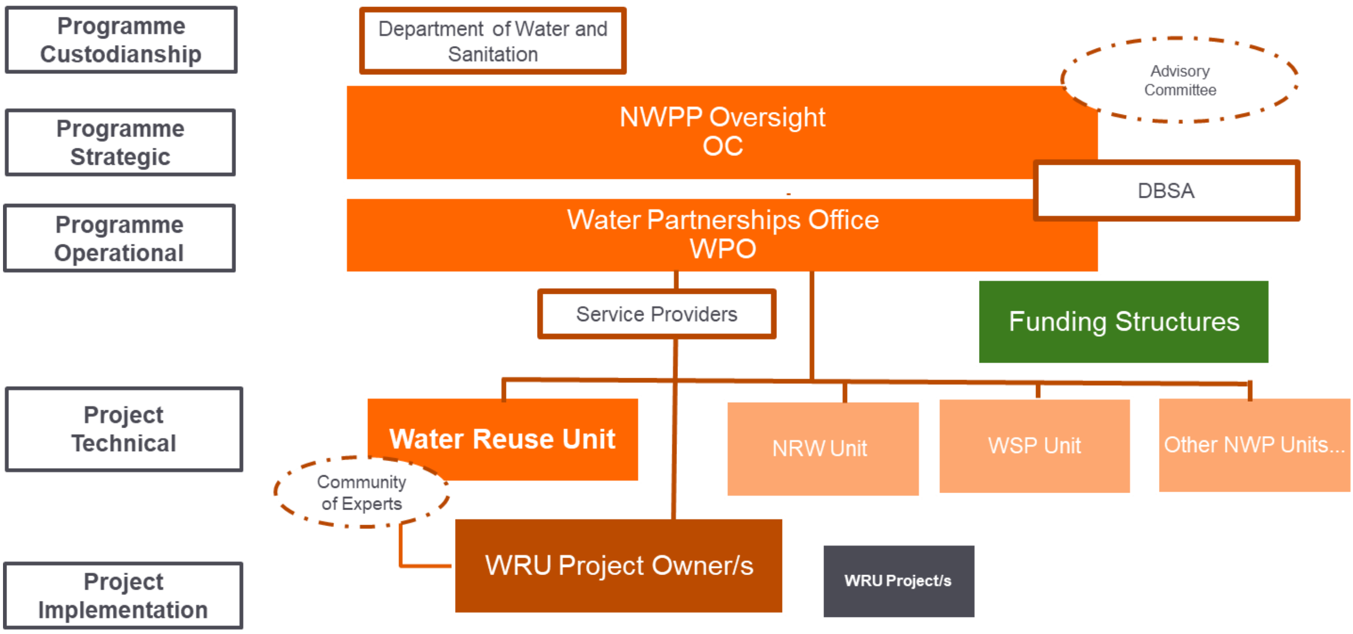


Figure 14: WRP Governance arrangements

Noting the progressive establishment and development of the WRP, the importance of ensuring that GCF is kept abreast of progress and will submit reports at key programme stage gates. In this regard the DBSA will:

- Provide a report to the GCF secretariat regarding the ongoing progress with Sub-component 1.1 Establishment and Operationalization of the WRP, outlining how the WRP has achieved operationality as presented in the FP package. The report will be submitted to GCF upon completion of key stage gates with Sub-component 1.1 noting that the WPO will be progressively strengthened with the development of the project pipeline. Recognizing the importance of this process of establishment and development in providing the basis for the implementation of the WRP, the DBSA will work closely with GCF in addressing gaps, in making amendments that may be required, as well as the process and timing for the disbursement of funds.
- Provide GCF with ongoing reports on project mobilisation for Sub-component 1.2 Projects Pipeline Preparation, presenting project mobilisation results, including potential PPP structures, financial engineering, and impact achievement potential. The detailed scope of the report should be agreed upon between DBSA and the GCF (specifically during the Terms Sheet negotiation).

Service Providers will be procured by DBSA and managed by the WPO to work closely with the WRU project preparation and implementation support processes. This is directly related to project implementation capacity and will be scalable and relevant to the WRP needs.

- Project Preparation Panel: a panel of service providers to provide the WRU and the municipalities with professional services for project identification and preparation. This will include supporting with the development of strategic cases, and full detailed business cases. The capacity could also be expanded to the procurement of implementation capacity if the delivery models and funding opportunities require that. Key activities of the Project preparation service providers will be to directly support municipalities to undertake their project preparation activities including scoping, feasibility and procurement. Although procured and contracted by the WPO (DBSA), the service providers will work directly with the project owners in the project preparation process. This structure will assist in the delivery of the key value proposition of the WRP, that the WRP will enable continuity of care between stages on the project preparation process. This capacity will further enable the achievement of the objective of becoming a centre of excellence.
- Independent Blended Capital Facilitator (IBCF): a contracted service provider to provide the WPO with professional blended capital and financing services in respect of each project. The IBCF will be highly skilled and should demonstrate financing and investment skills across numerous capital sources. Furthermore, the IBCF should be independent, meaning not conflicted in providing its own capital into projects in the programme, to allocate capital to each project to optimise the climate finance objectives and long-term financial sustainability of each sub-project.

Mandate documents

Mandates and decision making of the various governance structures will be regulated by the following instruments in **Error! Reference source not found.**15.

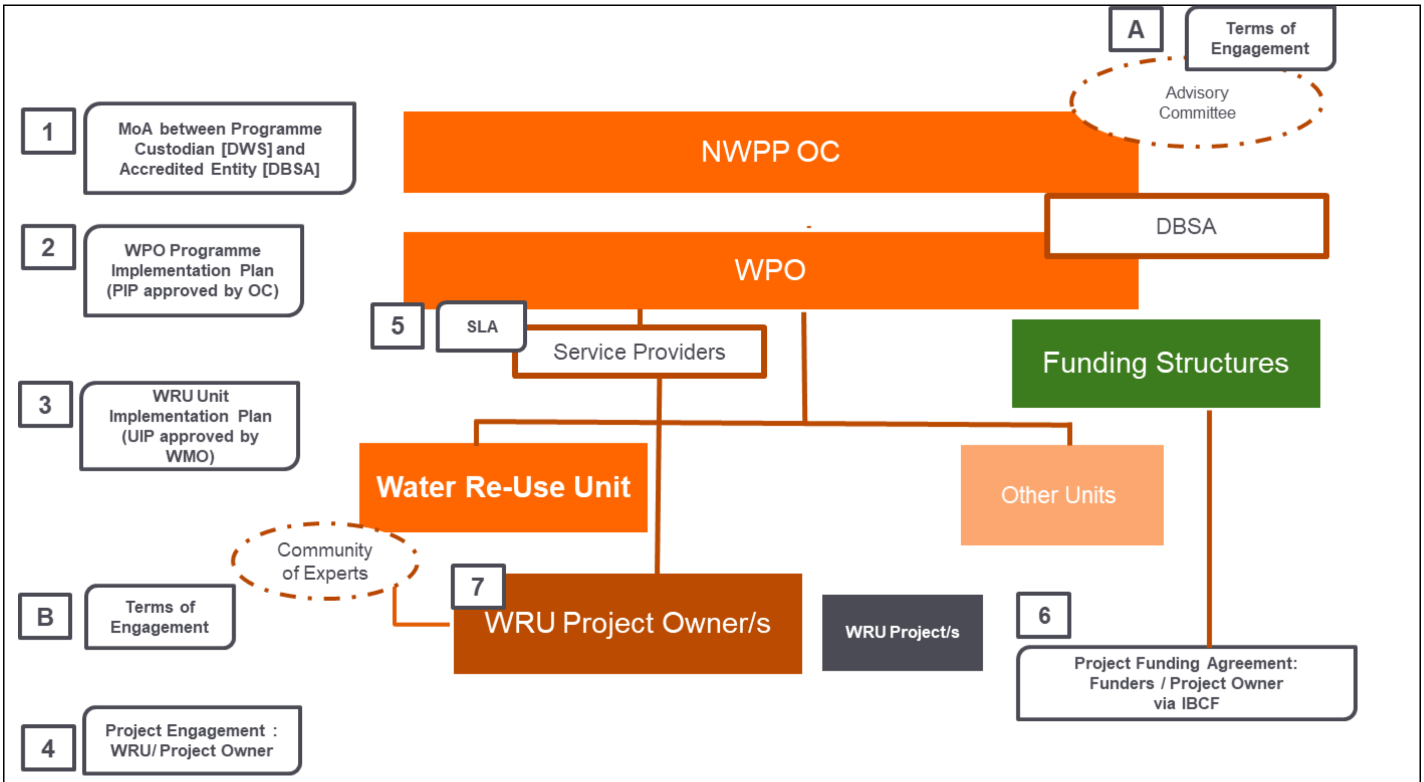


Figure 15: WRP Mandate Documents

Financial Flows

The WRP is proposing the use of significant GCF concessional capital, which is US Dollar denominated (USD). This concessional capital will be managed by the DBSA as the AE, which is a South African domiciled entity. Furthermore, the WRP's underlying water reuse sub-projects are financed in SA Rand (ZAR), which is overseen by the DBSA. Therefore, all debt repayments from sub-projects will be in ZAR, and the currency exposure from this funding required to repay the GCF concessional capital will be managed at a DBSA balance sheet level. The GCF does not directly finance the underlying sub-projects. The GCF provides loans to the Accredited Entity (the DBSA), who facilitates the GCF loan(s) into the underlying sub-projects, as required. The GCF provides loans for the DBSA's approved programme design (incl. project preparation approach), and the DBSA on-lends to the underlying sub-projects, per their Funded Activity Agreement (FAA). The DBSA shall transfer to the GCF all loan repayments in US Dollars and manage the GCF funding's USD exposure at a DBSA balance sheet level. This will ensure that no losses will be incurred by the GCF due to foreign exchange rate variations as this will be managed by the DBSA's treasury division.

Assuming that an underlying sub-project has been approved and allocated, only then would a pro-rata portion be drawn from line of credit from the GCF proceeds. Upon receipt of USD, it would be placed in the DBSA's general USD CFC account and used as per normal. A deal will be booked on treasury system (Quantum) capturing repayment and interest rate details, counterpart bank account, SARB approval number etc. Soon thereafter, it is assumed that a utilisation request would be received from sub-borrower.

The WRP's funding flows are provided in Figure 16.

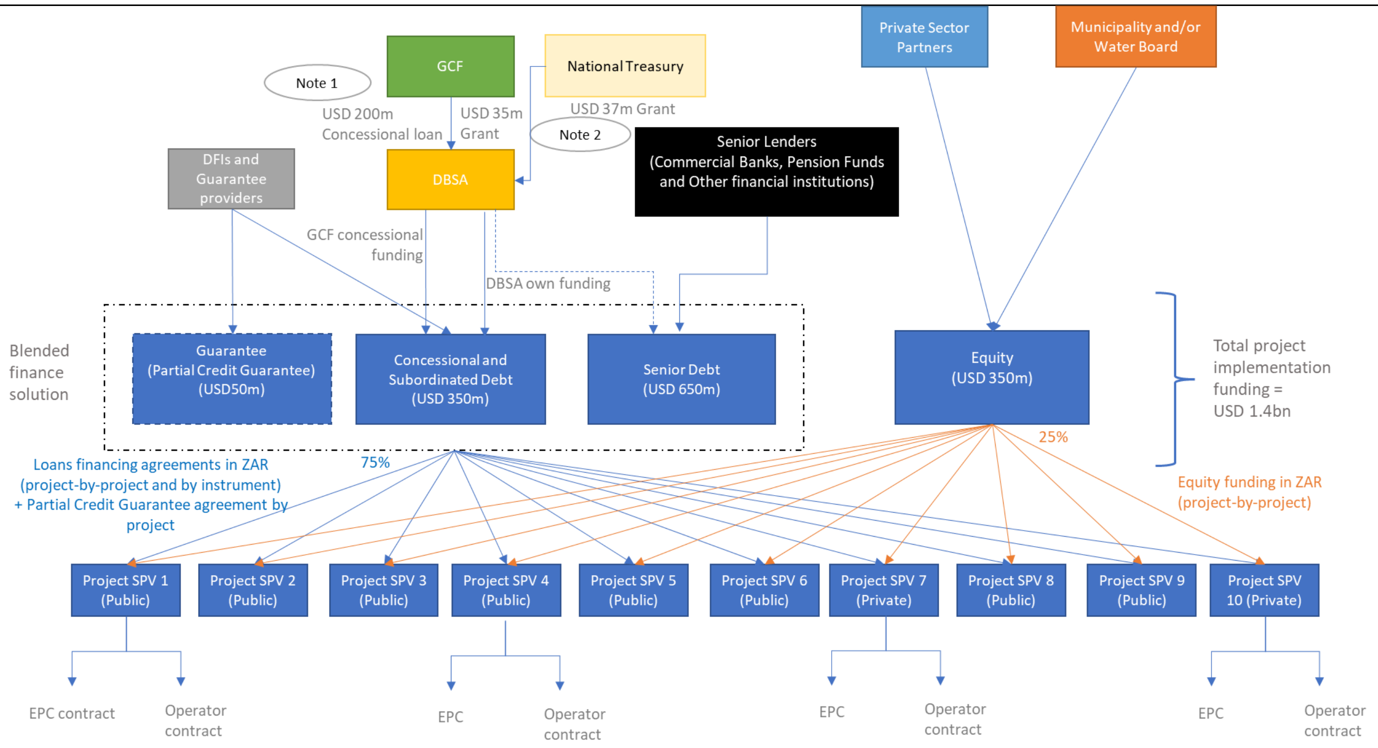


Figure 16: WRP Funding Flows

It is important to note that:

1. The GCF does not directly finance the underlying water reuse sub-projects. The GCF provides loans (at a Programme Level) to the Accredited Entity (the DBSA), who facilitates the GCF Concessional loan(s) into the underlying sub-projects, as required. The GCF will enter into a Funded Activity Agreement (FAA) with the DBSA, in the form of a loan agreement, with limited recourse with respect to the repayment risk of Loans. The GCF will provide the Concessional Loans in USD to DBSA in two tranches (i) a public sector tranche amounting to USD130m and ii) a private sector tranche amounting to USD 70m. These two tranches will be priced separately. The WRP's underlying water reuse sub-projects are financed in SA Rand (ZAR), which is overseen by the DBSA. The DBSA will obtain the ZAR equivalent swap rate of GCF concessional pricing to be passed onto sub-projects. The swap equivalent pricing will be obtained from a commercial bank, but no hedging agreements will necessarily be concluded. Therefore, all debt repayments from sub-projects will be in ZAR, and the currency exposure from this funding required to repay the GCF concessional capital will be managed by DBSA. The DBSA shall transfer to the GCF all loan repayments in US Dollars and manage the GCF funding's USD exposure at a DBSA balance sheet level.
2. A GCF grant of USD 35 million is primarily intended for project preparation (USD 30m) and capacity develop (USD 5m). The South African National Treasury and other providers of project preparation funding will co-finance the ZAR-equivalent of USD 37m through Grant/Budget allocations for project preparation (USD 30m), capacity develop (USD 5m) and WRP office core operating costs (USD 2m).

The legal and contractual arrangements for the WRP are provided in Figure 17, below.

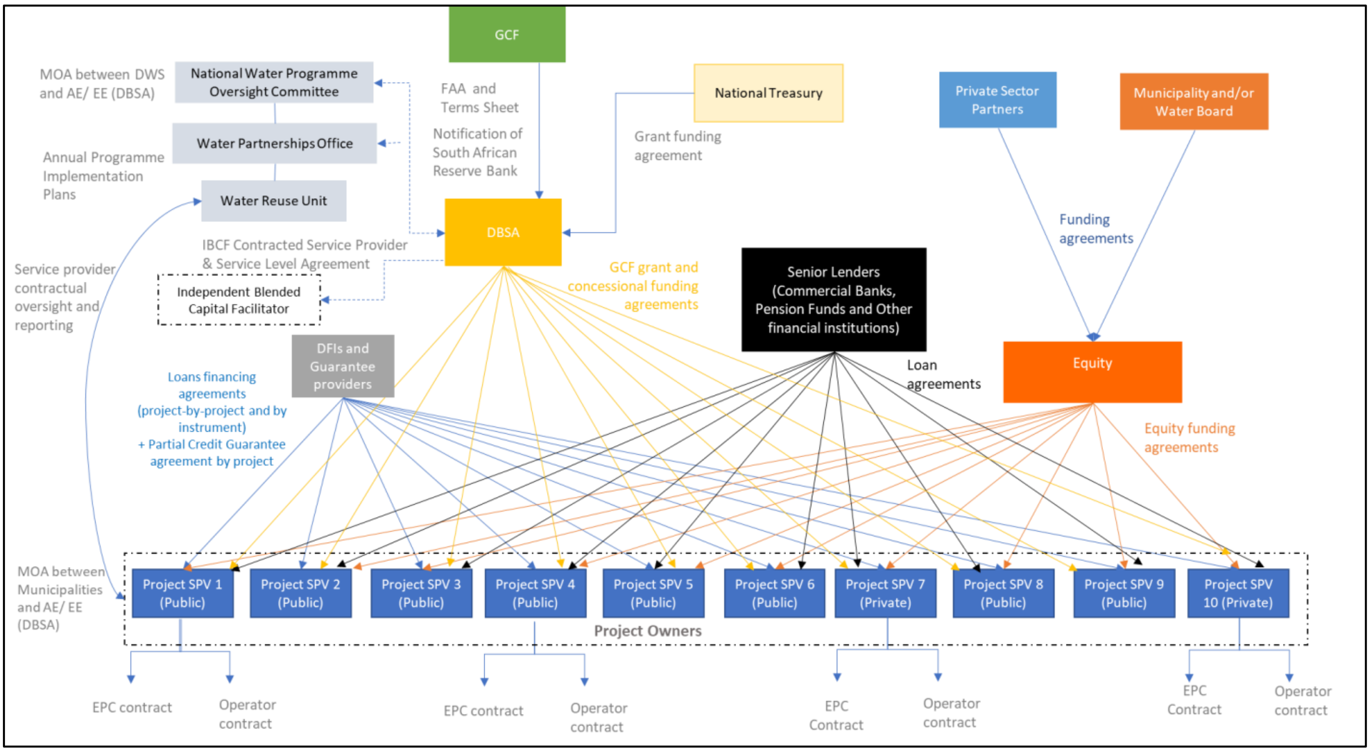


Figure 17: Legal and contractual arrangements

There is likely to be significant risk from fluctuating currency, between the USD and ZAR, over the repayment periods of the debt capital (envisaged to be 10 years or longer). Nonetheless, based on the DBSA's extensive experience in managing hard currency denominated funding provided by International Finance Institutions, the DBSA has the operational and institutional capacity in managing the USD exposure at a balance sheet level. A shortfall in USD may arises as a result of the Accredited Entity having to unwind relevant currency management arrangements and having to convert relevant ZAR amounts recovered from or paid by the relevant Sub-Borrower into USD at the then prevailing spot rate relating to: (a) a payment default by the relevant Sub-Borrower, and (b) and prepayment by the relevant Borrower, in each case under a Sub-Loan, with the relevant shortfall being calculated by reference to the amount of USD that the Accredited Entity would have received pursuant to the relevant currency management arrangements had the Sub-Borrower paid relevant amounts when due under the relevant Sub-Loan. The following foreign exchange solution will apply:

- If, during the first nine (9) years (i.e. end of year 9) following the effectiveness of the FAA (the "Reserve Period"), a Residual foreign exchange shortfall materializes, the relevant amount shall, in accordance with an appropriate mechanism to be set out in the FAA, be deducted from the Available Reserve Amount to the extent that the Available Reserve Amount is sufficient (each such deduction being a "Reserve Utilization") and be deemed to have been repaid to GCF by DBSA so as to satisfy the Residual foreign exchange shortfall (to the extent that the Available Reserve Amount is sufficient). Each Reserve Utilization shall reduce the available commitment under the GCF Loan by an equivalent amount.
- If at the end of the Reserve Period, the Available Reserve Amount is more than zero US Dollars (USD 0), such amount shall be made available for disbursement to the Accredited Entity until the closing date of the GCF Loan (closing date to be defined in the FAA).
- If the Available Reserve Amount falls to zero USD (USD 0), the risk of any subsequent Residual foreign exchange shortfall shall be borne by GCF.

The terms of the public sector sub-loans shall comply with the following provisions given in Table 12.

Table 12: Terms of public sector sub-loans

1.	**Facility Type**	Public Sector Sub-Loans (Component 2)
2.	**Sub-Project Approval Process**	• Governance structures will be set up to include the WRP Climate Screening Committee which will assess applications; the committee will include internal (DBSA) and external members. • The committee will utilize criteria and a screening tool to appraise and score applications for their climate change basis, based on the relevant climate change information alongside other drivers of water scarcity, as the first step in the screening process. • Projects which are able to demonstrate that climate change is a significant factor in water scarcity within their domain will proceed to due diligence with the view of funding; projects with insufficient climate change or water demand information will be reviewed for potential project preparation support should the screening tool support this. • Projects that complete the project preparation phase (PP) will be re-evaluated based on the available information to take them to implementation • All proposed sub-projects will first be reviewed and assessed by the WPO Oversight Committee • The Accredited Entity's Investment Committee will have the final decision on the approval of Sub-Projects and will not be bound by the recommendations of the Steercom. • The Accredited Entity will act as the lender of record of the Loans.
3.	**Amount**	The aggregate amount of the GCF Loan to be on lent by the Accredited Entity for the Public Sector Sub-Loans will be up to one hundred and thirty million Dollars (USD 130,000,000). For any Sub-Project, the committed concessional debt, including from DFIs, shall not exceed the committed senior debt. In instances where DBSA or other Co-financiers will provide credit enhancement into Sub-Projects, the DBSA loan and/or other subordinated loans will rank pari-passu with GCF. The amount of each individual Sub-Loan, at the time of its commitment, shall not exceed the lessor of: (1) fifteen per cent (15%) of the total Sub-Project cost; and (2) USD 30 million The credit exposure (loan co-financing ratio) of each investor in the Facility (i.e. GCF, the Accredited Entity and other Co-Financiers) under each individual Loan shall be proportional to the total committed contribution of each investor in the Facility, and **such ratio shall be maintained throughout the availability period of the Loans.**
4.	**Purpose of the facility**	Financing of public sector Sub-Projects that meet the Eligibility Criteria of the Programme.
5.	**Sub-Borrower**	Public Sector Sub-Borrowers that meet the Eligibility Criteria of the Programme.
6.	**Lender**	DBSA
7.	**Currency**	ZAR
8.	**Repayment Schedule**	Quarterly or semi-annually for all reflows during the relevant tenor of the Public Sector Sub-Loan.
9.	**Pricing**	Pricing shall comply with the principles indicated in item 14(e)(ii) of this Term Sheet and the principle of minimum concessionality.
10.	**Tenor**	In each Sub-Project, the tenor of the Public Sector Sub-Loan shall not exceed the lesser of: 1) Twenty (20) years from the effectiveness of the relevant Public Sector Sub-Loan agreement; 2) The tenor of the Water Purchase Agreement (WPA) provided to the relevant Sub-Project minus a three (3) year tail period and 3) The remaining tenor of the FAA.
11.	**Grace Period**	Principal repayment of the Loans will be subject to a maximum grace period of three (3) years, to be decided on case by case basis by the Accredited Entity taking into account the needs of the Sub-Project. For the avoidance of doubt the grace period does not apply for interest payments and fees. Interest repayment of the Loans, to be decided on case by case basis by the Accredited Entity taking into account the needs of the Sub-Project shall not exceed the lesser of: • a maximum grace period of three (3) years, • The Sub-Projects Commercial Operations date

12.	**Currency conversions for the disbursement and payment of GCF Tranches**	At each disbursement the DBSA will enter into a currency management solution converting the USD received from GCF to ZAR, securing an exchange rate for a particular disbursement to be used for the duration of the facility.
13.	**Pre-payment**	The Accredited Entity shall decide on any applicable prepayment penalty or fee. Any prepayment penalties or fees charged by the Accredited Entity shall apply proportionally to each co-financed tranche under the Loans.
14.	**Security structure**	Standard and customary for this type of international project finance transactions of this nature, including but not limited to second ranking security / charge over project assets and sponsor(s)' holding in the project, assignment of project contracts, documents and approvals.
15.	**Fees**	• Commitment fee will be of minimum 0.25% (twenty five basis points) accrued starting from financial close of the relevant Public Sector Sub-Loan, on the undrawn commitment of the relevant Public Sector Sub-Loan, to be payable on each payment date, together with interest rate payment. • Upfront fees, in the amount to be decided by the Accredited Entity, which shall not exceed one per cent 1% of the sub-Loan amount. **Additional Fees:** In case any additional fees are charged by the Accredited Entity under the Loans, the Accredited Entity shall distribute such payments to each sub-loan tranche (i.e. GCF, the Accredited Entity) proportionally to the amount of their relevant co-financed tranche under the Loans. Upfront fees and any other fees, if any to be shared pro-rata between the DBSA and GCF.
16.	**Waterfall provisions**	Any cash surplus after meeting all (i) operating and statutory expenses, (ii) senior debt obligations due to LFIs, and (iii) debt obligations to the Accredited Entity under the sub-Loans, will be deployed by the Borrowers in the following order of priority: (1) Creation and/or top up of reserve accounts (including but not limited to DSRA, any other reserves as may be applicable); and (2) Cash sweep in the following manner: Pro rata between the debt providers (i.e. LFIs and the Accredited Entity, as the lender of record of the Loans) and the Equity Providers in the proportion of the debt: equity ratio of the relevant Sub-Project.
17.	**Eligibility criteria**	The Accredited Entity shall only approve Sub-Projects that comply with the Eligibility Criteria.
18.	**Structuring criteria**	The structuring criteria for each sub-project shall include customary requirements for international project finance transactions or balance sheet lending transactions Total Borrower debt (i.e. senior and subordinated) to be sized based on the following considerations: (a) The amount that results in a minimum Total Debt Service Coverage Ratio ("**TDSCR**") (in relation to total senior and subordinated debt)) of **1.20x-1.25 x** for all Borrowers, provided that: (i) A minimum DSRA of six (6) to nine (9) months of total debt service for the Loan until twenty-five per cent (25%) of the relevant outstanding amount of the total debt is repaid at which point the DSRA requirement may reduce to a minimum of six (6) months debt service (for the residual life of the relevant Loan) depending on the prevailing financial and economic outlook of the Sub-Project at that point in time. (b) Other appropriate requirements will be added by the Accredited Entity at the due diligence stage of each Sub-Project, including among others appropriate level of contingencies to cover for delay and cost overruns risk.
19.	**Equity & equity support**	Minimum equity of twenty per cent (20%) of total Sub-Project costs contributed by the relevant Equity Provider(s) in the form of base equity at the Sub-Project level. The base equity will be either: (i) fully contributed upfront to the Sub-Project as a condition precedent to the first disbursement under the relevant Loan; or (ii) contributed pro-rata with disbursements under the relevant Loan, subject to the balance supported by the provision of a Letter of Credit from a financial institution acceptable to the Accredited Entity. Such letter of credit shall be without recourse to Sub-Project's assets In addition to the base equity, on a case-by-case basis the Accredited Entity may assess the need to have contingent equity commitments from the Equity Provider(s) to cover construction and/or operational risks.

20.	**Share retention**	The participating private sector investors and/or participating Municipality will together, directly or indirectly, maintain ownership of one hundred percent (100%) of all equity interests in the Sub-Project, till the completion of construction phase of the Sub-Project, it being agreed that private sector investors shall in no case hold 50% or more of the equity in the Sub-Project. The Accredited Entity shall ensure appropriate provisions are in place such that participating private sector investors and /or participating Municipality retain their economic interest in the Sub-Project for a reasonable period of time, compulsorily covering the operations stabilisation period. Any sale of the economic interest by the participating private sector investors and/or participating Municipality be subject to a prior written approval of the Accredited Entity and shall be to such a party which is acceptable to the Accredited Entity and subject to satisfactory KYC, AML, CFT and any appropriate checks. For the avoidance of doubt, sale of economic interest by a participating private sector investor, shall only be permitted with the prior written approval of the Accredited Entity.
21.	**Restricted payments**	The Accredited Entity shall include restricted payment provisions in financing agreements and such provisions shall be exercised in a manner which is not prejudicial to the interests of the Accredited Entity (as lender of record of the Sub-Loan). No dividends, bonuses or extraordinary compensation shall be permitted to be declared or paid unless the following conditions have been complied with (**Restricted Payment Conditions**): (i) the construction phase completion has occurred; (ii) the twelve-month historical and forward Sub-Project DSCR are not less than 1.25x; (iii) DSRA and other reserve accounts filled in as stipulated; (iv) the first two scheduled principal repayments of GCF Loans have been made according to the financing documents; (v) no event of default or of non-compliance has occurred and is continuing under any agreements; (vi) any other conditions to be added based on the specific structure of each Sub-Project; and (vii) the payment is in line with the cash sweep provisions under item 16.
22.	**Pre-execution conditions**	The Accredited Entity shall include pre-execution conditions in financing agreements and such provisions shall be exercised in a manner which is not prejudicial to the interests of the Accredited Entity (as lender of record of the GCF Loan. Prior to or at the time of execution of loan documentation for a Sub-Project, all the required debt shall be fully and legally committed to the Sub-Project.
23.	**Pre-disbursement conditions**	The Accredited Entity shall include pre-disbursement conditions in financing agreements and such provisions shall be exercised in a manner which is not prejudicial to the interests of the Accredited Entity (as lender of record of the GCF Loan. Disbursement of the Public Sector Sub-Loans will be conditioned upon the Sub-Project(s) complying with customary conditions precedent, including among others: (i) contribution of equity as per the sponsors' equity contribution provisions; (ii) payment of all fees and expenses due; (iii) execution and delivery of all financing documents and legal opinions; (iv) receipt of lenders' advisor reports (environmental, engineering, insurance, etc.), as applicable; (v) all Sub-Project documents, implementation agreement(s), water purchase agreement(s), distribution contracts and arrangements, insurance contracts and hedging contracts are in full force and effect; (vi) the required security for the lenders to have been created and perfected where possible; (vii) receipt of all material permits, licenses and government approvals related to the construction of the Sub-Project in a final and non-appealable form; (viii) no material adverse change; (ix) no event of default / potential event of default; (x) receipt of organizational documents and officer's certificates; (xi) schedule of permits required for construction and operation of the Sub-Project including but not limited to land rights, environmental permits, governmental and statutory approvals, as deemed necessary by the Accredited Entity; (xii) satisfactory Sub-Project budget and Sub-Project schedule; and (xiii) Creation of reserve accounts, as applicable.
24.	**Other financing tranches**	The Accredited Entity will engage with the other financial institutions, including but not limited to multilateral development banks, development finance institutions and LFIs to secure financing for any co-financing gap. Such financing - may rank senior to the Public Sector Sub-Loan in terms of security. Such financing may be provided in local currency.

25.	**Currency Risk**	Currency risk shall be managed inline with section 2b(1), 1(d)(ii) and 16 of this term sheet
26.	**Insurance**	The Accredited Entity shall include insurance conditions (project insurance against loss, damage and liability in a manner and with insurers satisfactory to the Accredited Entity during the tenor of the Public Sector Sub-Loan) in financing agreements and such provisions shall be exercised in a manner which is not prejudicial to the interests of the Accredited Entity (as lender of record of the Sub-Loan. The Sub-Projects shall maintain project insurance against loss, damage and liability in a manner and with insurers satisfactory to the Accredited Entity during the tenor of the Public Sector Sub-Loan, ensuring that all the lenders are beneficiaries of any loss proceeds under the insurance.
27.	**Other terms**	Other requirements customary for international project finance transactions to be added during the due diligence stage taking into account the characteristics of the contractual framework.
28.	**Subordination of payments**	The GCF Tranche shall rank *pari passu* with the DBSA Tranche and/or the lenders providing subordinated loans' tranche, if any, in all respects, including the same repayment schedule.
29.	**Treatment of losses**	To the extent losses occur in excess of reserve (i.e. the Debt Service Reserve Account ("**DSRA**"), Maintenance Reserve Account (where applicable) and any other reserves as may be applicable), those losses will be allocated to all Facility investors (i.e. GCF, the Accredited Entity and, if any, other Co-Financiers) proportionally to the relevant investor's tranche in the Loans.
30.	**Reserve Accounts**	The Debt Service Reserve Account ""**DSRA**" to cover at least 6 months of total debt service for loan, Maintenance Reserve Account (where applicable), to cover operation and maintenance costs of the Sub-Project.
31.	**Payment of Interest Frequency**	Interest over the outstanding amount under the Loans shall begin to accrue as of the date of first disbursement under the relevant Loan and shall be payable on semi-annually, quarterly or monthly basis.
32.	**Taxation**	The Accredited Entity shall include gross-up clauses or other equivalent provisions in the relevant legal agreements for the Loans, so that it receives the full amount of Reflowed Funds under the GCF Tranches clear and net of any applicable Taxes.

The terms of the private sector sub-loans shall comply with the following provisions given in Table 13.

Table 13: Terms of public sector sub-loans

1.	**Facility Type**	Private Sector Sub-Loans (Component 2)
2.	**Sub-project Approval Process**	• Governance structures will be set up to include the WRP Climate Screening Committee which will assess applications; the committee will include internal (DBSA) and external members. • The committee will utilize criteria and a screening tool to appraise and score applications for their climate change basis, based on the relevant climate change information alongside other drivers of water scarcity, as the first step in the screening process. • Projects which are able to demonstrate that climate change is a significant factor in water scarcity within their domain will proceed to due diligence with the view of funding; projects with insufficient climate change or water demand information will be reviewed for potential project preparation support should the screening tool support this. • Projects that complete the project preparation phase (PP) will be re-evaluated based on the available information to take them to implementation • All proposed sub-projects will first be reviewed and assessed by the WPO Oversight Committee • The Accredited Entity's Investment Committee will have the final decision on the approval of Sub-Projects and will not be bound by the recommendations of the Steercom. • The Accredited Entity will act as the lender of record of the Loans.
3.	**Amount**	The aggregate amount of the GCF Loan to be on lent by the Borrower for the Private Sector Sub-Loans will be up to seventy million Dollars (USD 70,000,000). For any Sub-Project, the committed concessional debt, including from DFIs, shall not exceed the committed senior debt. In instances where DBSA or other Co-financiers will provide credit enhancement (through first loss/subordinated loan, tenor extension) into private sector Sub-Projects, the DBSA loan and/or other subordinated loans will rank pari-passu with GCF. The amount of each individual Sub-Loan, at the time of its commitment, shall not exceed the lessor of: (1) fifteen per cent (15%) of the total Sub-Project cost; (2) USD 30 million

		The credit exposure (loan co-financing ratio) of each investor in the Facility (i.e. GCF, the Accredited Entity and other Co-Financiers) under each individual Loan shall be proportional to the total committed contribution of each investor in the Facility, and such ratio shall be maintained throughout the availability period of the Loans.
4.	**Purpose of the facility**	Financing of private sector Sub-Projects that meet the Eligibility Criteria of the Programme.
5.	**Sub-Borrower**	Private Sector Borrowers that meet the Eligibility Criteria of the Programme.
6.	**Lender**	DBSA
7.	**Currency**	ZAR
8.	**Repayment Schedule**	Quarterly or semi-annually for all reflows during the relevant tenor of the Private Sector Sub-Loan.
9.	**Pricing**	Pricing shall comply with the principles indicated in item 14(e)(ii) of this Term Sheet and the principle of minimum concessionality.
10.	**Tenor**	In each Sub-Project, the tenor of the Private Sector Sub-Loan shall not exceed the lesser of: 1) Twenty (20) years from the effectiveness of the relevant Private Sector Sub-Loan agreement; 2) The tenor of the Water Purchase Agreement (WPA) provided to the relevant Sub-Project minus a three (3) year tail period and 3) The remaining tenor of the FAA.
11.	**Grace Period**	Principal repayment of the Loans will be subject to a maximum grace period of three (3) years, to be decided on case by case basis by the Accredited Entity taking into account the needs of the Sub-Project. For the avoidance of doubt the grace period does not apply for interest payments and fees. Interest repayment of the Loans, to be decided on case by case basis by the Accredited Entity taking into account the needs of the Sub-Project shall not exceed the lesser of: • a maximum grace period of three (3) years, • The Sub-Projects Commercial Operations date
12.	**Currency conversions for the disbursement and payment of GCF Tranches**	At each disbursement the DBSA will enter into a currency management solution converting the USD received from GCF to ZAR, securing an exchange rate for a particular disbursement to be used for the duration of the facility.
13.	**Pre-payment**	The Accredited Entity shall decide on any applicable prepayment penalty or fee. Any prepayment penalties or fees charged by the Accredited Entity shall apply proportionally to each co-financed tranche under the Loans.
14.	**Security structure**	Standard and customary for this type of international project finance transactions of this nature, including but not limited to second ranking security / charge over project assets and sponsor(s)' holding in the project, assignment of project contracts, documents and approvals.
15.	**Fees**	• Commitment fee of minimum 0.25% (twenty five basis points) accrued starting from financial close of the relevant Private Sector Sub-Loan, on the undrawn commitment of the relevant Private Sector Sub-Loan, to be payable on each payment date, together with interest rate payment. • Upfront fees, in the amount to be decided by the Accredited Entity, which shall not exceed one per cent 1.5 % of the sub-Loan amount **Additional Fees:** In case any additional fees are charged by the Accredited Entity under the Loans, the Accredited Entity shall distribute such payments to each sub-loan tranche (i.e. GCF, the Accredited Entity) proportionally to the amount of their relevant co-financed tranche under the Loans. Upfront fees and any other fees, if any to be shared pro-rata between the DBSA and GCF.
16.	**Waterfall provisions**	Any cash surplus after meeting all (i) operating and statutory expenses, (ii) senior debt obligations due to LFIs, and (iii) debt obligations to the Accredited Entity under the Loans, will be deployed by the Borrowers in the following order of priority: (1) Creation and/or top up of reserve accounts (including but not limited to DSRA, any other reserves as may be applicable); and (2) Cash sweep in the following manner: Pro rata between the debt providers (i.e. LFIs and the Accredited Entity, as the lender of record of the Loans) the Equity Providers in the proportion of the debt: equity ratio of the relevant Sub-Project.

17.	**Eligibility criteria**	The Accredited Entity shall only approve Sub-Projects that comply with the Eligibility Criteria.
18.	**Structuring criteria**	The structuring criteria for each sub-project shall include customary requirements for international project finance transactions or balance sheet lending transactions. Total Borrower debt (i.e. senior and subordinated) to be sized based on the following considerations: (a) The amount that results in a minimum Total Debt Service Coverage Ratio ("**TDSCR**") (in relation to total senior and subordinated debt)) of **1.20x-1.25 x** for all Borrowers, provided that: (i) A minimum DSRA of six (6) to nine (9) months of total debt service for the Loan until twenty-five per cent (25%) of the relevant outstanding amount of the total debt is repaid at which point the DSRA requirement may reduce to a minimum of six (6) months debt service (for the residual life of the relevant Loan) depending on the prevailing financial and economic outlook of the Sub-Project at that point in time. (b) Other appropriate requirements will be added by the Accredited Entity at the due diligence stage of each Sub-Project, including among others appropriate level of contingencies to cover for delay and cost overruns risk.
19.	**Equity & equity support**	Minimum equity of twenty per cent (20%) of total Sub-Project costs contributed by the relevant Equity Provider(s) in the form of base equity at the Sub-Project level. The base equity will be either: (i) fully contributed upfront to the Sub-Project as a condition precedent to the first disbursement under the relevant Loan; or (ii) contributed pro-rata with disbursements under the relevant Loan, subject to the balance supported by the provision of a Letter of Credit from a financial institution acceptable to the Accredited Entity. Such letter of credit shall be without recourse to Sub-Project's assets In addition to the base equity, on a case-by-case basis the Accredited Entity may assess the need to have contingent equity commitments from the Equity Provider(s) to cover construction and/or operational risks.
20.	**Share retention**	The participating private sector investors and /or participating Municipality will together, directly or indirectly, maintain ownership of one hundred percent (100%) of all equity interests in the Sub-Project, till the completion of construction phase of the Sub-Project, it being agreed that private sector investors shall in all cases hold 50% of more of the equity in the relevant Sub-Project. The Accredited Entity shall ensure appropriate provisions are in place such that participating private sector investors and /or participating Municipality retain their economic interest in the Sub-Project for a reasonable period of time, compulsorily covering the operations stabilisation period. Any sale of the economic interest by the participating private sector investors and/or participating Municipality be subject to a prior written approval of the Accredited Entity and shall be to such a party which is acceptable to the Accredited Entity and subject to satisfactory KYC, AML, CFT and any appropriate checks. For the avoidance of doubt, sale of economic interest by a participating private sector investor, shall only be permitted with the prior written approval of the Accredited Entity
21.	**Restricted payments**	The Accredited Entity shall include restricted payment provisions in financing agreements and such provisions shall be exercised in a manner which is not prejudicial to the interests of the Accredited Entity (as lender of record of the GCF Loan. [No dividends, bonuses or extraordinary compensation shall be permitted to be declared or paid unless the following conditions have been complied with (**Restricted Payment Conditions**): (i) the construction phase completion has occurred; (ii) the twelve-month historical and forward Sub-Project DSCR are not less than 1.25x; (iii) DSRA and other reserve accounts filled in as stipulated; (iv) the first two scheduled principal repayments of GCF Loans have been made according to the financing documents; (v) no event of default or of non-compliance has occurred and is continuing under any agreements; (vi) any other conditions to be added based on the specific structure of each Sub-Project; and (vii) the payment is in line with the cash sweep provisions under item 16.
22.	**Pre-execution conditions**	The Accredited Entity shall endeavour to include pre-execution conditions in financing agreements and such provisions shall be exercised in a manner which is not prejudicial to the interests of the Accredited Entity (as lender of record of the GCF Loan. Prior to or at the time of execution of loan documentation for a Sub-Project, all the required debt shall be fully and legally committed to the Sub-Project.

23.	**Pre-disbursement conditions**	The Accredited Entity shall include pre-disbursement conditions in financing agreements and such provisions shall be exercised in a manner which is not prejudicial to the interests of the Accredited Entity (as lender of record of the GCF Loan. Disbursement of the Private Sector Sub-Loan will be conditioned upon the Sub-Project(s) complying with customary conditions precedent, including among others: (i) contribution of equity as per the sponsors' equity contribution provisions; (ii) payment of all fees and expenses due; (iii) execution and delivery of all financing documents and legal opinions; (iv) receipt of lenders' advisor reports (environmental, engineering, insurance, etc.), as applicable; (v) all Sub-Project documents, implementation agreement(s), water purchase agreement(s), distribution contracts and arrangements, insurance contracts and hedging contracts are in full force and effect; (vi) the required security for the lenders to have been created and perfected where possible;; (vii) receipt of all material permits, licenses and government approvals related to the construction of the Sub-Project in a final and non-appealable form; (viii) no material adverse change; (ix) no event of default / potential event of default; (x) receipt of organizational documents and officer's certificates; (xi) schedule of permits required for construction and operation of the Sub-Project including but not limited to land rights, environmental permits, governmental and statutory approvals, as deemed necessary by the Accredited Entity; (xii) satisfactory Sub-Project budget and Sub-Project schedule; and (xiii) Creation of reserve accounts, as applicable.
24.	**Other financing tranches**	The Accredited Entity will engage with the other financial institutions, including but not limited to multilateral development banks, development finance institutions and LFIs to secure financing for any co-financing gap. Such financing - may rank senior to the Private Sector Sub-Loan in terms of security. Such financing may be provided in local currency.
25.	**Currency Risk**	Currency risk shall be managed inline with section 2b(1), 1(d)(ii) and 16 of this term sheet
26.	**Insurance**	The Accredited Entity shall endeavour to include insurance conditions (project insurance against loss, damage and liability in a manner and with insurers satisfactory to the Accredited Entity during the tenor of the Public Sector Sub-Loan) in financing agreements and such provisions shall be exercised in a manner which is not prejudicial to the interests of the Accredited Entity (as lender of record of the Sub-Loan. The Sub-Projects shall maintain project insurance against loss, damage and liability in a manner and with insurers satisfactory to the Accredited Entity during the tenor of the Private Sector Sub-Loan, ensuring that all the lenders are beneficiaries of any loss proceeds under the insurance.
27.	**Other terms**	Other requirements customary for international project finance transactions to be added during the due diligence stage taking into account the characteristics of the contractual framework.
28.	**Subordination of payments**	The GCF Tranche shall rank *pari passu* with the DBSA Tranche and/or the lenders providing subordinated loans' tranche, if any, in all respects, including the same repayment schedule.
29.	**Treatment of losses**	To the extent losses occur in excess of reserve (i.e. the Debt Service Reserve Account ("**DSRA**"), Maintenance Reserve Account (where applicable) and any other reserves as may be applicable), those losses will be allocated to all Facility investors (i.e. GCF, the Accredited Entity and, if any, other Co-Financiers) proportionally to the relevant investor's tranche in the Loans.
30.	**Reserve Accounts**	the Debt Service Reserve Account "**DSRA**" to cover at least 6 months of total debt service for loan, Maintenance Reserve Account (where applicable), to cover operation and maintenance costs of the Sub-Project;
31.	**Payment of Interest Frequency**	Interest over the outstanding amount under the Loans shall begin to accrue as of the date of first disbursement under the relevant Loan and shall be payable on a semi-annual, quarterly or monthly basis.
32.	**Taxation**	The Accredited Entity shall include gross-up clauses or other equivalent provisions in the relevant legal agreements for the Loans, so that it receives the full amount of Reflowed Funds under the GCF Tranches clear and net of any applicable Taxes.

To conceptualise the financing of numerous water reuse sub-projects in a programme, a standard project finance approach is first conceptualised. Importantly, the standard approach hinges on the type of delivery model being applied for each sub-project. Delivery models are wholly dependent on the asset owner or sponsor; however, in the absence of knowing these preferences, a simplified project finance approach is provided (**Error! Reference source not found.**).

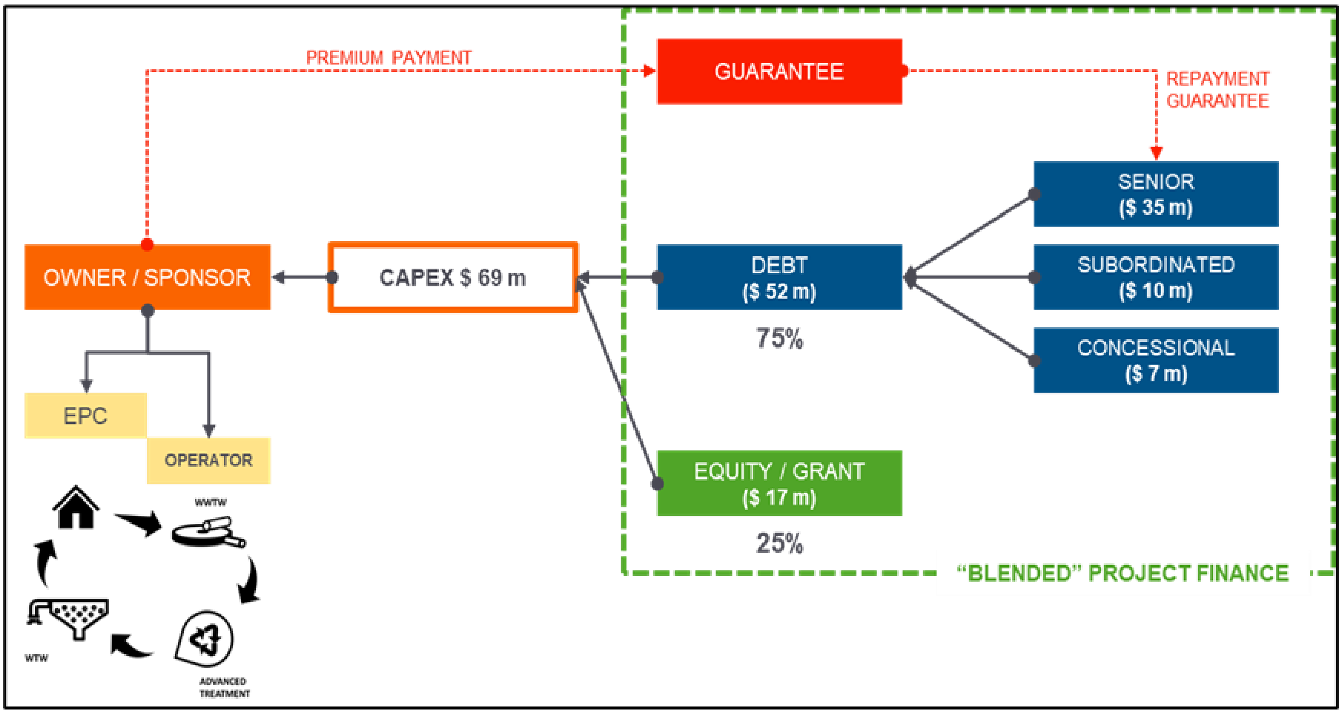


Figure 18: Simplified project finance approach

The simplified approach has the following key features:

- **Ownership:** The ownership model will be either a local government authority (municipality) or special purpose vehicle (SPV) – the latter comprising municipalities, the private sector, and/or possibly a national government entity (depending on size). In both instances, the municipalities will be meaningfully involved in the sub-project, providing resources such as integrated planning approval, land, in-kind human resources, and possibly capital budgets. It is also assumed that the owner will be the primary revenue collector, on behalf of the sub-project.
- **Engineering Procurement Contract (EPC) / Operator:** The construction and operations of the Water Reuse sub-projects will be undertaken by the private sector. This can either be undertaken through contract with a municipality or the project SPV, which is dependent on the ownership structure of the sub-project. The operations of the plant, against a strict performance specification, will be tantamount in securing private capital.
- **Capex:** A capex value of USD $69 million has been used to illustrate the splits between different capital sources. The simplified sub-project financing approach assumes that the opex will be covered through revenue collection, via water consumption tariffs.
- **Equity / Grant:** The split between debt and equity based upon desktop research and informal market soundings with private capital financiers, reflects a 75/25 split. This assumption applies in the instance when the project owner is likely to be an SPV. However, when the sub-project is wholly owned and delivered by a municipality it is assumed that 25% of the Capex will be covered through a budgeted grant, in keeping with their integrated development planning. This ensures that the same split is applied in the simplified project finance approach.
- **Debt:** The debt components of the simplified sub-project finance approach is where there is a significant opportunity to apply blended finance principles. A distinction is made between three types of debt: (a) Senior, (b) Subordinated, and (c) Concessional. These can be applied differentially,

without conflating the ranking of debt (senior vs subordinated) with its financial repayment expectations (commercial vs concessional). This introduces a blending narrative, which demonstrates that significant senior debt can be leverage when one intentionally blends with subordinated and/or concessional debt. This is illustrated computationally in the Programme Financial Model.

- **Guarantee:** A guarantee instrument to facilitate private sector investment into water reuse sub-projects forms part of the financial structure of the WRP. Potential funders and guarantee providers are being approached to fund the guarantee instrument.
- **Blended Project Finance:** The diagram illustrates that a blended finance approach can be achieved by using the various equity, debt and guarantee instruments. Importantly, the instruments are not mutually exclusive – their individual characterisation is wholly dependent on other instruments, i.e. bringing in subordinated debt will likely reduce the interest rate of senior, commercial debt, thereby increasing the amount of senior, commercial debt. The approach will be to maximise private capital investment in the WRP and ensure that the principle of least concessionality is applied. Therefore, the senior, commercial debt is prioritised higher than all other debt types, and the Programme Financial Model has been designed to solve for a higher private capital split in each underlying sub-project.

A co-funding contribution for project preparation from a municipality will be determined on a project-by-project basis but a minimum co-funding contribution towards project preparation costs will be 10%. For more financially strong municipalities a higher percentage will be required (up to a maximum of 50%).

<u>Proposed Partial Guarantee</u>

Partial credit guarantees (PCG) are widely used internationally to assist in alleviating funding constraints faced in public infrastructure development, specifically in environments where short to medium term liquidity risk is deemed to be significant, or there is concern around default risk. The nascent status of water reuse sub-projects in South Africa means project owners have limited or no access to affordable, unsecured funding from banks to implement public sector projects. Delivering water reuse sub-projects at a local and provincial level creates an added level of complexity and risk for funders who will look to the municipality/province to assess the risk of repayment.

The PCG conceptualised for the WRP will lower the risk to the lenders by substituting part of the risk of the project owner and public sector contracting party with that of the issuer of the PCG. The guarantee is envisaged to be for adverse market events only, during the operations period of the underlying sub-projects, whereas construction risk will be covered by the sponsors as per the norms for transactions of this nature. In addition, the guarantee fund envisaged for the WRP also seeks to lower the risk to both the lender and project owner in respect of delayed payments by the public sector (short term liquidity risk). Lastly, the partial credit guarantee also substitutes a portion of the lender's risk in the event of default.

Structure

It is envisaged that the guarantee will be a fully cash collateralized guarantee. The security of the PCG is provided to the senior debt issued via the blended finance solution. It will be administered and managed by the DBSA, and the DBSA will be the guarantor on record for each of the sub-projects that require a guarantee. This will be determined on a project-by-project basis, as is also the case for finance from the Blended Finance Solution. The PCG's cash collateral should only be reduced under one of the following adverse "events":

- A liquidity event;
- A public sector default event; or

- A default event by the project owner.

Figure 19 illustrates a conceptual structure for the proposed PCG.

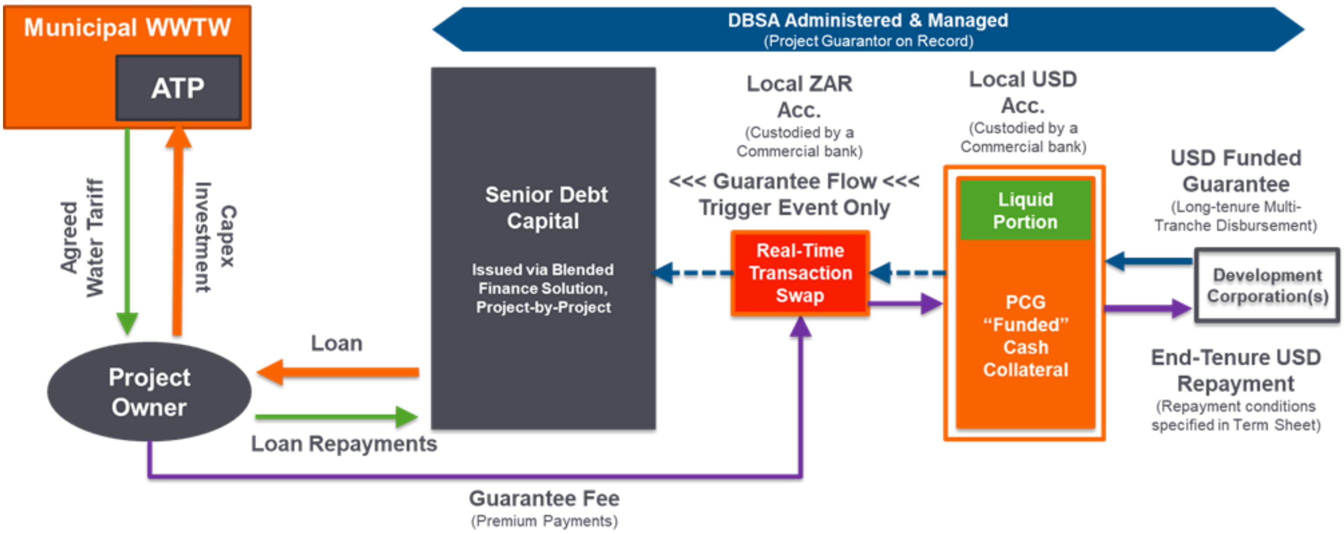


Figure19: Proposed early partial credit guarantee concept

With reference to the PCG concept, the DBSA will open a separate USD account with a reputable commercial bank. The guarantee shall be cash collateralized and provided through a letter of credit from the selected commercial bank. Importantly, the DBSA will not provide a corporate guarantee as lenders may require that additional due diligence (DD) is conducted on the DBSA, which should be avoided to ensure the process is simpler. The cash collateral is supplied by a development corporation , according to pre-agreed tranches.

DBSA will administer and manage the guarantee, and require a fee from the guarantee funder to do so. The selected commercial banks costs to custody the guarantee cash collateral will be included in DBSA's cost. The cash collateral will be held in USD until such time as it is required. When it is called upon, the portion that is called upon is swapped into ZAR and disbursed to according to the guarantee agreement at a sub-project level.

Furthermore, it is envisaged that the guarantee fund will support project owners during a "liquidity event". The Liquid Portion is a small portion of the total cash collateral that is available to cover a "liquidity event". The liquidity event is when a municipality or other off-taker does not fulfil its tariff payment to the Project Owner on time, meaning the Project Owner cannot make his/her/their loan repayment on time. In this instance the liquid portion covers this monthly loan repayment and recovers this exact cost from the Project Owner when they can make payment.

The Liquid Portion is shown as distinct from the total cash collateral, because it needs to be available to be paid as soon as it is required (i.e within 48hrs). We anticipate that the cash collateral will be custodied in fixed-income, notice deposit investment, as a default event typically takes many months to materialise. This is to ensure the cash collateral generates meaningful interest when it is not being called upon. The cash collateral should generate income over its tenure and provided that the returns earned on the collateral are positive in real terms, the fund's collateral should be protected against the erosive effect of inflation.

Adverse Events

The capital for the PCG will be exposed to a number of risks, including credit risk in respect of the project owner that is entering into a loan; credit risk of the municipality/province that will be contracting with the project owner; and liquidity risk in respect of temporary non-payment by a municipality or province.

Ideally the capital for the PCG should be managed to diversify its portfolio as much as possible by avoiding significant exposures to the following categories:

- A single municipality; and
- A single project owner.

The PCG will take on the following obligations in the event of a liquidity event or default (Table 14).

Table 14: PCG obligations

Event	Description	Obligation of the PCG	Project Owner's obligation
Liquidity event	Payment from public sector entity to the project owner is overdue by a pre-agreed time (for example 60 days)	The DBSA can access the PCG meet capital and interest repayments until the municipality/ province resumes payments. The total aggregate payments will be capped at the value of the guarantee.	Project owner to reimburse capital and interest upon receipt of overdue payment(s) to the PCG account
Public sector default event	Payment from the public sector entity to the project owner becomes impossible.	The full amount guaranteed, minus any payments made in respect of a liquidity event, can be transferred from the PCG account to reduce the portfolio loan.	The project owner remains liable to repay outstanding debt and interest amount less amount transferred from the collateral account. The GCF takes credit risk on this portion.
Project Owner default event	The project owner fails to meet debt repayments in the absence of a liquidity event or public sector default event.	The full amount guaranteed can be transferred from collateral account to reduce the portfolio loan.	The project owner remains liable to repay outstanding debt and interest amount less amount transferred from the collateral account. The GCF takes credit risk on this portion.

According to the Programme Financial Model, by combining GCF concessional loans with a partial credit guarantee, the minimum GCF funding requirement decreases by 7.0% to R2,950 or US$ 211 million, representing 20.77% of total project-related debt funding or 15.58% of total project-related capital at the programme level. The WRP will also be structured in a way that no one sub-project receives more than 30% concessional loans as a proportion of the total implementation capital; however, in exceptional cases, up to 50% will be reviewed where a particular sub-project requires it to ensure the ongoing viability of the entire WRP.

AE/ Executing Entity Experience

DBSA is a national entity based in South Africa with regional activity (www.dbsa.org). It is a development finance institution, with a mandate to finance both private and public sector activities at national and regional levels in Africa. DBSA provides sustainable infrastructure project preparation, finance and implementation support in order to improve the population's quality of life, accelerating the sustainable reduction of poverty and inequity, and promoting broad-based economic growth and regional economic integration. Since 2012 it has been implementing the Green Fund, with a focus on the climate financing.

From an infrastructure development and service delivery perspective, DBSA has a track record in programme and fund management and development which has facilitated the acceleration and realization of development impact from projects. It has experience in conceptualising, establishing and implementing large scale national and regional programmes, including the Student Housing Infrastructure Programme.

DBSA was accredited by the GCF in 2016, for large projects.

From a climate finance perspective, it is currently implementing as Accredited Entity and Executing Entity for two GCF approved proposals, including the Imbedded Generation Investment Programme (FP106 – 2019) and the DBSA Climate Finance Facility (FP098 – 2018). (Development Bank of Southern Africa (DBSA) | Green Climate Fund)

DBSA as accredited entity will ensure monitoring and reporting as required by the GCF, including the participation of Climate Finance Unit and the Monitoring and Evaluations Unit of the DBSA. The DBSA will offer programme management support and provide backroom management services as they have successfully done for other programmes such as the successful Renewable Energy Independent Power Producer Programme. The internal DBSA systems and policies will provide support to the execution of the WRP and will guide such aspects as recruitment and supply chain management and procurement. Likewise, the DBSA's policies with regards to gender, sexual harassment and sexual exploitation and abuse, amongst others, will guide the functioning of the WPO.

DBSA will also coordinate the work of third-party auditors engaged in the Programme to monitor and evaluate results. During the programme implementation DBSA will ensure that any subsidiary agreement that they enter into with the Programme custodian DWS in respect of a Funded Activity shall reflect or incorporate the terms and conditions of the FAA that may be applicable or relevant to the Executing Entity and the National Designated Authority.

Sub-project Compliance with Environment and Social Monitoring Framework (ESMF)

The ESMF which has been prepared for the WRP aims to avoid and minimise negative environmental and social (E&S) impacts and to enhance positive aspects of the Programme. The ESMF contains environmental and social baseline information at the national level and provides the framework and guidelines that ensure that the DBSA and the project owners (municipalities) are committed and able to comply with the revised GCF Environmental and Social Policies(B.BM-2021/18) and DBSA Environmental and Social Safeguard Standards. The ESMF is developed at a programme level and provides the framework within which specific ESMFs will be prepared at each sub-project level. The WPO will provide support to Municipalities to prepare projects for financing as well as to provide technical and financial support in project implementation. As such, the ESMF will be a key component of all project processes, with the WPO and the DBSA overseeing the use of the ESMF. In all sub-projects, an ESIA will be undertaken and an ESMP will be prepared.

The ESIA and ESMP will be subject to information disclosure requirements per GCF Revised Environmental and Social Policy and the Information Disclosure Policy. In order to develop the ESIA and ESMP required, the DBSA's own due diligence approaches will be applied. The ESIA and ESMP along with the ESS Disclosure Form will be submitted to GCF for clearance following the disclosure and aligned to the ESS requirements in the relevant GCF policies.

At sub-project levels the Social and Gender Oversight Officer and the Environmental Control Office (Figure 20) will report regularly on adherence and compliance with the ESMF requirements and safeguards at both programme and sub-project levels.

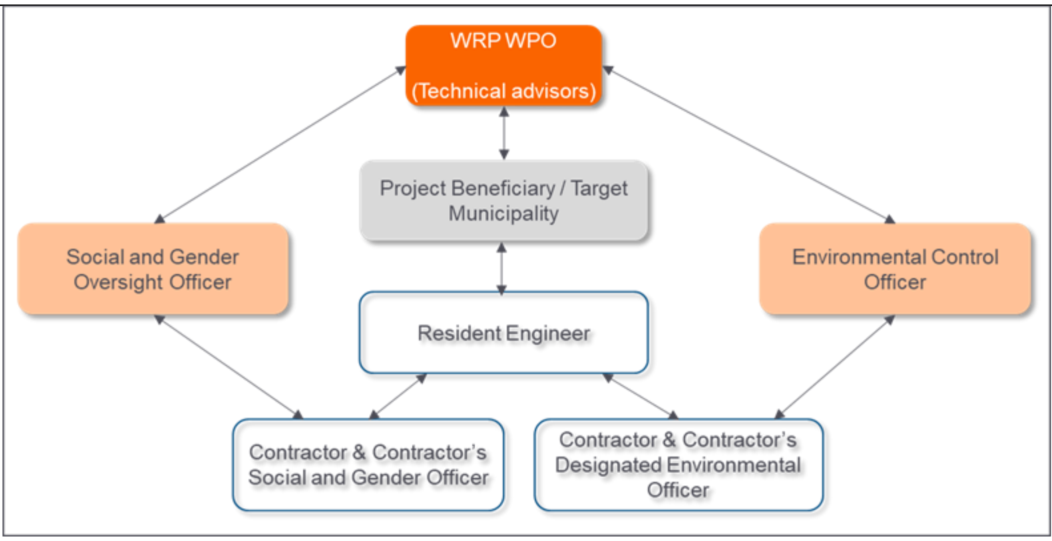


Figure 20: Sub-project arrangements to ensure compliance with the ESMF

B.5. Justification for GCF funding request (max. 1000 words, approximately 2 pages)

According to a study commissioned by Business Leadership South Africa, infrastructure investment in South Africa has fallen sharply over the past six years, from above 20% of Gross Domestic Product (GDP) in 2015 to below 18% in 2019, from the National Development Plan's target of 30% of GDP. Further, although availability (and access) to local commercial funding for infrastructure project development is present, the focus of continued investment has been primarily on renewable energy projects in South Africa through the success of the Renewable Independent Power Producer Programme (REIPP) Programme. Outside of that Programme, the development of bankable projects, especially in the water and water reuse sector, has been challenging.

The Government of South Africa through local implementing municipalities have little financial capacity to meet the initial costs of establishing a programme of this nature. Municipal-scale projects are capital-intensive and access to funding can be a major constraint, particularly for small, less financially stable municipalities. These often lack capacity to develop bankable projects and struggle to raise funds. Also, capital costs range significantly depending on effluent quality, treatment requirements and the reuse application. For example, local projects focussing on organic effluent treatment cost in the order of R 20 - R120 million per Ml/d produced, while those for inorganic effluent cost in the order of R 10-15 million per Ml/d produced (GreenCape, 2018). GCF grant and reimbursable grant funding, in additional to the concessional loan portion of the requested GCF funding is necessary to overcome this combination of financial, technical, institutional and market barriers that together prevent climate-vulnerable locations in South Africa from adequately building resilience to increasing climate hazards.

Existing capital pools in SA is a sign of buoyant market potential for infrastructure investing. Collectively, the deployment of private sector balance sheets is sized at over ZAR 20 trillion[3]. More than 80% of invested capital pools are concentrated with Banks (38%), Pension Funds (28,5%), and Life Insurers (17,5%). While the duration of assets held may vary, opportunities exist for this capital to find new assets as infrastructure

[3] SAVCA, National Treasury and ASISA (2022)

projects come to market. Figure 21 shows the quantum of funding deployed per sub-sector in the financial services industry.

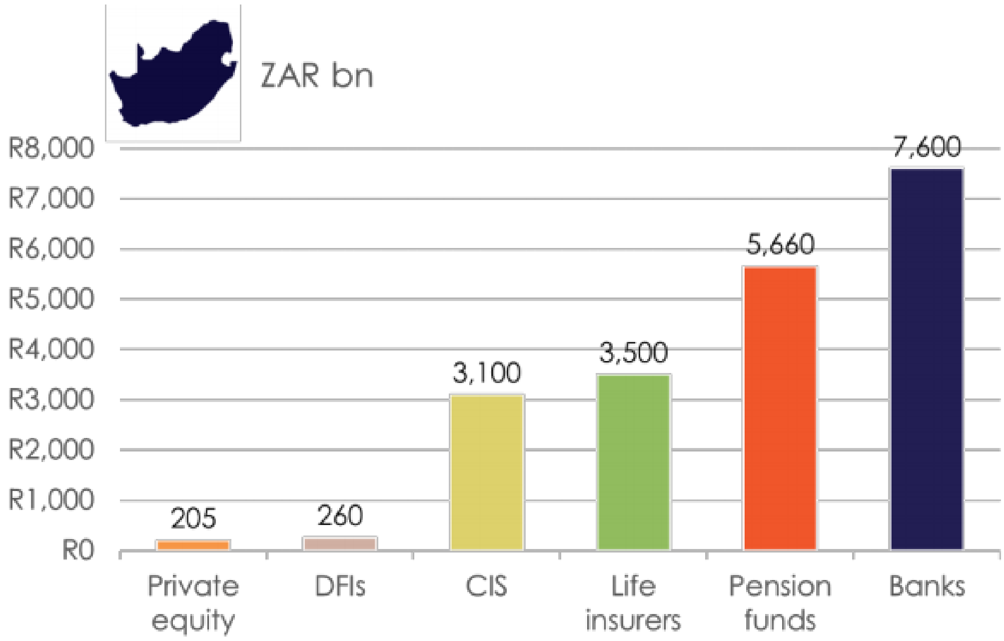


Figure 21. South African Funding Pools (Source: SAVCA, National Treasury, ASISA)

Blended finance in the form of concessionary funding with private sector finance will enhance government policy to increase investment into the water sector. This intervention is required at a time when climate change hazards impacting water quality and supply is undoubtedly worsening. According to the World Bank[4], National Treasury (NT) of South Africa is in the process of reviewing municipal and national PPP frameworks to leverage greater public-private partnerships (PPPs). In its latest 2022 Budget Review[5], NT affirms financial commitments to projects in water, reinforcing policy aimed at increased levels of private sector participation. According to NT, the Medium Term Estimates Framework (MTEF) investment in water over the next 3 years will surpass ZAR 131 billion. This funding will require substantial gearing from private sources and concessions if it is to maximise impact on the sector.

Water PPP projects in South Africa have struggled to attract investment from the private sector, as confirmed in Figure 22. The industry's exposure to the sector is substantially lower when compared with other sectors. In future, the sector will require more early-stage project development support from DFIs to de-risk projects, paving the way for sound bankable assessments. This constraint is further compounded by the lack of concessional funding to bring down the Weighted Average Cost of Capital (WACC) on projects, making projects more economically viable.

[4] World Bank Group Water. Term of Reference. South Africa: Country assessment for unconventional water resources and resource recovery from wastewater (2022)

[5] Budget 2022 Review, National Treasury (2022)

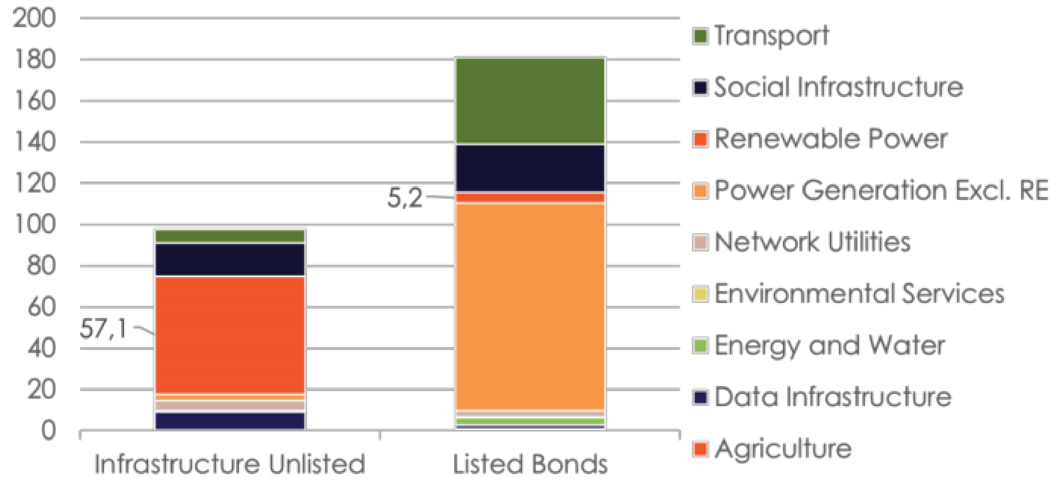


Figure 22. SA Private Sector Financial Exposures to Infrastructure Sectors (Source: ASISA, 2021).

In the case of the WRP's proposed Blended Finance Solution, GCF concessionary funding will coexist with increased levels of private sector finance. Private financiers will be able to leverage increased exposures to climate change adaptation projects on the back of strong anchor investors, such as the GCF and DFIs. The economics of blending private capital with concessionary funding is a positive development towards boosting confidence and raising the profile of water projects amongst willing investors. A number of letters of support for the WRP by potential investors has been appended in the Annexures.

Through WRP, the Bank's approach will be to plug funding gaps using GCF concessionary funding without crowding out the private sector. Lessons learnt from the Renewable Energy Independent Power Producer Programme (REIPPP) has demonstrated that infrastructure projects that overcome construction phase risk to achieve revenue generation from off-takers (commercial combined with fiscal support) are likely to be refinanced in both primary and secondary markets. Figure 23 illustrates the transition of typical infrastructure projects over the investment lifecycle. Thus, assets are likely to be taken off-balance sheet into green bonds and similar listed instruments, crowding in other institutional investors, hereby recycling capital for further use in other projects. Figure 24 provides a listing of recent green bonds issued by both the private and public sector in South Africa. Further introducing other instruments such as concessionary funding will create an ecosystem that is self-reinforcing, with new instruments and different sources of capital, creating opportunity where there is currently a market failure.

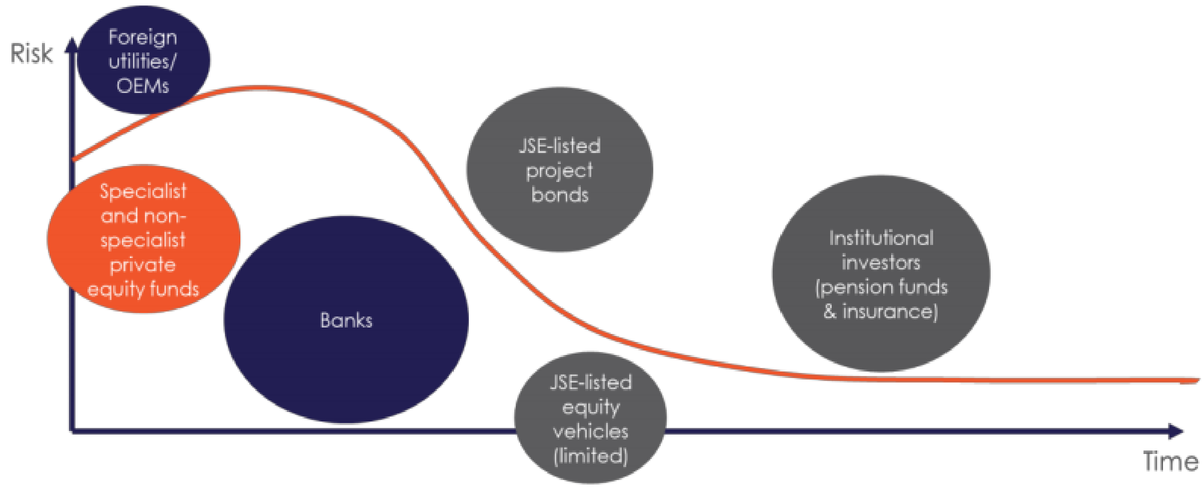


Figure 23. Infrastructure Investor Hierarchy (ASISA, 2021).

Date	Issuer	Instrument	Project	Value (ZAR)
2012	Industrial Development Corporation	Green bond	Clean energy infrastructure	5.00bn
2014	City of Johannesburg	Green bond	Biogas to Energy Project and the Solar Geyser Initiative	1.46bn
2017	City of Cape Town	Green bond	Various green projects, electric buses; energy efficiency in buildings; water resilience initiatives; sanitation treatment; and the coastal structure protection and rehabilitation	1.00bn
2017	Growthpoint	Green bond	Green buildings and green initiatives	1.10bn
2019	Nedbank	Green bond	Renewable energy	1.70bn
2022	Barloworld	Social (gender-linked) bond	Women empowerment goals, including reaching or exceeding 15% procurement spend on black women suppliers and reaching or exceeding 50% female representation in leadership by 2025	1.10bn
2022	Redefine Properties	Green bond	Improve environmental sustainability of eligible buildings in its portfolio	1.50bn
2022	Investec	Green bond	Renewable projects (477MW solar & wind)	1.00bn

Figure 24. SA Private Sector Financial Exposures to Infrastructure Sectors (ASISA, 2021).

Furthermore, on the regulatory front, recent developments in SA supporting further investment in infrastructure is encouraging. The market regulator has taken steps to increase investment in infrastructure. On 01 January 2023, pension funds will be able to invest 45% of Assets Under Management (AUM) into infrastructure. Water projects are expected to compete for this funding, as they are currently a significantly under-funded sector. Water infrastructure therefore requires a compelling investment thesis if it is to attract financing from the private sector. Water has been given elevated status in SA's national policy framework, given that SA is a water scarce country, with amplified risks brought by climate change hazards. Demand for secondary market infrastructure assets is expected to increase, supporting the business case for further supply.

According to the World Bank, the Build Operate, Transfer (BOT) model is a preferred financing vehicle favoured by the private sector in the circular economy6. As per the GCF Funding Proposal, we expect to see the use of SPV structures supporting this claim, as the Water Reuse Programme increases the supply of unconventional water projects in the market.

Overall, the private sector market in SA is well regulated with financial service providers requiring licenses and regulatory approval in order to legally operate. Capital markets possess the depth and track record to raise funding for infrastructure projects where risk-return parameters are justified. Policy changes in SA is showing support for a growing infrastructure investing market to come. The existing market has room for further expansion, but current size far outweighs the total capital requirement of the WRP.

The development of this Programme is thus critical as it would allow the creation of a new asset class and allow for the diversification and augmentation of South Africa's water mix. In response to the impacts of climate change on the water resources in South Africa, only through the leveraging of GCF funding will the realisation of the WRP be achieved, enabling the scaled development of water reuse sub-projects nationally and ultimately improving South Africa's resilience.

Broadly, GCF funding will allow for three key enablers to be realised, namely:

- **Enabling an environment providing enhanced access to private capital:** By blending the GCF funding with private investor funding, the WRP will enable private funders and municipalities to lower their risk exposure on a project-by-project basis. Increased participation from the private sector in this newly created asset class would, over time, allow for private sector support to be sustainable without continued GCF participation. However, GCF funding is critical to initially mobilise the Programme and create the new asset class for future participation by the private sector
- **Enabling an environment that enhances access to more affordable finance:** In long term infrastructure financing, the cost of finance is amongst the biggest obstacles to capital formation. GCF funding (in its proposed form of concessional loans) for the Programme would allow for the overall cost of financing on a project level to be lowered, and these lower financing costs (and potentially longer tenors) may help in the facilitation of tariffs that are more
- **Enabling a programmatic approach to water reuse implementation:** The establishment of a replicable programmatic approach to the development, finance and implementation of water reuse infrastructure in South Africa through that enables the application of water reuse in South Africa's most vulnerable municipalities to enhance water security and climate resilience.

The quantum of GCF funding for the Programme will be managed by the AE on a project-by-project basis. GCF Funding will be used to implement initial proof of concept projects that enable the programme to lead to transformative change in the long term. i.e. overcoming both cultural objection and market failures around water reuse, where GCF funding will be needed more for the first few sub-projects, and eventually these investments work without GCF funding. These sub-projects demonstrate the concept that enables the programme as a whole to make a material contribution towards the achievement of climate adaptation benefits in areas where climate induced drought and water stress are forecasted to have a significant impact.

Although the resources requested from GCF are in the region of 30% of the initial total WRP implementation costs, they are expected to be catalytic in enabling a public perception shift in relation to water reuse and bringing about a transformation in the manner in which South Africa manages and utilizes its wastewater. While South Africa is currently classified as an upper middle-income country (World Bank, 2020), South

[6] World Bank Group Water. Term of Reference. South Africa: Country assessment for unconventional water resources and resource recovery from wastewater (2022)

Africa continues to face increasing levels of socio-economic pressures due to growing unemployment, political instability, and corruption.

As a result, South Africa's macroeconomic indicators have weakened (including the downgrading of national credit rating to sub-investment grade by major rating agencies), the fiscal deficit rose, and financing of programmes such as the WRP has become increasingly challenging for government. GDP growth, which averaged 3.5% percent during 2001 and 2010, has since fallen to an average of circa 1.5% between 2011 and 2019. With the unabated continuation of the COVID-19 pandemic, South Africa is likely to continue to face considerable economic pressure, reducing overall public finance available to support the planning and execution of climatic interventions such as the WRP. The investment by GCF to is thus viewed as being critical in the establishment of this Programme.

The proposed WRP is critical for enabling municipalities to better manage wastewater and for enabling the crowding-in of private finance to facilitate investments into water reuse infrastructure and increase water resilience in South Africa. Further, the Programme may include the incorporation of irrigation reuse and potentially ecosystem based approached, thus directly (positively) impacting on the agricultural sector of South Africa. The agriculture sector is of critical importance in South Africa, for its socio-economic fabric, as well as its central role in food security, rural development and employment creation.

South Africa also has a high proportion of urban and rural communities living in informal settlements that have limited access to clean water and basic sanitation services. Within these vulnerable communities, women and children are often most vulnerable to the impacts of climate change as they are often more reliant on access to natural resources that are required to support livelihoods. These communities are often also situated in areas that are exposed to hazards such as floods, which place both lives and livelihoods at extreme risk. The request for financing to GCF is premised on the understanding that an increasing number of South Africans are vulnerable to reduced access to potable water as a result of climate change related aspects.

It is well noted that national governments, through the fiscus, have a critical role to play in supporting investment into climate adaptation programmes such as the WRP. However, traditionally, investments have been lacking in sectors such as the water and water reuse, as these sectors may be viewed as unprofitable, and are seen as ''a public good'. This lack of investment over time significantly contributes towards the development of funding gaps for such sectors despite national commitments and (apparent) private sector innovation. Thus, GCF funding, coupled with state action will play an extremely important role in catalysing private sector finance for the WRP, which will otherwise not be possible.

The AE is seeking to secure funding through grants and concessional loans from GCF. Without GCF involvement to catalyse this Programme and enable co-funding, South Africa will be unable to undertake the adequate steps to help diversify its existing water supply sources, enhance water productivity, and enhance water access in the drier, more climate vulnerable South-Western regions of the country. GCF support for this WRP will enable additional investments that will allow the scaling up of other parallel existing efforts, contributing towards a stronger national water programmatic level intervention. GCF funding will also enable the wider participation and involvement from South Africa's relatively well-established private sector (through the debt capital markets), thus increasing collaboration between sectors and ultimately creating a new sustainable asset class for continued future investment.

In the absence of GCF grants, 7 out of the proposed 10 initial sub-projects were assessed to have target minimum senior debt DSCR of below 1.5x and were thus deemed to be 'at risk' for bankability. This not only impacts the financial viability of the individual sub-projects but also the overall viability of the WRP. Hence, even with this initial high-level assessment, it can be seen that GCF funding brings additionality to establishing this Programme and water reuse as a new asset class.

The Government of South Africa through local implementing municipalities have little financial capacity to meet the initial costs of establishing a programme of this nature. GCF grant and reimbursable grant funding, in additional to the concessional loan portion of the requested GCF funding is necessary to overcome this combination of financial, technical, institutional and market barriers that together prevent climate-vulnerable locations in South Africa from adequately building resilience to increasing climate hazards. For example, these barriers include the inability of many regions in the south-western portions of South Africa to manage and sustain their water usage during increasingly more frequent drought periods (such as the droughts experienced in the Western Cape between 2016 and 2018).

B.6. Exit strategy (max. 500 words, approximately 1 page)

The longer-term programme sustainability is important to address climate vulnerabilities in the longer term. To ensure programme sustainability the design of the WRP has addressed the aspects outlined below.

Political support and ownership

The WRP will be a sub-programme of the overarching NWPP with the establishment of the NWPP being approved by national government. The ownership of the NWPP and the WRP will be by DWS and has the approval of the National Designated Authority (DFFE). DWS is the water sector lead and as such has the mandate to ensure sustainable water resource management and development, as well as oversee/ regulate that municipalities provide water services according to national norms and standards. With the WRP supporting the implementation of various climate and water sector strategies the ongoing support of national government is secure.

Partnerships

DCOG and MISA play a critical role to support municipalities in providing effective services. With local government having an important mandate to drive local socio-economic development and the provision of services, DCOG has a key role to facilitate across different sectors of government in facilitating support provision to municipalities. MISA provides support and develops technical capacity to facilitate sustained accelerated municipal infrastructure and service delivery. Through this support MISA aims to provide leadership, strategic direction, management and administration services to the organisation; and provide technical support to targeted municipalities, thereby improving infrastructure planning, implementation, as well as operations and maintenance. MISA have been actively engaged in the development of the WRP and are supportive of the programme.

Importantly, the National Treasury and DFFE are both supportive of undertaking projects that underpin the ability of the country to ensure resilience against the impacts that climate is having on the national ability to ensure the sustainable provision of services, whether ecological or built. Noting the importance of these institutions in realising long-term implementation of water reuse sub-projects across South Africa to address climate induced water insecurity, it has been seen as essential to ensure that these actors are partners to the development and the implementation of the NWPP and WRP.

Private Sector Engagement

While public sector partnerships are an essential part of developing a sustainable policy, regulatory and institutional framework to be supportive of ongoing water reuse sub-projects, the private sector are an important part of ongoing project implementation. While the blended finance solutions will provide a mix of public and private sector finance, the private sector is equally critical in guiding and providing technological solutions and innovations as well as undertaking implementation focused support through various contractual arrangements. A key element of the WRP will be the ability of the programme to ensure that this engagement is initiated, improved and secured for future projects. There has been to some degree a level of mistrust between the Government and the private sector and as such the development of this partnership during the roll-out of the WRP will be a key outcome.

Sustainable and innovative technologies

The National Water Reuse Strategy (DWS, 2011) indicates the importance of selecting and implementing appropriate treatment technologies as a critical precursor to successful water reuse sub-project implementation. The WPO will develop itself as a centre of excellence that will plan and implement technologies that are functional, sustainable and have achieve the required levels of performance and will take these factors into consideration in providing support to projects. Furthermore, in the planning phases of this project, a Community of Experts has been established to provide for engagement and knowledge exchange between key sector experts towards continued improvement.

Operations and maintenance

The structuring of the programme has been such as to incorporate clarity as to operational aspects across the programme's lifecycle including programme development and set-up, implementation, evaluation, maintenance, as well as ongoing communications and information dissemination. These aspects have been outlined in an operational case that provides for the governance modalities including aspects of oversight and accountability supported by monitoring and reporting as well as aspects that enable the operational implementation of the WRP through the WPO.

Organisational and institutional capacity

A key element of the programme's sustainability is the development of sufficient capacity and competencies to support the management and implementation of the programme as well as that of the projects themselves at municipal level. The building of the necessary staff cohorts with the appropriate skills (such as technical, financial and project management) within the WPO as well as at project level has been recognised as key part of the ability to ensure project implementation as well as taking such projects to scale, across the country.

Funding stability

The key tenets, structures and arrangements of the WRP's Financial Architecture are developed to ensure the necessary long term financial needs of the programme. The design and establishment of the WRP is premised on a phased approach, whereby, during the first 3 years of establishment, the DBSA will be the Implementing Agent, and the WRP's Core Costs will be covered by grants via the National Treasury and the GCF. However, over the medium-term, the WRP's Core Costs will be covered through a repayment mechanism as part of the financing of project implementation. This enables the programme to have a secure financial basis for the establishment and development phase prior to transitioning into a more self-sustaining model as the pipeline of projects develops over time.

Monitoring and evaluation

The operational case for the WRP sets out a clear programme governance regime with an OC that will oversee the NWPP at a strategic level and administer management support and guidance to the WRP. The OC will ensure there are the necessary operational checks and balances, effective systems of internal control and risk management, as well as promoting a 'governance culture' of transparency and accountability that will include regularised and periodic monitoring and reporting. The hosting of the WPO by the DBSA and the utilization of the internal systems of the DBSA will further underpin the programme's sustainability

Programme exit strategy and longer-term ownership

A successful exit strategy will ensure continuity of projects under the WRP, the implementation of best practice and the realisation of impact: Given the positioning of the WRP under the banner of the NWPP and the strategy to undertake phased development of the programme to enable the scale-up of these infrastructure projects through a developing pipeline of bankable projects, the WRP will transition into a sustainable, long-term programme. This has the full support of national Government though the PICC. The exit strategy for GCF financing will ultimately be though repayment of the debt, equity and guarantee from project cash flows up to maturity.

C. FINANCING INFORMATION

C.1. Total financing

(a) Requested GCF funding (i + ii + iii + iv + v + vi + vii)	**Total amount**			**Currency**
	235			million USD ($)
GCF financial instrument	**Amount**	**Tenor**	**Grace period**	**Pricing**
(i) Senior loans	-	-	-	-
(ii) Subordinated loans	200	Determined on a project by project basis	Determined on a project by project basis	Determined on a project by project basis
(iii) Equity	-	-	-	Not applicable
(iv) Guarantees	-	-	-	-
(v) Reimbursable grants	-	-	-	-
(vi) Grants	35	Not applicable	Not applicable	-
(vii) Results-based payments	-	-	-	-

(b) Co-financing information	**Total amount**	**Currency**
	1237	million USD ($)

Name of institution	**Financial instrument**	**Amount**	**Currency**	**Tenor & grace**	**Pricing**	**Seniority**
Options	Grant	2	million USD ($)	Not applicable	-	-
Options	Grant	30	million USD ($)	Not applicable	-	-
Private Development Partners	Equity	200	million USD ($)	Determined on a project by project basis	Determined on a project by project basis	- Determined on a project by project basis
Private Debt Capital Partners	Senior Loans	650	million USD ($)	Determined on a project by project basis	Determined on a project by project basis	Determined on a project by project basis
DBSA, DFI Partners	Subordinated Loans	150	million USD ($)	Determined on a project	Determined on a project	Determined on a project

				by project basis	by project basis	by project basis
Participating Municipalities, SA Infrastructure Fund	Grant	155	million USD ($)	Not applicable	-	-
Development Corporation(s)	Guarantees	50	million USD ($)	Determined on a project by project basis	Determined on a project by project basis	Determined on a project by project basis
(c) Total financing (c) = (a)+(b)	**Amount**			**Currency**		
	1472			million USD ($)		
(d) Other financing arrangements and contributions (max. 250 words, approximately 0.5 page)	Due diligence assessments of co-financiers is not possible at this stage of the programme development and will be undertaken during the first year of the programme and then will be undertaken on an annual basis as projects move up the implementation pipeline					

C.2. Financing by component

While there has been some engagement with co-financiers in the development of the WRP there has been no formal due diligence. Broadly speaking, co-financiers will take on three forms:

- **Equity / budget allocation grants**. These are likely to come from participating local authorities, who own and/or are mandated to run a particular WWTW. Municipalities have been engaged numerous times and there is significant interest in the larger metropolitan municipalities to engage with the WRP. These municipalities have a range of financial instruments available to provide these grants.
- **Concessional loan**. DBSA will undertake market soundings with other DFIs to provide support to sub-projects, as project preparation support is being provided to ensure the production of bankable sub-projects.
- **Senior debt**. Commercial banks have been engaged throughout the preparation of the WRP, and have informed the structure of the Blended Finance Solution, how project bonds could be used, and what magnitudes of debt could be allocated by private capital providers. These commercial banks have indicated interest in providing debt and equity to support WRP sub-projects.
- **Guarantees**. Guarantees will be provided, as required on a project-by-project basis, to ensure the financial viability of a project by improving the senior debt terms.

In terms of process risk assessments and due diligence on co-financiers, this is a requirement of the IBCF and this will be undertaken in the design of each blended finance solution.

Component/Outcome	**Indicative cost (USD)**	**GCF financing**		**Co-financing**		
		Amount (USD)	**Financial Instrument**	**Amount (USD)**	**Financial Instrument**	**Name of Institutions**
Component 1A: Establishing a WRP and Project Pipeline: Establishment and Operationalisation of the WRP	2 mil	Nil	-	2 mil	Grant	SA National Treasury
Component 1B: Establishing a WRP and Project Pipeline:	60 mil	30 mil	Grants	30 mil	Budget allocations, grants, and in-kind support	SA National Treasury, Other Participating Organisations

Project pipeline preparation [B]						(Municipalities, DFIs, etc.)
Component 2: Implementing Water Reuse Projects:[C] **Provision of Technical Assistance for Project Procurement Development of the Blended Financing Solutions**	1.400 bil	Nil	-	200 mil	A. Equity	Private Development Partners
		Nil	-	650 mil	B. Senior Debt	Private Debt Capital Partners
		Nil	-	150 mil	C. Subordinated Debt	DBSA, DFI Partners
		200 mil	D. Concessional Loans	Nil	-	-
		Nil	-	150 mil	E. Budget Allocations, Grants	Participating Municipalities, SA Infrastructure Fund
		Nil	-	50 mil	F. Guarantee [D]	Public Development Corporation(s)
Component 3: Building capacity and awareness [E]	10 mil	5 mil	Grants	5 mil	F. Budget Allocations, Grants	Participating Municipalities, SA Infrastructure Fund
Indicative total cost (USD)	**1.472 billion**	**235 million**		**1,237 million**		

Note [A] **Outcome 1A – Core Costs:** Costs represent the establishment and operational cost of the Water Reuse WPO, which are estimated at USD 2 mil, over the first 3 years of establishment. The SA National Treasury is covering these costs for the first 3 years, only; after which, the WPOs ongoing core costs for operations will be covered through the levying of a financial-close premium, which will be proportioned and capitalised as part of the implementation costs of each sub-project.

Note [B] **Outcome 1B – Project Preparation and Transaction Advisory (TA) Costs**: The WPO's primary focus is the preparation of bankable water reuse sub-projects. The proposed USD 30 mil Grant from the GCF will be allocated alongside USD 30 mil of capital from other participating organisations, to leverage USD 850 mil of private capital for the Programme. It is noted that the USD 60 mil Grant will be allocated to sub-projects and is aimed to be transferred for the account of the winning bidder, thus allowing for this facility to be replenished through the life of the programme. There will therefore be reimbursable and non-reimbursable elements to this component.

Note [C] **Outcome 2 – Implementation Costs**: Costs will be covered by a Blended Finance solution comprising the following:

A. Equity of approx. USD 200 mil from private development partners (based on a D:E ratio of 75:25).

B. Senior Debt of approx. USD 650 mil from private debt capital partners such as commercial banks and institutional investors, currently representing 50% of the total implementation cost cover.

C. Subordinated Debt of approx. USD 150 mil from the DBSA and DFI Partners with the discreate purpose of enhancing the creditworthiness of individual sub-projects.

D. Concessional Loans of approx. USD 200 mil from the GCF to target the gap financing requirements and ensure the financial viability of individual sub-projects across the programme.

E. Budget Allocations & Grants of approx. USD 150 mil from participating authorities and the SA Infrastructure Fund.

F. Guarantee of approx. USD 50 mil by a national public development corporation. This does not form part of the direct Implementation Costs, but is used to enhance the creditworthiness of individual sub-projects, to catalyse USD 650 mil Senior Debt from private debt capital partners.

In instances where DBSA will provide credit enhancement into sub-projects, the DBSA will rank pari-passu with GCF.

Note D **Guarantee** of approx. USD 50 mil by a national public development corporation, does not form part of the direct Implementation Costs, but is used to enhance the creditworthiness of individual sub-projects, to catalyse USD 650 mil Senior Debt from private debt capital partners.

Note E **Outcome 3** - **Capacity Development and Information Communication Education Program** will be funded through an equal contribution by the GCF (through grants) and national budget allocations.

In respect of the GCF Loan, with limited recourse, the GCF will bear the credit and project risk of the sub-projects.

The GCF Loan to be provided by GCF to the Accredited Entity in accordance with the FAA will reflect, among others, the relevant terms and conditions. The terms and conditions of the Sub-Loans will be differentiated between Public Sector Sub-Loans and Private Sector Sub-Loans.

A sub-project is considered a public sector sub-project if the Sub-Borrower is more than 50 percent owned and/or controlled by the public sector; a project is considered a private sector sub-project if the Sub-Borrower is more than 50 percent owned and/or controlled by private shareholders.

The Parties may agree to reduce the GCF Loan amount five (5) years after the effectiveness of the FAA based on the results of the mid-term independent evaluation report and on the performance of the Programme.

For the implementation of Component 2 and the financing of relevant sub-projects, the Accredited Entity shall on-lend the GCF Loan to ultimate borrowers in the public and private sector in accordance with the provisions in the term sheet.

C.3 Capacity building and technology development/transfer (max. 250 words, approximately 0.5 page)

C.3.1 Does GCF funding finance capacity building activities?	Yes ☒ No ☐
C.3.2. Does GCF funding finance technology development/transfer?	Yes ☐ No ☒

Component 3 of the WRP provides for national and regional public awareness and education processes that will support the implementation of existing national, provincial and local climate adaptation strategies and plans. These knowledge exchange interventions will underpin the development of awareness regarding climate change impacts at local levels and will support the successful scaling and implementation of water reuse sub-projects in South Africa as an adaptation response. These interventions will focus upon developing improved awareness across a broader swathe of stakeholders, while providing more focused technical and financial capacity building interventions at national, provincial and local government levels. These governmentally focused interventions will deepen the understanding of the water sector's climatic risks while taking into consideration updated analysis, knowledge, and insights. Strengthened by this experience and evidence, the Government of South Africa through the NWPP, can successfully and actively develop an updated climate strategy to better engage with water security risk. This will then support the implementation of a longer-term pipeline of projects as well as assist in addressing various barriers to implementation such as institutional and regulatory frameworks.

Financing: Technical assistance for these capacity building activities will be financed by a GCF grant of USD 5,0 million and USD 5,0 million from government budget allocations.

D. EXPECTED PERFORMANCE AGAINST INVESTMENT CRITERIA

This section refers to the performance of the project/programme against the investment criteria as set out in the GCF's Initial Investment Framework.

D.1. Impact potential (max. 500 words, approximately 1 page)

A priority element of the success of the WRP will be the ability to realise impact and as such is a priority investment criterion.

The WPO as the lead entity for the WRP will play a central role in initiating, procuring, and overseeing interventions undertaken by the WRP. As such the WPO will require a minimum threshold of capacity and competency, supported by a suite of operational procedures and instruments. This will position the WPO as being recognized for its technical expertise, planning ability, project management capability, financial competencies, trusted delivery and accepted by stakeholders as a reliable organisation. Noting the technical and institutional complexity of water reuse the WPO must be able to demonstrate the capability to champion and facilitate the implementation of water reuse sub-projects. This will require the WPO to have capability to collaborate with the qualifying agencies and organisations. As such the WPO will be the core advocate for adaptation working closely with project owners to ensure catalytic change.

The WRP will be developed in phases whereby the first phase targets sub-projects with the highest probability of being implemented. There may be only a few sub-projects within the first 3 years to prove the concept and build confidence in the WRP as well as the technology. Clearly the initial list of sub-projects is likely to come from those Municipalities that have already started planning for reuse and have initiated feasibility studies. There are also a few that have a strong wastewater and water treatment capability who may also be interested but may not yet have initiated a project. These first phase Municipalities could thus come from the following list, which is purely indicative:

- City of Cape Town Metropolitan Municipality;
- Nelson Mandela Metropolitan Municipality;
- City of Ekurhuleni Metropolitan Municipality;
- City of Johannesburg Metropolitan Municipality;
- City of Tshwane Metropolitan Municipality;
- eThekwini Metropolitan Municipality;
- Mangaung Local Municipality;
- Drakenstein Local Municipality;
- Indicative Project A; and
- Indicative Project B

Of these, 6 are Metropolitan Municipalities and the most significant growth nodes of the country, all of which face significant water security challenges. Of these projects 80% face climate induced vulnerability (see Tables 1 and 4).

The value of water resilience to businesses can be proxied by the vulnerability of economic output to a curtailment in water supply. That is, the value to a business from reducing the risk of interruptions to water supply is a function of its reliance on water supply to produce economic output, defined here by gross value added. The economic value of each sector varies from one municipality to another, while the exposure of economic value to water restrictions varies from one sector to another, in function of the climate stress in each municipality. The estimated value in Gross Value Added (GVA) reduction is provided in Table 15.

Table 15: Beneficiaries and Employment Impact of the WRP

Municipality	Contribution to National GDP (%)	Value at risk per day (NEU)* (USD)	Value at risk per day (EDO)* (USD)
Nelson Mandela Metropolitan Municipality (40 Ml)	3.35%	517 826	5 353 870
City of Ekurhuleni (60 Ml)	6.21%	863 577	9715852
City of Johannesburg Metropolitan Municipality (50 Ml)	16.71%	1 929 006	24 774 581
City of Tshwane Metropolitan Municipality (30 Ml)	10.03%	1 012 430	14 644 945
eThekwini Metropolitan Municipality (100 Ml)	10.71%	1 550 421	17 290 141
Indicative Project A (75 Ml)	0.57%	98 736	951 216
City of Cape Town Metropolitan Municipality (40 Ml)	11.18%	1 452 118	16 968 657
Drakenstein Local Municipality (10 Ml)	0.45%	76 850	661 916
Indicative Project B (15 Ml)	0.68%	93 024	944 872
Mangaung Metropolitan Municipality (25 Ml)	1.72%	226 139	2 302 925
Total	61.61%	7 820 125	93 608 975

* Water usage restrictions at a lower non-essential use (NEU) and upper emergency drought order (EDO) estimate

Based on the per capita consumption of potable water in South Africa, direct beneficiaries are counted as people who benefit from the additional water supplied into the municipal systems by the WRP first phase (i.e. the pilot portfolio of reuse sub-projects delivering an additional 445 Ml/day to select municipalities. Per capita water consumption ranges from 87 litres/person/day to 285 litres/person/day across municipalities). The WRP first phase could reach a total of ***3 424 737 direct beneficiaries***. Indirect beneficiaries are counted as those reached with water supply through the full scale WRP (i.e. the full set of 27 water reuse sub-projects delivering 1067 Ml of water across the country). At scale the WRP could reach a further ***3 877 805 indirect beneficiarie***s. Based upon the first phase selection of 10 water reuse sub-projects, the WRPs indicative impact in terms of direct beneficiaries as well as jobs created and saved is outline in Table 16, below.

Table 16: Beneficiaries and Employment Impact of the WRP

MUNICIPALITIES with Climate Rationale (included in economic component)	Combined Treated Water Output from WRP Ml/d	Water Consumption per capita (litres on average)	Number of direct beneficiaries disaggregated by gender		Total number of direct beneficiaries per municipality	Municipal Population	Proportion of unserved population per municipality	Number of unserved population per municipality
			%Male	%Female				
EC								
Nelson Mandela Metropolitan Municipality	40	87	222 069	237 701	459 770	1254000	3,20%	40 128
FS								
Mangaung Metropolitan Municipality	25	178	67 837	72 612	140 449	893749	5,70%	50 944
GP								
City of Ekurhuleni	60	90	322 000	344 667	666 667	3894000	4,25%	165 495
City of Johannesburg Metropolitan Municipality	50	285	84 737	90 702	175 439	5600000	4,40%	246 400
City of Tshwane Metropolitan Municipality	30	240	60 375	64 625	125 000	2300000	9,40%	216 200
KZN								
eThekwini Metropolitan Municipality	100	100	483 000	517 000	1 000 000	3158000	9,50%	300 010
Indicative Municipality A	75	182	199 038	213 049	412 088	410465	10,70%	43 920
NC								
Indicative Municipality B	15	115	63 000	67 435	130 435	285000	4%	11 400
WC								
City of Cape Town Metropolitan Municipality	40	156	123 846	132 564	256 410	4618000	2,70%	124 686
Drakenstein Local Municipality	10	171	28 246	30 234	58 480	284475	12%	34 137
TOTALS			**1 654 148**	**1 770 589**	**3 424 737**	**22 697 689**		**1 233 319**

The economic Internal Rate of Return (IRR) is a function of the contribution of the WRP to reducing the risk of water shortages in the event of a drought. The WRP contributes to minimising the water supply deficit and the chance of water restrictions in response to water shortages. This forecasted contribution of the WRP is applied to the value at risk to households and businesses across all municipalities to generate an indication of the value of the WRP in terms of the avoided cost of water shortages. This is summarised in Figure 25 below.

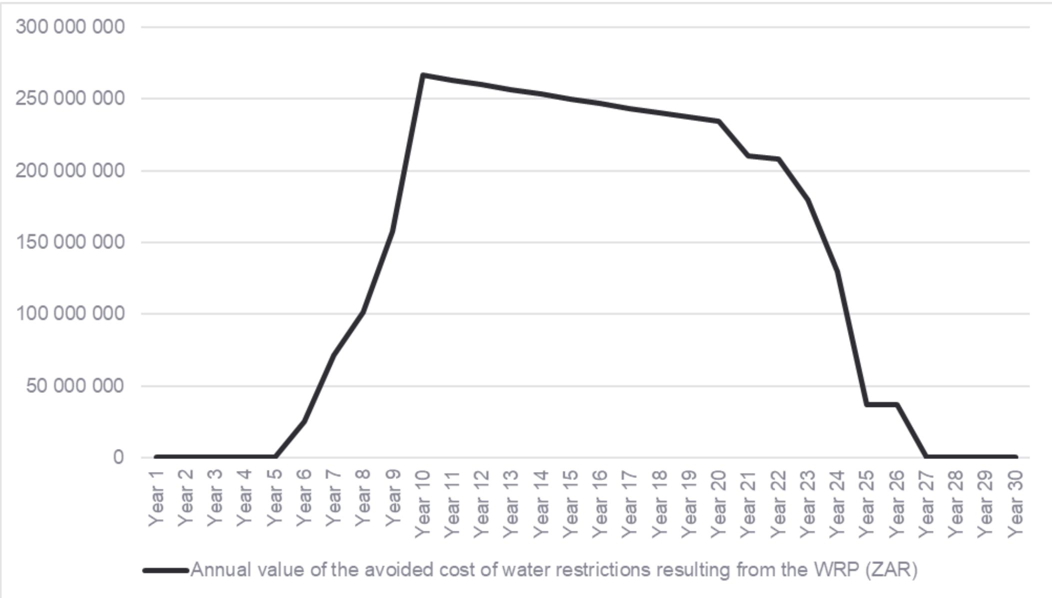


Figure 25: Annual value of the avoided cost of water restrictions resulting from the WRP (ZAR)

The average annual avoided cost to households and businesses that would have resulted from water restrictions in the absence of the WRP is over ZAR 130 million (approx.. USD 8.1 million). The annual value of avoided water restrictions to households and businesses is summed with the net financial cashflows for the programme and discounted over the period to 2050.

- The Economic IRR for the WRP, as a whole is 28%, where the Financial IRR for the programme as a whole is around 26%.
- The Net Present Value for the programme as a whole factoring in economic co-benefits ranges from ZAR 13 billion to ZAR 34 billion (USD 817 million to USD 2.1 billion), justifying the investment from an economic perspective.

D.2. Paradigm shift potential (max. 500 words, approximately 1 page)

The development of water reuse sub-projects has been limited to date with climate related hazards influencing impacted municipalities to explore these options to address increasing vulnerability and increasing water stress. To date only 14% of the water supply mix comes from water reuse demonstrating the opportunity to upscale this to improve water security. As such, the development of a pipeline of water reuse sub-projects towards the establishment of a new asset class of water reuse sub-projects that result in improved water security as well as the development of improved levels of adaptive capacity that enable climate responsive planning will enable the WRP to have a transformational impact on the water sector.

The WRP is designed to operate as an ongoing, long-term programme that supports the preparation, financing, and technical support to suitable water reuse sub-projects in South Africa in multiple phases, supporting the country's 8 metropolitan municipalities as well as 44 district municipalities (and, based on the need and opportunity, the 226 local municipalities). Within the broader structure of the WRP, a key portion of the GCF's funding will be directed towards the project preparation function, to ensure that the projects identified optimize climate change adaptation benefits and are designed with climate resilience considerations embedded from the very inception. Thus, GCF's support would continue to drive climate resilience in South Africa's water sector through every project preparation process.

Noting the importance of introducing the paradigm of a circular economy, the WRP has the potential to embody this shift and introduce waste-water reuse as the norm for municipalities across the country. It will do so through demonstrating viable business models for water reuse amongst industrial, commercial, and residential users, and by bringing <u>advanced water reuse technologies</u> and ecosystem-based approaches to the South African market.

By developing typologies of water reuse infrastructure interventions that match certain context-specific characteristics (e.g. population served, projected growth in local water demand, predominant water use, governance structures, environmental factors etc.), the WRP will facilitate easier identification of comparable municipal contexts and sites where suitable project types could be replicated, not only in terms of technology but also governance measures, financial structuring, and implementation arrangements. This will enable the programme to <u>scale-up the roll-out of projects over time</u>, with weaker municipalities benefitting from the knowledge creation and experience from early pathfinder projects.

The WRP will therefore lead to market transformation through enabling mechanisms and stakeholder-driven processes that provide an impetus to behaviour change by water users. Further, the WRP will, through a centre of excellence and its WPO, enable municipalities to have increased access to a pool of expert technical, financial and legal advisors for advice on project selection, project preparation and project implementation.

The WRP will demonstrably change incentive structures for climate change adaptation in water management, by making wastewater treatment and reuse <u>financially viable and cost-effective for municipalities</u>. The WRP will facilitate the <u>creation of a new asset class</u>, which will allow for investors to initially invest in a number of pathfinder projects following which subsequent projects will follow a more standardized contractual and procurement process, thereby gaining efficiencies and iteratively mitigating inherent project related risk, thus enhancing the likelihood of increased investment into such subsequent sub-projects.

Through the management of strategic governmental partnerships, the WRP will be in a position to engage with key government agencies to <u>address a range of policy, strategy, institutional, and regulatory aspects that will support the strengthening of integrated and adaptive responses to climate risks</u>, through the improvement of water security. The implementation of water reuse sub-projects will require a number of regulatory instruments to be improved. With DWS owning this programme and providing oversight, it will be an important sponsor in addressing these aspects.

There is a legacy of resistance to charging for ecosystem maintenance and restoration that underpins a large component of the water service sector. South Africa recognizes that it needs to fix the economies of water pricing to drive sustainability at all levels of the water value chain. In this regard, the Raw Water Pricing Strategy is currently under review with the aim to finalise this by March 2022. The WRP will help raise awareness in this regard and align with work being undertaken by various agencies and departments to address water pricing and ecosystem services. It will not replace such work but it will work to promote financial flows to ecosystem services and it will support the changing the policy landscape wherever strong potential linkages exist. This integration of ecosystem services and infrastructure finance is a new paradigm for South Africa.

D.3. Sustainable development (max. 500 words, approximately 1 page)

Under the South African governance framework, municipalities have a critical role in driving local socio-economic development. The WRP is designed to deliver a range of benefits to water users such as households and industries, and to municipal governments, and thereby to South Africa. The intended co-benefits include:

- **Environmental co-benefits:** through the reduction in water pollution, improvements in water quality and improvements in soil and water quality management in agricultural areas and urban farms. The sub-projects will help strengthen South Africans' resilience and adaptive capacity to climate-related hazards and natural disasters (SDG 13) by building on a distributed model of water treatment and reuse. There will also be mitigation co-benefits from greenhouse gas (GHG) emissions reductions and improved water security by mainstreaming ecosystem adaptation and restoration and through improved ecosystem maintenance in upper catchments. Further environmental co-benefits may also be achieved through the incorporation of ecosystem-based approaches to sustain environmental flows and increase climate resilience.
- **Economic co-benefits:** through the expansion of markets for treated wastewater, revenue-generation through tariffs for reused water, beneficiation of sludge, avoided costs to the municipalities of creating alternate water supply from freshwater sources to meet the same demand, and improved industrial and commercial productivity in targeted municipalities due to more reliable water supply and security of supply.
- **Gender-sensitive development impact:** through a reduction in household water insecurity, which disproportionately affects women, and through deliberate structuring and implementation of the WRP to increase women's participation in the programme itself. Women direct beneficiaries are estimated at 1,77 million, and this is doubled when including indirect women beneficiaries.
- **Contribution to the achievement of the Sustainable Development Goals (SDGs):** through support to a number of SDGs, the WRP will primarily help South Africa make progress towards:
    - SDG 6 – ensure availability and sustainable management of water and sanitation for all (especially targets 6.1,[lxv] 6.2,[lxvi] 6.3,[lxvii] 6.4,[lxviii] and 6.b[lxix]);
    - SDG 11 – make cities more inclusive, safe, resilient, and sustainable (especially target 11.6[lxx]);
    - SDG 12 – ensure sustainable consumption and production patterns (especially target 12.5[lxxi]); and
    - SDG 13 – take urgent action to combat climate change and its impacts (especially target 13.2[lxxii]).

It is important when dealing with water and ecosystem services to ensure sub-projects focus on the inter-connectivity of all SDGs and to ensure no significant trade-offs occur between goals. WRP investments will therefore focus on measures of return on investment – i.e., where the likelihood of positive returns for both people and the environment are highest and as such modelling coupled with adaptive management approaches will likely be necessary

D.4. Needs of recipient (max. 500 words, approximately 1 page)

South Africa is facing rising temperatures together with declining rainfall and a significant increase in spatial and temporal variability of rainfall resulting in more frequent floods and droughts. South Africa is warming at more than twice the rate of global temperature increase [lxxiii].Due to the variability of rainfall and rates of evapotranspiration across the country some parts of the country will experience these vulnerabilities differentially, however, due to the dependency of some areas on bulk water transfers these vulnerabilities can have far wider levels of impact. Overall, there is robust evidence that slow-onset climate hazards like drought, in particular, are expected to become far more frequent, with 'once in a 100-year drought events' likely to occur multiple times in a century[lxxiv] . South African settlements are susceptible to the effects of climate variability, and since 1980 have recorded 86 noticeable weather-related disasters that have affected more than 22 million South Africans and have cost the economy in excess of USD 6.81 billion in economic losses[lxxv]

As in most under-resourced countries, climate change will disproportionately burden the poor and most vulnerable communities in South Africa. Large populations living in impoverished circumstances, in informal settlements that are located in sites vulnerable to extreme weather and natural hazards, further perpetuating socio-economic fragility through the lack of adequate infrastructure and services, including access to water. Nearly 20% of the country's population lives on less than USD 2.00 a day, and almost 60% of the entire country subsists on USD 5.50 a day. South Africa has been the most unequal country in the world in recent years, based on the Gini coefficient. Compared to all other emerging markets, income inequality has remained high in South Africa, decoupled markedly from the average for such markets.

Unemployment levels in South Africa are exceptionally high, at 28%, but even higher at a staggering 53% for youth. South Africa has been one of the most negatively affected emerging economies by COVID-19 related economic contraction, with a GDP growth rate reduction of 7.2%. The country's budget deficit is at an estimated 11.2% of GDP. Its credit rating has been reduced to less-than investment grade, or speculative (junk status) by all major ratings agencies, with a negative outlook. Its debt-to-GDP-ratio of 81% is regarded as approaching unserviceable levels. This has severely constrained alternative sources of financing – especially for climate change adaptation measures, which are regarded as having high risk and negligible returns on investment – from commercial or private lenders. Therefore, South Africa must focus its very limited public financial resources on COVID-19 public health measures, post-COVID economic recovery, and continued support to social protection programmes.

While South Africa has retained its institutional capacity in some areas, over the last decade there has also been a steady decline in institutional capacity, efficacy, and governance. The National Planning Commission recognized that in many areas service delivery has fallen dramatically short of expectations. As municipalities face challenges in meeting even basic service-delivery needs, they are not well-placed to develop and roll-out a market-transforming initiative such as the WRP, with advanced technology and sophisticated financial structuring requirements. This requires national government to strategically guide the WRP, with the WRP itself adequately resourced to provide the institutional strengthening and capacity-building to beneficiary municipalities to introduce these important climate adaptation and resilience building interventions. In support of this, the DWS is also currently undertaking a review of the national water supply reconciliation strategies to develop an up-to-date understanding of the current and future water supply challenges as well as the impacts of these upon the national and local economies. These studies will further inform the prioritisation of water reuse and other supporting interventions.

It will be imperative to fully understand the contextual needs at sub-project level, noting that these can be distinctive. The provision of preparation support, both from a technical and financial perspective, to ensure that sub-projects are appropriate, well-structured in terms of risk, and financially viable will be a critical success factor for the WRP.

D.5. Country ownership (max. 500 words, approximately 1 page)

The Constitution of South Africa Water guarantees the right of access to sufficient water, and this is supported through the National Water Act (Act 36 of 1998) and the Water Services Act (Act 108 of 1997). These laws provide a range of strategic and planning instruments that enable the sustainable development of water resources to support national development imperatives and include the NWRS, Catchment Management Strategies, and Water Services Development Plans and Integrated Development Plans. The second Edition of the NWRS in 2013, outlined a National Water Reuse Strategy to support the diversification of the national water mix. [lxxvi] Noting the increasing levels of concern regarding the declining status of water resources the National Water and Sanitation Master Plan was developed in 2018 to consolidate priority interventions to address the projected supply deficit, with water reuse being a key priority for the country. Furthermore, water reuse has been outlined as one of the key approaches to adaptation in the NCCAS, which is South Africa's *de facto* National Adaptation Plan. The NCCAS urges water management institutions to adopt more adaptive responses and more water-wise practices.[lxxvii] South Africa's updated draft NDC[lxxviii] reinforces this commitment further through its Goal 3: Implementation of the NCCAS interventions for the period 2021-2030, including as a priority adaptation within the water sector.[lxxix] The implementation of the WRP is therefore well aligned to the NCCAS and the recently updated NDC.

The WRP will fall under the auspices of the NWPP that will be hosted and supported by the DBSA as Executing Entity. The NWPP will be centrally managed by and implemented through a WPO, established, and initially housed by the DBSA and will supports the achievement of South Africa's National Development Plan (NDP), the country's primary socio-economic development planning framework. The NDP notes the impact of climate change and explicitly directs the investigation of water reuse, through a regional programme (the WRP) to support focused investment in such sub-projects to address climate risk and improve water security.

The NWPP was endorsed at a meeting of the PICC in May 2021 and the PICC recognised and approved that the NWPP will become the "enabler" of various government planning and coordinating instruments such as the NDP, NWSMP and the District Development Model.

It should be noted that although the NWPP will ultimately comprise of priority sub-programme initiatives, at present only the WRP has progressed materially, and as such the WRP and its success will play a crucial initiating and pilot role for the success of the larger NWPP that will develop in a phased and progressive manner. This provides a pragmatic approach to the development of the NWPP. Noting the importance of the WRP in supporting these national strategies and planning instruments, the engagement of national government departments in the design and implementation of the WRP is important and is described in Section B.4. The DWS is the owner of the NWPP and the WRP and will be engaged in the oversight committee for the WRP.

The DBSA is well-placed to drive the hosting, financing and structuring of the WRP, for the Government of South Africa, as the accredited entity and executing entity. The DBSA has a strong track record with the GCF, and is suited to steering a large, nationwide, infrastructure programme requiring coordination amongst public and private actors at the national, provincial, and local levels. The DBSA has a strong track record in the management of programmatic risk and the effective use of safeguards to support this.

The WRP has been discussed with other key and supportive organisations who have interest in participating in the design and implementation of the WRP. These organisations include:

- GTAC and the National Treasury;
- Lead Government Departments including DFFE as the Nationally Designated Authority, as well as DWS and DCOG;
- Public institutions including Infrastructure South Africa and MISA;

- Water Research Commission (WRC) and selected Water Service Authorities (WSAs);
- SALGA;
- Private banks (and other financial services institutions), such as Nedbank and International Finance Corporation; and
- Civil society organisations.

The DFFE is the country's lead agent for climate change adaptation and has overseen the development of a range of strategies including the NCCAS and the NDC, and its support of the WRP has been endorsed as a key intervention to support adaptation in the water sector.

D.6. Efficiency and effectiveness (max`. 500 words, approximately 1 page)

GCF intervention into the WRP is critical, as outlined above in Section B.5. While it is apparent that there will be government and other private co-funding that will be made available, current municipal level appetite and budget for a programme of this nature appears to be limited. This municipal-level budget is insufficient to capacitate existing WWTW's and develop new ATP plants to enable the potable reuse of water. According to the International Monetary Fund, as of end-2020, the Government of South Africa had a total gross government debt of circa 80% of GDP, increased steadily every year from 2014 onwards. Therefore, the Government of South Africa is unable to increase investments into climate resilient water programmes, which not only impact growth and development, but could also hamper or constraint future economic growth prospects.

Value for Money

The efficiency of the WRP can be assessed using several Value for Money metrics. The GCF traditionally seeks to understand how the programme under consideration might perform against similar programmes in the region, and for which the GCF has provided funding. To this end the financial architecture can provide preliminary Value for Money metrics outlined in Table below.

Table 17: Summary of Value for Money Metrics

Value for Money Metric	Cost/Beneficiary
GCF Cost per Beneficiary (Public Sector Project)	USD 58.54
GCF Cost per Beneficiary (Private Sector Project)	USD 57.59
Total cost per beneficiary	**USD 411.05**

These metrics demonstrate that the WRP supports adaptation impact with more efficient use of GCF capital than other similar programmes. For example, the recently awarded Programme in Jordan - Building resilience to cope with climate change in Jordan through improving water use efficiency in the agriculture sector (BRCCJ) – represented a GCF cost per beneficiary of USD 117.70.

Least Concessionality

The scenarios in which debt funding cost implications for GCF support show that Option 2 (GCF concessional loan with guarantee) enable the WRP to be viable with the least amount of concessional finance. Table 18 below summarises the assessment of the cost of debt against the three scenarios.

Table 18: Debt Funding Options for the WRP7

	Debt Funding Options		Default: No GCF Funding/ guarantee)	Option 1: With GCF Concessional Loan only	Option 2: With GCF Concessional Loan + guarantee
1	**Senior Debt - Commercial Term Loan (Nominal)**				
	Indicative Debt Terms				
	Term Loan Tenor	*years*	10.0	10.0	10.0
	Grace period	*years*	3.0	3.0	3.0
	Cost of Debt				
	Base rate (Fixed rate swap)	*per annum*	7.18%	7.18%	7.18%
	Liquidity cost	*per annum*	1.00%	1.00%	1.00%
	Statutory cost	*per annum*	1.00%	1.00%	1.00%
	Credit margin	*per annum*	4.10%	3.50%	2.90%
	Total cost of Debt	*per annum*	**13.28%**	**12.68%**	**12.08%**
	Finance Fees				
	Upfront fees	*Once off*	1.00%	1.00%	1.00%
	Commitment Fees	*per annum*	0.75%	0.75%	0.75%
2	**Senior Debt - Inflation-linked Term Loan**				
	Indicative Debt Terms				
	Loan Tenor	*years*	10.0	10.0	10.0
	Grace period	*years*	3.0	3.0	3.0
	Cost of Debt				
	Base rate (Real)	*per annum*	3.97%	3.97%	3.97%
	Liquidity cost	*per annum*	1.00%	1.00%	1.00%
	Statutory cost	*per annum*	1.00%	1.00%	1.00%
	Credit margin	*per annum*	2.25%	1.65%	1.05%
	Total cost of Debt	*per annum*	**8.22%**	**7.62%**	**7.02%**
	Finance Fees				
	Upfront fees	*Once off*	1.00%	1.00%	1.00%
	Commitment Fees	*per annum*	0.75%	0.75%	0.75%
3	**Senior Debt - Concessional Term Loan**				
	Indicative Debt Terms				
	Loan Tenor	*years*	13.0	13.0	13.0
	Grace period	*years*	3.0	3.0	3.0
	Cost of Debt				
	Base rate (Fixed rate swap)	*per annum*	7.20%	7.20%	7.20%
	Liquidity cost	*per annum*	1.00%	1.00%	1.00%
	Statutory cost	*per annum*	1.00%	1.00%	1.00%
	Credit margin	*per annum*	4.10%	3.50%	2.90%
	Total cost of Debt	*per annum*	**13.28%**	**12.68%**	**12.08%**
	Finance Fees				
	Upfront fees	*Once off*	1.00%	1.00%	1.00%
	Commitment Fees	*per annum*	0.75%	0.75%	0.75%
4	**Senior Debt - Commercial Bond Facility**				
	Indicative Debt Terms				
	Loan Tenor	*years*	10.0	10.0	10.0

[7] Option 1 and 2 allows for min GCF capital to meet min Target DSCR of 1.5x. The credit margin is reduced by 50 bps for option 1 and a further 50 bps for Option 2. GCF Guarantee only applies to Senior Debt holders (1 - 4) and this is likely to be a partial guarantee

	Grace period	*years*	3.0	3.0	3.0
	Cost of Debt				
	Base rate (Fixed rate swap))	*per annum*	7.18%	7.18%	7.18%
	Liquidity cost	*per annum*			
	Statutory cost	*per annum*			
	Credit margin	*per annum*	5.60%	5.00%	4.40%
	Total cost of Debt	*per annum*	**12.78%**	**12.18%**	**11.58%**
	Finance Fees				
	Upfront fees	*Once off*	1.00%	1.00%	1.00%
	Commitment Fees	*per annum*	0.75%	0.75%	0.75%
5	**Subordinated Loan**				
	Indicative Debt Terms				
	Loan Tenor	*years*	10.0	10.0	10.0
	Grace period	*years*	3.0	3.0	3.0
	Cost of Debt				
	Base rate (Fixed rate swap)	*per annum*	7.18%	7.18%	7.18%
	Liquidity cost	*per annum*	1.00%	1.00%	1.00%
	Statutory cost	*per annum*	1.00%	1.00%	1.00%
	Credit margin	*per annum*	4.85%	4.25%	3.65%
	Total cost of Debt	*per annum*	**14.03%**	**13.43%**	**12.83%**
	Finance Fees				
	Upfront fees	*Once off*	1.00%	1.00%	1.00%
	Commitment Fees	*per annum*	0.75%	0.75%	0.75%
6	**GCF Concessional Loan**				
	Indicative Debt Terms				
	Loan Tenor	*years*	10.0	10.0	10.0
	Grace period	*years*	3.0	3.0	3.0
	Cost of Debt				
	Base swap rate	*per annum*	6.73%	6.73%	6.73%
	Liquidity cost	*per annum*			
	Statutory cost	*per annum*			
	Credit margin	*per annum*	1.27%	1.27%	1.27%
	Currency swap (USD-ZAR)		0.45%	0.45%	0.45%
	Transaction Cost		0.15%	0.15%	0.15%
	Total cost of Debt	*per annum*	**8.60%**	**8.60%**	**8.60%**
	Finance Fees				
	Upfront fees	*Once off*	1.00%	1.00%	1.00%
	Commitment Fees	*per annum*	0.75%	0.75%	0.75%

It is noted that the grace periods for the different instruments for underlying projects vary between 2 and 4 years, based on the construction period. The table above reflects the mid-point for illustrative purposes only. The credit margin for senior debt instruments decreases by roughly 50-60 bps when GCF funding, without a partial credit guarantee, is introduced into the funding mix and a further 50-60 bps, when GCF funding is accompanied by a partial credit guarantee.

It is envisaged that the GCF will provide concessional loans to both public sector and the private sector project owners (see Section above). In this instance, it applies different terms and conditions for outgoing concessional loans, depending on, among other factors, whether the borrower is public or private.

The WRP will include sub-projects in its portfolio with both public sector and private sector project owner. Whether a project is classified as either publicly owned or privately owned, is based on the majority equity ownership in the underlying project legal form (such as a PPP-SPV, in the case of privately owned projects). Based on the initial proposed portfolio of ten projects, it is estimated that two distinctly priced tranches of GCF concessional loans will be required, comprising 65% public and 35% private, of the total GCF concessional loan amount. As a result, the GCF concessional loans applicable to the WRP, will be differentiated by applying the following proposed interest rates, based on the respective public sector and private sector GCF concessional loan tranches (Table 19).

Table 19: Proposed interest rates, based on the respective public sector and private sector GCF concessional loan tranches.

Concessional Tranche	Amount (USDm)	Indicative Pricing (in USD terms)
Public sector	130	Interest rate – 75 bps p.a. Service fee – 50 bps p.a **All-in rate = 1.25%,** plus Commitment fee – 25 bps
Private sector	70	Interest rate – 100 bps p.a. Service fee – 50 bps p.a. **All-in rate = 1.50% p.a.**, plus Commitment fee – 25 bps
Total Amount	**200**	-

It is important to note that the underlying information in the application (and particularly the Programme Financial Model and related financial analyses) is not based on these different rates applied to the public and private concessional loan tranches. Instead, the Programme Financial Model and related financial analyses are based on a single tranche with pricing that is more reflective of privately owned projects. However, the climate adaptation beneficiary impact metrics do reflect this differentiation in GCF concessional loan pricing based on the estimated split of the project portfolio between publicly and privately owned projects.

Impact of GCF funding on water reuse charges

Table 20, below, provides an indication of the lower Water Reuse charges, made possible by the introduction of GCF funding.

Table 20: Potential for lower water charge due to GCF funding.

Project #	Municipality	Water charges with GCF Funding ZAR/Kℓ	Water charges without GCF Funding ZAR/Kℓ	Difference ZAR/Kℓ
1	Nelson Mandela Metropolitan Municipality	13.82	30.28	(16.46)
2	City of Ekurhuleni	26.19	31.32	(5.13)
3	City of Johannesburg Metropolitan Municipality	32.61	32.61	-
4	City of Tshwane Metropolitan Municipality	27.75	32.30	(4.55)
5	eThekwini Metropolitan Municipality	30.87	30.87	-
6	Indicative Project A	24.01	39.85	(15.84)
7	Indicative Project B	25.38	36.59	(11.21)
8	City of Cape Town Metropolitan Municipality	27.76	30.22	(2.46)
9	Drakenstein Local Municipality	25.25	35.62	(10.38)
10	Mangaung Metropolitan Municipality	25.38	33.33	(7.96)

Best Available Technology

The Programme will require the use of the latest international best practice in water reuse design approaches and technology selection. By ensuring that the reuse treatment train designs are vetted by a panel of experts, the WRP will hold the individual sub-projects to the strictest reuse practices, follow the World Health Organization's (WHO's) Potable Reuse guidelines (as well as SANS 241), achieve a minimum pathogen log reduction across the treatment train and have sufficient micropollutant removal. This can only be achieved if the latest technologies are selected for each of the sub-projects.

For Direct and Indirect potable reuse, the selection and agreement of the treatment train will be a key component of the project vetting process and a clear Hazard Analysis Critical Control Points (HACCP) will be required for the selected treatment technologies to ensure a minimum pathogen log reduction and micropollutant removal. South Africa doesn't have a regulated potable reuse minimum standard and it is proposed that the programme adopts the California State Department of Health potable reuse standards for pathogen reduction (12/10/10) and micropollutant (Chemicals of Emerging Concern) removal.

E. LOGICAL FRAMEWORK

*This section refers to the project/programme's logical framework in accordance with the **GCF's Integrated Results Management Framework** to which the project/programme contributes as a whole, including in respect of any co-financing.*

E.1. Project/Programme Focus

☐ Reduced emissions (mitigation)

☒ Increased resilience (adaptation)

E.2. GCF Impact level: Paradigm shift potential (max 600 words, approximately 1-2 pages)

Assessment Dimension	Current state (baseline)		Potential target scenario (Description)	How the project/programme will contribute (Description)
	Description	Rating		
Scale	Climate induced water stress is impacting on local water supplies that are largely dependent on surface water resources supplemented with groundwater supply. Increasing water supply deficits require alternative sources of supply. Water reuse sub-projects have been limited to date with only 12 identified sub-projects in operation to date, some of which are small scale, and in total are producing about 172Ml/day[lxxx]. Of the total water supply mix only 14% is from water reuse[lxxxi].	<u>Medium</u>	A paradigm shift would Involve less reliance on surface water and groundwater resources that are often over-allocated and also have an important role to play in maintaining ecological infrastructure. The upscaling of water reuse sub-projects would require programmatic planning and preparation support be provided to municipalities (Outcome 1) together with improvement in the awareness and understanding of stakeholders of water reuse as an important intervention to support climate adaptation (Outcome 3).	The provision of programmatic support and planning through a Water Partnership Office (Outcome 1) will assist municipalities to develop prioritised investments in large-scale water reuse sub-projects with a potential market of 27 potential sub-projects with a total reuse flow of 1067Ml/day. This would attain 46% of the projected 2040 target for municipal and industrial water reuse as set out in the National Water and Sanitation masterplan.

Replicability	The existing reuse projects are generally (with some exceptions) small scale, but it is evident that the planned reuse sub-projects appear to be much larger. An economy of scale can be achieved with such projects , but there is a reluctance to invest a large amount if the technology appears to be unproven or there is inadequate skills at municipal level to drive such projects. This is to be expected, as initially the sub-projects would need to be based on proven technology and would need to not be excessively large (in terms of capacity and investment costs required) in order to prove a 'proof of concept' and achieve the public's acceptance.	Medium	The paradigm shift would be realized when innovative project implementation is ensured through technical support (Outcome 2), and knowledge and technology transfer on water reuse is provided (Outcome 3), whereafter a move to larger full-scale projects may be made to realise the economy of scale benefits of this programmatic approach.	The provision of technical and financial support for the design and implementation of water reuse sub-projects (Outcome 2) will not only assist municiplaities to ensure the roll-out of innovative projects, but will also enable the transfer of lessons learned to other municiplaities and water reuse sub-projects (Outcome 3).
Sustainability	The realization of Government of the need to address increasing climate vulnerabilities and water insecurity across the country has been outlined in the various climate adaptation and water sector strategies. However, there is still much to be achieved in strengthening institutional and regulatory frameworks and capacity to develop the sustainable solutions required. Unfortunately, there are also limited financial resources available to drive such initiatives.	Medium	The paradigm shift would be realized when sustainable government interventions are also supported by the crowding-in of private capital, other commercial funding and development finance to provide a range of financing options that will unlock water reuse sub-projects (Outcome 2).Additionally, the improvements in capacity and understanding at municipal scales and within the private sector will support the effective and sustainable management of these interventions.	While the WRP will work with government and municiplaities to build their capacity with regards to climate adaptation and water reuse as an approach to improve water security (Outcome 3), the programme will work with private sector partners who can support the sub-projects technically, operationally and financially (Outcome2).

E.3. GCF Outcome level: Reduced emissions and increased resilience (IRMF core indicators 1-4, quantitative indicators)

GCF Result Area	IRMF		Baseline	Target	Assumptions / Note

	Indicator	Means of Verification (MoV)		Mid-term	Final[8]	
Total Beneficiaries		Municipal customer databases and municipal population data and statistics Household and affected stakeholder surveys undertaken at Municipal level Progress reports (annual) and final report to be completed at project level, as well as at WPO level	0 males and females reached	Direct Beneficiaries Total = 1369895 Male=661659 Female=708236	Direct Beneficiaries Total = 3424737 Male=1654148 Female=1770589	Based on the per capita consumption of potable water in each municipality, and the additional water that will be supplied by the WRP across its sub-projects
ARA1 Most vulnerable people and communities	*Core 2: Direct and indirect beneficiaries reached*	Municipal customer databases and municipal population data and statistics Household and affected stakeholder surveys undertaken at Municipal level Progress reports (annual) and final report to be completed at sub-project level, as well as at WPO level	0 males and females reached	Direct Beneficiaries Total = 1369895 Male=661659 Female=708236 Direct	Direct beneficiaries Total = 3424737 Male=1654148 Female=1770589	Based on the per capita consumption of potable water in each municipality, and the additional water that will be supplied by the WRP across its sub-projects, direct beneficiaries are counted as those additional people who benefit from the additional water supplied into the municipal systems (per capita water consumption * additional water supplied = additional people receiving water). This calculation is then disaggregated by male and female beneficiaries using South Africa's national proportional split between

[8] The final target means the target at the end of project/programme implementation period. However, for core indicator 1 (GHG emission reduction), please also provide the target value at the end of the total lifespan period which is defined as the maximum number of years over which the impacts of the investment are expected to be effective.

						males and females in the population.
	Supplementary 2.1: Beneficiaries (female/male) adopting improved and/or new climate-resilient livelihood options	Municipal customer databases and municipal population data and statistics Household and affected stakeholder surveys undertaken at Municipal level Progress reports (annual) and final report to be completed at sub-project level, as well as at WPO level	0 males and females benefitting from the adoption of diversified, climate resilient livelihood options	Direct Beneficiaries Total= 273979 Male = 132332 Female= 141647	Total= 684 948 Male = 330830 Female= 354118	Improved water security provides the basis for the use of water for local productive purposes 20%[9] of the direct beneficiaries are the most vulnerable and will use improved water security for improved livelihoods choices. Municipalities prioritize local community level livelihood initiatives through integrated development planning
ARA2 Health, well-being, food and water security	*Core 2: Direct and indirect beneficiaries reached*	Municipal customer databases and municipal population data and statistics Household and affected stakeholder surveys undertaken at Municipal level Progress reports (annual) and final report to be completed at sub-project level, as well as at WPO level	0 males and females reached	Direct Beneficiaries Total = 1369895 Male=661659 Female=708236 Direct	Direct Beneficiaries Total = 3424737 Male=1654148 Female=177058 9	Based on the per capita consumption of potable water in each municipality, and the additional water that will be supplied by the WRP across its sub-projects, direct beneficiaries are counted as those additional people who benefit from the additional water supplied into the municipal systems (per capita water consumption * additional water supplied = additional

[9] http://www.statssa.gov.za/?p=12135

						people receiving water). This calculation is then disaggregated by male and female beneficiaries using South Africa's national proportional split between males and females in the population.
	Supplementary 2.3: Beneficiaries (female/male) with more climate-resilient water security	Municipal customer databases and municipal population data and statistics Household and affected stakeholder surveys undertaken at Municipal level Progress reports (annual) and final report to be completed at sub-project level, as well as at WPO level	0 males and females with year-round access to reliable and safe water supply despite climate shocks and stresses	Direct Beneficiaries Total = 1369895 Male=661659 Female=708236 Direct	Direct Beneficiaries Total = 3424737 Male=1654148 Female=1770589	Based on the first phase selection of 10 water reuse sub-projects Absence of extreme natural disasters that significantly reduce catchment yields, or economic shocks affecting household economy; and migration patterns
ARA3 Infrastructure and built environment	Core Indicator 3 Value of physical assets made more resilient to the effects of climate change and/or more able to reduce GHG emissions	For target (ex-ante) estimate values, project financial data from pre-feasibility studies, feasibility studies, economic /financial analysis; For actual (ex-post) results, actual cost incurred during the reporting period	0 support provided	USD 392m (aligned with the budget plan)	USD 1400m (aligned with the budget plan)	Project preparation pipeline develops progressively; Co-financiers support blended finance solutions. Project financial data is monitored, reported and reviewed. Project evaluation studies are undertaken to assess progress.

E.4. GCF Outcome level: Enabling environment (IRMF core indicators 5-8 as applicable)					
Core Indicator	**Baseline context (description)**	**Rating for current state (baseline)**	**Target scenario (description)**	**How the project will contribute**	**Coverage**
Core Indicator 5: Degree to which GCF investments contribute to strengthening	NWRS and the National Climate Change Adaptation Strategy exists and provides direction in terms of the	medium	The regulatory frameworks are in place to incentivize interventions to improve	One of the three programme components is dedicated to the supporting	Multiple sub-national areas within a country

institutional and regulatory frameworks for low emission climate-resilient development pathways in a country-driven manner	importance of climate adaptation and water security. This is supported by national, catchment and local scale water management and integrated development plans that provide technical and operational guidance. While these instrurments and the institutional frameworks are in place, there is a need to improve the regulatory frameworks and institutional capacities to address barriers and to support water reuse upscaling.		local level water use efficiency and improve water security. This is supported by improvements in the Green Drop and Blue Drop regulatory incentive system results. The institutional capacity is in place within 10 municipalities to support the development, procurement, oversight and operational implementation of local water reuse sub-projects.	the development of institutional capacity and the improvement of regulatory frameworks in order to facilitate the upscaling of water reuse sub-projects. Project outcomes will include enhanced institutional arrangements and stakeholder capacity to support the implementation of transformative water reuse sub-projects	
Core Indicator 6: Degree to which GCF investments contribute to technology deployment, dissemination, development or transfer and innovation	Direct Potable reuse is the preferred water reuse archetype, but most operational DPR projects are small in scale. However, there are several planned DPR projects indicating that Potable Reuse (over industrial reuse) is gaining popularity. This is understandable given the potable reuse technology is relatively new and the capacity at municipal levels to support the upscaling of new technologies is low.	medium	The WRP has enabled the use of the latest international best practice in water reuse design approaches and technology selection. By ensuring that the reuse treatment train designs are vetted by a panel of experts, the Programme has held the individual sub-projects to the strictest reuse practices, follow the WHO Potable Reuse guidelines (as well as SANS 241), have achieved a minimum pathogen log reduction across the treatment train and have ensured sufficient micropollutant removal.	The WRP will finance the design and implementation of Advanced Treatment Plants supported by technical experts that will provide oversight in the delivery of these sub-projects to ensure the use of the latest international best practice in water reuse design approaches and technology selection.	Multiple sub-national areas within a country
Core indicator 7: Degree to which GCF Investments contribute to market development/transformation at the sectoral, local, or national level	While there are numerous drivers, there are also significant barriers to investment in municipal-scale water reuse sub-projects in the South African context. In general, water reuse not only requires additional treatment capacity to reach potable standards for DPR or IPR, but also often requires upgrading local	medium	Municipal uptake for the development and implementation of water reuse sub-projects has been significantly improved with a pipeline of projects using innovative technical, operational and financial arrangements.	The programme will work at national and local scales to build the capacity and incentivize the development of water reuse sub-projects and will through project outcomes share knowledge and lessons learned that will assist the catalytic	Multiple sub-national areas within a country

	municipal WWTW to meet quality discharge standards. This creates a barrier to water reuse as the required capacity (technical, institutional and financial) to facilitate these upgrades is not always available in the public sector.			implementation of new reuse sub-projects. The programme will provide easy access to expertise to assist project owners to unlock new projects.	
Core indicator 8: Degree to which GCF investments contribute to effective knowledge generation and learning processes, and use of good practices, methodologies and standards	There is limited technical understanding of climate change impacts and the water reuse alternatives to address the associated water insecurity. This is often exacerbated by the poor acceptance of water reuse as an alternative to water supply by stakeholders.	medium	Water reuse sub-projects are being implemented across the country to support climate change adaptation, with associated knowledge and lessons learned being shared across projects to ensure best practice and the attainment of standards.	The programmes focus upon developing adaptive capacity (component 3), underpinned by the establishment of a WPO that will provide access to a panel of experts, will ensure that best available practice and lessons learned will be shared with new and ongoing sub-projects.	Multiple sub-national areas within a country

E.5. Project/programme specific indicators (project outcomes and outputs)

Project/programme results (outcomes/ outputs)	Project/programme specific Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final	
Outcome 1: Climate resilient water reuse sub-project solutions at municipal levels are taken to scale and are being replicated	Number of bankable water reuse sub-projects identified and structured in the programme project pipeline	Progress reports (annual) and final report to be completed at sub-project level, as well as at WPO level	0	4	10	Municipalities prioritise water reuse sub-projects. Sector institutions support in promoting water reuse interventions. Technical and financial support ensures sub-projects are viable and bankable.

Output 1.1: The WRP is established and operationalized	WRP WPO and WRU established and strategic plan and annual workplan and budgets developed	Progress reports (annual) and final report to be completed at sub-project level, as well as at WPO level	0	WPO fully operational. First order strategic plan developed and annual workplans completed	Second order strategic plan developed and implemented. Monitoring and evaluation reports completed.	The WRP Oversight Committee will monitor progress and collate reports
Output 1.2: Identified and transformative large-scale water reuse sub-projects are supported, structured and prepared.	Number of sub-projects Implemented	Project monitoring and completion reports	0	4	10	Municipalities prioritise water reuse sub-projects. Sector institutions support in promoting water reuse interventions. Technical and financial support ensures sub-projects are viable and bankable
Outcome 2: : A new asset class of sustainable water reuse sub-projects have improved the water security of municipalities and communities	Number of people with increased access to water of permissible quality standards due to water reuse interventions	Project monitoring and completion reports Household surveys, disaggregated by sex	0	Direct Beneficiaries Total = 1369895 Male=661659 Female=708236 Direct	Direct Beneficiaries Direct Beneficiaries Total = 3424737 Male=1654148 Female=177058 9	Based on the first phase selection of 10 water reuse sub-projects. Absence of extreme natural disasters that significantly reduce catchment yields, or economic shocks affecting household economy; and migration patterns.
	Number of people adopting improved and/or new climate-resilient livelihood options	Project monitoring and completion reports Household surveys, disaggregated by sex	0	Direct Beneficiaries Total= 273979 Male = 132332 Female= 141647	Total= 684 948 Male = 330830 Female= 354118	Based on improved water security, will support communities to take up new livelihood options. Results may be delayed as water reuse sub-projects need to realize improved water security.

Output 2.1: Innovative project implementation is ensured through technical support, and knowledge and technology transfer on water reuse	Volume of additional water treated for reuse	Project monitoring and completion reports	0	200Ml/day	440 Ml/day	Based on the first phase selection of 10 water reuse sub-projects
	Number of knowledge exchange / technical support sessions held per sub-project	Project monitoring and completion reports, Agenda and support materials/ knowledge products	0	16	40	On average 4 work sessions will be held per sub-project
Output 2.2: Mobilized capital from a range of sources has de-risked and supported sub-project bankability and the replication of water reuse investments for climate adaptation	Number of innovative blended finance mechanisms piloted and established	Project monitoring and completion reports	0	4	10	At least 1 per sub-project
	Total value of WRP funds disbursed	Project monitoring and completion reports	0	900	1472	Million USD value of based on pipeline projection as per impact model. All approved funds will be dispersed
	Number of new projects applications utilising developed belnded finance solutions	Project monitoring and completion reports, Record of project applications	0	5	10	Additional projects beyond the current pipeline that apply for support based on the financing approaches developed
Outcome 3: : Adaptive capacity of the water sector has been improved to support climate responsive planning and development	Number of people who report increased resilience to climate risks due to improved awareness and understanding	Project monitoring and completion reports Household surveys, sex disaggregated. Municipal statistics.	Limited systematic effort to create awareness and ensure knowledge exchange regarding climate threats	Total= 2190763 Male = 1050666 Female= 1139197	Total= 7302542 Male= 3505220 Female= 3797322	Awareness creation and knowledge exchange reaches both direct and indirect beneficiaries supporting taking the sub-projects to scale No perverse incentives (policies, regulations, prices etc) are introduced in the sub-project areas
	Number of senior municipal staff trained with regards to climate adaptation	Project monitoring and completion reports	0	12	30	3 senior staff per municipality
Output 3.1: Enhanced institutional and	Number of policy, strategy and regulatory	Project monitoring and completion reports	0	At least 4 municipalities	At least 10 municipalities	Identification of policies, strategies, and regulatory

stakeholder capacity has supported the implementation of transformative water reuse sub-projects	instruments improved to strengthen climate adaptation			have climate responsive plans and regulatory frameworks	have climate responsive plans and regulatory frameworks	instruments requiring strengthening at programme inception At least one instrument per sub-project.
	Number of local level public functionaries trained and capacitated in climate change management planning and financing.	Progress reports (annual) and final report to be completed at sub-project level, as well as at WPO level	0	At least 40	At least 100	An agreement will be signed with each sub-project and will incorporate staff to be trained, this included staff from various spheres of government
	Number of males and females made aware of climate threats and related appropriate responses	Project assessment reports. Hits on information hub/ website	Male=0 Female=0	Total= 102742 Male = 49625 Female= 53117	Total= 342474 Male = 165415 Female= 177059	Projects at local level will drive a need amongst people to develop their understanding of climate risk and water reuse as an adaptation response
	Number of working groups established to support water reuse sub-projects	Project monitoring and completion reports	0	4	10	1 working group per municipality to support water reuse sub-projects and facilitate knowledge exchange
Output 3.2: Strengthened institutional and regulatory frameworks are realized to embed water reuse as a key component of climate adaptation	Institutional reforms undertaken to support the implementation of water reuse	Project monitoring and completion reports	0	2 policy adjustments	8 policy adjustments	At least one adjustment per to water resource management policy and one to relevant municipal policy by mid-term.
	Water quality standards for water reuse are amended	Project monitoring and completion reports Government Gazette	0	Draft	Finalised and gazetted	The support of the South African Bureau of Standards and DWS are essential
Project/programme co-benefit indicators						
Co-benefit 1: Environmental	Reduced pollution into local water resources	Catchment sampling and infrastructure design reports	Baseline sampling per sub-project	10% reduction	20% reduction	Baseline sampling and water resources reports will inform on local pollution issues
	Increased volume of flow in local watercourses	Catchment sampling and infrastructure design reports	Baseline sampling per sub-project	5% increase	10% increase	Baseline sampling and water resources reports will inform on local water resource issues
	Reduced GHG Emissions	Project monitoring and completion reports	0	TBD for each sub-project	TBD for each sub-project	The mitigation impact will be estimated based upon the technology scope on each

						sub-project (e.g. biogas, sludge management, sludge beneficiation, etc.) using an appropriate carbon methodology.
Co-benefit 3: Economic	Number of jobs created	Project monitoring and completion reports Household surveys sex disaggregated.	0	Total= 300 (220 male, 80 female)	Total= 585 (460 male, 125 female)	Jobs saved through improved water security= 205 (161 male, 44 female)
	Number of private sector actors/companies that are trained and that will start providing climate resilience technologies and inputs introduced by the sub-project in target areas.	Project monitoring and completion reports	0	4	10	At least 1 per sub-project
Co-benefit 4: Gender	Positive impact on local women livelihoods as a result of improved water security	Household surveys sex disaggregated.	Baseline survey	25% improvement	50% improvement	Water reuse sub-projects will result in improved water security and water services

E.6. Project/programme activities and deliverables

Activities	Description	Sub-activities	Deliverables
Component 1 – Establish a WRP and project pipeline			
Sub-component 1.1 Establishment and Operationalisation of the WRP	Facilitate the establishment of the WRP within the DBSA and clarify operational aspects	Activity 1.1.1: Establish the WRP within the DBSA	• Operational WRP and supporting staffing and systemic competencies. • MOA between the Programme owner DWS and, DBSA • Final organizational design and operational case for the WRP and WPO • Draft Terms of reference for the OC
	Facilitate the establishment of the OC and establish the protocols for programmatic governance and oversight	Activity 1.1.2: Establish Oversight Committee and develop governance modalities	• Operational OC • Finalised OC Terms of Reference and governance modalities report • Regular meeting minutes (quarterly) • Monitoring and evaluation reports (bi-annually)

	Appointment of staff to the WPO and the Water Reuse Unit (WRU)	Activity 1.1.3: Appoint staff and operationalise the WPO and technical WRU	• Final organizational design report including organograms, job specifications, salaries and staff benefits, the contractual arrangements to support staff appointments. • Phased staffing plan aligned to the PIP
	Development of key planning instruments to support the implementation of the WRP	Activity 1.1.4: Develop PIP and annual workplan and budgets	• WRP five-year PIP • WPO/ WRU annual workplan and budget
	Preparation of the final operational manual and supporting tools based on the draft manuals and tools prepared as part of the programme design	Activity 1.1.5: Finalise operational manual and supporting tools	• WRP operations manual • Standardised contracting documents and procurement procedures for professional service providers
	Formalization of the financial management arrangements and the necessary reporting that would underpin this based upon the financial architecture developed during programme design	Activity 1.1.6: Establish financial arrangements and reporting regimes.	• Financial management manual • Clarified business processes and standard operating procedures • Finalise the eligibility criteria for sub-projects
	Finalise programme level monitoring, reporting and evaluation protocols in compliance with the internal systems of the DBSA and aligned to GCF requirements	Activity 1.1.7: Develop and implement monitoring and evaluation and reporting regimes	• Final monitoring, evaluation, earning and reporting framework and protocol • Regular programmatic reporting aligned to the monitoring, evaluation, learning and reporting framework and protocol • Annual programme review and evaluation reports
Sub-component 1.2 Project pipeline preparation	Development of the project pipeline as well as engage with already identified priority sub-projects to initiate support from the WRP	Activity 1.2.1: Design and launch an open-ended RFP.	• Call for projects including key project selection criteria • Ongoing market research and pipeline development. • Short-list of initial sub-projects for support • Long-list of longer-term sub-projects to support
	Establishment of the panel of service providers to provide support for the preparation of sub-projects and their implementation, this will include the finalization of procurement procedures drafted during programme design	Activity 1.2.2: Procure and manage a panel of project preparation service providers	• Finalsed procurement procedures • Database of service providers • Drawdown of technical, financial, legal service providers to support project preparation • Progress reports
	Provision of support to the sub-projects for their preparation including scoping, feasibility and procurement	Activity 1.2.3: Provide technical assistance for sub-project structuring and preparation	• Project management reports • Project readiness reports • Suite of bankable sub-projects
Component 2 – Implementing water reuse projects			

Sub-component 2.1 Provision of Technical Assistance for Sub-project Implementation	Utilising the findings of the feasibility study, technical support will be provided to undertake the detailed design of the water reuse sub-projects including the development of an implementation plan as well as providing support in sub-project implementation where required/ appropriate.	Activity 2.1.1: Provide technical assistance for detailed design and implementation of sub-projects	• Sub-project detailed design and implementation plans • Project regulatory compliance reviews
	Preparation of procurement plans and contracting arrangements for each sub-project, developed in conjunction with the Project Owner	Activity 2.1.2: Facilitate procurement and contracting	• Procurement plans • Contractual arrangements and supporting legal documents
	Development of project management and monitoring plan and undertake monitoring of progress with regards to sub-project development	Activity 2.1.3: Monitor, evaluate and report on sub-project development.	• Project management and monitoring plan • Progress reports • Tracking of number of sub-projects under construction or commissioned
Sub-component 2.2 Development of the Blended Financing Solutions	Clarification and formalization of protocols that will outline the approaches to blending capital and how GCF funding will be blended with private investor funding, to enable private funders and municipalities to lower their risk exposure on a project-by-project basis.	Activity 2.2.1: Develop principles and approved protocols for developing the BFS	• Blended finance principles and protocols report
	Design of BFS to provide blended project finance to a portfolio of sub-projects in the WRP, on a project-by-project basis. Each sub-project will be prepared with its own financial analysis and due diligence. This will inform the type and corresponding magnitude of finance that will be allocated to the sub-project through the BFS. The types of financial instruments that could be allocated by the BFS in a blended fashion, include: equity, senior loans, subordinated loans, and local government budget allocations. In addition to these project finance instruments, a partial credit guarantee instrument is used, to attract a greater portion of senior debt, and reduce the portion of subordinated debt.	Activity 2.2.2: Project finance support through the BFS.	• Project finance analysis, and blended finance allocation report, per sub-project; • Term Sheet of instrument pricing, per sub-project; • Fund deployment and reporting.
	Undertaking the necessary steps to establish the Independent Blended Capital Facilitator functions as well as appointing an	Activity 2.2.3: Establish the IBCF function and appoint specialist independent service provider	• Terms of Reference for the IBCF function • Contractual arrangements to support the appointment of a service provider

	independent specialist service provider to perform this function.		
	Providing support to the WRP and Project Owners in developing Blended Financial Solutions on a project-by-project basis	Activity 2.2.4: Provide financial structuring services on a project-by-project basis	• Functional protocol and due diligence procedures • Innovative blended financing solutions developed and documented on a project-by-project basis
	Coordinating the continual measurement, verification and reporting of Impact Metrics, against the Impact & ESG Framework, with the support of the WRU and WPO. The IBCF will be responsible to report on all of its undertaking to the WPO, as well as provide an overview of the BFS's use, status and efficacy.	Activity 2.2.5: Audit and report to measure impact of the BFS use and efficacy	• Impact & ESG monitoring framework • Quarterly progress reports and documenting of lessons learned
Component 3 – Building capacity and creating awareness			
Sub-component 3.1: Communications and awareness creation	Development of communications strategy and brand as well as establish an information hub that will underpin the establishment the WRP as a centre of excellence.	Activity 3.1.1: Develop, manage, and maintain the WRP brand	• Corporate identity and branding guideline • Programme communications strategy • Information and knowledge exchange hub
	Development of knowledge products and collateral that supports the development of awareness and improvement of understanding of water reuse as a climate adaptation response.	Activity 3.1.2: Development of knowledge products and collateral	• Communication materials and products.
	Undertaking a range of engagement interventions to create awareness regarding climate vulnerability and water security and the role of the WRP and water reuse to support climate resilience	Activity 3.1.3: Undertake outreach and awareness creation events	• WRP events • Technical workshops • Market surveys and sector assessments • Perception surveys
	Establishing and maintaining an advisory forum that will be supportive of developing innovation and supporting knowledge exchange	Activity 3.1.4: Facilitate and manage forum	• Terms of Reference for the Community of Experts • Establishment of the Community of Experts forum • Records of meetings and inputs • Information and knowledge exchange
	Providing independent technical support and review in the preparation, design and structuring of water re-use sub-projects.	Activity 3.1.5: Provide strategic review and guidance on specific sub-projects and interventions	• Protocols for expert review and inputs into sub-projects • Review reports and records of recommendations for sub-projects

Sub-component 3.2: Strengthen institutional and regulatory frameworks	Undertaking a comprehensive stakeholder analysis and developing a strategic approach to ensure ongoing and appropriate stakeholder engagement across the programme	Activity 3.2.1: Develop a key stakeholder engagement framework	• Stakeholder analysis • Stakeholder engagement framework
	Undertaking focused engagements with Government, developers, commercial banks, DFI and other key stakeholders to support the needs of sub-projects, to unlock financial and technical aspects of sub-projects, or in introducing innovative approaches to specific sub-projects	Activity 3.2.2: Undertake targeted strategic outreach	• Technically based events • Sub-project level engagements
	Maintaining key strategic partnerships such stakeholders as Government departments; utilities; business and industry associations, financiers and developers, amongst others, to work collectively at building longer-term institutional capacity as well as addressing various regulatory issues and other barriers to the development of adaptive capacity and the roll-out of water reuse sub-projects.	Activity 3.2.3: Establish and maintain strategic partnerships with key actors and institutions	• Governmental and key sector institutional engagements • Institutional and regulatory reviews and recommendations • Capacity building interventions at varying levels of government and public utilities

E.7. Monitoring, reporting and evaluation arrangements (max. 500 words, approximately 1 page)

Monitoring and Evaluation of the WRP will be aligned to the DBSA Development Results Reporting Framework and the requirements of GCF as detailed GCF master accreditation agreement. This supports 'inwardly' focused reporting to strengthen processes and the achievement of outputs and outcomes, whilst also enabling 'outwardly' focused reporting to the GCF and other key stakeholders on the attainment of impact.

The broad monitoring and reporting requirements are as follows (and in Figure 26):

- **Project level monitoring**
  - Undertaken as per the project specific log frame that will be compiled for each sub-project funded by the WRP.
  - The project teams will be required to report to the WRP biannually based on the project log frame.
  - As part of the monitoring the WRP will establish baselines for each paradigm shift dimension to develop the paradigm shift scorecard using the approach outlined by GCF in the IRMF Handbook.
- **Programme level monitoring**
  - Information from sub-projects will be collated into the Annual Project Report (APR) and submitted to GCF at the end of the calendar year.

- The WRP WPO will compile the APR based on information received on individual sub-projects, together with the broader programmatic objectives towards realizing the identified impact and goals.
- The APR will provide a brief qualitative report on the contribution to paradigm shifts and the enabling environment.

- **Project level evaluation**
  - Mid-term and terminal project evaluations will be undertaken as per the GCF requirements. This will utilize the paradigm shifts and enabling environment scorecards
  - The evaluations will be undertaken by an independent evaluator/ consultant at the cost of the Programme.
- **WRP level evaluation**
  - A mid-term evaluation will be undertaken on the performance of the WRP as per the indicators identified in the programme's log frame.
  - At the end of the term of the WRP a terminal evaluation will be undertaken on the programme's performance as per the log frame indicators and other qualitative information.
  - The midterm and terminal evaluations will be undertaken by an independent consultant appointed by the DBSA.
  - Both evaluations will incorporate the paradigm shift and enabling environment scorecards that would have been developed.

In addition, the DBSA independent evaluation function, the Operations Evaluation Unit (OEU) will provide advisory and oversight services to the WRP. The OEU provides a framework for the monitoring and evaluation of all the DBSA's operations.

On a month-to-month basis the WRP WPO will report to the OC on the day-to-day and operational business of the WPO and the technically focused water reuse unit. Noting the phased established and development of the WRP, this reporting will be particularly important in the sustainable set-up of the programme, the finalization of standardised documentation and protocols. This more operationally focused reporting will be against annual work plans that the WPO have developed and agreed upon with the OC.

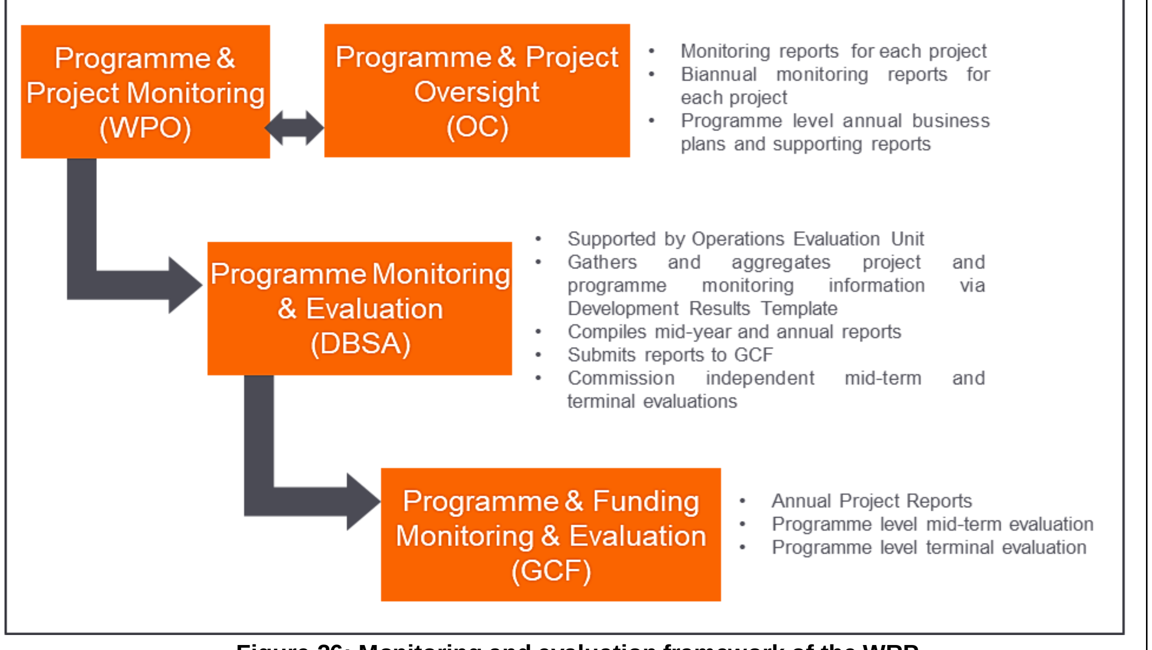


Figure 26: Monitoring and evaluation framework of the WRP

F. RISK ASSESSMENT AND MANAGEMENT

F.1. Risk factors and mitigations measures (max. 3 pages)

The risks that exist in South Africa's water reuse sector currently undermine the country's ability to use water reuse effectively as part of its measures to respond to water security challenges as a result of climate change. The WRP will manage these risks and mitigate their impacts as tabulated below. In addition, it must be noted that:

- No sub-projects will be undertaken in any jurisdiction which is subject to or affected by United Nations Security Council Resolutions (UNSCRs).
- No individual or entity that is listed on any UN Security Council sanctions list, including the UN Consolidated Sanctions list will be involved in any manner with the sub-project or its activities, either as a counterparty, implementer, or beneficiary.
- There are no intentions to directly distribute or disburse funds to beneficiaries.

Selected Risk Factor 1: Generating project pipeline			
Category	**Probability**	**Impact**	**Priority**
Technical and Operational	Low	Medium	Low
Description			
Municipal capacity is often limited at technical and operational levels, with these staff often being challenged in the ability to conceptualise and prepare water reuse sub-projects that are bankable, this is further exacerbated by weak capacity to support the procurement and contracting required to undertake these infrastructural solutions. As such, these innovative projects that are important to address local climate and water sector vulnerabilities are often not prioritised for municipal investment. The 2018 Municipal Water Services Authority Business Health assessment found 78% of municipalities to be extremely and highly vulnerable in their abilities to provide water services. Generating a sufficient project pipeline to support the WRP, under these circumstances, will be challenging.			
Mitigation Measures			
The WRP has undertaken a market study that provides a strong initial pipeline that will enable the WRP to be initiated. Through these initial sub-projects the WRP will develop and strengthen its competencies as well as develop the longer-term project pipeline. The WPO will provide support to Municipalities to conceptualise and prepare bankable projects. The WPO will be supported by the procurement competencies of the DBSA thereby enabling the provision of contracted technical and financial support for project preparation.			

Selected Risk Factor 2: Public Perception			
Category	**Probability**	**Impact**	**Priority**
Technical and Operational	Medium	Medium	Medium

Description			
Despite water re-use being increasingly accepted as a technical response to climate vulnerability and water insecurity, public perception relating to the reuse of wastewater effluent is still a significant obstacle to the projects. This is exacerbated by limited trust in the levels of competency with municipalities to develop, operate and maintain infrastructure. Local case studies quote several instances where this poor public perception has resulted in the failure of direct water reuse sub-projects.			
Mitigation Measures			
Given the scale of South Africa's water challenges and the investment required to diversify the current water mix, it is crucial that the WRPs implementing partners communicate clearly, consistently, and effectively about South Africa's water challenges, the need for specific water reuse sub-projects and how such projects will be rolled out. Water re-use at a policy level is already recognized as part of the water mix (NWSMP, DWS (2018)) and already successfully implemented in some municipalities. This will be leveraged by the WRP WPO, using strategic communications in partnership with DWS, DCOG, MISA and ISA to make stakeholders aware of the benefits of water reuse. The WRP has developed a robust Communication Strategy, with a clear public outreach campaign as a key mitigation measure to manage public perception of the WRP and to develop an improved awareness of the importance of these interventions to support ongoing growth and development.			
Selected Risk Factor 3: Partner risk			
Category	**Probability**	**Impact**	**Priority**
Financial	Medium	High	High
Description			
While municipalities often have significant capacity constraints at the technical level, that hinders the conceptualisation and preparation of water reuse sub-projects, there are often significant challenges in terms of municipal leadership in obtaining the required support for such interventions. This can be due to insufficient capacity as well as political interference that results in weak governance that can have impact on the financial status of the municipality as well as its ability to adhere to formal agreements. This is often exacerbated by Municipalities often facing significant financial pressures due to weak financial circumstances and many competing demands on limited financial resources.			
Mitigation Measures			
The WRP will be initiated with partners that are committed to the project outcomes and have experience in implementing similar projects. This will support the WRP in strengthening municipal competencies with time, as well as enabling the WPO to embed its protocols and practices. The Project owners will be the municipalities and as such will engage the WRP for support through the WPO for specified water reuse interventions. Ensuring alignment with municipal Integrated Development Plans (IDP) will be a key element of ensuring strategic commitment at the municipal level, noting the progress of IDP implementation is monitored, measured, and reported. Engagement will be through formal written correspondence, signed by the required authority, as required at various stages and gates of the project preparation process, and will			

allow for the sub-project to be stopped at any stage of project preparation, if appropriate in the circumstances. The municipalities will sign project funding agreements, as formal agreements between the project owners and the lenders. The structure of these agreements will depend on the structure of the funding arrangements and the specifics of the municipal context. In addition, the use of performance guarantees will be used to underpin finance arrangements to mitigate these partner risks.

At this stage no due diligence assessments have been undertaken as this would only be appropriate during the project preparation phase. It is important to note that the municipalities do undergo an annual audit by the national Audit-General and this does provide for the level of due diligence needed for these sub-projects. These assessments will also be supported by the findings from the Municipal Strategic Self Assessment analyses and the Green Drop and Blue Drop performance assessments, all of which are led by the DWS.

Selected Risk Factor 4: Project implementation risks

Category	**Probability**	**Impact**	**Priority**
Financial / technical	Low	High	Medium

Description

There are a range of risks associated with project implementation that can have financial implications as well as implications for the achievement of programmatic targets. As such there could be delays in project delivery due to contracting challenges, construction delays, operational challenges, supplies risks as well as delays and risks associated with municipal support and buy-in. While the initial stages of project preparation should ensure that these risks are well outlined with a clear mitigation strategy, these risks could nevertheless surface.

Mitigation Measures

Project risk assessments will be undertaken as a key element of the project preparation phase and these risks will be outlined with clear mitigation measures during that stage, and in conjunction with the Municipalities. As sub-projects move through the project pipeline, formalised agreements and contractual arrangements will be established with the project owners as well as the service providers to ensure that these risks are not only managed and minimised, but also that appropriate penalties are in place should these delays occur.

Selected Risk Factor 5: Sufficient co-investment to support GCF investment

Category	**Probability**	**Impact**	**Priority**
Financial	Low	High	Medium

Description

In response to the impacts of climate change on the water sector, only through the leveraging of GCF funding will the realisation of the WRP be achieved; enabling the scaled development of water reuse sub-projects nationally and ultimately improving South Africa's climate resilience. While GCF funding will be specifically considered on a project-by-project basis within the larger programme context, the WRP will also be dependent upon a range of co-investments that will underpin the differing components of the programme and provide the blended finance solution to support project implementation costs for Component 2 of the WRP. Approximately 60% (USD 600 mil of the initial total programme costs of USD 1,5 bn) of the financial requirement will be dependent on securing senior debt from private debt capital partners and equity from private development partners. However, it is well noted that private sector involvement will be highly dependent on the WRP's ability to lower their risk exposure, especially on a sub-project level.

Mitigation Measures

The design and establishment of the WRP will be premised on a phased approach, whereby, during the first 3 years of establishment, the DBSA will be the Management Agent, and the WRP's Core Costs will likely be covered by grants. However, over the medium-term, the WRP's Core Costs will be covered through a repayment mechanism as part of the financing of sub-project implementation. Broadly, GCF funding, supplemented by grant funding from National Treasury to support the WPO, will allow for key enablers that will support the diversified co-investment required. The programme including sub-programmes will be subjected to an external audit on an annual basis and this mechanism will ensure that there is no cross-subsidisation of GCF funds.

Selected Risk Factor 6: Foreign Exchange

Category	**Probability**	**Impact**	**Priority**
Financial	**Medium**	**Medium**	**Medium**

Description

The WRP will use GCF concessional finance, which is US Dollar denominated (USD). This concessional capital will be provided to the DBSA who will therefore manage the USD exposure. Furthermore, the WRP's underlying water reuse projects will be financed in South African Rands (ZAR). Therefore, all debt repayments from sub-projects will be in ZAR with DBSA managing the foreign exchange risk of the GCF funding on a sub-project basis.

Mitigation Measures

With the GCF funding being provided through limited recourse loans, the currency mismatch between the funding from GCF and the loan to sub-projects will be managed by the DBSA on a sub-project basis. .

The concept will only be finalised after review by the GCF, with the finalised concept and terms to be included in the FAA, after the Full Funding proposal has been submitted and approved by the GCF.

Risk Factor 7: Compliance (AML/CFT) risk

Category	**Probability**	**Impact**	**Priority**
Financial	Low	Medium	Medium

Description

Instances of fraud and other corrupt practices, as outlined in the GCF's Policy on Prohibited Practices, will undermine the programme's due diligence efforts as well as negatively impact upon the reputation of the GCF, the DBSA as Accredited Entity, and the Executing Entity as well as the creditability of the WRP. Such

issues could ultimately impact upon the levels of investment in the WRP, particularly impacting upon the blended finance solutions that the programme aims to develop.
Mitigation Measures
The South African legislation, through the PFMA (Act 1 of 1999) and Municipal Finance Management Act (Act 56 of 2003) provide for regulatory oversight and procedures to counter prohibited practices. These are supported by various national and provincial level audit processes. In the development of the WPO, the procurement processes, standardised procedures and supporting guidelines will ensure alignment with the GCF's Policy on Prohibited Practices, as well as build upon the systemic checks and balances that are provided by the DBSA, as Accredited Entity. The Accredited Entity shall apply its own fiduciary principles and standards relating to any "know your customer" checks, Anti-Money Laundering (AML) /Combating the Financing of Terrorists (CFT), financial sanctions imposed by the United Nations Security Council which should enable it to comply with the Policy on Prohibited Practices and the principles of the AML/CFT Policy. The relevant policies of the Accredited Entity are substantially consistent with the principles of the AML/CFT Policy. In this regard, the DBSA Whistle Blowing Policy will be utilised to enable staff members to raise concerns where they have reasonable grounds for believing that there is fraud, corruption, corporate crime or impropriety. Additionally, the WPO will develop thorough risk management protocols linked to a comprehensive governance arrangement that will ensure effective oversight and reporting.

G. GCF POLICIES AND STANDARDS

G.1. Environmental and social risk assessment (max. 750 words, approximately 1.5 pages)

Noting the importance of water reuse to diversifying the 'water mix' in South Africa, and the challenges and barriers to entry that exist in the development of these water reuse sub-projects at scale, the development of a focused programme to address these challenges and ultimately implement pathfinder sub-projects is critical to contributing towards building a more climate-resilient water future. The design of a National WRP aimed at enhancing water security and improved resilience to climate change through the scale-up of water reuse approaches and infrastructure in municipalities and through improved management of Ecological Infrastructure (EI).

At the highest level, the WRP will encourage and support municipalities to implement water reuse and reclamation initiatives that would also include nexus issues, such as combined heat and power (CHP) installations and sewage sludge reuse. The WRP would need to be highly effective, technically advanced, and ready-to-launch towards demonstrably achieving climate change resilience objectives (by strengthening the country's adaptive capacity against water stress and scarcity), and measurably maximize climate change mitigation co-benefits, in a manner that explicitly meets the criteria and requirements of the Green Climate Fund.

Four water reuse archetypes that may be considered as potential project implementation options within the WRP and each water reuse sub-project that will be supported by this Programme will follow a rigorous appraisal procedure and will require an Environmental and Social Impact Assessment as well as a Gender, Sexual Abuse, Exploitation and Sexual Harassment and Indigenous Peoples Assessments. The core aims of the programme are to:

- Ensure Climate adaptation as a principle objective of the programme.
- Mainstreaming climate resilience into the water use and reuse sector.
- Assist municipalities to counter the adverse effects of climate change in the water and wastewater services sector.
- Encourage the scaled development of water reuse sub-projects at municipal level;
- Create a new asset class around water reuse infrastructure.
- Improve EI for water security and to meet climate mitigation strategic objectives.

An Environmental and Social Management Framework (ESMF) has been prepared for the Programme with the aim of avoiding and minimising negative environmental and social impacts and enhancing positive aspects of the Programme (Annex 6). The ESMF contains environmental and social baseline information at the national level and provides the framework and guidelines that ensure that the DBSA and the programme beneficiaries are committed and able to comply with GCF Environmental and Social Policies, and DBSA E&SSS. The ESMF is developed at a programme level and provides the framework within which project specific ESMFs will be prepared. In addition, a generic Environmental and Social Management Plan (ESMP) has been developed for the Programme to assist in the compilation of project specific ESMPs which will set out the project specific measures and actions required to comply with the E&SSS over a specified timeframe.

This ESMF has identified the generic high level environmental and social negative impacts and will serve as a guiding document for sub-projects. The ESMF has determined that the archetypes may have negative environmental and social impacts, and proposed sub-projects should follow a minimum due diligence procedure as follows:

- Identify the environmental and social conditions in the targeted municipality along with the applicable ESS safeguards.
- The due diligence for the project must include a dedicated environmental and social risk assessment.

- Establish the conditions and measures required to ensure that all the negative environmental and social impacts are properly and effectively mitigated.

A checklist of Environmental and Social Eligibility Criteria for Potential Projects has been developed to be completed for each water reuse sub-project that will be supported by the Programme and be used to determine the environmental and social eligibility of the sub-projects. If the answer to at least one of these questions is yes, then the sub-project would be classified as a Category 1 sub-project (with potential significant adverse social and/or environmental impacts that are diverse, irreversible, or unprecedented).

A checklist of Environmental and Social Conditions to be fulfilled during Project Design and Implementation has been developed for completion during the Initial Screening of sub-projects as a first step to evaluate the applicability of the DBSA / GCF environmental and social standard safeguards. Planning Phase considerations common to all the archetypes have been included for consideration during the site selection and planning phase and provides various environmental aspects and impacts that may be applicable. Should an Environmental Authorisation and/or water use application (WUA) be required, the sub-projects will be subject to the legislative requirements as well as those of the DBSA E&SSS. It is recommended that as part of the project specific Environmental and Social Impact Assessment, risks are rated in accordance with accepted DBSA risk rating methodology.

The Environmental and Social Risks and potential impacts during the Construction and Operational Phases associated with each type of water reuse archetype have been provided along with potential mitigation measures. The impacts are presented based on the sub-project's phases, construction, operation, and decommissioning of the facility. The construction and operational phase impacts are similar for each type of technology. A number of the impacts can be mitigated/prevented by careful consideration of various aspects during the planning and site selection phases and therefore these impacts and risks must be considered in conjunction with the planning phase considerations. Potential high-level impacts and mitigation measures that are to be considered during the operational stage of any of the selected archetypes/ sub-projects are also provided.

In terms of the safeguards, the sub-projects to be undertaken would be categorised as Category 2 projects and the WRP will establish a dedicated WPO to manage the programme level compliance to GCF environmental and social standards. The WPO will be responsible for implementation of current ESMF. The WPO will be supported by a team of Implementing Consultants that will be responsible for development of feasibility studies and implementing each sub-project including the Environmental and Social Due Diligence components, the ESMPs and the Gender and Stakeholder Engagement Plans.

The Programme will aim to avoid affecting indigenous communities in the implementation of sub-projects. However, should the situation arise where future sub-projects impact on indigenous communities, an Indigenous Peoples Plan (IPP) will be developed. In this regard, it is noted that one sub-project that has been identified in the potential sub-project pipeline is situated in Sol Plaatjie Municipality where indigenous KhoiKhoi and San peoples do reside. The proposed project will be assessed during preparation to assess the level of impact on these communities. At this scoping stage, the impacts are deemed to be only beneficial. Noting this the programme will develop and IPP framework for the entire programme to ensure that such aspects are addressed at project preparation stage. Noting the ongoing marginalisation of indigenous peoples, engagement with Indigenous Peoples Organisations will be undertaken as part of sub-project implementation and to ensure that the proposed IPP framework is adequate and inclusive.

Additional exclusions to the DBSA Exclusion List of projects that should not be funded have also been drafted.

The aim of the ESMF is to avoid and minimise negative environmental and social impacts and to enhance positive aspects of the Programme. The proposed Programme will have an overall positive impact on the environment in terms of more sustainable water resource management, climate adaption and provide municipalities with a more resilient water and climate future. The Programme will also provide indirect positive social impacts including job creation. The negative impacts, which have been described, will be mitigated in the planning, design, planning, construction and operation of the facilities by implementation and compliance with the requirements of the ESMF and ESMP.

G.2. Gender assessment and action plan (max. 500 words, approximately 1 page)

In addressing the current water security issue, a gendered lens to address the broader challenges was required, that not only aligned to global and GCF measures, but also strengthened the country's approach to water, gender and equity. The assessment undertaken demonstrates a sound legal, regulatory and policy framework that will underpin the approach of the WRP to achieve gender equality in the undertakings of the programme. While there is a national policy framework for women's empowerment and gender equality[lxxxii], this is supported by the Women's Charter for Accelerated Development[lxxxiii] and the Codes of Good Practice on Broad Based Black Economic Empowerment (BBBEE).[lxxxiv]

The BBBEE codes address affirmative action both in terms of race and gender equity and address this through assessing Ownership, Management Control, Employment Equity, Skills Development, Preferential Procurement, Enterprise Development and Social-Economic Development.

South Africa in 2008 signed the Southern African Development Community (SADC) Protocol on Gender and Development and ratified it in 2012. Article 12 of the protocol commits States Parties to achieve **50% representation of women** in decision-making bodies in the public and private sectors by 2015. The National Framework for Women's Empowerment provides a target of 30% of new recruits to the middle and senior management echelons must be women. This national framework has provided Government with its gender representation targets for a number of years[lxxxv] with this being supported by racial representation targets of 50% and more, by black and historically disadvantaged peoples (those who did not have the democratic right to vote prior to April 1994), as outlined in the BBBEE codes.

With the development of the sub-project pipeline for the WRP being only at scoping level, detailed baseline assessments within these municipal contexts have not been undertaken and would form a key part of the detailed vulnerability assessments that would be undertaken by the WRP for each sub-project during project inception and preparation phases. The recent Green Drop report (2022) while assessing the performance of wastewater treatment works does look at technical skills available at each plant, it does not gather data in a gendered manner. Likewise, the 2018 Municipal Strategic Self Assessment study that assesses performance in terms of providing water services does assess management skills, staff skills and technical staff capacity. However, this again does not provide gendered data. The development of gender baselines will therefore be a key part of the project preparation phase.

Efforts to drive gender equity is supported a number of mechanisms for transforming gender relations collectively known within government and civil society as the "National Gender Machinery". Together, these mechanisms aim to promote and protect gender equality, both by mainstreaming it and by dealing with it separately. The National Gender Machinery has structures at different levels in national government, legislature, and statutory bodies (Figure 27) and the DWS, DBSA and the WRP will work closely with this machinery to not drive gender equity within the programme and its sub-projects, but also to support national and sectoral imperatives.

While the targets and approaches to achieve gender equity and inclusion are outlined through these various instruments, there are still lingering patriarchal norms, practices, and customs; poverty; historical legacies, gender-based violence and lack of employment opportunities in senior leadership positions that continue to hamper women from achieving economic parity. The water sector has significant impact on gender disparities, and the challenges identified are complex and inter-linked. These are elaborated below.

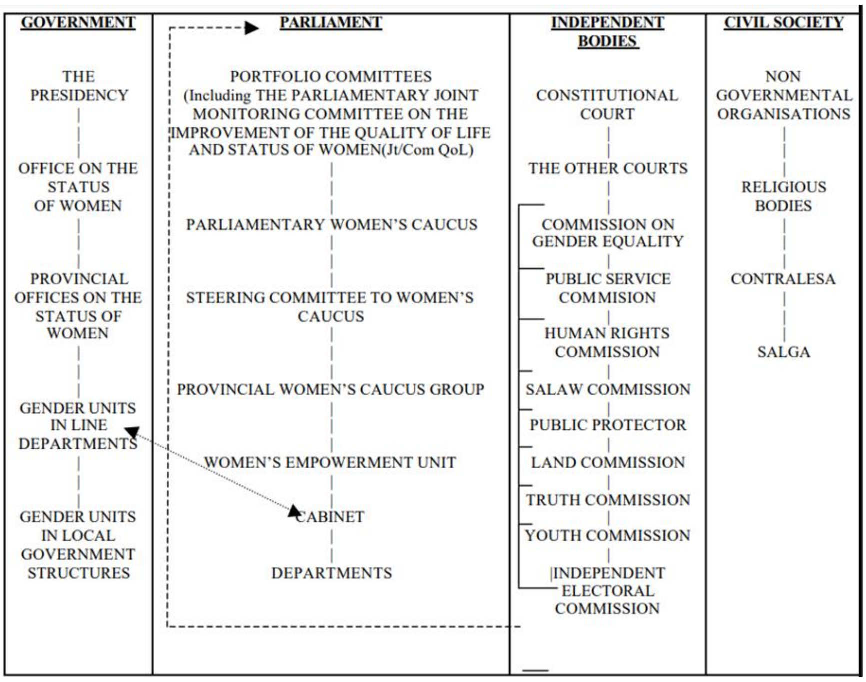


Figure 27: Structure of the national machinery for the advancement of gender equality in South Africa

- **Gender and Climate Change**: Women and girls in developing countries are often more vulnerable than men and boys to the impacts of climate change and have less opportunity to effect change due to pre-existing gender inequalities regarding political leadership, access to information and resources, and mobility and voice. Equally important, due to economic marginalization, political disenfranchisement, and differentiated labour responsibilities, women face the consequences of the climate crisis first and worst, and women lack equitable representation and power in crafting climate policies to address their needs.
- **Gender and Water, Sanitation and Hygiene (WASH)**: Menstrual hygiene continues to receive limited attention in policies, research priorities, programmes, and resource allocation. Most sanitation programmes fail to consider women's need to manage menstruation. Furthermore, latrine design often overlooks the specific needs of women and girls. In cases where hygiene promotion programmes exist, many fail to include the issue of menstrual hygiene, focusing instead principally on hand washing practices. Lack of access to private, safe, and hygienic facilities for managing their menstruation in the workplace can also lead to loss of earnings and present barriers to economic participation.
- **Gender and rural development:** The provision of hygiene and sanitation are often considered women's tasks – especially so in rural contexts. Women and girls are most often the primary users, providers, and managers of water in their households and tend to have the main responsibility for household hygiene. If a water system falls into disrepair, women and girls are usually the ones forced to travel long distances over many hours to meet their families' water needs. Women and children often bear the brunt of the lack of toilets and other sanitation facilities.
- **Gender and the Labour Market**: Historically, from high to low skilled women, women are less likely to seek employment in the utilities sector (for instance, energy, water, gas, electricity etc.). Notably in 2016, high-skilled women accounted for 0.9% of jobs in the utility sector. Similarly in the construction sector, low-skilled women comprise 3.2% of jobs. in 2015, South Africa recorded 45% of total researchers in STEM were women however, leadership and power positions were predominantly held by men.**Invalid source specified.**.

The stakeholder engagement process involved consultations at different levels, not only through the review of documents, but through targeted engagements as well. International, National and local-level discussions were had to glean insights on how to ensure a robust gender-sensitive programme design.

Despite these broader challenges and progressive achievements through policy development in South Africa, gaps still persist. This National WRP provides a pivotable moment to challenge and change the trajectory towards improved gender equality and women's empowerment. Through the various activities as outlined in the Gender Action Plan, the WRP will aim to address the current social and gender inequalities that will impact upon the success of the programme, such as:

- The programme ensures an equitable development undertone that requires an approach that recognises that characteristics such as gender, race, class, and physical ability create overlapping and interdependent vulnerabilities in the face of climate change. The process of incorporating these aspects into the WRP will cement the systematic consideration of the differences between the conditions, situations, and needs of women and men, and the integration of gender equality concerns into the programme design.
- Addressing various stereotypes and norms that lead women to internalise and doubt their ability to assume positions of leadership, and men to question this competency, often resulting in a void of mentorship and leadership in the field to guide and encourage women to navigate through gendered barriers and take these leadership roles. This will be tackled through the Communication Plan and through empowerment approaches at sub-project levels.
- Applying quotas for female and vulnerable groups to participate in water committee, Boards and agencies.
- Ensuring a 30-40% target for the beneficiaries of the program to be women.
- Integrating a monitoring system with gender monitoring indicators and appropriate gender budgets.
- Raising public awareness campaigns aimed at creating an appreciation and understanding of the benefits of water reuse.
- Enabling all water stakeholders—from the implementing agencies to the beneficiaries - to build requisite skills and knowledge for gender-sensitive services and management.

The programme will thus endeavour to explore a multitude of complex gendered issues and offer activities and objectives that support sustainable and equitable outcomes. It will actively engage diverse perspectives and experiences to ensure that the product, services, and systems are designed to address gender differences and barriers. Understanding gender differences along the waste and water value chain will influence decisions, and responsibilities, and will be critical to ensuring access to improved water supply services for all. To identify and design for these gender differences, the programme will be established with a strong intent to transform the water sector and promote positive gender-related impact. As the WRP scales and evolves, greater in-depth research will need to be conducted to complement this current gender assessment report. Each sub-project under the programme will be required to develop context specific gender action plans, ensure protection from sexual abuse, exploitation and sexual harassment, as well as being addressing issues relating to indigenous people and cultural heritage.

The WRP is committed to applying a gender lens to its project design and execution. The team is also acutely aware of the possible failures that may occur when limited consideration is applied to understanding the gender inequalities in a country and at a programme level. These inequalities will hamper the effective implementation of the programme and reap unintended consequences for those most vulnerable in society if they are not adequately addressed.

Based on the above, Gender integration needs to happen at two levels:

- **Programme Level -** to provide the broader overarching framework and mechanisms for gender integration.
- **Project Level** – to address the specific gender gaps in the localities identified and sub-projects implemented.

The outline of the Gender Action Plan is therefore as follows.

PROGRAMME LEVEL

A. Planning and Governance

- A.1. Embed gender considerations in the establishment and operationalisation of the WPO
- A.2 Engendered procurement
- A.3. Budget for gender activities

B. Capacity Building and Awareness

- B.1. Engender WRP brand development and communications strategies
- B.2. Develop gender-focussed knowledge products
- B.3. Strengthen institutional capacity and awareness

C. Monitoring, Evaluation and Reporting

- C.1. Design of gender framework and monitoring and reporting strategies
- C.2. Appoint a gender specialist
- C.3. Measure gender disaggregated outcomes
- C.4. Undertake continuous monitoring, evaluation, and learning

PROJECT LEVEL

A. Planning and Governance

- A.1. Alignment to programme-level gender monitoring framework
- A.2. Map institutional/governance gender elements at sub-project-level
- A.3. Budget for gender activities
- A.4. Align to Programme Procurement

B. Capacity Building and Awareness

- B.1. Share Repository of resources, templates, and tools
- B.2. Create Awareness
- B.3. Implement training activities

C. Monitoring, Evaluation and Reporting

- C.1. Determine the Baselines

- C.2. Report on M&E activities (with disaggregated data, where available)

An Independent Grievance Redress Mechanism process is established to resolve all complaints that arise from the programme. The mechanisms will be aligned with the GCF's independent redress mechanism and experiences.

The full gender assessment and sub-project level gender action plan is attached as Annex 8

G.3. Financial management and procurement (max. 500 words, approximately 1 page)

The implementation arrangements outlined in section B.4. gives clarity as to the roles and responsibilities for the WRP, linked to the NWPP. Annex 10 provides a procurement plan for the first five years, noting that the mid-term review process will guide the implementation of the next five year period.

The financial management between the AE and EEs of the WRP will be governed by DBSA's finance manual and policy and procedure on procurement of Goods and Services. The DBSA is the Accredited Entity of the GCF and is a Programme developer (through its Programme Development Unit) and provides project preparation support (through its Project Preparation Unit). DBSA will participate in the oversight committee as the implementing agent of the NWPP and will act as management agent of the WPO for a period of 5 years or until such time that the WPO is legally institutionalized. DBSA will provide all administrative services such as procurement of goods and services through DBSA supply chain management processes, procurement of panels of service providers, entering into legal contracts with service providers and employees. It will also operate a ring-fenced bank account for the WPO function and make payments on behalf of the WPO. Key instruments that will guide the financial management are as follows.

National Water Partnerships Programme Memorandum of Agreement

An agreement between the Programme owners DWS, SALGA and DBSAhave been signed. This sets out the objective of the NWPP, the roles and responsibilities of the respective parties, funding arrangements, and the obligations regarding the WPO. It includes a term of reference for the OC. DBSA facilitated this contracting process and will act as secretariat of the OC to facilitate contract implementation until the WPO has been capacitated and resourced.

Water Partnership Office Implementation Plan, including budget

The WPO PIP will be developed by the WPO annually and submitted to the OC for approval. Once approved, the WPO will implement and report against it. It will set out the roles and budget of DBSA in programme implementation, including NWPP and WRP development and water reuse sub-project preparation, as well as the capacity requirements and how they will be met. It will include the budgeting of the various units, differentiating between the sub-programme budgets (i.e., each sub-programme will be ring-fenced with its own costing and funding structures).

Water Reuse Unit Implementation Plan, including budget

The WRP implementation plan will be developed by the WRU and submitted to the WPO for consideration and approval. It will set out all project information and capacity and budget requirements for the WRP. The WPO will include this detail in the WPO's PIP.

Project Owner agreements with WPO

The WRP project owners are primarily the municipalities as water services authorities for their areas of jurisdiction. The project owners will engage the WRP for support through the WPO and for specific water reuse support through the WRU. Engagement will be through formal written correspondence including on-boarding contractual commitments, signed by the required authority, as required at various stages and gates of the project preparation process.

Service provider service level agreements

The WPO, through the DBSA, will procure two key service providers, namely the panel of service providers to support project preparation and the IBCF. Both will be competitively procured and contracted by the DBSA to address the specific performance requirements of the WRP. The contracts will set out scope of anticipated work and remuneration schedules. Service providers will only be engaged and paid when there is specific work to be done for the WRP. When project owners request project preparation support, the service providers will be allocated from the panel and an addendum to their framework contract will be concluded setting out scope ad remuneration. They will be allocated to work with a specific municipality. They will be paid by the WPO as set out in the addendum scope of works tailored for the project preparation phase and associated performance requirements. The WPO (via DBSA Supply Chain Management Unit) will manage the procurement and contracting process of the panel of service providers.

Project Funding Agreements

Project funding agreements will be concluded between the project owners and the funders. The funding agreements will be formal agreements between the project owners and the lenders. The structure of the documents will depend on the structure of the funding arrangements. The WPO and the IBCF will facilitate the process of arranging and securing project funding.

Advisory Terms of Engagement

The OC will identify and invite members to participate in the Advisory Committee. The OC will develop and issue a term of reference with engagement requirements and roles and responsibilities.

Sub-projects will be owned and implemented by the Project Owners. Sub-projects can be procured in several ways. They can be procured following traditional procurement strategies, including compliance with supply chain management requirements of the Municipal Finance Management Act. In this instance the municipality procures separate service providers to design, contractors to build, and operators to operate, the project. All sub-projects will need to go through the project preparation stages of identification, scoping and preparation, before the procurement strategy is determined. It does however still require a proper project preparation process including scoping and feasibility. The decision as to which procurement strategy to follow will be determined through the various project preparation stages (see Figure 28). The Procurement strategy followed will inform the contracting structures.

The WPO will adopt DBSA's accounting systems and will be audited independently (auditors selected through a competitive bidding process where Terms of Reference are approved by the DBSA Finance Unit) on a yearly basis.

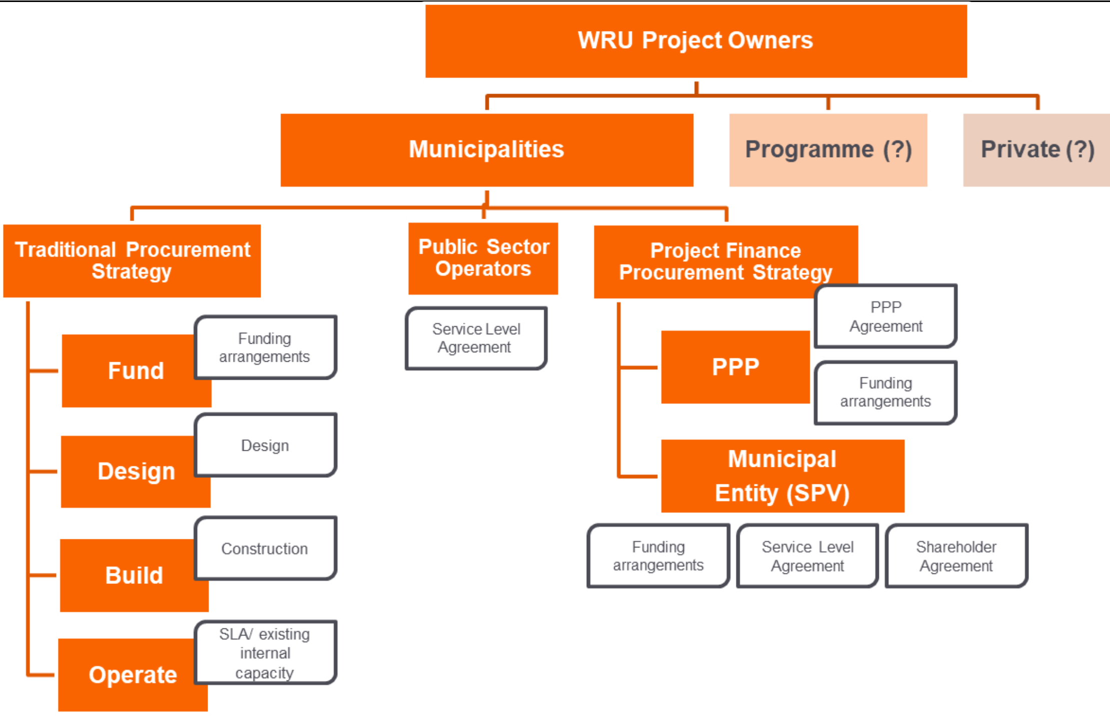


Figure 28: Project Procurement Strategies

Equipment and improper usage

The use of the DBSA's supply chain system, protocols and contractual arrangements will ensure that all equipment, materials or technology are to be used appropriately for the purpose for which they were intended or purchased. This system and the contractual arrangements that support procurement provide clarity as to the use, purpose and ownership of such equipment, including the return of this to the DBSA and WPO upon sub-project completion. All equipment will be registered in a project database. The sub-project's contractual arrangements together with these equipment management protocols will guide the management of this equipment and technology, as well as its return after sub-project completion.

G.4. Disclosure of funding proposal

☐ No confidential information: The accredited entity confirms that the funding proposal, including its annexes, may be disclosed in full by the GCF, as no information is being provided in confidence.

☒ With confidential information: The accredited entity declares that the funding proposal, including its annexes, may not be disclosed in full by the GCF, as certain information is being provided in confidence. Accordingly, the accredited entity is providing to the Secretariat the following two copies of the funding proposal, including all annexes:

- full copy for internal use of the GCF in which the confidential portions are marked accordingly, together with an explanatory note regarding the said portions and the corresponding reason for confidentiality under the accredited entity's disclosure policy, and
- redacted copy for disclosure on the GCF website.

The funding proposal can only be processed upon receipt of the two copies above, if containing confidential information.

C. ANNEXES

H.1. Mandatory annexes

☒	Annex 1	NDA no-objection letter(s) **(template provided)**
☒	Annex 2	Market study
☒	Annex 3	Financial Architecture Report
☒	Annex 4	Detailed budget plan **(template provided)**
☒	Annex 5	Implementation timetable including key project/programme milestones **(template provided)**
☒	Annex 6	Environmental and social document corresponding to the environmental and social category (A, B or C; or I1, I2 or I3): **(ESS disclosure form provided)** ☐ Environmental and Social Impact Assessment (ESIA) or ☒ Environmental and Social Management Framework and Plan (ESMFP) or ☐ Environmental and Social Management System (ESMS) ☐ Others (please specify – e.g. Resettlement Action Plan, Resettlement Policy Framework, Indigenous People's Plan, Land Acquisition Plan, etc.)
☒	Annex 7	Summary of consultations and stakeholder engagement plan
☒	Annex 8	Gender assessment and project/programme-level action plan **(template provided)**
☒	Annex 9	Legal due diligence (regulation, taxation and insurance)
☒	Annex 10	Procurement plan **(template provided)**
☒	Annex 11	Monitoring and evaluation plan **(template provided)**
☒	Annex 12	AE fee request **(template provided)**
☒	Annex 13	Co-financing commitment letter, if applicable **(template provided)**
☒	Annex 14	Term sheet including a detailed disbursement schedule and, if applicable, repayment schedule

H.2. Other annexes as applicable

☒	Annex 15	Evidence of internal approval **(template provided)**
☒	Annex 16	Map(s) indicating the location of proposed interventions
☐	Annex 17	Multi-country project/programme information **(template provided)**
☐	Annex 18	Appraisal, due diligence or evaluation report for proposals based on up-scaling or replicating a pilot project
☐	Annex 19	Procedures for controlling procurement by third parties or executing entities undertaking projects financed by the entity
☐	Annex 20	First level AML/CFT (KYC) assessment
☒	Annex 21	Governance and Operations Report
☒	Annex 22	Economic and Beneficiary Assessment
☒	Annex 23	Climate Screening Procedure
☒	Annex 24	Communications Strategy and Plan
☒	Annex 25	WRP Screening Scoresheets

☒	Annex 26	Economics Assessment Spreadsheet
☒	Annex 27	Investor Letters of Interest in the Water Reuse Programme
☒	Annex 28	Climate Change Risk and Vulnerability Indicative Analysis Report

** Please note that a funding proposal will be considered complete only upon receipt of all the applicable supporting documents.*

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[i] World Bank Climate Change Knowledge Portal. (n.d.). South Africa - Climate Projections: Mean Projections. Retrieved November 2021, from Climate Change Knowledge Portal: https://climateknowledgeportal.worldbank.org/country/south-africa/climate-data-projections

[ii] World Bank Climate Change Knowledge Portal. (n.d.). South Africa - Climate Projections: Mean Projections. Retrieved November 2021, from Climate Change Knowledge Portal: https://climateknowledgeportal.worldbank.org/country/south-africa/climate-data-projections

[iii] World Resources Institute (WRI), Aqueduct, Country Rankings. Available at https://www.wri.org/applications/aqueduct/country-rankings/?indicator=bws (last accessed June 2021).

iv DWS. 2013. National water resource strategy, Edition 2. Department of Water and Sanitation. Pretoria

[v] University of Notre Dame – Global Adaptation Initiative (ND-GAIN) Index. Available at https://gain.nd.edu/our-work/country-index/rankings/ (last accessed June 2021). Note that the ND-GAIN Index's water score captures a country's vulnerability of fresh water supplies to climate change, based on the following specific indicators: projected change of annual runoff, projected change of annual groundwater recharge, freshwater withdrawal rate, water dependency ratio, dam capacity, and access to reliable drinking water.

[vi] David Eckstein et al., Germanwatch, *Global Climate Risk Index 2020* (2020). Available at https://www.germanwatch.org/sites/default/files/20-2-01e%20Global%20Climate%20Risk%20Index%202020_14.pdf (last accessed June 2021).

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[x] World Bank Climate Change Knowledge Portal, South Africa, Vulnerabilities. Available at https://climateknowledgeportal.worldbank.org/country/south-africa/vulnerability (last accessed November 2022).

[xi] The study's output is The Green Book, one of South Africa's most highly regarded and well-respected sources of climate change risk analysis. Given the Green Book's stature in South Africa, our climate assessment has directly relied on the Green Book's existing analysis and municipal risk level estimations. The Green Book is especially apt as a climate information source for South Africa because:

- It downscaled climate change projections for the country into 8 km x 8 km grids, providing high spatial resolution and reflecting local conditions;
- It integrated municipality-specific data on vulnerability, profiling each individual municipality in terms of demographic and socio-economic vulnerability factors;
- It was developed principally to assist municipalities with evaluating their climate change risks – and the methodology was consistent nationally, which enables comparisons to be made in terms of relative risks across the country;
- It provides readily available data and information on specific climate related hazards and vulnerabilities at the appropriate scale for this analysis;
- It integrates significant additional analysis to convert the primary climate variables into critical hazard levels for several climate hazards including drought.; and
- The ability of the underlying climate models to realistically simulate present-day southern African climate has been extensively demonstrated through a plethora of peer-reviewed publications.

The regional climate model used by the Green Book is the conformal-cubic atmospheric model (CCAM), a variable-resolution general circulation model (GCM), a global climate model developed by the Commonwealth Scientific and Industrial Research Organisation. CCAM runs coupled to a dynamic land-surface model CABLE (CSIRO Atmosphere Biosphere Land Exchange model).

For the Green Book, six GCM simulations of the Coupled Model Intercomparison Project Phase Five (CMIP5) and Assessment Report Five (AR5) of the Intergovernmental Panel on Climate Change (IPCC), obtained for the emission scenarios described by Representative Concentration Pathways 4.5 and 8.5 (RCP4.5 or high mitigation / moderate emissions scenario, and RCP 8.5, or a low-mitigation / high emissions scenario) were first downscaled to 50 km resolution globally. The simulations span the period 1960-2100 (Francois Engelbrecht et al., 2019).

The specific GCMs downscaled for the Green Book include the Australian Community Climate and Earth System Simulator (ACCESS1-0); the Geophysical Fluid Dynamics Laboratory Coupled Model (GFDL-CM3); the National Centre for Meteorological Research Coupled Global Climate Model, version 5 (CNRM-CM5); the Max Planck Institute Coupled Earth System Model (MPI-ESM-LR); the Norwegian Earth System Model (NorESM1-M) and the Community Climate System Model (CCSM4) (Francois Engelbrecht et al., 2019).

As noted in the Green Book's technical report, which clarifies the methodology,

> *"…the simulations were performed on supercomputers of the Centre for High Performance Computing (CHPC) of the Meraka Institute of the CSIR in South Africa. In these simulations CCAM was forced with the bias-corrected daily sea-surface temperatures (SSTs) and sea-ice concentrations of each host model, and with CO_2, sulphate and ozone forcing consistent with the RCP4.5 and RCP8.5 scenarios…*
>
> *…Most current coupled GCMs do not employ flux corrections between atmosphere and ocean, which contributes to the existence of biases in their simulations of present-day sea-surface temperatures (SSTs) – more than 2°C along the West African coast.*
>
> *An important feature of the downscalings performed for the Green Book is that the model was forced with the bias-corrected SSTs and sea-ice fields of the GCMs. The bias is computed by subtracting for each month the Reynolds (1988) SST climatology (for 1961-2000) from the corresponding CGCM climatology. The bias-correction is applied consistently throughout the simulation. Through this procedure the climatology of the SSTs applied as lower boundary forcing is the same as that of the Reynolds SSTs. However, the intra-annual variability and climate-change signal of the CGCM SSTs are preserved.*
>
> *A multiple-nudging strategy was followed to obtain the 8 km resolution downscalings. After completion of the 50 km resolution simulations described above CCAM was integrated in stretched-grid mode over South Africa, at a resolution of about 8 km (0.08° degrees in latitude and longitude). The high resolution part of the model domain was about 2000 x 2000 km2 in size. The higher resolution simulations were nudged within the quasi-uniform global simulations, through the application of a digital filter using a 600 km length scale. The filter was applied at six-hourly intervals and from 900 hPa upwards."* (Francois Engelbrecht et al., 2019)

On the matter of model validation (i.e., testing models against observed climate, to see if they are able to accurately recreate observed climatology, including by using the models to simulate current-day climatology of a 30-year period to the present), the CCAM model used for the Green Book does replicate South African climate conditions. The Green Book's technical methodology clarifies, in terms of the model's ability to replicate current climate:: "***The model's ability to realistically simulate present-day southern African climate has been extensively demonstrated (e.g. Engelbrecht et al., 2009; Engelbrecht et al., 2011; Engelbrecht et al., 2012; Malherbe et al., 2013; Winsemius et al., 2014; Engelbrecht et al., 2015)***." (page 8)

A number of times in the methodology document https://pta-gis-2-web1.csir.co.za/portal/sharing/rest/content/items/718369b1f453455e8febdf4f25f68249/data the GreenBook technical document reaffirms that the baseline conditions for each variable were simulated ("model-simulated and bias corrected" for the "baseline of 1961-1990.").

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[xv] DWS. 2018. National Water and Sanitation Masterplan. Volume2: A plan to action. Department of Water and Sanitation, Pretoria, South Africa.

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[xvii] University of Notre Dame – Global Adaptation Initiative (ND-GAIN) Index. Available at https://gain.nd.edu/our-work/country-index/rankings/ (last accessed May 2022).

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[lxvi] Target 6.2: By 2030, achieve access to adequate and equitable sanitation and hygiene for all and end open defecation, paying special attention to the needs of women and girls and those in vulnerable situations. Indicator 6.2.1: Proportion of population using safely managed sanitation services, including a hand-washing facility with soap and water.

[lxvii] Target 6.3: By 2030, improve water quality by reducing pollution, eliminating dumping and minimizing release of hazardous chemicals and materials, halving the proportion of untreated wastewater and substantially increasing recycling and safe reuse globally. Indicator 6.3.1: Proportion of wastewater safely treated.

[lxviii] Target 6.4: By 2030, substantially increase water-use efficiency across all sectors and ensure sustainable withdrawals and supply of freshwater to address water scarcity and substantially reduce the number of people suffering from water scarcity. Indicator 6.4.2: Level of water stress: freshwater withdrawal as a proportion of available freshwater resources.

[lxix] Target 6.b: Support and strengthen the participation of local communities in improving water and sanitation management. Indicator 6.b.1: Proportion of local administrative units with established and operational policies and procedures for participation of local communities in water and sanitation management.

[lxx] Target 11.6: By 2030, reduce the adverse per capita environmental impact of cities, including by paying special attention to air quality and municipal and other waste management. Indicator: 11.6.1: Proportion of urban solid waste regularly collected and with adequate final discharge out of total urban solid waste generated, by cities.

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[lxxii] Target 13.2: Integrate climate change measures into national policies, strategies and planning. Indicator 13.2.1: Number of countries that have communicated the establishment or operationalization of an integrated policy/strategy/plan which increases their ability to adapt to the adverse impacts of climate change, and foster climate resilience and low greenhouse gas emissions development in a manner that does not threaten food production (including a national adaptation plan, nationally determined contribution, national communication, biennial update report or **other**).

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[lxxviii] South Africa is finalizing its updated NDC submission in June/July 2021, after a series of nationwide consultations in April and May 2021. Reference has been made to the draft NDC shared for public consultations, noting that this may be subject to change before the cabinet approves it and prior to UNFCCC submission.

[lxxix] Department of Environment, Forestry, and Fisheries, Draft Proposed Updated Nationally Determined Contribution – South Africa's First Nationally Determined Contribution under the Paris Agreement (2021). Available at https://www.environment.gov.za/sites/default/files/reports/draftnationalydeterminedcontributions_2021updated.pdf (last accessed June 2021).

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No-objection letter issued by the national designated authority(ies) or focal point(s)

environmental affairs

Department:
Environmental Affairs
REPUBLIC OF SOUTH AFRICA

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The Green Climate Fund
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Email: fundingproposal@gcfund.org

Dear Sir/Madam

PROPOSAL FOR THE GCF PROJECT PREPARATION FACILITY BY THE DEVELOPMENT BANK OF SOUTHERN AFRICA REGARDING THE SA WATER REUSE PROGRAMME

We refer to the Project Preparation Facility proposal regarding the SA Water Reuse Programme ("WRP") as included in the concept note submitted by Development Bank of Southern Africa to us on 25 September 2019.

The undersigned is the duly authorised representative of the Department of Environmental Affairs, the National Designated Authority/focal point for South Africa.

Pursuant to Green Climate Fund (GCF) decisions B.08/10 and B.13/21, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the Project Preparation Facility activities as included in the PPF Proposal. By communicating our no-objection, it is implied that:

(a) the government of South Africa has no-objection to the Project Preparation Facility request as included in the Project Preparation Facility (PPF) Proposal;
(b) the PPF Proposal is in conformity with South Africa's national priorities, strategies and plans; and
(c) in accordance with the GCF's environmental and social safeguards, the PPF activities as included in the PPF Proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the PPF Proposal has been duly followed.

PROPOSAL FOR THE GCF PROJECT PREPARATION FACILITY BY THE DEVELOPMENT BANK OF SOUTHERN AFRICA REGARDING THE SA WATER REUSE PROGRAMME

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the Project.

We acknowledge that this letter will be made publicly available on the Green Climate Fund (GCF) website.

Yours sincerely

Ms Nosipho Ngcaba
DIRECTOR-GENERAL
DATE: 25/10/2019

Environmental and social safeguards report form pursuant to para. 17 of the IDP

Basic project or programme information	
Project or programme title	South Africa's Water Reuse Programme
Existence of subproject(s) to be identified after GCF Board approval	Yes
Sector (public or private)	
Accredited entity	Development Bank of Southern Africa (DBSA)
Environmental and social safeguards (ESS) category	Category B
Location – specific location(s) of project or target country or location(s) of programme	South Africa specific municipalities to be determined on a case-by-case basis subject to meeting programme criteria (including climate).
Environmental and Social Impact Assessment (ESIA) (if applicable)	
Date of disclosure on accredited entity's website	Monday, September 12, 2022
Language(s) of disclosure	English
Explanation on language	English is an official language of South Africa and language understandable to affected peoples/stakeholders
Link to disclosure	https://www.dbsa.org/sites/default/files/media/documents/2022-09/Annexure%206%20DBSA%20WRP_ESMF%20and%20plan_12092022.pdf
Other link(s)	N/A
Remarks	An ESIA consistent with the requirements for a category B project is contained in the "Environmental and Social Management Framework and Plan".
Environmental and Social Management Plan (ESMP) (if applicable)	
Date of disclosure on accredited entity's website	Monday, September 12, 2022
Language(s) of disclosure	English
Explanation on language	English is an official language of South Africa and language understandable to affected peoples/stakeholders
Link to disclosure	https://www.dbsa.org/sites/default/files/media/documents/2022-09/Annexure%206%20DBSA%20WRP_ESMF%20and%20plan_12092022.pdf
Other link(s)	N/A
Remarks	An ESMP consistent with the requirements for a category B project is contained in the "Environmental and Social Management Framework and Plan".
Environmental and Social Management (ESMS) (if applicable)	
Date of disclosure on accredited entity's website	N/A
Language(s) of disclosure	N/A
Explanation on language	N/A
Link to disclosure	N/A

Other link(s)	N/A
Remarks	N/A
Any other relevant ESS reports, e.g. Resettlement Action Plan (RAP), Resettlement Policy Framework (RPF), Indigenous Peoples Plan (IPP), IPP Framework (if applicable)	
Description of report/disclosure on accredited entity's website	N/A
Language(s) of disclosure	N/A
Explanation on language	N/A
Link to disclosure	N/A
Other link(s)	N/A
Remarks	The outline of Resettlement Action Plan and Indigenous Peoples Plan have been included in the "Environmental and Social Management Framework and Plan".
Disclosure in locations convenient to affected peoples (stakeholders)	
Date	Friday, June 2, 2023
Place	All projects will require environmental due diligence, and most will require a Basic Assessment/ EIA and authorisation. This legal process requires disclosure in a suitable language and place for interested and affected parties. Therefore, at the time of implementation the related parties will be notified in the relevant language amongst South Africa's 11 official languages. Currently the disclosure is shared with the major stakeholders and can also be found at the DBSA offices: **Department of Water and Sanitation (DWS)** *Address*: Ndinaye House, 178 Francis Baard St, Pretoria Central, Pretoria, 0001 **South African Local Government Association (SALGA)** *Address:* Block B, Cnr Garsfontein and Corobay, Menlyn Corporate Park, ext11, 175 Corobay Ave, Waterkloof Glen, Pretoria, 018 **Department of Forestry Fisheries and Environment (DFFE)** *Address*: 473, Steve Biko Rd & Soutpansberg Rd, Arcadia, Pretoria, 0083 **Development Bank of Southern Africa (DBSA)** *Address*: 1258 Lever Road. Headway Hill, Midrand 1685, Gauteng
Date of Board meeting in which the FP is intended to be considered	
Date of accredited entity's Board meeting	N/A
Date of GCF's Board meeting	Monday, July 10, 2023

Note: This form was prepared by the accredited entity stated above.

Secretariat's assessment of FP209

Proposal name:	Climate Change Resilience through South Africa's Water Reuse Programme ("WRP")
Accredited entity:	Development Bank of Southern Africa
Country/(ies):	South Africa
Project/Programme size:	Large

I. Overall assessment of the Secretariat

1. The funding proposal is presented to the Board for consideration with the following remarks:

Strengths	Points of caution
The programme will support the structuring of blended finance options and the creation of a new asset class for a water-stressed country that is ranked among the countries most vulnerable to climate change. The aim is to transform the water sector, with a focus on wastewater, from a predominantly public investment model to a private sector investment model.	GCF will be exposed to high credit risk due to its seniority in the loan structure. However, this is balanced by the high impact potential, paradigm shift potential and high leverage of private sector financing. This risk is also shared with the accredited entity and other development financial institutions, because they will co-finance in the same junior tranche with predetermined targeted ratios.
The introduction of a water reuse programme (WRP) serves as a dedicated asset for investing in climate adaptation measures, integrating climate risk assessment and management tools with local water security approaches, and providing key stakeholders with technical assistance and capacity-building, thereby contributing to a paradigm shift in South Africa's financial and water sector that will spur the prioritization of climate adaptation measures at scale and encourage private sector investment.	Capacities of municipalities for implementing the projects, including designing, may be a risk. However, the education programme/activities in this project would minimize this risk.
The project could demonstrate the benefits for future innovative financing structures through public–private partnerships, which could be expanded to water and sanitation services, off-grid sanitation services and rural/community water services.	The proposal identified the first 10 locations with preliminary technical and environmental analysis. The project will be developed in phases, and it is critical to efficiently generate knowledge and lessons learned from earlier phases to inform the subsequent phases. This innovation comes with project implementation risk; however, potential delays (especially in the early stage) may be expected.

2. The Board may wish to consider approving this funding proposal with the terms and conditions listed in the term sheet and addendum XVII, titled "List of proposed conditions and recommendations", respectively.

II. Summary of the Secretariat's assessment

2.1 Project background

3. South Africa is a water-scarce country, and the sustainable provision of water is among its most significant challenges. Basic water-related services, including the provision of potable water and access to sanitation and stormwater drainage, are not readily available to a significant proportion of the population. More than three-quarters of municipalities have high to extreme vulnerability to water security and the situation continues to worsen with the growing population and urbanization.

4. The challenges in water provision in the country are compounded by extreme weather events induced by climate change, especially the observed increased frequency and severity of droughts and floods in recent years. South Africa is ranked as one of most vulnerable countries to climate change and, in particular, to water-related climate change vulnerability. Since the 1960s the average temperature of South Africa has risen by 1.5°C, and there is an observed tendency towards an increase in the intensity of rainfalls, and prolonged dry seasons. It is projected that the country could face a 17 per cent water deficit by 2030.

5. To avoid the projected water deficit, the country needs to diversify its water supply sources, reduce water demand and increase water supply through, among other things, the "re-use of effluent from wastewater treatment plants, water reclamation as well as desalination and treated acid mine drainage". Considering the water scarcity issues in Africa and in South Africa in particular, this project is a high priority for South Africa to address water scarcity and transform and scale up the wastewater system and treating water as a new asset class that would attract private sector investment in the country, with potential for replication elsewhere.

6. The objective of the proposed programme is to establish and operationalize a national water reuse programme (WRP) to support the development of water reuse projects in South Africa and to attract private sector investment, paving the way for future growth by treating water reuse infrastructure as a new asset class, and stimulating and activating the water reuse market. More specifically, the programme aims to:

(a) Encourage the scaled development of water reuse projects at the municipal level;

(b) Support municipalities with scaling up their water reuse projects by providing support for identifying, conceptualizing and prioritizing large-scale water reuse projects, and during project preparation and in the development of blended finance options to fund implementation; and

(c) Assist municipalities to develop diversified projects that not only support water reuse but have extended benefits from aspects such as water reclamation or remediation through nature-based solutions and sludge management, as well as energy generation from biogas.

7. This project will therefore address the following key barriers:

(a) Fragmentation and inconsistency across the sector, and often involving duplication of effort on water security initiatives;

(b) The inability of municipalities to technically conceptualize and prepare water reuse projects;

(c) Limited municipal capacity to procure and structure contracts to support new business models;

(d) Limited access to funding for adequate project preparation;

(e) Lack of private sector engagement in the financing, development and ownership of climate resilience interventions; and

(f) Limited understanding of climate impacts on water security and the interventions required.

8. This programme will deliver outcomes related to each of the GCF investment criteria, as follows: (1) mitigation of climate change via expanding private investment in climate-friendly infrastructure; (2) providing a replicable model for building nation-specific capacity to scale up climate finance in support of Paris climate goals; (3) supporting at least eight of the United Nations Sustainable Development Goals (SDGs) (i.e. access to clean water, renewable energy generation, economic growth, development of sustainable infrastructure, reduced inequality in terms of energy access and access to clean water, sustainable cities and communities, and taking climate action); (4) providing necessary financial capacity to meet socioeconomic needs related to climate-friendly and clean water infrastructure; (5) expanding national capacity to achieve established climate goals working through local stakeholders; and (6) increasing market efficiency through a leverage approach to scale up private investment.

9. The indicative total budget value of the programme is estimated to be USD 1.472 billion, comprising USD 62 million to support the establishment of a WRP and project pipeline preparation, USD 1.4 billion that provides blended financing solutions to actual project investments and USD 10 million dedicated for communications and awareness creation among the stakeholders to understand climate vulnerability and its impacts on water situation. Of this total, GCF is requested to provide USD 35 million in grants to share the cost of project preparation and create broader awareness, and an additional USD 200 million in concessional loans to allow for a lower overall cost of financing on a project level, thereby leading to a more attractive and acceptable tariff.

10. Co-financing will take four forms: (1) equity/budget allocation grants coming from participating municipalities that own and/or are mandated to run a particular wastewater treatment works; (2) senior debt from commercial banks and financial institutions; (3) concessional loans from Development Bank of Southern Africa (DBSA) and other development partners; and (4) partial risk guarantee for credit enhancement aiming to attract a greater portion of senior debt and reduce the portion of subordinated debt.

11. The programme will only invest in projects that meet climate eligibility criteria with environmental and social risk under category B and C. This is a funding proposal that has received GCF financial support through the Project Preparation Facility (completed on 14 July 2022, after 24 months including extension).

2.2 Component-by-component analysis

12. The programme will shift the development pathway of South Africa's water sector by supporting integrated and transformational planning (component 1), catalysing innovative technologies and mobilizing finance at scale (component 2), and sharing knowledge of successful innovations at scale and developing climate expertise (component 3).

Component 1: Establishing a WRP project pipeline (total cost: USD 62 million; GCF cost: USD 30 million)

13. Component 1 will support the establishment of the WRP with clear governance arrangements. The component aims at establishing a programmatic approach to the

development and implementation of water reuse infrastructure by supporting municipalities to identify, prepare and structure bankable water reuse projects. This component will establish, operationalize and resource a WRP that will provide an opportunity to support municipalities as primary custodians of local climate adaptation plans to undertake innovative and targeted projects that will underpin climate resilience interventions towards improved water security. The component will also include project pipeline preparation, and the establishment of a national Water Partnerships Office (WPO), which will support and facilitate a programmatic approach that will guide project identification and prioritization; drive project preparation; facilitate funding solutions; and monitor project implementation.

14. In addition, the financial arrangements and reporting regimes will be established under this component. This will formalize the financial management arrangements and the necessary reporting that would underpin them. The programme will apply the financial management policies and guidelines for capital that are already used by the accredited entity (AE) through loans from the DBSA itself (provided from GCF concessional capital). An Independent Blended Capital Facilitator would manage other capital in alignment with DBSA processes. The financial management of disbursements will be primarily managed under the DBSA Loan Management Unit. Its purpose is to ensure the continuous monitoring of all relevant covenants throughout the lifetime of the facility (this relates to both financial and non-financial covenants).

Component 2: Implementing the WRP, providing technical assistance for project procurement and developing the blended financing solutions (total cost: USD 1.4 billion; GCF cost: USD 200 million)

15. This component will provide technical and financial transaction advisory services during the implementation of projects through procured and contacted service providers; and will support this through the deployment of blended finance solutions such as concessional loans, subordinated debt, partial credit guarantees and grants to de-risk private sector investments.

16. This component is essential to support the implementation of projects. The approach will be to create an enabling blended financing environment through alternative funding solutions. It will identify and promote opportunities to attract financing through traditional methods (e.g. through municipal budgets and commercial loans) and more innovative approaches (e.g. municipal bonds, public–private partnerships (PPPs), project bonds and impact investing). This mechanism is important to achieve the key outcomes, one of which is to create a new asset class for water reuse by using credit enhancement to crowd-in private sector funding targeted towards developing debt capital market instruments around a specific asset class. The asset class will offer acceptable financial returns but remain in line with environmental, social and governance impacts and help to meet the SDGs (specifically SDG 6 – safe and clean water and sanitation; SDG 11 – more sustainable cities; SDG 12 – more sustainable consumption and production; and SDG 13 – combat climate change).

17. In addition, the use of credit enhancement will support the blended finance approach through the employment of first loss/subordinated facilities, loan tenor extensions, or the employment of guarantees (which is an essential financial instrument to provide security and mobilize the private sector in the context of South Africa). The GCF concessional loans, together with the subordinated loans provided by DBSA and other development/financial institutions, would reduce the cost of borrowing on the project level, which further makes the tariffs more attractive and affordable to water users.

18. A set of eligibility criteria has been developed and agreed to make sure the selected projects address climate-related vulnerabilities; are well-structured and bankable; and that the use of any concessional loan corresponds to the actual de-risking needs and not to distorting the market. The selected projects need to demonstrate a clear water supply vulnerability to climate change, comply with technical standards (i.e. capacity, type of technology, environmental and social safeguards risk, chemical), and need to be financially viable with certainty around offtake

agreements. GCF concessional loans will only be deployed when the target debt-service coverage ratio for an individual projects falls under the minimum range on a principal of minimum concessionality. The projects will demonstrate this through financial analysis to be conducted for each project, with and without GCF concessionality.

Component 3: Building capacity and awareness (total cost: USD 10 million; GCF cost: USD 10 million)

19. This component aims to drive communications and capacity-building initiatives that target institutions and wider society with the aim of improving understanding of climate change and its impact on the water situation. It will also help to improve the understanding of water reuse and its benefits, which is critical noting the often-negative views on recycled water. The component will be implemented at the national and regional scale for public awareness and education processes that will support the implementation of existing national, provincial and local climate adaptation strategies and plans.

20. DBSA will take the lead in the rollout of this component due to its strong institutional position in the water sector, and the WPO will undertake key supportive actions. It is important to note the role of the Department of Water and Sanitation (DWS) in supporting these important processes, particularly as the water sector leader.

21. This component will develop a strategy on how to educate and communicate the details of water reuse as a means of improving the public perception of the approach. The key drivers affecting water reuse choices include climate change impacts and the importance of water security, water quality, the cost relative to other water supply options and the social, cultural and religious perceptions. In support of this, as well as informing components 1 and 2, a “community of experts” will be established as an advisory forum as a way of developing innovation and supporting knowledge exchange. Equally, for specified projects this group will be asked to provide strategic review and guidance. Lastly, it is essential to maintain strategic partnerships that will underpin and inform the approach towards climate adaptation, and this will assist in working towards improved institutional and regulatory frameworks that support adaptation and resilience building. Two key functions will be conducted under this component: communications and awareness creation; and strengthen institutional and regulatory frameworks.

22. These knowledge-exchange interventions will underpin the development of awareness regarding climate change impacts at the local level and will support the successful scaling up and implementation of water reuse projects in South Africa as an adaptation response. These interventions will focus on developing improved awareness across a broader swathe of stakeholders, while providing more focused technical and financial capacity-building interventions at the national, provincial and local government level. These governmentally focused interventions will deepen the understanding of the water sector’s climatic risks while taking into consideration updated analysis, knowledge and insights.

23. Project management cost is not specifically indicated in the budget and DBSA will not seek GCF financing for that work.

III. Assessment of performance against investment criteria

24. Overall, the proposal is well aligned with the six GCF investment criteria, particularly on its high paradigm shift potential, because the successful implementation of the project could change the water reuse sector of the country.

3.1 Impact potential *Scale: Medium to high*

25. The funding proposal has a medium–high impact potential with a solid and well-articulated climate rationale. South Africa is facing rising temperatures together with declining rainfall and a significant increase in spatial and temporal variability of rainfall. As a water-scarce country that is already under climate change impact, this programme will directly benefit 3.42 million people in areas where the water supply is to be made more accessible and reliable. Another 3.87 million people will indirectly benefit from the successful demonstration of the water reuse concept that will lead to full scale up of the WRP.

26. There will also be mitigation co-benefits from reduced greenhouse gas emissions by utilizing sludge as a potential source of energy and nutrients. However, such a reduction will be determined at a later stage when technical designs for each project are set during the implementation phase. This is also included as a secondary eligibility criterion encouraging such interventions in the projects. The WRP contributes to minimizing the water supply deficit and the chance of water restrictions in response to water shortages. The average annual avoided cost to households and businesses that would have resulted from water restrictions in the absence of the WRP is over 130 million South African rand (ZAR) (approx. USD 8.1 million).

3.2 Paradigm shift potential *Scale: High*

27. The establishment and successful implementation of this programme has a paradigm shift potential because it will pave the way for fully utilizing reused water in water-scarce municipalities and can be replicated in other developing countries where water shortage has become a challenge driven by climate change, and where countries are seeking to address market barriers to bring in more private sector participation. The blended finance structure, together with the dedicated technical assistance, serves as a one-stop-shop model that is designed to drive this transformational shift in this relatively new sector. This model is replicable for building nation-specific capacity to scale up climate finance in support of Paris climate goals, including treating water as a new asset class (i.e. it is an innovative mechanism) to mobilize private sector investment in the most dominated of public invested sectors (i.e. the water sector).

3.3 Sustainable development potential *Scale: High*

28. In addition to the considerable increased resilience of municipalities and populations to climate change, successful creation of this new asset class of sustainable water reuse projects will also deliver environmental benefits to South Africa, mainly through the reduction of water pollution in rivers and groundwater because of the consequential reduction of discharge from wastewater treatment works as they will be redirected to water reuse plants. Improved water quality brings in improvements in soil and water quality management in agricultural areas and urban farms. Incorporation of ecosystem-based approaches to sustain environmental flows would also result in environmental co-benefits.

29. Enhanced water security would enable further expansion of economic activities in sectors with considerable water consumption, such as agriculture and industry, as well as other businesses. These will naturally lead to higher income and job creation locally.

30. The proposal also has gender co-benefits by engaging women, youth and marginalized groups in planning, investment and decision-making throughout the project design and implementation. Such engagement will ensure that the projects are sensitive to their needs and concerns. Improvements in water security will also benefit local economic activities that typically involve women, such as agriculture.

31. Finally, this project is supporting at least eight of the SDGs, in that it involves access to clean water, renewable energy generation, economic growth, development of sustainable

infrastructure, reduced inequality in terms of energy access and access to clean water, sustainable cities and communities, and taking climate action.

3.4 Needs of the recipient *Scale: High*

32. The programme is expected to catalyse greater levels of private sector financing for water-related projects in South Africa where the participation of the private sector has been very limited to date, and where the water sector is currently dominated by public finance. This is critical given the urgent and substantial infrastructure funding gap in South Africa's water sector. It is providing necessary financial capacity to meet socioeconomic needs related to climate-friendly and clean water infrastructure.

33. The dedicated support to the municipalities will also address the current inability of municipalities to technically conceptualize, prepare, structure and implement water projects. The introduction of the programmatic technical, financial and procurement support by this project will help municipalities to develop the required capacity to drive paradigm shift in this sector and scale up the wastewater and water reuse projects.

3.5 Country ownership *Scale: High*

34. The interventions proposed are well in line with the national climate policies and priorities of South Africa. Strengthening climate resilience of water security at the municipal level is noted as its planned contribution to the global adaptation goal during the nationally determined contribution period. Water reuse is also consistent with South Africa's National Development Plan that aims to improve its national water access and water security.

35. DBSA has significant experience with projects in South Africa as well as projects in climate finance. It has a strong track record in the management of programmatic risk and the effective use of safeguards to support the implementation of a WRP in the country.

36. DBSA has engaged with a wide range of stakeholders in the preparation and development of this programme, including DWS, water utilities, private sector water developer and service providers, civil society organizations (residents' associations, community-based organizations, etc.), and potential financiers. A comprehensive stakeholder plan is also prepared to ensure that the key stakeholders continue to be engaged and their inputs considered, thereby creating the foundation for achieving stakeholder buy-in and effective implementation of the WRP. Finally, this project is expanding national capacity to achieve established climate goals by working through local stakeholders.

3.6 Efficiency and effectiveness *Scale: Medium to high*

37. Without doubt, GCF's concessionality and intervention in the WRP are crucial to the success of this WRP. GCF support will increase market efficiency through a leverage approach to scaling up private investment to address the current municipal-level appetite and budget for a programme of this nature that appears to be limited to enable the reuse of (potable) water.

38. The GCF cost per beneficiary (public sector project) is USD 58.54 and the GCF cost per beneficiary (private sector project) is USD 57.59 which is very attractive, considering the total cost per beneficiary is USD 411.05. The efficiency of the WRP is quite high, and these costs indicate that the WRP supports adaptation impact with more efficient use of GCF capital than other similar programmes. For example, the recently awarded programme in Jordan (Building resilience to cope with climate change in Jordan through improving water use efficiency in the agriculture sector (BRCCJ)) represented a GCF cost per beneficiary of USD 117.70. In addition,

the economic internal rate of return for the WRP as a whole is 28 per cent, while the financial internal rate of return for the programme as a whole is around 26 per cent. The net present value for the programme as a whole, factoring in economic co-benefits, ranges from ZAR 13 billion to ZAR 34 billion (USD 0.817 billion to USD 2.1 billion), justifying the investment from an economic perspective.

39. GCF will provide concessional loans to both public sector and private sector project owners (with different terms and fees for outgoing concessional loans, depending on, among other factors, whether the borrower is public or private). The project will be classified as either publicly owned or privately owned, based on the majority equity ownership in the underlying project legal form (such as a PPP special purpose vehicles, in the case of privately owned projects). Based on the initial proposed portfolio of ten projects, it is estimated that two distinctly priced tranches of GCF concessional loans will be required, comprising 65 per cent public and 35 per cent private, of the total GCF concessional loan amount.

40. The WRP will ensure the use of the latest international best practice in water reuse design approaches and technology selection through guaranteeing (1) designing the best reuse treatment training by a panel of experts; (2) holding the individual projects to the strictest reuse practices; (3) following the World Health Organization Potable Reuse guidelines (as well as SANS 241); (4) achieve a minimum pathogen log reduction across the treatment train; and (5) having sufficient micropollutant removal. This will only be achieved by selecting the latest technologies for each of the projects for direct and indirect reuse of (potable) water.

41. Finally, the programme is expected to be effective in overcoming key market barriers such as (1) fragmentation and inconsistency across the sector, and often involving duplication of effort water security initiatives; (2) inability of municipalities to technically conceptualize and prepare water reuse projects; (3) limited municipal capacity to procure and structure contracts to support new business models; (4) limited access to funding for adequate project preparation; and (5) lack of private sector engagement in the financing, development and ownership of climate resilience interventions.

IV. Assessment of consistency with GCF safeguards and policies

4.1 Environmental and social safeguards

42. **Programme background.** The programme will finance the development and implementation of water reuse infrastructure identified at the municipal level as part of the WRP and scale up a pipeline of wastewater reuse projects to enhance water security and climate resilience at the local and regional scale in South Africa. The programme will also support institutional capacity-building, knowledge exchange and information sharing. The key environmental co-benefit of the programme is the improvement in the quality of rivers and groundwater resulting from the reduction in water demand and reducing as well reusing wastewater discharges from wastewater treatment plants. With improved water services and security, the programme will contribute to more resilient livelihoods, the expansion of local economic activities and job creation as the main economic and social co-benefits.

43. **Environmental and social risk category.** The AE has assigned the programme medium environmental and social (E&S) risk category, which is equivalent to category B as per GCF environmental and social risk categorization and within the AE's E&S risk accreditation level. The programme itself, being focused on environmental and social enhancement, is expected to improve sustainable water resource management and benefit targeted municipalities with climate adaptation technologies and enhanced infrastructure, including ecological infrastructure, in the water sector. The proposed activities are expected to have potential limited adverse environmental and/or social risks and impacts that are few in number,

generally site-specific, largely reversible, and can readily be addressed through good management practices and adaptive strategies during implementation. Focusing on water reuse infrastructure development, the programme will exclude financing the AE's category 1 (high and substantial risk category) investments, equivalent of GCF category A investments, and no dam or impoundment infrastructure will be considered eligible.

44. **Safeguard instruments.** The AE has provided an environmental and social management framework and plan (ESMF) that presents the general context of the E&S risk assessment; an analysis on the potential E&S risks and impacts likely to be associated with the programme components and with four water reuse archetypes targeted; and information on mitigation measures to address such risks and impacts with specification of roles and responsibilities for appraisal and supervision. The ESMF has laid down a due diligence process that requires all potential investments to be screened on the basis of a checklist of environmental and social eligibility criteria and in compliance with national laws and regulations as well as the AE's environmental and social safeguards (ESS) standards. All subprojects will undertake site-specific environmental and social impact assessment (ESIA) that informs the development of a fit-for-purpose environmental and social management plan (ESMP), along with an environmental and social scoping report and a safeguard report. Indicative outlines of an Indigenous Peoples plan, a resettlement action plan, a biodiversity protection plan and a cultural heritage management plan are annexed to the ESMF.

45. **Compliance with GCF's environmental and social safeguards (ESS) standards.** The paragraphs below describe the programme's compliance with GCF's ESS standards.

46. **ESS1: Assessment and management of environmental risks and impacts.** The programme is categorized as medium risk against the AE's ESS standards. Further to the environmental and social baseline information at the national level, the ESMF delineates the potential environmental and social impacts per programme component and separately in terms of the water reuse archetypes to be operationalized at subproject level. The ESMF also provides an initial E&S risk screening of the first ten subprojects in the current pipeline and establishes the E&S risk due diligence process, including eligibility criteria for financing and a checklist for subproject screening and site selection to be applied to the entire portfolio. All subprojects will be required to satisfy the legislative requirements and comply with the AE's ESS standards. The ESMF requires a thorough ESIA and site-specific ESMP for each subproject, also incorporating information on the E&S risks and impacts of associated facilities and third parties. This is because the subprojects are likely to be sited adjacent to existing wastewater treatment works/plants, and may be dependent on other infrastructural requirements, such as networks for distribution, reticulation, storage, and brine and sludge management. The ESMF further details the E&S risk assessment related to the construction and operational phase for each archetype with mitigation measures. The ESMF also provides standardized templates of ESIA and ESMP and structure and content guidance regarding safeguarding instruments to be applied across different phases of subproject implementation. As the programme may not be limited to supporting only water reuse, but may include other types of subprojects, such as water reclamation, sludge management, biogas to energy, upgrading wastewater treatment processes, the Secretariat suggests that the AE consistently include associated facilities and third-party impacts in the E&S risk assessment on a case-by-case basis on all activities to be financed by the programme based on but not limiting to the proposed archetypes.

47. **ESS2: Labour and working conditions.** The programme has identified risks regarding labour conditions during the construction and operational phases of subprojects and has triggered the AE's safeguard on labour and working conditions to manage such risks. Subprojects will be required to develop detailed labour plans to address identified challenges including mitigation and management options, as well as reporting structures to ensure responsive management. Recognizing the importance of employment creation and income generation in pursuit of poverty reduction and inclusive economic growth, the ESMF will

exclude subprojects that involve harmful or exploitive forms of forced and/or child labour. All subprojects will be required to source their labour force locally and to include specific requirements in the procurement and contracting processes to promote fair treatment, non-discrimination and equal opportunity of subproject workers; protect workers, especially vulnerable workers such as women, people with disabilities, migrant and contract workers; promote health and safety in the workplace; prevent the use of all forms of forced and/or child labour; and provide workers with accessible means to raise workplace concerns. The ESMF also requires the programme to comply with the national Occupational Health and Safety Act to provide health and safety measures in the workplace and extend the protection and safety measures to those who may be exposed to the hazards arising out of or in connection with the activities of people at work.

48. **ESS 3: Resource efficiency and pollution prevention.** The programme is expected to improve water use efficiency and contribute to the long-term water security in South Africa through supporting a pipeline of subprojects at the local and municipal level. This standard is triggered because potential waste/effluent streams are planned for treatment that will allow reuse and conversion into a resource, which has potential environmental benefits but at the same time requires robust environmental management to prevent and control possible pollution incidents. The discharges from the wastewater treatment facilities and associated facilities may contain complex chemical compounds that are of environmental and human health concern. The ESMF requires the AE and all implementing partners to ensure that this standard is applied to safeguard environmental integrity and human health, and this applies to emissions to air, discharges to water, greenhouse gas emissions, soil contamination, subproject-associated waste, and environmentally sound management of hazardous substances following international best practices and standards. As part of the subproject due diligence, the programme will require subproject ESIAs and ESMPs to emphasize pollution prevention and to put in place specific mitigation measures. The ESMF also requires all subprojects to consider the sensitivity and vulnerability of the receiving environments and local capacities in enforcing relevant policies and standards of environmental management. As part of the eligibility criteria, subprojects that generate significant amounts of waste, including hazardous waste that could harm the communities or impair the quality of the receiving environment, will be categorized as high risk and considered ineligibility for the programme. The ESMF has noted management of localized brine and sludge as an important issue and requires that options for sludge beneficiation be explored for each subproject. As part of the minimum requirements for decommissioning and closure, the ESMF requires that, should the land be contaminated as a result of the programme activities, the owner of the facility must comply with all relevant legislation dealing with remediation of contaminated land. In line with the precautionary principle of the ESMF, the AE is requested to thoroughly assess the potential risks and impacts related to the wastewater treatment processes and potential discharges (including sludge) and reuse, both for the infrastructure directly invested by the programme and the facilities and processes of third parties associated with programme. It is suggested that the AE provide specific guidance and capacity-building support to ensure local capacity on the proper environmental management of the wastewater discharges related to the assets invested.

49. **ESS 4: Community health, safety and security.** The ESMF has identified an initial set of risks to the community and vulnerable members of communities during different phases of the programme, such as exposure to abuse at the early planning and stakeholder engagement stages by programme personnel, safety issues emerging during construction and caused by an influx of workers, and the risk of violence and gangsterism surfacing relating to ensuring access to work or associated contracts. Recognizing that subproject activities, equipment and infrastructure supported by the programme can increase community exposure to risks and impacts, the programme will exclude high social risk subprojects that will potentially result in significant socioeconomic impacts, potential for loss, or cause conflict or societal instability. The programme will also exclude subprojects that negatively affect specific socioeconomic groups,

such as minority ethnic groups and women, in terms of their basic needs (e.g. reducing income and food supply) or strategic needs (e.g. limiting agency and decision-making power). The ESMF further requires due diligence to be conducted at the subproject level in order to: anticipate and avoid adverse impacts on the health and safety of affected communities during the whole subproject lifecycle; promote quality and safety in the infrastructure design and construction; avoid or minimize community exposure to subproject-related traffic and road safety risks, diseases and hazardous materials; put in place effective measures to address emergency events and avoid disasters; and ensure the safety of the programme personnel and property. Detailed action plans will be devised for each subproject through the subproject ESIA and ESMP to manage identified and emergent health and safety issues relating to the community and vulnerable groups. On-site social and gender oversight officers and environmental control officers will be deployed to oversee and ensure these risks are managed.

50. **ESS 5: Land acquisition and involuntary resettlement.** The programme expects to develop all subprojects with new and existing facilities on land owned by the municipalities. Hence no acquisition of land or the resettlement of communities is expected. Land acquisition and resettlement is a key consideration regarding subproject eligibility. The eligibility criteria in the ESMF include a stipulation that subprojects involving displacement or relocation of more than 50 homes or 200 people will be considered as high risk; hence they will not be eligible for financing by the programme. The ESMF also requires that the site selection of subprojects will not negatively impact land use and that there will be no restricted access to resources. The programme will also raise awareness and ensure transparency of land rights in consultations and engagements with communities. The AE has provisionally triggered its own safeguard on land acquisition and involuntary resettlement where limited impact cannot be avoided. Further assessment will be conducted for every subproject, and detailed action plans are required where it is proven that there is no feasible alternative; and the ESMF provides an outline of a resettlement action plan to be developed at subproject level under such circumstances.

51. **ESS 6: Biodiversity conservation and sustainable management of living natural resources.** The programme is not expected to have high impacts on biodiversity because it will seek to apply ecosystem-based approaches in the design of water reuse subprojects. The programme will exclude subprojects with high biodiversity risk and the ESMF requires all subprojects to demonstrate positive impacts on biodiversity and ecosystem services over and above legal or AE safeguard requirements. It also requires such impacts to be integrated into subproject co-benefit objectives with clear targets and indicators in order to align with emerging global targets, standards and good practices on biodiversity conservation. The programme will avoid any sites that have been set aside for conservation purposes, and the AE's related safeguard will be applied for sites selected for subprojects that may fall within critical biodiversity areas or any other natural areas in line with the mitigation hierarchy. The safeguard requirement is to demonstrate that there is no viable alternative within the region; no measurable adverse impacts on those biodiversity values and the underpinning ecological processes for which the critical habitat was designated; no net reduction in the global and/or national/regional population of any critically endangered or endangered species over a reasonable period of time; and that a robust, appropriately designed, and long-term biodiversity monitoring and evaluation programme must be integrated into the client's management programme. Upon meeting all these requirements, the programme may support such a subproject with the relevant mitigation strategy encapsulated in a biodiversity action plan as required by the ESMF. If degraded sites are selected for subprojects, the programme will require rehabilitation or restoration of the site to be included in the subproject design. Where biodiversity offsets will be used to compensate for any impacts on natural habitats that cannot be avoided or mitigated in any other way, the ESMF will require these offsets to be used only as a last resort, with the biodiversity action plan outlining the necessary management and mitigation measures needed.

52. **GCF Indigenous Peoples Policy and ESS 7: Indigenous Peoples.** The AE has noted that impacts to Indigenous Peoples are not expected in the implementation of the projects and to that effect has included project exclusions on: projects that contravene the constitutional rights of South Africans and in particular Indigenous Peoples' rights and projects that undermine Indigenous Peoples' community rights to land, natural resources, language and indigenous knowledge. Noting that there are a few areas where San and Khoi-Khoi peoples may be present, the AE has included screening guidance in the ESMF and included provisions to guide the development of Indigenous Peoples plans where required, including the need for free, prior and informed consent. In line with their roles and functions, the GCF Indigenous Peoples Advisory Group is available to provide advice to the AE and executing entities (EEs).

53. **ESS 8: Cultural heritage.** The programme will avoid displacing, modifying or rendering inaccessible cultural heritage sites or structures by applying the eligibility criteria for subproject screening. However, the associated pipelines of the assets invested by the programme may trigger the requirements of the National Heritage Resources Act. The AE has therefore provisionally triggered its relevant safeguard, which will be applied when a subproject involves excavations, demolition, movement of earth, flooding or other changes in the physical environment; is located within a legally protected area or a legally defined buffer zone; is located in or in the vicinity of a recognized cultural heritage site; is designed to support cultural heritage conservation, management and use; may materially impact on intangible cultural heritage or if a subproject intends to use such intangible cultural heritage for commercial purposes; or may impact or depend on cultural heritage including human capital, natural capital or institutional capital. In such a case, the programme will deploy a heritage specialist to support the permitting process required by national regulations and include relevant requirements in the subproject ESMP. The ESMF provides an indicative outline for a cultural heritage management plan to guide further assessment and risk mitigation at the subproject level.

54. **Sexual exploitation, sexual abuse and sexual harassment (SEAH) safeguarding.** The ESMF has identified SEAH risks at the programme level relating to the proposed archetypes, primarily during the construction and operation of subprojects. Local community members and vulnerable groups may be exposed to SEAH risks due to the influx of workers. Female (but also male) construction workers are vulnerable to SEAH, exacerbated by the traditionally male-dominated working environment. The AE will focus on SEAH safeguarding in relation to the workplace, as well as community health and safety. The ESMF has incorporated SEAH-related questions in the subproject eligibility screening checklist, focusing on the client's capacity to address and redress gender-based violence and incidences of SEAH. As part of the subproject due diligence, the ESMF will require SEAH risks and impacts be identified following a four-step approach (i.e. assessing, addressing, responding and communicating) to ensure protection against SEAH. The process will result in an SEAH management plan to be implemented and monitored in each subproject. The ESMF provides an indicative outline of the SEAH management plan to guide its development at the subproject level. The AE will also address SEAH risks through its safeguard on gender mainstreaming. That is, the gender-responsive social assessment required at the subproject appraisal stage will incorporate SEAH safeguarding in a structural manner, examining SEAH risk assessment, organization commitment, mitigation measures, training and awareness-raising, and mechanisms for SEAH grievance redress in a gender-sensitive and survivor-centred approach.

55. **Implementation arrangements.** The ESMF specifies the roles and responsibilities among the AE, programme clients, interested and affected parties, national authority, and regional and local authorities. As the ESMF is developed at a programme level to provide a framework and processes within which site-specific ESIAs and ESMPs will be prepared at the subproject level, the programme will establish a WPO to support targeted municipalities that are preparing and implementing subproject due diligence. Both the AE and the WPO will oversee the implementation of the ESMF, with the WPO playing a central role in initiating,

procuring and overseeing interventions undertaken by the programme. The WPO will be responsible for implementation of the ESMF, supported by a team of implementing consultants who will be tasked with developing feasibility studies and implementing due diligence, ESMPs, and gender and stakeholder engagement plans at the subproject level. The programme will secure gender specialists and environmental and social experts at the implementation stage. All 27 subprojects envisioned in the longer-term project pipeline will be required to develop subproject ESIAs and ESMPs. Once a subproject ESMP is developed, the responsibilities for ensuring its implementation will be shared among the programme office, implementing consultants, local municipality, resident engineer, contractors and the environmental control officer. The AE and the WPO, with the support of a monitoring and evaluation unit will track the programme's progress and ensure its compliance with the AE's safeguard standards and the ESMF requirements. This monitoring and evaluation will take place across the subproject lifecycle from initiation, development, construction and operations, through to subproject decommissioning and closure. All subprojects will require annual reviews and mid-term reviews with supporting reports.

56. **Stakeholder engagement and information disclosure.** Following the AE's own safeguard standard on stakeholder engagement, the programme has undertaken a wide range of stakeholder engagement processes to inform the design of the programme, as well as to create alignment among key strategies and plans on water reuse and management at the national and subnational level. The stakeholder engagement conducted so far has helped improve the level of understanding on strengthening climate resilience through water reuse as the main intervention proposed by the programme. A communications strategy and implementation plan has been developed along with a consultation and stakeholder engagement plan at the programme level. As part of the programme component (component 3), a stakeholder engagement, learning and communication strategy will be developed to support scaling up the interventions across the local, provincial and national levels. All subprojects will be required to develop a context-specific stakeholder engagement plan in each targeted municipality, incorporating the regulatory requirements for public participation and consultation where listed activities by national regulations are identified. The ESMF provides a draft structure for the stakeholder engagement plan to guide subproject consultations. The programme's E&S risk documents, including the subproject ESIAs and ESMPs, will be subject to information disclosure requirements set out in the ESMF. The subproject ESIA and ESMP will be submitted to GCF along with the ESS disclosure form for clearance following the disclosure requirements in accordance with the relevant GCF policies.

57. **Grievance redress mechanism (GRM).** Having triggered the AE's own standard on grievance redress, the programme will establish an independent GRM to resolve all complaints arising from the programme and ensure transparent, fair, equitable and prompt resolution of complaints/grievances from aggrieved parties. The ESMF specifies criteria for handing complaints through the independent GRM to ensure the scope, scale and type of grievance mechanism required will be proportionate to the nature and scale of the potential subproject risks and impacts. At the subproject level, individual GRMs will be established to complement the independent GRM and to work with other redress mechanisms, such as public participation community meetings at the municipal office and through their ward councillors, the Municipal Help Desk and a suggestion box located at central offices. The subproject GRMs will be required to follow the same approach as the independent GRM incorporating procedures to ensure SEAH incidents will be addressed in a gender-responsive and survivor-centred manner. The programme will encourage the use of local or subproject level GRMs, recognizing that the local GRMs can provide an effective and direct remedy to complainants. The programme will also make sure that stakeholders are informed about the GCF's Independent Redress Mechanism (IRM) and that complainants will have direct access to the IRM. In line with the GCF Indigenous Peoples Policy, the GCF Indigenous Peoples focal point will be available for assistance at any stage, including before a claim has been made.

4.2 Gender Policy

58. The AE provided a gender assessment and action plan with the funding proposal and therefore complies with the requirements of the GCF Gender Policy. The gender assessment submitted by the AE provided an analysis of the status of gender equality and the empowerment of women in South Africa, and as it relates to the various themes in the environmental sector – water and sanitation in particular – in order to align the assessment with the proposed programme. The gender assessment involved secondary review of data including internal and external reports, the most recent literature on gender-responsive and transformative approaches and guidelines, good practices on gender mainstreaming, and the national gender policy direction and development plans. Some consultations on the gender aspects were also conducted to inform the gender assessment, albeit the coronavirus disease 2019 pandemic contributed to the limitations of the consultations.

59. The 1994 Constitution of South Africa guarantees equal rights and opportunities for women and men, and the right to sanitation is encompassed in Section 24 of the Constitution, guaranteeing that everyone has the right to an environment that is not harmful to their well-being, as well as a right to have the environment protected. Section 9 of the same Constitution provides for equality of all individuals including the full and equal enjoyment of all rights and freedoms. This provision prohibits direct or indirect discrimination based on several grounds, including gender and/or sex. The country has a national strategy on gender mainstreaming in the environment sector (2016–2022). This strategy aims at ensuring that initiatives in the sector support the creation of inclusive policies, and overall provides strategic guidance in the environment sector. The international, continental and regional protocols related to gender equality and the empowerment of women, such as the Convention on the Elimination of All Forms of Discrimination against Women (CEDAW), the Protocol to the African Charter on the Rights of Women (Maputo Protocol) and the Southern African Development Community, have been domesticated in the country – emphasizing the need to address any gender-related disparities in the water sector, and the need for concerted efforts to develop programmes that are not only gender sensitive but gender responsive and gender transformative.

60. There are various intersections of water and gender equality, particularly because water insecurities are rapidly growing in South Africa especially in urban areas (women, girls and poor communities being disproportionality affected), with pressure on water resources as demand increases. There are significant gender and social inclusion dimensions of water access, use and control. Climate change worsens existing vulnerabilities and adds to the pressures on the environment and natural resources. Moreover, the gender differential impacts of climate change are attributed to existing inequalities between men and women in South Africa, such as unequal access to resources and the gendered division of labour and decision-making power that affects the ability to respond to the effects of climate change.

61. Meeting WASH priorities and needs requires recognition of gender barriers and development of a gender-mainstreaming approach that includes socioeconomic co-benefits and ensures adequate empowerment of marginalized individuals and communities. The gender assessment for the programme therefore called for an expanded approach to gender mainstreaming to include gender and social inclusion perspectives that would identify vulnerable populations as part of the design and implementation of the WASH infrastructure projects and more specifically in water reuse initiatives and programmes. The gender assessment further affirms that programmes such as WRP require coordinated actions across sectors to drive progress on gender equality and empowerment of women, collaboration between infrastructure service delivery and public works programmes, water and gender agencies and stakeholders responsible for climate change, disaster risk management and urban development.

62. The gender action plan (GAP) of the programme was built on the gender assessment by looking at key entry points and stand-alone gender activities that will be implemented during the programme in order to have gender-responsive and gender-transformative WRP. The AE is required to have a dedicated gender specialist to facilitate implementation of the GAP, and the DBSA Gender Forum will provide overall support and accountability. The GAP impact statement seeks "improved economic empowerment for women through the establishment of a national water reuse programme that integrates gender-responsive initiatives and interventions into the design and implementation of projects as well as in the procurement, management and operational functions". The project envisages achieving this through (1) enhancing the capacity of women in senior leadership positions; (2) continuously learning and adapting the programme to gender-sensitive practices; (3) improving the working conditions and procurement processes, and creating gender-specific contracting provisions and appointment of staff; (4) improving the protective clothing, equipment and sanitary facilities on site; (5) selecting projects that are gender-responsive and that do not exacerbate existing social inequalities; and (6) ensuring that there are appropriate budget allocations for gender-related activities.

63. The GAP has categorized its activities in three main intervention areas: (1) planning and governance; (2) capacity-building and awareness; and (3) monitoring, evaluation and reporting. Under planning and governance, activities to be delivered will include developing a gender mainstreaming policy for the WPO programme as part of the adoption of gender-responsive and inclusive human resources policies by the WRP Oversight Committee; adoption of gender-related quotas in leadership positions; development of gender-sensitive working conditions; and reviewing existing contracts and inclusive water and sanitation infrastructure through a gender-related lens. In addition, the project will guarantee that procurement processes are inclusive (targeting women in the value chain and/or women-led companies), and will ensure that all projects have terms of reference and requests for proposals that take gender into consideration. With regards to women's representation, the project aims at reaching 40 per cent representation for women as per the national laws/targets; however, as it evolves the project will monitor this target with the aim of reaching gender parity. Regarding capacity-building, the project will develop a gender communications strategy; publish at least two gender knowledge projects per year that will identify barriers to women's senior leadership participation in managing utilities; research initiatives on women in science, technology, engineering and mathematics; and identify the main skills women workers require in order to enhance their skills and expertise in the sector. The project will further strengthen its institutional capacity on gender and will facilitate training/workshops on inclusion and diversity, gender policy, SEAH, gender monitoring and inclusive procurement. For the third intervention area of monitoring, evaluation and reporting, the project will design a gender monitoring and evaluation framework and a tracking tool for the project and facilitate various activities that will help inform the project from a gender perspective. The AE is requested to disaggregate the data for all the training sessions and other outputs of the GAP and overall project activities, and periodically monitor the gender targets that have been included in the GAP and funding proposal. In addition, the AE is requested to ensure that the grievance mechanisms of the programme are accessible to both women and men.

4.3 Risks

4.3.1. Overall programme assessment (medium risk)

64. This is an adaptation programme investing in water reuse infrastructure in South Africa. The project will establish a blended finance structure that enables investors to participate in individual water reuse projects.

65. GCF is requested to provide a subordinated loan of USD 200 million and a grant of USD 35 million. The DBSA and/or other development finance institution (DFI) partners will co-finance a total of USD 150 million in subordinated debt, and participating municipalities and the country's Infrastructure Fund will provide USD 155 million in grant/budget allocations. The AE is expected to identify a co-financier providing funds to execute a guarantee of USD 50 million that will help to mobilize an additional USD 650 million in senior debt and USD 200 million in equity from private investors. The National Treasury of South Africa will provide a grant of USD 2 million, and other investors will provide a grant or recoverable grant of USD 30 million to establish the WRP and pipeline. The expected total programme financing is approximately USD 1.4 billion.

66. The success of the project hinges on the AE's ability to attract commercial investors to capitalize the WRP and ensure the underlying assets in subprojects realize the project's aims.

4.3.2. Accredited entity/executing entity capability to execute the current programme (low risk)

67. DBSA (rated Ba3 by Moody's) is the AE and EE for the programme. It will work closely with DWS. The AE is a national entity and fully owned by the Government of South Africa. DBSA has a track record in investing in public and private sector projects in the country.

4.3.3. Programme-specific execution risks (high risk)

68. Credit risk: GCF will assume the credit risk of the borrowers under the blended financing structure (component 2). The GCF loan will be subordinated to other senior debts and rank pari passu with the DBSA and/or DFI partners. The public sector (e.g. municipalities, utilities, water user associations) and private sector borrowers that benefit from the project will be selected by the AE on the basis of the eligibility criteria described in the term sheet. Due to the financial industry's low exposure to and high risk perception of the water reuse sector, the creditworthiness of the borrowers and their capacities for implementing this kind of project is unknown or they have limited track records. The comfort can be derived from the fact that the GCF commitment in each subproject is capped at 15 per cent of the total subproject cost or USD 30 million per subproject, whichever is lower. The AE's due diligence on the borrowers and its capacity to identify and finance pipeline projects will be critical to the success of the WRP.

69. Co-financing and impact risks: the AE needs to mobilize and facilitate the following co-financing amounts:

(a) Senior debt of USD 650 million;

(b) Equity of USD 200 million;

(c) Subordinated debt of USD 150 million;

(d) Grant/budget allocation of USD 150 million; and

(e) Guarantee of USD 50 million.

70. The GCF loan takes a junior position along with other DFIs and/or DBSA, which would help to unlock the private capital in a senior position. The guarantee will also provide additional comfort for commercial investors. A low appetite for financing such projects from other investors may lower the intended impact of the programme. To avoid co-financing risks, which will lead to the impact risk, the term sheet includes: (1) the co-financing ratio of subordinate loans between GCF and other DFIs/DBSA; (2) the overall co-financing ratio at the programme level; (3) confirmation from a co-financier to finance the guarantee; (4) commitment from municipalities for their budget allocation or confirmation of grant financing; and (5) a minimum equity ratio as part of the eligibility criteria of the subloans.

71. Foreign exchange risk: the AE would convert the GCF loans in United States dollars into South African rand and provide that to the sub-borrowers. The AE will enter into an appropriate hedging mechanism to cover the tenor of the loan. Depreciation of the local currency may adversely affect the cash flow in the subprojects and borrowers' ability to repay. To partially manage this risk during the first 9 years, the GCF loan amount includes a reserve of up to 5 per cent or USD 10 million. This reserved amount may be used to cover a residual foreign exchange shortfall (i.e. a shortfall in USD which occurs from a payment default or prepayment by the borrowers). At year 10, if the available reserve amount is more than zero, the remaining balance will be made available for disbursement; while if the amount falls below zero, the risk of any subsequent residual foreign exchange shortfall will be assumed by GCF.

72. Concessionality: the GCF concessional loans will be provided for both public and private borrowers up to the total amounts of USD 130 million and USD 70 million respectively. The interest rate to be provided for the public sector shall never be higher than the rate for the private sector. The pricing for the private sector will be determined by the AE based on the creditworthiness of the borrowers and the AE's internal policy. The provision of passing down the GCF concessionality to the sub-borrowers is included in the term sheet, but there is no reference rate or margin cap in the pricing calculations. The AE is requested to provide confirmation that the sub-borrowers receive the most optimal blended rate, along with the calculation of the prices of subloans for each sub-borrower at the disbursement request and annual reporting during implementation.

4.3.4. Compliance risk (medium risk)

73. South Africa is not subject to any current United Nations Security Council (UNSC) resolutions. DBSA, as AE, confirmed that:

(a) No projects will be undertaken in any jurisdiction which is subject to or affected by UNSC resolutions; and

(b) No individual or entity that is listed on any UNSC sanctions list, including the United Nations Consolidated Sanctions list will be involved in any manner with the project or its activities, either as a counterparty, implementer or beneficiary.

74. DBSA acknowledged that instances of fraud and other corrupt practices, as outlined in the GCF Policy on Prohibited Practices, will undermine the project's due diligence efforts as well as negatively impact upon the reputation of the GCF, and DBSA as the AE and EE, as well as the creditability of the WRP. Such issues could ultimately impact upon the levels of investment in the WRP, particularly impacting the blended finance solutions that the programme aims to develop.

75. In this regard, DBSA has conducted an assessment of money laundering and terrorist financing risks in the project and has determined a low probability of risk, medium impact and medium priority for risk mitigation.

76. The South African legislation, through the Public Finance Management Act 1 of 1999 and Municipal Finance Management Act 56 of 2003, provides for regulatory oversight and procedures to counter prohibited practices. These statutes are supported by various audit processes at the national and provincial level. In the development of the WPO, the procurement processes, standardized procedures and supporting guidelines will ensure alignment with the GCF Policy on Prohibited Practices, as well as build upon the systemic checks and balances that are provided by the DBSA, as AE.

77. DBSA confirmed that it will apply its own fiduciary principles and standards relating to any know-your-customer checks, anti-money laundering and countering the financing of terrorism (AML/CFT), and financial sanctions imposed by the UNSC, which should enable it to

comply with the GCF Policy on Prohibited Practices and the principles of the GCF AML/CFT Policy.

78. The relevant DBSA policies are substantially consistent with the principles of the GCF AML/CFT Policy. In this regard, the DBSA Whistle Blowing Policy will be utilized to enable staff members to raise concerns where they have reasonable grounds for believing that there is fraud, corruption, corporate crime or impropriety. Additionally, the WPO will develop thorough risk management protocols linked to a comprehensive governance arrangement that will ensure effective oversight and reporting.

79. DBSA acknowledged that, while municipalities involved in the project often have significant capacity constraints at the technical level that hinder the conceptualization and preparation of water reuse projects, there are often significant challenges in terms of municipal leadership in obtaining the required support for such interventions. This can be due to insufficient capacity as well as political interference that results in weak governance that can have impacts on the financial status of the municipality as well as its ability to adhere to formal agreements. This is often exacerbated by municipalities facing significant financial pressures due to weak financial circumstances and many competing demands on limited financial resources.

80. In this regard, DBSA has conducted a partner risk assessment of the municipalities and has determined a medium probability of risk, high impact and high priority for risk mitigation.

81. The WRP will be initiated with partners that are committed to the project outcomes and have experience in implementing similar projects. This will support the WRP in strengthening municipal competencies with time, as well as enabling the WPO to embed its protocols and practices. The project owners will be the municipalities and, as such, will engage the WRP for support through the WPO for specified water reuse interventions. Ensuring alignment with municipal integrated development plans will be a key element for ensuring strategic commitment at the municipal level, noting that progress on implementing such plans is monitored, measured and reported.

82. Engagement will be through formal written correspondence, signed by the required authority, as required at various stages and gates of the project preparation process, and will allow for the project to be stopped at any stage of project preparation, if appropriate in the circumstances. The municipalities will sign project funding agreements, as formal agreements between the project owners and the lenders. The structure of these agreements will depend on the structure of the funding arrangements and the specifics of the municipal context. In addition, the use of performance guarantees will be used to underpin finance arrangements to mitigate these partner risks.

83. DBSA confirmed that, at this stage, no due diligence assessments have been undertaken because this would only be appropriate during the project preparation phase. DBSA expressed its understanding of the importance for first-level due diligence assessments of counterparties involved in the project and its activities. However, DBSA advised that the municipalities involved in the project undergo an annual audit by the national Audit-General and this provides the required level of due diligence. These assessments will also be supported by the findings from the Municipal Strategic Self-Assessment analyses and the Green Drop and Blue Drop performance assessments, all of which are led by DWS.

84. DBSA informed GCF that, with the use of a bespoke blended finance solution being developed for each project, it is not possible to undertake a due diligence assessment at this stage. While there has been some engagement with co-financiers in the development of the WRP, there has been no formal due diligence process. Broadly speaking, there will be three forms of co-finance:

(a) ***Equity/budget allocation grants***. These are likely to come from participating local authorities, which own and/or are mandated to run a particular wastewater treatment works. Municipalities have been engaged numerous times and there is significant interest among the larger metropolitan municipalities on engaging with the WRP. These municipalities have a range of financial instruments available;

(b) ***Concessional loans***. DBSA will undertake market soundings with other development finance institutions to provide support to projects, as project preparation support is being provided to ensure the production of bankable projects; and

(c) ***Senior debt***. Commercial banks have been engaged throughout the preparation of the WRP, and this has informed the structure of the blended finance solution, how project bonds could be used, and what magnitudes of debt could be allocated by private capital providers. These commercial banks have indicated interest in providing debt and equity to support WRP projects.

85. In terms of processing risk assessments and due diligence on co-financiers, DBSA has confirmed that this is a requirement of the Independent Blended Capital Facilitator and this will be undertaken in the design of each blended finance solution.

86. DBSA informed GCF that, although due diligence assessments of co-financiers are not possible at this stage of programme development, assessments will be undertaken during the first year of the programme and will then be undertaken on an annual basis as projects move up the implementation pipeline.

87. DBSA confirmed that, during project preparation and procurement, it will undertake that agreements with service providers and arrangements with project owners address any risks associated with prohibited practices. The use of the DBSA supply chain system, protocols and contractual arrangements will ensure that all equipment, materials or technology are to be used appropriately for the purpose for which they were intended or purchased. This system and the contractual arrangements that support procurement provide clarity as to the use, purpose and ownership of such equipment, including the return of this to the DBSA and WPO upon project completion. All equipment will be registered in a project database. The project's contractual arrangements together with these equipment management protocols will guide the management of this equipment and technology, as well as its return after project completion.

88. It is a key objective of the WRP to drive efficiency by standardizing processes and documentation. There will need to be development and, where appropriate, standardization of WRP documentation and water reuse project documentation regarding project preparation (scoping, feasibility and procurement) and implementation (contracting and monitoring and reporting). A key value that DBSA brings to the programme development process is its existing experience and capacity in developing and implementing programmes and, as such, has indicated that it has a suite of standard documentation.

89. As part of the monitoring and evaluation process, controls will be used to ensure that any materials or technology procured under this project are used only for the purposes intended and are not diverted or misused for unauthorized, improper or illicit purposes.

90. DBSA confirmed that there will be no distribution or disbursements to individual beneficiaries other than through contractual arrangements for service provision on projects. This will use the DBSA supply chain management policies and systems that provide for effective financial management.

91. DBSA confirmed that the project will make use of the DBSA internal policies and systems to support the reporting of complaints and allegations of impropriety, wrong-doing or other related issues in the project and its activities, and it has a whistle-blowers policy in place. The DBSA will provide support to the whistle-blower protection programme in terms of ensuring implementation.

92. **Condition:** Although DBSA has acknowledged the importance of due diligence on co-financiers, DBSA has advised the GCF that it is unable to produce such assessments at this stage of the project. DBSA has, however, indicated that due diligence will be undertaken and provided to GCF during the project preparation phase (i.e. during the first year of the programme). Therefore, as a prerequisite to the second disbursement, DBSA has agreed to a condition precedent which has been included in the term sheet, and obligates DBSA to submit evidence in form and substance satisfactory to the Secretariat, confirming that AML/CFT due diligence has been undertaken on the co-financiers, and that no such co-financier is subject to financial sanctions imposed by the UNSC.

4.3.5. Risk rating

93. The Secretariat has conducted a review of the project in accordance with relevant GCF Board-approved policies and does not find any material issue or deviation with respect to compliance issues. Based on available information for this funding proposal, the Secretariat has determined a risk rating of "medium".

4.3.6. GCF portfolio concentration risk (low risk)

94. In the case of approval, the impact of this proposal on the GCF portfolio concentration in terms of results area and single proposal is not material.

4.3.7. Recommendation

95. It is recommended that the Board consider the above factors in its decision.

Summary risk assessment	
Overall programme	**medium**
Accredited entity (AE)/executing entity (EE) capability	**low**
Project-specific execution	**high**
GCF portfolio concentration	**Low**
Compliance	Medium

4.4 Fiduciary

96. DBSA will play the dual role of AE and EE. DBSA as AE will ensure monitoring and reporting as required by GCF and will offer programme management support and provide backroom management services.

97. DBSA will coordinate the work of third-party auditors engaged in the programme to monitor and evaluate results. During programme implementation DBSA will ensure that any subsidiary agreement that it enters into with the programme custodian, DWS, in respect of a funded activity will reflect or incorporate the terms and conditions of the funded activity agreement that may be applicable or relevant to the EE and the national designated authority.

98. DWS will contract with DBSA, in its capacity as GCF AE and EE, to establish and support implementation of the governance arrangements to give effect to the National Water Programme, including WRP. At a strategic level, an oversight committee will be established by DWS and DBSA and, at an operational level, DWS and DBSA will establish a programme management office (i.e. the WPO).

99. The proposed blended finance solution is designed to finance water reuse projects in a way that optimizes the use of climate finance, and enhances the use of private capital, thereby

taking this critical infrastructure to market. Applying a developed and agreed upon suite of blended capital principles for the WRP, an Independent Blended Capital Facilitator will act as a specialist service provider in the WPO, working alongside the Water Reuse Unit and the Project Preparation Panel to source and appropriately blend capital for water reuse infrastructure projects as approved by the Water Reuse Unit and WPO.

100. The financial management between the AE and EE of the WRP will be governed by the DBSA finance manual and its policy and procedures on procurement of goods and services. DBSA will participate in the oversight committee as the implementing agent of the National Water Programme and will act as management agent of the WPO for a period of 5 years or until such time that the WPO is legally institutionalized. DBSA will provide all administrative services such as procurement of goods and services through DBSA supply chain management processes, procurement of panels of service providers, entering into legal contracts with service providers and employees. It will also operate a ring-fenced bank account for the WPO function and make payments on behalf of the WPO.

101. The WPO will adopt DBSA's accounting systems and will be audited independently on a yearly basis.

102. Given that financing is expected to be in the local currency while GCF extends the facility in USD, the AE will arrange a hedge. In the event that the AE would need to unwind a hedge (typically in the event of default), a reserve of USD 10 million is set aside for the first 9 years to meet residual foreign exchange shortfalls.

4.5 Results monitoring and reporting

103. This is an adaptation project focusing on strengthening socioeconomic resilience at the community, regional and national level through improved water security by implementing water reuse projects through innovative public–private blended financing arrangements. The project will reach and increase the adaptive capacity of 7.3 million beneficiaries, of which 3.42 million are direct beneficiaries and 3.87 million are indirect beneficiaries. The economic co-benefits are foreseen to result in a total of 585 new jobs and a 20 per cent reduction in pollution into local water resources, as reported under the correspondent co-benefit indicators in the funding proposal (section E.5).

104. The theory of change adequately illustrates how the results chain will cascade from the goal statement to the project activities and clearly articulates the relevant interlinkages between logic levels, barriers and risks. The "if, then, because" logic is adequately formulated, and the proposed outputs and outcomes are defined in a manner that adequately supports meeting the ultimate programme goal.

105. The logical framework has been designed with relevant details, including the inclusion of solid means of verification, and selecting the appropriate core and supplementary indicators for the targeted results areas as per the GCF integrated results management framework.

106. The secretariat, has identified the two-fold high-level risks for this programme, which are:

(a) The WRP, being an actual vehicle to process projects and finance, has not yet been established, and GCF funds will be dedicated for setting it up; and

(b) There are projects ready in the pipeline, and pipeline generation depends on the WRP being operation and its effectiveness in mobilizing projects.

107. Therefore, the secretariat has proposed the following two conditions and agreed with the AE in the term sheet accordingly, :

(a) Delivery to the GCF by the AE of a written confirmation, in a form and substance satisfactory to the GCF Secretariat, that the implementation of subcomponent 1.1 (excluding activity 1.1.7 in reference to monitoring, evaluation and reporting regimes structure and scope) has been completed and that all the elements required for the set up and operationalization of the WRP, as set out in the funding proposal, have been implemented; and

(b) Delivery to the GCF by an authorized officer of the AE of the indicative pipeline of subprojects expected to start implementation following six (6) months from the date of the requested disbursement, which correspond to the requested disbursement amount, detailing the proposed blended finance package (including the financiers), technical specification, budget breakdown, envisaged PPP structures, projects financial engineering, and impact/indicators to be achieved per project.

108. Section E.3 of the logical framework has been designed with relevant details, including the inclusion of robust means of verification and reporting on the appropriate core and supplementary indicators for the targeted results areas as per the GCF integrated results management framework.

109. The monitoring and evaluation plan in annex 11 of the funding proposal has been found to follow the appropriate requirements, as it includes the entire list of the logical framework indicators while ensuring consistency between the data/sources and set means of verification. The monitoring and evaluation budget provided in the monitoring and evaluation plan accounts for 2.5 per cent of the total programme budget, ensuring compliance with the GCF Evaluation Policy provisions.

4.6 Legal assessment

110. The original accreditation master agreement was signed with the AE on 8 September 2016, and it became effective on 12 January 2017. The AE's first term of accreditation expired on 11 July 2022. The AE was re-accredited by the Board pursuant to decision B.31/12. The first amendment and restatement agreement amending and restating the accreditation master agreement was signed on 29 August 2022, and it became effective on 13 October 2022.

111. The AE has provided a legal certificate confirming that it has obtained all internal approvals and it has the capacity and authority to implement the programme.

112. The proposed programme will be implemented in the Republic of South Africa, a country in which GCF is not provided with privileges and immunities. This means that, among other things, GCF is not protected against litigation or expropriation in this country, which risks need to be further assessed. The Secretariat submitted the first draft of the privileges and immunities agreement and the background note to the Government of the Republic of South Africa on 9 March 2016.

113. The Heads of the IRM and the Independent Integrity Unit have both expressed that it would not be legally feasible to undertake their redress activities and/or investigations, as appropriate, in countries where the GCF is not provided with relevant privileges and immunities. Therefore, it is recommended that disbursements by the GCF are made only after the GCF has obtained satisfactory protection against litigation and expropriation in the country, or has been provided with appropriate privileges and immunities.

114. In order to mitigate risk, it is recommended that any approval by the Board is made subject to the following conditions:

(a) Signature of the funded activity agreement in a form and substance satisfactory to the GCF Secretariat within 180 days from the date of Board approval; and

(b) Completion of the legal due diligence to the satisfaction of the GCF Secretariat.

Independent Technical Advisory Panel's assessment of FP209

Proposal name:	Climate Change Resilience through South Africa's Water Reuse Programme ("WRP")
Accredited entity:	Development Bank of Southern Africa
Country/(ies):	South Africa
Project/Programme size:	Large

I. Assessment of the independent Technical Advisory Panel

1.1 Impact potential *Scale: Medium*

1. South Africa is a water-scarce country, ranked fifth in terms of water scarcity among the sub-Saharan countries, with at least half of the country classified as arid or semi-arid.[1] The World Resources Institute's water risk index defines South Africa's baseline physical water risk as "extremely high".

2. The proposal's climate rationale argues that, because of anthropogenic climate change, average temperatures have already increased by 1.5 °C in South Africa, causing increased evapotranspiration, aridity and drought, and worsening water-related vulnerability in general.

3. Regarding precipitation, the proposal states that observed changes since the 1960s include a marginal reduction in rainfall during the early rainfall season, and potentially significant decreases in the number of rain days across almost all hydrological zones, which would indicate an increase in the intensity of rainfall events, coupled with prolonged dry spells.[2] In earlier assessments, the independent Technical Advisory Panel (TAP) noted that the funding proposal did not reflect on scientific analysis of the spatial dimensions of rainfall occurrence, its seasonality and trends. The current edition of the funding proposal and additional materials provided extra clarity on how the accredited entity (AE) for the project intends to incorporate regional variability and non-climate impacts in its subproject assessments and investment decision-making.

4. To address water scarcity, the proposed project seeks GCF co-financing to design and implement South Africa's national Water Reuse Programme (WRP), as part of the country's National Water Programme (NWP). The proposal has a programmatic approach, whereby under the WRP, a project pipeline would be established, including a detailed selection process, and GCF funding would be allocated to grant financing of water reuse projects.

5. The main project objectives are to:

(a) Establish and operationalize the WRP;

(b) Support the preparation of the first 10 water reuse projects;

(c) Provide financial resources for water reuse projects and support their implementation;

[1] Oluwasanya, G et al. 2022. Water Security in Africa: A Preliminary Assessment. Issue 13: p.15. United Nations University Institute for Water, Environment and Health, Hamilton, Canada. Available at https://inweh.unu.edu/water-security-in-africa-a-preliminary-assessment/

[2] Id. p.6.

(d) Develop a "blended financial solution" to enhance access to private funding; and

(e) Provide institutional capacity-building and awareness and education campaigns.

6. The proposal identifies four archetypes of water reuse: (i) direct potable reuse (DPR); (ii) indirect potable reuse; (iii) irrigation reuse; and (iv) industrial reuse. Indirect potable reuse involves treating wastewater, including advanced treatment, and injecting the treated water into an environmental buffer such as an aquifer or surface water reservoir, to increase water availability in those sources.

7. Subprojects may consider all types of opportunities related to water reclamation, sludge "beneficiation" (see para. 8 below), electricity cogeneration with biogas, and wastewater treatment process upgrades.

8. The tertiary screening criteria include a criterion called "beneficiation", which aims at prioritizing potential projects that include mitigation measures such as solar power and energy generation with biogas and sludge beneficiation. The AE, the Development Bank of Southern Africa (DBSA), states in its proposal that potential mitigation co-benefits would be evaluated on a case-by-case basis. The independent TAP suggests that, to ensure alignment with the Paris Agreement, subprojects should be assessed in terms of greenhouse gas (GHG) emissions under a business-as-usual situation and the potential mitigation benefits of avoided and/or reduced GHGs. This would enhance the capacity of the project to enable the host country to achieve its mitigation commitments and potentially attract finance from climate change finance providers, especially around GHG reduction and/or avoidance.

9. Direct beneficiaries have been calculated based on the estimated beneficiaries for the 10 selected water reuse projects. The total treatment capacity of these projects is 445 million litres (l) per day which, assuming an average consumption of 130 l per person per day would result in approximately 3.4 million direct beneficiaries.[3] This strikes the independent TAP as being overly optimistic considering that the number of indirect beneficiaries is calculated as the total beneficiaries of the other 17 projects that are expected to be implemented through the full-scale WRP, and would deliver 1,067 million l/day across the country. It should be noted that these estimations assume that 100 per cent of the projects would result in DPR, although in reality the programme would also fund irrigation and industrial reuse. Notwithstanding the above, it is clear to the independent TAP that there can be substantial direct and indirect beneficiaries even if below the calculation proffered by the AE.

10. The adjustments and additional information provided by the AE are the basis upon which the independent TAP has increased the impact potential assessment of this proposal to Medium, while noting that there remains considerable risk around subproject selection. However, the independent TAP is comfortable that the AE's revised approach can mitigate some of these risks. The independent TAP encourages the GCF Secretariat to implement ongoing oversight to ensure that GCF funding is primarily used to address climate risks.

1.2 Paradigm shift potential *Scale: Medium*

1.2.1. Innovation

11. The proposal identifies 13 water reuse projects already in operation in South Africa. Most of these are pilot and small-scale projects, resulting in a total capacity of 200 million l/day, or 2.3 m^3/s, which represent approximately 1–2.5 per cent of the country's total residential water consumption. Of these 13 projects, 8 are for DPR but, given their rather small scale, they account for a total capacity of only 36 million l/day. Moreover, there are also nine water reuse

[3] Please note that the number of beneficiaries could be considerably less, since water losses are not considered in this calculation.

projects in the planning and preparation phase, with a total capacity of 420 million l/day. Some of these (uMhlathuze, Drakenstein, eThekwini and Cape Flats wastewater treatment works) would be part of the 10 shortlisted projects of the present programme, adding up to 225 million l/day. Given the relatively low coverage of water reuse projects in South Africa, and mainly for DPR, the proposed programme can be considered innovative.

12. The paradigm shift of the programme would consist of introducing wastewater reuse at a larger scale for municipalities across the country, through the demonstration of viable business models for water reuse among industrial, commercial, and residential users, and by applying advanced water reuse technologies.

1.2.2. Potential for knowledge and learning

13. Under component 3, the proposal includes a series of knowledge exchange interventions, aimed at enhancing awareness of climate change impacts at the local level, and supporting the successful scaling and implementation of water reuse projects. These include: (i) the development of a communications strategy to improve public knowledge of South Africa's water-related climate change vulnerability; (ii) communication materials and products; (iii) awareness-raising campaigns; (iv) management of a "community of experts" to provide strategic guidance and technical support for the preparation, design and structuring of water reuse projects; and (v) specific market engagement sessions to provide information and create awareness regarding the WRP.

14. Monitoring, reporting, and evaluation arrangements are adequately described in the proposal, and include project-level monitoring and evaluation, programme-level monitoring, and WRP-level evaluation.

1.2.3. Contribution to the creation of an enabling environment

15. Given the programmatic approach of the proposal, there are no specific arrangements concerning post-project operation and maintenance. On the other hand, the programme seeks to ensure sustainability by applying an "efficiency and effectiveness" criterion in the primary selection process, defined as "…the degree to which the project is persuasive of its financial and economic soundness, including the extent to which the project or programme's funding model fosters cost-effectiveness and private sector funding mobilization". However, cost recovery represents a current barrier to the sustainability of sanitation projects in South Africa. As stated in the funding proposal, the projected gap between water demand and supply is driven by several factors, including low tariffs and inadequate cost recovery.

16. All water reuse subprojects would be assessed for eligibility within the investment decision-making structures of the DBSA, by applying an integrated environmental, social, economic, financial and sector assessment. Projects would be screened according to the DBSA Environmental and Social Safeguards Standards Framework.

17. The proposal (annex 2, p.15) recognizes that South Africa experiences high water losses in most municipal water networks, which result in loss of revenue and, consequently, reduced financial resources for operation and maintenance. The proposed advanced treatment processes would involve higher operation and maintenance costs per m^3 than the current water supply systems, hence the long-term financial sustainability of water reuse projects would be highly dependent on achieving a high level of cost recovery. Information on the coverage of micro-metering in the targeted municipalities was not found in the funding proposal. When questioned by the independent TAP, the proponent submitted a description of the current coverage of micro-metering and of the state of non-revenue water (NRW) at the national and municipal level, and stated that "The Department of Water and Sanitation has been supporting municipalities through its War on Leaks programme and the country is planning to initiate a NRW programme as part of the National Water Programme," and that "The metering in most

South African cities, particularly, the large metropolitan centres tends to be in the 75th percentile and above, however, in the more rural and informal settings this is much lower." The independent TAP considers 75 per cent metering coverage to be a reasonable level of coverage that may prevent excessive demand.

18. Subprojects should be obliged to carry out an assessment of NRW and micro-metering coverage and, if considered necessary, the assessment should include actions to reduce NRW and increase the micro-metering coverage as a pre-requisite for project approval.

19. The programme seeks to create a new asset class (reuse water) by defining the investment value of water reuse infrastructure for a range of context-specific characteristics, and by developing an enabling institutional and financial environment that supports the marketability of this new asset.

1.2.4. Contribution to the regulatory framework and policies

20. The proposal recognizes that the implementation of water reuse projects would require the improvement of several regulatory instruments. Under activity 3.2.3, the programme seeks to establish and maintain strategic partnerships with key actors and institutions, including government departments, to help build longer-term institutional capacity and address various regulatory issues.

21. The programme's targets, as identified under the specific indicators (funding proposal, p.84) include: (i) at least 10 municipalities having climate responsive plans and regulatory frameworks; (ii) at least one adjustment to water resource management policies and one to relevant municipal policies by the mid-term; and (iii) amendment of water quality standards for reuse water.

1.2.5. Scalability and replicability

22. The project would help create a database of context-specific water reuse infrastructure interventions that would facilitate the replication of the implemented subprojects. Having proven new water treatment techniques in the subprojects would help to reduce investment risk, thus promoting replication.

23. The provision of programmatic support and planning through a water partnership office (WPO) would assist municipalities in developing large-scale water reuse projects.

1.3 Sustainable development potential *Scale: High*

1.3.1. Environmental co-benefits

24. Water reuse projects would reduce the flow of untreated wastewater into the environment, hence reducing water pollution.

25. Withdrawal from natural water sources would also be reduced, thereby increasing water availability to supply ecological water requirements.

26. Sludge management should include electricity cogeneration with biogas and, potentially, co-digestion of industrial waste in anaerobic digestors, to reduce the increase in the carbon footprint of these new facilities. The increase in GHG emissions is related not only to potential methane emissions from sludge management, but also nitrous oxide emissions from nitrification–denitrification processes and scope 2 emissions from energy demand (additional pumping for tertiary treatment and additional oxygen demand in biological nutrient removal systems). An adequately planned programme for biosolids (i.e. sludge of a quality that allows its beneficial use) should be in place to reduce the amount of sludge being disposed of in landfills,

and its related emissions. Plans for agricultural/urban use as fertilizer or for the improvement of degraded soils should be clearly described and be part of the programme.

27. There will also be mitigation co-benefits from GHG emission reductions and improved water security by mainstreaming ecosystem adaptation and restoration, and through improved ecosystem maintenance in upper catchments. Further environmental co-benefits may also be achieved through the incorporation of ecosystem-based approaches to sustain environmental flows and increase climate resilience.

28. On the other hand, water reuse projects that include reverse osmosis would face the problem of disposing of the saline brine waste stream generated in the process. Coastal sites typically dispose of brine into the sea. However, given that most municipalities do not permit the direct discharge of brine into sewerage lines, the disposal of brine in inland sites would be more complicated and costly. Inadequate treatment of the brine stream may result in negative environmental impacts. The present submission explains that inland plants might be more suited to water reuse processes that would not necessarily need reverse osmosis, such as indirect potable reuse, industrial reuse, and irrigation reuse. It also explains that, in instances where brine disposal is required at inland sites, this would be incorporated in the feasibility assessment of each project.

1.3.2. Social co-benefits

29. The main social co-benefits would be the increased availability of water for agricultural and residential uses, with the consequent improvements in the health of the population and increased food production.

30. However, if not adequately designed and managed, DPR projects could involve significant health risks to the population, mainly arising from the presence of contaminants of emerging concern, which may be difficult to control.

31. *Economic co-benefits* - The expansion of the market for treated wastewater, sludge beneficiation and biogas and energy generation would have a positive economic impact in the municipalities where reuse projects are implemented. In addition, a more reliable water supply system would improve industrial and commercial productivity.

1.3.3. Gender-sensitive development impact

32. The WRP would incorporate gender-related aspects, to allow for the systematic consideration of the differences between the conditions, situations and needs of women and men, and the integration of gender-equality concerns into the programme design. Specific actions within the programme include: (i) applying quotas for female and vulnerable groups to participate in water committees, boards, and agencies; (ii) ensuring a 30–40 per cent target for the beneficiaries of the programme to be women; (iii) development of gender monitoring indicators and appropriate gender budgets; and (iv) enabling all water stakeholders to build the requisite skills and knowledge for gender-sensitive services and management.

1.4 Needs of the recipient *Scale: Medium*

33. *Vulnerability of the country and vulnerable groups* - South African settlements are susceptible to the effects of climate variability. The proposal's climate rationale is mostly based on a literature review and on results from the general circulation model multi-model ensembles of the Climate Model Inter-comparison Project Phases 5 and 6 (CMIP5 and CMIP6, respectively). Additional information provided by the AE clarifies some of the ambiguity in prior editions of the funding proposal regarding climate trends in South Africa. However, the core concern of past independent TAP assessments was understanding whether the water scarcity is caused by

change in the climate system or other human-centric activities. The current proposal provides an enhanced process at the subproject level to make this assessment and ensure that limited GCF funds and concessionality are in fact being used to address climate impacts and not simply compensating for lack of planning, investment, revenues, and growth management that exist regardless of climate projections. Under the current proposal the risk of misapplication of GCF funds is not completely removed but the independent TAP considers that the current proposal provides a mechanism which, if well managed and overseen, can mitigate the above risk on a project-by-project basis. Outcomes 1 and 2 will address these needs.

34. *Economic and social development* - The unemployment level in South Africa is exceptionally high, at 28 per cent, but is even higher for youth – at a staggering 53 per cent. In South Africa, nearly 20 per cent of the population lives on less than USD 2.00/day, and almost 60 per cent of the entire country subsists on less than USD 5.50/day.

35. *Absence of alternative sources of financing* - The country's budget deficit is estimated at 11.2 per cent of its gross domestic product (GDP) and its debt-to-GDP ratio is 81 per cent, which severely constrains alternative sources of financing from commercial or private lenders. Activities relating to outcomes 1 and 2 will address these needs.

36. *Need for strengthening institutions and implementation capacity* - The proposal recognizes the limited technical and management capacity of the municipalities regarding the adaptive responses required to address future water stress, and to procure and structure contracts to support new business models. The establishment of the WRP as a centre of excellence, and of the WPO within the WRP, would provide the municipalities with technical support for project development, and assistance in structuring the necessary arrangements. It would also build longer-term capacity to oversee the implementation of projects. Outcome 3 will address these needs.

1.5 Country ownership *Scale: High*

37. *Alignment with national climate strategy and policies* - The implementation of the WRP is clearly aligned with the national climate strategy. The second edition of the National Water Resource Strategy outlines the need to support the diversification of the national water mix. The National Water and Sanitation Master Plan, which was developed in 2018 to define priority interventions to address the projected supply deficit, sets water reuse as a key priority for the country. The National Climate Change Adaptation Strategy, which is South Africa's national adaptation plan, urges water management institutions to adopt more adaptive responses and more water-wise practices.

38. *Capacity of accredited entities or executing entities to deliver* - The executing entity is the same as the AE, namely DBSA. The executing entity would manage the GCF funds and work closely with the NWP and the South African national Department of Water and Sanitation.

39. The national designated authority is the national Department of Forestry, Fisheries and Environment. The NWP will report to this department.

40. DBSA has a set of tools and procedures for classifying green operations and for assessing and monitoring environmental and social impacts, which are supported by internationally accredited environmental and social safeguards standards and are aligned with the legislation in all the jurisdictions in which it operates.

41. *Engagement with civil society organizations and other relevant stakeholders* - Annex 7 of the funding proposal presents a clear description of the stakeholders' engagement plan, which includes key engagements with the Project Steering Committee and other key stakeholders, as well as an action plan with a schedule of key engagements, and a list of the engagements held during the programme inception phase.

1.6 Efficiency and effectiveness *Scale: Medium*

1.6.1. Cost-effectiveness and efficiency

42. Previous versions of this funding proposal (annex 26) included an economic assessment of the 10 indicative water reuse projects. However, capital expenditure (CapEx) and operational expenditure (OpEx) were presented as overall values for the 10 projects and lacked a detailed description of the assumptions made to estimate the specific CapEx and OpEx values for each project. For this reason, it was not possible to evaluate whether the proposed financial structure was adequate to achieve the proposal's objectives, so the independent TAP requested the AE to clarify the process by which each subproject investment would be evaluated on a climate and financial basis. The current funding proposal provides greater clarity that the DBSA will undertake the following assessments:

(a) A detailed climate risk and vulnerability assessment that will indicate the various climate scenarios and the resultant impacts on water insecurity and the need to address GHG emissions;

(b) A project feasibility study that will outline in detail the most appropriate technical options (direct, indirect, irrigation and/or industrial water reuse) to address the specific water insecurity risks, while also providing mitigation solutions that include GHG emission reductions and sludge beneficiation. At this stage, the feasibility study would include the financial implications for the project options, linked to the climate scenarios; and

(c) A detailed financial bankability analysis that would then link these options, under a business-as-usual scenario and under the various climate scenarios (i.e., RCP 4.5 and 8.5,[4] with and without adaptation measures), to the blended finance solution, which would assess the optimal structuring of the investment against aspects of affordability, while enhancing project sustainability and bankability.

1.6.2. Amount of co-financing

43. The proposal requests USD 235 million of GCF funding, with USD 35 million being in the form of grants and USD 200 million in subordinated loans. The total co-financing from private partners and municipalities amounts to USD 1,237 million, resulting in a co-financing ratio (total/GCF) of 6.3.

44. As outlined in the GCF funding agreement and term sheet, the ratio of GCF grant to co-financing would be required to be at least 3:1 for five subprojects, and then the remaining projects would be co-financed to achieve the targeted programme level ratio of 1:1, for component 1 and component 3. The proponent commented in the term sheet that "The reasoning behind this is to use a higher portion of GCF funding initially to prepare projects, which will assist in demonstrating to other preparation funding providers that the WRP is working and for them to make project preparation funding available for subsequent stages/projects, utilizing a lower portion of GCF grant funding."

45. Asked about the risk of GCF being a sole or overrepresented grant funder of subprojects, the proponent responded that there is no such risk because of the conditions placed in the term sheet that prohibits this, and stated that:

> "… the Department of Water and Sanitation through the South African National Treasury has committed funds and its support to the programme. As such, the National Treasury (or other project preparation providers) will provide the committed funds and will work

[4] Representative concentration pathways RCP 4.5 and RCP 8.5 in the Fifth Assessment Report of the Intergovernmental Panel on Climate Change (AR5).

with the Department of Water and Sanitation, the Department of Cooperative Government to ensure that the grants at Municipal level will be provided, noting that each Municipality does receive funding through the Municipal Infrastructure Grant."

46. The WPO intends to seek other providers of grants and project preparation funding. Fund recovery mechanisms would be considered on a case-by-case basis.

47. The independent TAP also notes that the funding proposal provides for a currency management solution that would be a cost and operational responsibility of DBSA as opposed to GCF. The independent TAP views this positively and regards it as an appropriate approach for scenarios where GCF's partners are sufficiently sophisticated (such as DBSA) to take on the associated costs and risk associated with currency management in their own national currency. The proposed currency management solution consists of DBSA entering a currency swap on a subproject-by-subproject basis with a DBSA-approved hedge counterparty and hedging the GCF United States dollar exposure (the United States dollar funding commitment to the sub-borrower) prior to the first disbursement to the sub-borrower. The proponent further described the sub-loans structure and the challenges that might arise while implementing the proposed currency management solution.

1.6.3. Financial viability

48. The economic assessment presented gives a financial internal rate of return (IRR) of 26 per cent, an economic IRR of 28 per cent, and a net present value of approximately USD 780 million, considering a discount rate of 8.35 per cent, and an asset life expectancy of 25 years.

49. The funding proposal (p.32) states that project-specific feasibility studies would be required to demonstrate the financial viability of the project and clear off-take arrangements.

50. Component 1.2 on project pipeline preparation includes the provision of technical assistance for project structuring and preparation, including the development of strategic cases and full detailed business cases, thereby supporting the creation of bankable water reuse projects.

51. WRP would also be structured in a way that no one project receives more than 30 per cent concessional loans as a proportion of the total implementation capital; however, in exceptional cases, up to 50 per cent would be reviewed where a particular project requires it to ensure the ongoing viability of the entire WRP.

52. In the event of a GCF increase beyond the 30 per cent threshold, the DBSA/development finance institution (DFI) portion would not necessarily remain the same and would need to be fully assessed on a case-by-case basis to determine the optimal solution.

53. Based on the information provided, the independent TAP noted that the funding proposal envisages subordinated debt of approximately USD 150 million from DBSA and DFI partners, and GCF concessional loans, subordinated to the DFI tranche, of USD 200 million. Asked about the rationale for subordinating GCF funds to the DFI tranche, the proponent responded that this is not correct and that it has been agreed through the term sheet that these will rank *pari passu*. Regarding the cost (interest rate) of the DFI debt, the proponent explained that it would only be known once a project has been prepared and assessed from a financial, due diligence and risk perspective and once a credit risk and pricing has been performed on the individual project.

1.6.4. Blended finance

54. Blended finance should support the initial stages to prove the viability and concept, but it should evolve, in the standardization and aggregation phase, to allow commercial options. In response to comments made by the independent TAP on the lack of indications of the financial

status of the selected municipalities in which the proponents intend to develop the water investments, and their financial viability to either take debt or equity, the proponent explained that the projects included in the funding proposal are indicative and, given that these projects have not yet been negotiated, providing information on the financial status of this list of municipalities would be inappropriate. Each project feasibility and bankability assessment would include an assessment of the financial status of each municipality.

55. The independent TAP also noted that the information in terms of a market analysis of possible private sector investors willing to invest in the proposed blended structure is not robust.

56. The funding proposal provides that an independent blended capital facilitator (IBCF) will be contracted to provide the WPO with professional blended capital and financing services in respect of each project.[5] The IBCF would be required to utilize DBSA's supply chain management policies and procedures to ensure that all processes are independent and managed in terms of South Africa's Public Finance Management Act (Act No. 1 of 1999). Ultimately, the DBSA Board and Investment Committee would make the final decisions regarding funding of projects by DBSA and GCF.

57. On the matter of GCF's interest in maximizing the impact of its funds by ensuring least concessionality, the proponent further explained that the WPO would oversee, review, and assess recommendations made by the IBCF, and the IBCF would use the programme financial model to ensure that the GCF concessional capital is being utilized efficiently across the programme. The IBCF would make recommendations regarding the capital structure and once these are approved, the IBCF would seek this capital from pre-approved lenders that meet the GCF investment criteria.

58. The funding proposal mentions that the parties may agree to reduce the GCF loan amount five years after the effectiveness of the funded activity agreement based on the results of the mid-term independent evaluation report and on the performance of the programme. In response to the independent TAP's comments, the proponent explained that although GCF grant funds are to be largely used for setting up the WPO, providing support to initial project preparation and to support capacity-building interventions, and that a significant portion of these funds would be spent during the first five years of the programme, the phased and progressive nature of the programme implementation process might result in savings in the use of the grants.

59. In determining the financial feasibility of a subproject, the level of GCF funding is estimated to be in the order of 15 to 20 per cent.

II. Overall remarks from the independent Technical Advisory Panel

60. Wastewater reuse programmes, when adequately designed and operated, represent a strong adaptation measure in areas where water stress is driven by climate change.

[5] Asked by the iTAP to elaborate on the process used for identifying the IBCF and the selection requirements that would be used to ensure competency and independence, the proponent responded that the IBCF would be appointed and contracted using the South African legislative and regulatory requirements pertaining to supply chain management, including the Preferential Policy Framework Act (Act No. 5 of 2000) and its associated Preferential Procurement Regulations of 2017 and the Broad-Based Black Economic Empowerment Act (Act No. 53 of 2003 as amended), and that this would take place in accordance with the DBSA Supply Chain Management Policy (March 2021). The WPO would prepare terms of reference that would stipulate the skills, competency and experience requirements, the roles and responsibilities, and the process and protocols for the IBCF functioning and reporting. The terms of reference will be reviewed for approval by the Oversight Committee of the WPO, as well as the Supply Chain Management Unit of DBSA. Proposals for providing these services would be adjudicated by the DBSA's Supply Chain Management Committee. The IBCF would be required to report monthly to the WPO's Oversight Committee.

61. In the assessment of earlier submissions, the independent TAP concluded that the technical information in the proposal was mostly based on a literature review and results from climate models, and the proposal failed to present historical temperature and precipitation data from weather stations and trend analyses to prove that the proposed projects were driven by climate change. This issue was particularly acute because it had an impact on subproject selection and decision-making. The indicative list of subprojects and the minimal analysis provided for each project was concerning.

62. The independent TAP suggested fortifying the overall climate impact information of the proposal and applying a reliable methodology to determine whether candidate projects are driven by projected water shortages that result, at least in significant part, from climate change. The project proponents have addressed this concern by providing for and elaborating a project screening and investment decision process with better governance and a clearer method of screening for climate impacts at the municipal and regional level. The funding proposal now contains a detailed screening scorecard, and a governance process has been developed to ensure that only projects that are underpinned by a clear climate change basis are provided with support by the WRP.

63. Similarly, the previous independent TAP assessments noted that the blended financial model was not well explained in the funding proposal and that this model was crucial for ensuring financial returns to the investors willing to take risks at the initial stages but expecting to see returns in the medium term. The subsequent AE submission provided greater clarity around the subproject selection process, which includes greater analysis of the financial returns and a short-, medium- and long-term analysis of the tariffs in each of the municipalities, and the impact on the expected returns in the project life cycle.

64. In the current funding proposal, a programme financial model was provided, which seeks to demonstrate the financial efficacy of the programme. The model looks at an indicative set of subprojects to determine the total projected funding requirement for set-up costs, project preparation costs including transaction advisory costs and total project-related capital costs of the WRP. The individual project-level cashflows (expenditures and revenues) were assessed over the life of the projects to determine the financial net present value of each project and the financial IRR on each project investment.

65. The current proposal provides the independent TAP with some comfort that DBSA intends to conduct robust screening and analysis and has a reasonable plan and process for doing so. However, the independent TAP notes that the plan and process are yet to be tested; therefore the independent TAP sees this proposal as another example of a programme that could benefit from ongoing GCF engagement and learning.

66. The WRP has great potential for GCF funding if managed responsibly to ensure that GCF funding is being applied to finance interventions that result in effective adaptation to climate change. Based on the adjustments and improvements to this proposal, the independent TAP endorses this funding proposal.

Response from the accredited entity to the independent Technical Advisory Panel's assessment (FP209)

Proposal name:	Climate Change Resilience through South Africa's Water Reuse Programme ("WRP")
Accredited entity:	Development Bank of Southern Africa
Country/(ies):	South Africa
Project/Programme size:	Large

Impact potential
AE does not have any additional comments.
Paradigm shift potential
AE does not have any additional comments.
Sustainable development potential
AE does not have any additional comments.
Needs of the recipient
AE does not have any additional comments.
Country ownership
AE does not have any additional comments.
Efficiency and effectiveness
AE does not have any additional comments.

Overall remarks from the independent Technical Advisory Panel: AE does not have any additional comments.

Gender documentation for FP209

National Water Reuse Programme:

Programme Design and Preparation of a Full Funding Proposal to the Green Climate Fund (GCF)

Annexure 8a: Gender Assessment and Action Plan

05 April 2023

This Deliverable has been prepared by Pegasys (Pty) Ltd (referred to as the ***'Service Provider'*** in the Service Level Agreement executed between Pegasys and the DBSA on 10 January 2021 and hereafter referred to as '***the Consultant***'), in association with:

- JG Afrika (Pty) Ltd;
- Amber Public Sector Consulting (Pty) Ltd;
- Clarity Global Strategic Communications; and
- Yubifin (Pty) Ltd.

JG Afrika (Pty) Ltd, Amber Public Sector Consulting (Pty) Ltd, Clarity Global Strategic Communication and Yubifin (Pty) Ltd may be referred to individually as 'the subconsultant**' or collectively as '**the subconsultants**'.*

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LIST OF ABBREVIATIONS

BBBEE	**Broad-Based Black Economic Empowerment**
CBD	Convention on Biological Diversity
CEDAW	Convention on the Elimination of all forms of Discrimination Against Women
CGE	Commission of Gender Equality
DBSA	Development Bank of South Africa
DCOG	Department of Cooperative Governance
DFFE	Department of Forestry, Fisheries and the Environment
DFID	UK Department for International Development
DWS	Department of Water and Sanitation
ESG	Environment, Social and Governance
GBV	Gender Based Violence
GBVCC	Gender Based Violence Command Centre
GCF	Green Climate Fund
GESI	Gender Equality and Social Inclusion
HDI	Human Development Index
IDP	Integrated Development Plan
KPI	Key Performance Indicator
LGBTIQ+	Lesbian, Gay, Bisexual, Trans, Intersex and Queer
MISA	Municipal Infrastructure Support Agent
NDP	National Development Plan
PDO	Predetermined Objective
PGDS	Provincial Growth and Development Strategy
POA	Programme of Action
SA	South Africa
SADC	Southern African Development Community
SAPS	South African Police Service
SIF	Strategic Implementation Framework on Gender And Development

SMMEs	Small Business Key to Job Creation Small, Medium and Micro Enterprises
SSA	Sub-Saharan Africa
STEM	Science, Technology, Engineering, And Mathematics
ToR	Terms of Reference
UIS	UNESCO Institute for Statistics
UNCCD	United Nations Convention to Combat Desertification
UNFCCC	United Nations Framework Convention on Climate Change
UWSEIMP	Urban Water Supply and Environmental Improvement Project
WASH	WASH Water, Sanitation and Hygiene
WPO	Water Partnership Office
WRP	National Water Reuse Programme
WWF	World Wide Fund

PROGRAMME SUMMARY

Programme	National Water Reuse Programme
Country	South Africa
Sectors	Water; Waste
Project Type	Green Climate Funding
Submission Date	08 December 2022
Version	Version 4
Compiled by	Traci Reddy (Pegasys)

1. INTRODUCTION

1.1 PROJECT CONTEXT

South Africa is a water-stressed country and consumptive water use is driving pressures on the national water resource system, with a potential 17% water deficit forecast by 2030. Several interventions have been initiated by national government already to avoid this projected water deficit with a key element of these interventions being to develop an enhanced level of diversification to the "mix" of water supply sources. **The South African National Water and Sanitation Master Plan (2019) makes a specific note of the need to reduce water demand and increase water supply through the "re-use of effluent from wastewater treatment plants, water reclamation, as well as desalination and treated acid mine drainage".**

At present, most effluent discharge and urban run-off are not reused and in light of The South African National Water and Sanitation Master Plan note, the opportunity to initiate a framework for the scaled development of water reuse infrastructure is evident. To this end, the Development Bank of Southern Africa (DBSA) has partnered with various government departments (including the Department of Water and Sanitation (DWS), the Department of Cooperative Governance (DCOG) through its agency the Municipal Infrastructure Support Agent (MISA), and the National Treasury for the development of a National Water Reuse Programme (WRP). In addition, as an Accredited Entity of the GCF, the DBSA also submitted a proposal to the GCF to support the design and implementation of the WRP in South Africa. Noting the importance of water reuse to diversifying the 'water mix' in South Africa, and the challenges and barriers to entry that exist in the development of these water reuse projects at scale, the development of a focussed programme to address these challenges and ultimately implement pathfinder projects is critical to contributing towards building a more resilient water future. **The following document is an annexure to the proposal and provides an outline of the status of gender equality in South Africa by exploring various key themes relevant to the programme's context.**

1.2 METHODOLOGY

<u>Document Review</u>

In close consultation with the client, the consulting team undertook a review of the following documents:

- Internal Bank Policies and Strategies (e.g., Gender Strategy, Human Capital and Investment Policies, Syndicate Group Investing Pillar Report, Environmental and Social Safeguard Standard 3, DBSA Gender Strategy Framework, DBSA Recruitment Policy and Process, DBSA Corporate Plan 2021/22 – 2023/24, DBSA draft Corporate Plan 2021/22 – 2024/25, DBSA New Employment Equity Plan 2021 – 2024, Gender Equality Subcommittee Plan 2021 - 2024 amongst others).
- reports and Board Presentations on DBSA's Gender Mainstreaming Programme;

- relevant National gender legislative requirements and protocols, such as the National Development Programme, SDGs, SADC's Protocol on Gender and Development, the Green Climate Fund requirements, and the Global Environmental Facility's gender requirements, UN Compact; and
- recent literature on approaches, toolkits and guidelines for carrying out Gender Analyses, Gender Assessments and Gender Action Plans of relevance.

The findings from this review have been collated into a succinct summary report, focused on capturing the relevance and key takeaways (per category of literature) to inform the development of the Gender Action Plan.

Stakeholder Engagement

The process of stakeholder engagements plays a vital role in any project and often is the most important tool available to obtain inputs and support from stakeholders while also promoting transparency and effective communication. The Gender Action Plan as well as the development of the Full Funding Proposal will benefit from the insights and inputs of different stakeholders, as such, there is a need to ensure that these engagement processes are rooted in core principles:

1. **Identify and understand:** Successful stakeholder engagements relies on identifying and prioritising all stakeholders to be engaged with as well as understanding their interests and concerns. This also means understanding which stakeholders are to be informed, consulted with, contribute towards a particular outcome, or collaborated with.
2. **Engage as early as possible:** Once stakeholders have been identified and prioritised, it is necessary to engage as early as possible so that stakeholders are informed and involved throughout the project.
3. **Communication and transparency:** Effective and ongoing communication underlies all engagements and sharing of information at engagements helps to contribute towards transparency and buy-in.
4. **Inclusivity:** Stakeholders vary across sectors and geographies and the engagement process must ensure inclusivity, so no group is excluded from the process.
5. **Follow up and report back:** Stakeholder engagements are ongoing and entails not just initiating engagements, but also following up with stakeholder on issues as well as reporting back to provide updates and outcomes of previous action items.

The COVID-19 pandemic has resulted in the need to develop adaptive approaches to stakeholder engagements that takes into consideration the absence of in-person interactions and the advantages thereof. This means exploring virtual platforms that allow for sufficient engagement while also ensuring a greater reach to stakeholder that spans multiple geographies. During consultations, South Africa was and still is dealing with the COVID-19 pandemic, and therefore stakeholder engagement approaches were cognisant of this and how to adapt to travel- and social gathering-related restrictions. Furthermore, all stakeholder consultations are according to current regulations with regards to social distancing, mask wearing etc.

The outcomes of the document review and the stakeholder engagement are presented in this report.

1.3 BACKGROUND TO THE ASSESSMENT

1.3.1 Wash Themes

Water insecurity is rapidly growing in urban areas, and heavy pressure on neighbouring water resources exacerbate access to adequate water, sanitation and hygiene (WASH) infrastructure and amplifies the complex relationship between gender, development and water demand. WASH has significant implications for gender equality. This is because women and girls tend to be disproportionally affected by a lack of access to adequate WASH infrastructure, in large part due to biological and social factors connected to the productive and reproductive roles that they play.

Meeting the WASH needs of women and girls requires recognising the gendered barriers that prevent access to WASH services and infrastructure as well as addressing women's and girls' specific WASH needs. These needs include reproductive health needs like menstrual hygiene and maternal and new-born health as well as those connected to their role as mothers, caregivers and income generators. At the same time, it is important to recognize that integration of gender considerations has impacts beyond good water and sanitation performance, including economic benefits, empowerment of women and benefits to children and other vulnerable populations. This calls for an expanded approach to gender mainstreaming that considers a gender and social inclusion (GESI) perspective to identify vulnerable populations as part of the design and implementation of WASH infrastructure projects, and more specifically in water reuse initiatives and programmes. Gender mainstreaming also requires finding ways to involve men to solicit their views on gender issues while engaging them as partners who can ensure that women can be equal beneficiaries and participants in all phases of project development and implementation.

1.3.2 CLIMATE CHANGE

Women, especially poor women, are one of the groups most vulnerable to the effects of climate change, and least likely to have the resources to cope with them. Climate change worsens existing vulnerabilities and adds to the pressures on the environment and natural resources. More so, gender differential impacts of climate change are attributed to existing inequalities between men and women in South Africa such as unequal access to resources, gendered division of labour and decision-making power which affects the ability to respond to the effects of climate change. A core example is found in the rural areas where women, usually the head of the household, are not readily mobile to move households. When climate impacts occur in prone areas, women are less likely to leave their household as their responsibility is to care for children and elderly while needing to rely on limited/reduced natural resources to sustain their families or earn an income. Because women are one of the most vulnerable groups, it is essential that they are equally represented and equal participants and agents of change in climate change solution development – and that this is reflected in South Africa's legislation and policy framework. This theme is essentially explored throughout the assessment.

1.3.3 Purpose of WRP and the Gender Assessment

Programmes, such as WRP, can improve efforts to help those vulnerable to climate change to anticipate, absorb and adapt to droughts, floods and other climate-induced shocks. Coordinated action is needed across sectors to drive progress on gender equality and female empowerment, as well as sustainable reductions in poverty, vulnerability and water insecurity. This requires thoughtful collaboration between infrastructure service delivery and public works programmes, water and gender agencies, alongside those responsible for climate change, disaster risk management and urban development.

This programme intends to be the nexus between addressing water insecurity exacerbated by the impacts of climate change and strengthening South Africa's adaptative capacity while considering the array of gender dimensions that will enhance, divert, impact and support the success of the programme.

This gender assessment will examine the roles and different needs of women and men in the context of water reuse and serve as a practical tool for identifying opportunities for promoting gender equality in the programme. The gender action plan will build on this assessment and include gender-responsive activities, indicators, timelines and responsibilities.

1.4 GENDER AND THE WATER REUSE PROGRAMME

The **gender assessment** is intricately weaved into the SA Water Reuse Programme. The Programme aims to optimise the water mix, which is currently strongly dominated by surface water, with some groundwater and return flows, to a water mix that includes increased groundwater use, re-use of effluent from wastewater treatment plants, water reclamation as well as desalination and treated acid mine drainage. This has been outlined as a key strategic intervention for the country as part of the National Water and Sanitation Masterplan (DWS, 2018). As shown in Figure 1-1 The gap between water supply and project water demand under two scenarios by 2032 in South Africa (Green cape 2018), if planned additional water supply is considered, water efficiency is achievable as the gap between supply and demand will narrow, optimally by 2035.

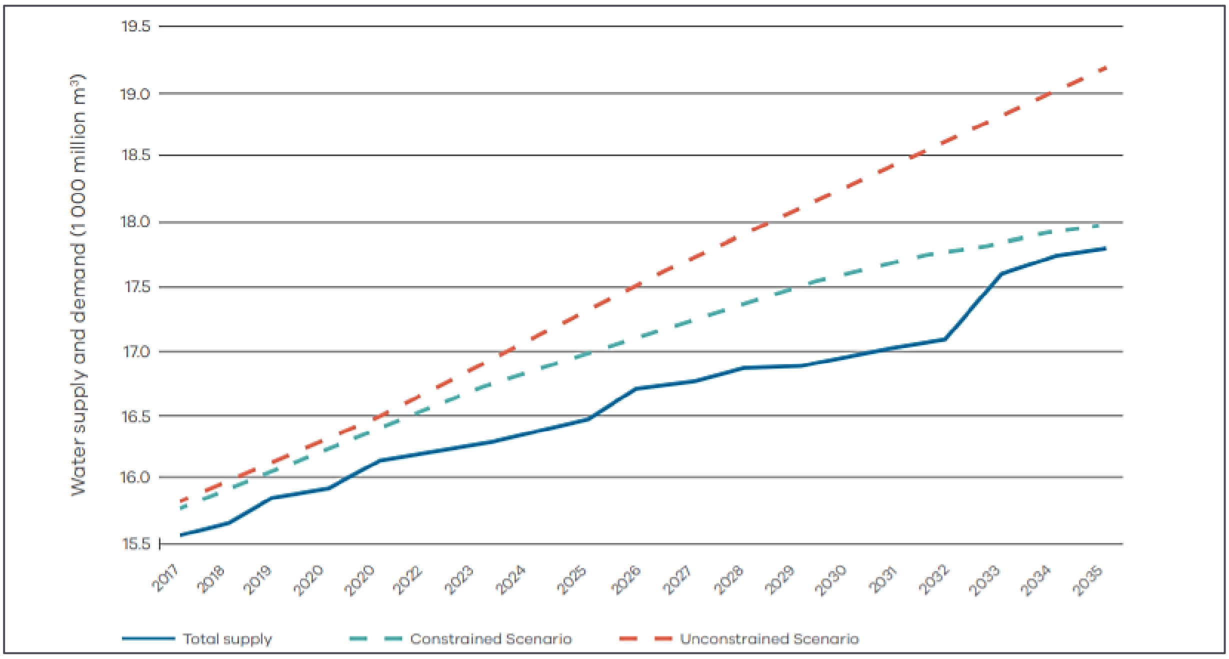

Figure 1-1 The gap between water supply and project water demand under two scenarios by 2032 in South Africa (Green cape 2018)

Applying an innovative finance lens, the programme will assist to design a pathway to mobilise resources to scale up water reuse as a key element of South Africa's approach to climate adaptation. More specifically, the programme will apply a blended finance approach, to increase climate-related water reuse investments in South Africa while establishing water reuse as a new asset class in South Africa. The team will also consider a programmatic approach to water reuse in South Africa to help cities and vulnerable communities in cities and rural areas adapt to climate change and more sustainably utilise scarce water resources.

Addressing the current water security issue will not only consider the financial component but will also require a **gendered lens** to address the broader challenges. This includes acknowledging the social inequalities prevalent in accessing water due to historical legacies of discrimination against gender. Additional issues relevant to the programme that will be explored include, gender-based violence, menstrual health, decisions making and representation, equipment design, safety, labour provisions, working conditions, etc.

The programme will thus endeavour to explore a multitude of complex gendered issues and offer activities and outcomes that support sustainable and equitable objectives. It will actively engage diverse perspectives and experiences to ensure that the product, services, and systems are designed to address gender differences and barriers. **Understanding gender differences along the waste and water value chain will influence decisions, and responsibilities, and will be critical to ensuring access to improved water supply services for all. To identify and design for these gender differences, the programme will be designed in a manner that will be intentional about bringing about positive gender-related impact.**

1.5 PRINCIPLES AND DEFINITIONS

The following guiding principles will assist to create a framework that influences the final activities, outputs and anticipated outcomes for the gender action plan:

- Relying on international and national good practices to gender equality and human rights to guide the process to mainstream gender and promote gender equality and empowerment into the programme.
- Ensuring that the allocated financed activities addressed do not exacerbate existing gender-based inequalities.
- Applying a gender-responsive approach throughout the identification, design, implementation, monitoring, and evaluation.
- Addressing the gender gaps and supporting the empowerment of women to help achieve global environmental benefits.

For ease of reference, the following definitions are relevant for the programme:

1. **GENDER ASSESSMENT**: is a systematic approach to understand differences between the development needs and priorities of men and women and the variable impact of development programs on them. It uses data to understand men's and women's different roles, responsibilities, decision-making power, incentives, and access to productive resources and basic services. Gender assessment includes contextual analysis of the socioeconomic, legal, and political environment as they affect gender-based roles and constraints in society (Asian Development Bank, 2021).

2. **GENDER ACTION PLAN (GAP):** is a tool to strategically plan and implement specific activities to enhance gender equity among a project's target groups. A GAP includes a summary of gender activities, action steps, timeframe, indicators and measures to be carried out by project implementing agencies (Asian Development Bank, 2021).

3. **GENDER EQUITY**: refers to the fair distribution of benefits and responsibilities between women and men according to their respective needs. This may involve equal treatment or treatment that is different but considered equivalent in terms of rights, benefits, and opportunities. In the development context, a gender equity goal may introduce special measures to compensate the disadvantaged women and men, to end inequality and foster autonomy (Asian Development Bank, 2021).

4. **SAFE AND AFFORDABLE DRINKING WATER**: is used for drinking, cooking, food preparation and personal hygiene. It is always free from pathogens and elevated levels of toxic chemicals and is suitable for use by women, men, girls, and boys including people living with disabilities.

5. **ADEQUATE AND EQUITABLE SANITATION:** is the provision of facilities and services for safe management and disposal of human urine and excreta (faeces). Facilities are close to home so that they can be

easily reached and used when needed. They are suitable for use by men, women, girls and boys of all ages including people living with disabilities.

6. **HYGIENE:** refers to behaviours that encourage the widespread adoption of safe hygiene practices to keep people and their environment clean, enhance dignity, prevent the spread of diseases, reduce under-nutrition and maintain health. Key hygiene behaviours include: i) handwashing with soap at critical moments, ii) safe and hygienic management of human excreta and cleanliness of sanitation facilities, iii) safe domestic water management from source to the point of consumption, iv) food hygiene, and v) menstrual hygiene and management.

1.6 DBSA'S APPROACH TO GENDER MAINSTREAMING

The Development Bank of South Africa (DBSA) has an integral link with the South African government, through its 100% ownership of the bank. DBSA is one of the country's primary vehicles for promoting infrastructure development in both the public and private sector and therefore strongly aligns to the country's national objectives on race, gender and equity redress.

The DBSA has long-recognised the critical role that women play in driving inclusive economic growth and development in the region. In 2018, gender mainstreaming was incorporated as part of 11 DBSA Environmental and Social Safeguard Standards.

DBSA launched their flagship Gender Mainstreaming (GM) Programme in 2019 that seeks to source finance and address barriers for large- and small-scale infrastructure in the energy, Information and Communications Technology (ICT), transport and water and sanitation sectors by promoting investment in women-owned or women-led projects. The four strategic pillars in the Gender Mainstreaming Programme that were developed with and approved by the DBSA Board are:

- Promoting investments in women-owned projects.
- Adapting strategies, policies, and procedures to enable gender mainstreaming across the Bank.
- Providing capacity building and knowledge sharing.
- Building partnerships with public and private partners who share our vision for gender equality.

As part of the GM Programme, the Bank's investment pillar has adopted a gender framework which embeds gender considerations in every stage of the project cycle and the monitoring and evaluation teams have implemented the DRT (development results tool) to track each project's targets. The Bank has also developed KPIs for women-led deals which have been incorporated into the balance scorecard.

Building on the on-going success of GM programme and based on findings from a Frankfurt School study on strengthening gender equality in the development banking sector, the DBSA is now looking to develop and implement a customised Gender Mainstreaming (GM) Policy and Framework for both their operational practices and (internal and external) deal-making activities. This is part of the ongoing refinement & fast-tracking implementation of GM processes and projects at the Bank.

Entrenching GM into DBSA'S day to day financing process, has been implemented and the Bank has shown a commitment to gender-sensitive investing and to gender equality in the infrastructure financing sector by the inclusion of the Gender Marker System as an Addendum to the DBSA's Environmental and Social Safeguard Standards. The Safeguard provides a framework for project sponsors or clients to ensure that their proposals align with the DBSA's expectations as they relate to gender mainstreaming. In the establishment of the Water Partners Office, the Water Reuse Programme will lean on the existing practices, instruments, and tools that the DBSA already has in place for gender mainstreaming throughout the entire project cycle.The Gender Marker System is used to categorise projects based on the extent to which they address GM issues. These four categories are:

1. **Category 1**: the objective of the project and its outcomes directly address gender mainstreaming.
2. **Category 2:** gender mainstreaming is one of the outcomes of the projects but it is not the principal one.
3. **Category 3:** one or more of the outputs of the project specifically focus on gender mainstreaming.
4. **Category 4**: marginal gender elements indicates that one or more of the activities specifically benefit both men and women.

Each of the categories then have a set of requirements, for instance, a gender action plan, gender assessment and analysis or both. The Water Reuse Programme will fall into category 3 as "*one or more of the outputs of the project specifically focusses on Gender Mainstreaming*" and the outcomes will not be the main key features of the programme. As such, the following gender assessment and action plan provides greater detail on how gender mainstreaming will be incorporated into the programme and the outputs that will specifically address gender mainstreaming during the design and execution of the programme. The DBSA has conducted training sessions for its frontline staff to cascade its GM definition, strategy, positioning and principles throughout the Bank. The GM Infrastructure Finance Training Pack introduces GM at the project level and introduces the Gender Marker System and its implementation in the DBSA appraisal process. The aim of the training sessions is to enhance the capacity of DBSA staff in GM knowledge and skills.

The DBSA has a gender forum that meets regularly, and a gender lead that participates in project selection and procurement. The Gender Policy being developed will help strengthen the way the DBSA approaches these activities, both internally within the institution and externally, in the clients/service providers it engages with. The DBSA implements a dynamic Development Results Reporting Framework (DRRF) to support effective development impact reporting on the DBSA Portfolio. The DRRF enables reporting on relevant impact metrics by project category and is premised on a Development Results Template (DRT) through which project objectives, indicators, metrics, and monitoring requirements are identified at the project appraisal stage. This forms the basis for reporting on the anticipated development results and impacts of all projects financed, including all projects selected for financing. The gender indicators that are reported on the DRT are shown in **Error! Reference source not found.**.

Table 1-1 Gender indicators on the DRT

Gender Indicators on the DRT

1.1 Total No. of direct construction jobs created
1.1.1 No. of women employed
1.2 Total No. of direct operational phase jobs created/sustained
1.2.1 No. of women employed
2. Broad Based Black Economic Empowerment (BBBEE)
2.1 Percentage of black women shareholder in the transaction
2.2 Client's BBBEE spend on empowering suppliers that are women-owned (%)/R '000
ENVIRONMENTAL SOCIAL AND GOVERNANCE:
Are the following effective environmental and social governance mechanisms in place?
Gender Action Plan?
Gender Marker Category (select from the list)
NQF credit-bearing training opportunities for women (No.)
Non-credit-bearing training opportunities for women (No.)
Improved gender diversity - % of women beneficiaries (direct)
Improved gender diversity - % of women at senior management level
ENVIRONMENTAL:
Number of females reached by [or total geographic coverage of] climate related early warning systems and other risk reduction measures established/strengthened
Number of females made aware of climate threats and related appropriate responses

The Bank's core mandate is financing large scale infrastructure projects. However, its experience in engaging with female-led enterprises and projects in its focus enterprise sectors (energy, ICT, transport and water and sanitation) is that women-owned or women-led businesses tend to operate smaller projects with lower financing needs. To achieve its objective of financing such smaller projects, the Bank's gender workstream operations team has developed new definitions and criteria for the Bank's lending and technical assistance processes directed at them. This has included redefining its KPIs from size of project to number of projects.

2. LEGAL AND REGULATORY REVIEW

The various instruments, laws, policies, and regulations currently in effect in South Africa create a legal framework for the provision of water and waste services as well as certain infrastructure consideration at several specific locales. The key documents are noted below and offer a guide on the current regulatory enabling environment that the project will be implemented in.

2.1 INTERNATIONAL AND REGIONAL LEGISLATION

2.1.1 Convention on The Elimination of All Forms of Discrimination Against Women (CEDAW) And African Charter on The Rights of Women (Maputo Protocol)

The right to 'adequate water' is specifically provided for in the International Convention on the Elimination of All forms of Discrimination against Women (CEDAW) and Protocol to the African Charter on the Rights of Women (Maputo Protocol).

CEDAW recognises the right of women in rural areas to enjoy adequate living conditions, particularly in relation to sanitation and water supply. CEDAW in the context of climate change and disasters highlights that States parties have both general obligations to ensure gender equality as well as specific obligations to guarantee rights that may be negatively affected by climate change and natural disasters. Article 15 of the Maputo Protocol requires states to ensure that women have the right to nutritious and adequate food and access to clean drinking water. These rights are reinforced in African Union's Agenda 2063 (Aspiration 6) and in the Sustainable Development Goals 5 and 8.

2.1.2 GREEN CLIMATE FUND

The GCF commits to (1) gender equality and equity; and (2) inclusiveness in all activities. Gender mainstreaming is central to the GCF's objectives and guiding principles, including through engaging women and men of all ages as stakeholders in the design, development and implementation of strategies and activities to be financed. The GCF Governing Instrument states that: "*The Fund will strive to maximize the impact of its funding for adaptation and mitigation… promoting environmental, social, economic and development co-benefits and taking a gender-sensitive approach*."

2.1.3 UNITED NATIONS (UN) SUSTAINABLE DEVELOPMENT GOALS

Goal 5 of the Sustainable Development Goals ('SDG's' or the 'Goals') encourages an end to all forms of discrimination against all women and girls everywhere. The Goals aims to eliminate all forms of violence against all women and girls, and harmful practices. It recognizes the value of unpaid care and domestic work through the provision of public services, infrastructure and social protection policies and the promotion of shared responsibility within the household and the family. Further, it ensures the universal access to sexual and reproductive health and reproductive rights. It encourages countries to undertake reforms to give women equal rights to economic resources, as well as access to natural resources.

SDG 6 is a core feature for the goals as shown below in Figure 2-1 Sustainable Development Goals At a programme level, SDG Target 6.3 states that *"by 2030, improve water quality by reducing pollution, eliminating dumping and minimizing release of hazardous chemicals and materials, halving the proportion of untreated wastewater and substantially increasing recycling and safe reuse globally*".

Figure 2-1 Sustainable Development Goals

Finally, the Goals encourage the adoption and strengthening of sound policies and enforceable legislation for the promotion of gender equality and the empowerment of all women and girls at all levels.

2.1.4 SADC FRAMEWORKS

The Southern African Development Community (SADC) has adopted a two-pronged approach to achieving gender equality (1) it has created equal opportunities for women and men (2) it has promoted a specific focus on women's empowerment. It has also demonstrated a commitment to integrating gender into its overall work. Its commitment is outlined in the Revised Regional Indicative Strategic Development Plan, reflected in the Strategic Implementation Framework on Gender and Development (SIF), backed by the SADC Gender Policy and demonstrated by the adoption of a SADC Protocol on Gender and Development.

2.2 NATIONAL LEGISLATION

2.2.1 THE CONSTITUTION OF SOUTH AFRICA (1994)

Section 27 of the Constitution of South Africa requires the State to take reasonable and other measures, within it available resources to achieve the progressive realisation of the right to water. The State is not obligated to provide water freely, but it is under an obligation to create mechanisms that enable people to have access to sufficient water. In the event of resource constraints, which limit the ability of the State to fulfil its obligations, the State is still obliged to provide a plan of action that demonstrates that the full realisation of the right shall be achieved over time (South African Human Rights Commission, n.d.). Similarly, the right to sanitation is encompassed in Section 24 of the Constitution whereby it guarantees that everyone has the right to an environment that is not harmful to their wellbeing as well as a right to have the environment protected. In terms of Section 9, the equality of all individuals is guaranteed, including the full and equal enjoyment of all rights and freedoms. This provision prohibits direct or indirect discrimination based on several grounds, including gender.

2.2.2 WATER SERVICES ACT (1997)

The South African Water Services Act legislates the municipal function of providing water and sanitation services. It defines basic sanitation as "*the prescribed minimum standard of services necessary for the safe, hygienic and adequate collection; removal, disposal or purification of human excreta; domestic waste-water and sewage from households, including informal households*". The Act also provides for the setting of national standards and norms regarding water services and creates an obligation for water services authorities (all metropolitan and district municipalities and authorised local municipalities) to "*ensure efficient, affordable, economical and sustainable access to water services*". It also provides for the rights of access to basic water supply and sanitation needs, a regulatory framework for water service providers, and financial assistance to water service providers.

2.2.3 NATIONAL SANITATION POLICY (2016)

The 2016 National Sanitation Policy highlights sanitation as a basic necessity and "*an essential pre-requisite for success in the fight against poverty, hunger, child deaths, gender inequality and empowerment*." The policy notes that the specific needs of women and the vulnerable must be considered in sanitation services provision and recognises that sanitation interventions that are designed and managed with the full participation of women are more likely to be sustainable and effective. It also recommends.

- Partnerships between local government, local women's groups and the private sector should be forged to overcome technical and financial barriers to women accessing urban sanitation.

- Gender mainstreaming approaches should be used in sanitation plans and sanitation services provision.
- Integrating women into existing sanitation development process through targeting their needs is essential.

2.2.4 THE NATIONAL NORMS AND STANDARDS FOR DOMESTIC WATER AND SANITATION SERVICES (2017)

The National Norms and Standards for Domestic Water and Sanitation Services, gazetted in 2017 by the DWS, and produced as mandated in Sections 9 and 10 of the Water Services Act (1997), set minimum standards for access to water and sanitation. The Norms and Standards stipulate the requirements with which sanitation facilities and infrastructure shall comply. They also set the minimum basic standard for the provision of water at 25 litres per person per day, or 6,000 litres per month to a household and the provision of "*uninterrupted access to an adequate, appropriate, sanitation facility*." The Standards emphasize health and hygiene education so that the provision of water and sanitation services will be accompanied by improvements in health and significant reductions in water-related diseases such as cholera and diarrhoea. It also considers that water services are sustainable when the management of the water service involves the consumers, is sensitive to gender issues, establishes partnerships with local authorities, and involves the private sector.

2.2.5 NATIONAL STRATEGY TOWARDS GENDER MAINSTREAMING IN THE ENVIRONMENT SECTOR (2016 – 2021)

The strategy, as documented by the Department of Forestry, Fisheries and the Environment (DFFE), outlines, how the sector can, and should, entrench values of gender mainstreaming and gender equality within the running of its environmental programmes. The purpose of this strategy is to ensure that initiatives in the sector are aimed to support the creation of policies that support gender mainstreaming and gender analyses. The objectives of the strategy are to provide strategic guidance for gender mainstreaming in the environment sector, provide direction on how gender mainstreaming for the environment sector can be put into practice and lastly to provide a framework for gender mainstreaming and outlining funding opportunities.

2.2.6 NATIONAL WATER RESOURCES STRATEGY SECOND EDITION

South Africa's National Water Resources Strategy Second Edition places significant emphasis on water re-use as a possible response to increasing demand and water scarcity (DWA, 2013). The Strategy also provides the water sectors' strategic framework for the implementation of legislation and policy towards sustainable water resource management and development. The strategy emphasises the importance of gender in water allocation reform and new water use license applications, noting that water allocation "*must contribute to broad-based black economic empowerment (BBBEE) and gender equity by facilitating access by black- and women-owned enterprises to water*".

2.2.7 WATER AND SANITATION GENDER POLICY (1996)

This sectoral policy was adopted in 1996 and provides guiding principles for gender integration in the water sector as a means of redressing of previous inequalities. Although comprehensive, the policy is antiquated and as a result it is being reviewed to address gaps and align it to current policies and objectives to ensure it effectively responds to current challenges. The Gender Policy tasks the water sector lead to (DWA, 2006):

- follow the Constitutional principle of gender equality, to recognise and address the current conditions that mitigate against women taking their full part in society.
- commit to a programme of action which recognizes present gender roles and works to counteract the gender inequities of the past; and
- end discriminatory practices and according to recognition and special treatment to women as a means towards redressing the imbalances of the past.

2.2.8 NATIONAL DEVELOPMENT PLAN (NDP)

The NDP addresses the challenges South African women face by prioritising several subsidiary goals which affect women, such as poverty and unemployment (South African Government, 2012). The NDP reinforces the notion that women's empowerment and participation in the economy is critical for economic transformation in South Africa. Furthermore, the NDP recommends that:

- specific focus should be placed on unemployed women in relation to public employment.
- all sectors of society should support the promotion of women leadership.
- measures should be put in place to provide women access to basic services.
- South Africa develops the South African Integrated Programme of Action (POA) Addressing Violence Against Women and Children, which has served as South Africa's action plan for addressing gender-based violence.
- South Africa develops the Draft Prohibition of Forced Marriages and Child Marriages Bill, which will provide even greater protection from these harmful practices.

2.2.9 LOCAL GOVERNMENT: MUNICIPAL SYSTEMS ACT 32 OF 2000

The Local Government: Municipal Systems Act (Act no. 32 of 2000) provides the fundamental principles, mechanisms and processes necessary for municipalities to ensure access to basic services, like water and sanitation services. Section 78 of the Municipal Systems Act (MSA) has a particular impact on the provision of water services as it requires municipalities to perform a rigorous process when determining whether to allow an institution to provide municipal services like water. The council of a municipality has the right to govern, on its own initiative, the local government affairs of the local community; and to exercise the municipality's executive and legislative authority without interference. Municipal councils must respect the rights of citizens in the way in which they exercise their powers. Municipal councils have duties as well as rights. These include the duties to:

- exercise their powers and use their resources in the best interests of the local community;
- provide, without favour or prejudice, democratic and accountable government
- encourage the participation of the local community;
- ensure that municipal services are provided to the local community in an equitable, financially and environmentally sustainable manner;
- promote development in the municipality;
- promote gender equity;
- promote a safe and healthy environment in the municipality; and
- contribute to the progressive realisation of the fundamental rights contained in the Constitution.

3. STAKEHOLDER MAPPING AND LESSONS LEARNED

The key stakeholders that are relevant for the WRP and are necessary for understanding the gender dimensions for implementing the programme are presented in Figure 3-1.

Figure 3-1 Stakeholder Map

Their roles are detailed below and throughout the programme, each of the stakeholders will be engaged with to help refine and enhance the gender action plan for the WRP. These can be categorised as key and supporting stakeholders and include the following.

- **GCF** has a governing instrument with a clear mandate to enhance a gender-sensitive approach in its processes and operations. The team will not only consult its Gender Policy but also, where possible, seek assistance for gender-responsive interventions from resources within the Fund.
- **DBSA** encourages gender mainstreaming in development projects to break down barriers for women and has a gender mainstreaming programme to drive this process to achieve success in the project lifecycle stages of preparation, development, monitoring and maintenance.
- **DWS** recognises the importance of gender across the water value chain both in terms of managing and developing water resources, as well as being the recipient of water services. DWS notes the importance of the development of women in relation to water management as women are the traditional custodians of

natural resources in rural areas, as well as in many urban contexts. In this role, women often suffer the most from the degradation of water and other natural resources. Noting the importance of gender dimensions in the water sector, the Department programmatically supports women in being represented at all levels and activities in the political, technical and management positions.

- **DCOG** has programmes related to gender equality, youth and women empowerment. A gender sensitive budget is one of their key initiatives. As DCOG play a key role is assisting Local Government to implement policy, they have significant impact on the approach used by Local Government to support socio-economic development and the gender interfaces with developmental initiatives.
- **National Treasury** sets clear stipulations in terms of gender responsive planning, budgeting, monitoring and evaluation.
- **Local governments** have a significant role in driving localised socio-economic development through such functions as the delivery of water and sanitation services. They must draw up plans, policies and strategies that support development and strive for optimal service delivery. In so doing, these plans must consider effective delivery of services, with gender equity provisions being a key dimension of delivery objectives and targets.
- **The Commission of Gender Equality (CGE) and the Office on the Status of Women** was established with a mandate of ensuring that gender equality becomes a reality in South Africa.
- **Catchment Management Agencies** are being established in six water management areas and give effect to water resource management through the development of a catchment management strategy. These strategies not only have a range of water management objectives and targets, but also provide for a range of socio-economic and environmental objectives within the context of climate impacts. As such, gender and equity dimensions linked to climate resilience are central to these strategies.
- **Water User Associations** are aimed at water users cooperatively using and managing a shared water resource for mutual benefit. There are meaningful opportunities for these associations to provide technical and mentoring support to women. The DWS provides clear requirements for these institutions in terms of ensuring improved gender representation in the management of these institutions. The business plans for these institutions are also required to provide clear gender objectives and targets.
- **Water Boards** are established to provide bulk water supply to municipalities and undertake an array of supporting studies to develop, operate and maintain infrastructure as well as catchment-based support studies where water resources may threaten bulk water supplies. The prioritisation of female entrepreneurs is often supported by these institutions.
- **Catchment Management Forums** are not legislated institutions but play a key role in supporting information exchange and the development of practices at the catchment scale. These forums often play a significant role in linking water related projects to various localised initiatives and interventions, particularly with an eye on beneficiation and impact.

- **SOUTH AFRICAN HUMAN RIGHTS COMMISSION** identifies, discusses and integrates strategic priority areas in line with the Human Rights Matrix including the environment, natural resources, rural development. The Commission notes the need to advance the equality of all within South Africa, particularly based on sex and gender.
- **SONKE GENDER JUSTICE** works to strengthen the capacity of governments, civil society and citizens to advance gender justice and women's rights in a way that contributes to social justice and the elimination of poverty.
- **WORLD WIDE FUND (WWF)** is a leading organisation on conservation in South Africa. WWF develops platforms and coalitions for environmental governance and monitoring, leading innovative projects to build political participation of civil society in support of sustainable development, empowering women and promoting youth leadership in environment and development to secure the rights of future generations.
- **WOMENG** provides awareness training for new career opportunities in the engineering and tech fields for women and girls; run employability training for female engineering students; support founders and innovators in Science, Technology, Engineering, and Mathematics (STEM) businesses through targeted incubator and accelerator programmes and train women in leadership intrapreneurship.
- **CRIDF** is funded by the UK Department for International Development (DFID) and supports regional integration and development in the Southern African Development Community (SADC).

3.1 Good Practice and Lessons Learned

3.1.1 NAMIBIA'S WASTEWATER RECLAMATION PLANT

The use of reclaimed water was the only affordable option for the city of Windhoek to cope with the water shortage caused by population growth, increased demand and declining rainfall following the water crisis of 1957. This has led to the establishment of the Wastewater Reclamation Plant in Windhoek, Namibia in 1969. The plant's continued success is attributable to several factors, including the vision and great dedication of the potable reclamation pioneers; the excellent information policy and education campaigns supporting buy-in; the absence of water related health problems; a multiple-barrier approach; reliable operation and online processes and water quality control; and the near absence of practicable alternatives (Lahnsteiner, 2013). **This initiative has embraced gender mainstreaming by providing provisions for (1) the development of gender-responsive training and promotional material (2) construction of climate-resilient inclusive sanitation facilities for vulnerable people who are not able to access or afford a sewerage connection; undertake WASH Friendly school campaign (hand washing and sanitation facilities, responsive to gender and disability), and integration of the "*Leave No-one Behind*" in the rural areas and finally to (3) mainstream gender into the Integrated Water Resource Management Plan that is implemented country wide.**

The case study highlights the need for a functional, sound and robust enabling environment that encourages strengthening governance and accountability as a mean to improving gender responsiveness in programmes.

A country without a robust and supported policy and regulatory framework that includes evidence of mainstreaming gender into its framework, will struggle to successfully implement programmes of this kind.

Figure 3-2 Wastewater Plant in Windhoek Namibia

3.1.2 INDIA: THE URBAN WATER SUPPLY AND ENVIRONMENTAL IMPROVEMENT PROJECT (UWSEIMP OR PROJECT UDAY)

UWSEIMP, implemented in India by the Asian Development Bank, was designed to rehabilitate and expand the urban water supply systems and improve the wastewater collection and treatment systems in four cities of Madhya Pradesh: Bhopal, Gwalior, Indore, and Jabalpur. The objective of the project, among other aspects, was to strengthen the capacity of the project cities to plan and manage their urban water supply and sanitation systems in a more effective, transparent, and sustainable manner. The project also included the improvement and expansion of municipal infrastructure and services, including urban water supply, sewage and sanitation, storm water drainage and solid waste management.

Some of the strategies included were to prepare municipal action plans for poverty reduction, introducing cost-sharing arrangements in infrastructure (sanitation) at the household level and ensuring public participation and awareness mechanisms are weaved into the programme. **From a gender perspective, it was noted that in India, women were often overlooked in planning and implementation of infrastructure projects and men traditionally have a greater role in decision making.** Further, institutions working on water supply and sanitation projects often lack both an understanding of gender issues and the capacity to take positive action. The gender analysis noted that water engineers were often disapproving of people making the case for a gender perspective, asking "*what do water tanks have to do with gender*?". **To address the gender-related issues, the project included certain elements such as (1) introducing evidence-based approaches and**

strategies, like a gender mainstreaming strategy (2) development of a gender action plan (3) formulation of a gender field manual and (4) community mobilisation (Asian Development Bank, 2015)**.**

3.1.3 BARBADOS: THE R'S (REDUCE, REUSE AND RECYCLE) FOR CLIMATE RESILIENCE WASTEWATER SYSTEMS IN BARBADOS

Barbados has been experiencing increased drought periods and reduced annual rainfall as a result of climate change. The Caribbean Community Climate Change Centre applied to the GCF for support to reduce the stress of Barbados' internal renewable water resources. The adaptation measures considered were to: (1) educate Barbadians and visitors about climate change as well as on wastewater reduction, reuse and recycling, (2) reduce CO_2 emissions from wastewater treatment (3) build climate resilience in the wastewater treatment process. Gender-related elements were also included in the project outline through a gender study. **The aim of the gender study was to ensure the development of gender sensitive outcomes and to identify the issues, needs and contextual factors affecting the male and female stakeholders.** The activities noted were as follows (Green Climate Fund , 2019):

- **Monitoring and Evaluation**: Tracking and assessing progress toward goals and objectives to improve gender sensitivity.
- **Targeting and Participation**: Meaningfully engaging beneficiaries and other stakeholders in gender-sensitive project design and implementation.
- **Public Awareness and Social Marketing**: Informing and effecting behavioural changes among water users in the way they gain access to and manage shared resources.
- **Capacity Building and Organizational Development**: Enabling all water stakeholders—from the implementing agencies to the beneficiaries—to build requisite skills and knowledge for gender-sensitive water services and resource management.

3.1.4 FIJI URBAN WATER SUPPLY AND WASTEWATER MANAGEMENT PROJECT

The urban water supply and wastewater management project methods to improve water supply and wastewater management to contribute to Fiji's sustainable development. The main feature of the project is to design and construct a new water intake by the River Rewa, with a pumping station, wastewater treatment plant, clear water reservoir and pipeline to increase water production. Woven into the programme were commendable gender mainstreaming activities including to:

- Conduct gender awareness training for staff, ministries, and service providers at all levels.
- Conduct socio-economic surveys, gender analysis, gender action plans, community consultations, and awareness training.
- Collect sex-disaggregated data to identify beneficiaries and ensure women headed households are given priority.

- Use existing health or relevant religious groups sub- committees and networks to assist in monitoring effectiveness of community engagement and awareness programs. These will include consultation and focus groups sessions organized by women’s committees to solicit feedback from community members.
- Ensure recruitment policies provides for retention of increasing percentage of female staff in all divisions from 10% to 20% by 2022 and set up leadership training for females.
- Ensure women promoted at rates equal to men in all levels and divisions, to management positions in three departments, and two to senior management posts by 2022.

4. CHALLENGES, OPPORTUNITIES, AND INTERVENTIONS

4.1 KEY CHALLENGES AND SOURCES OF VULNERABILITIES FOR WOMEN

4.1.1 GENDER AND CLIMATE CHANGE

Although climate change negatively impacts all members of the community, in more recent years, there is an increasingly acknowledged link between gender equality and climate change. Climate change necessitates complex community decision-making. Women and girls in developing countries are often more vulnerable than men and boys to the impacts of climate change and have less opportunity to effect change due to pre-existing gender inequalities regarding political leadership, access to information and resources, and mobility and voice. In contrast, women in Sub-Saharan Africa communities are generally subjected to unequal decision-making power, lack of access to resources, gender-based discrimination and a general lack of capacity to enjoy an equal social, cultural, economic, and political status in their communities.

Equally important, due to economic marginalization, political disenfranchisement, and differentiated labour responsibilities, women face the consequences of the climate crisis first and worst, and women lack equitable representation and power in crafting climate policies to address their needs. While South African society has been becoming increasingly urbanised, with statistics estimating that currently 33% of the population is rural[1], within these urban contexts some settlements (approximately 13%) only receive a limited or basic service. In rural settings this equates to 15% having no service or an unimproved service and 19% receiving a limited service (Statistics-SA, 2019). Whether in these urban or rural settings women and girl children are most often responsible for ensuring households have water to meet household requirements. Urban and rural agriculture to support livelihoods are important for food security, noting that 49% of the adult population lives below the upper bound poverty line. The impact of poverty on adult women is greater than men, regardless of the poverty line used[2]. Rural communities are not spared from these harsh realities as they rely on natural resources for survival, with agriculture being one of the most vital sectors in their economies (Georgetown Journal of International Affairs, 2020). Women in the rural areas are also burdened by gender norms and roles as they are required to fulfil key tasks such as water collection and smallholder farming that are impacted directly by droughts or disasters.

Women tend to have less access to information and technology about climate change and its effects, such as early warning systems, than men. This is because many rural women do not have their own cell phones and

[1] https://data.worldbank.org/indicator/SP.RUR.TOTL.ZS?locations=ZA
[2] http://www.statssa.gov.za/?p=12075

literacy remains a challenge amongst older women. However, women are resilient, and many are the holders of traditional conservation strategies and knowledge, which makes them a valuable part of the solutions and responses to climate change and waste management strategies (Gender Women for Climate Justice Southern Africa, 2020).

Rural women farmers face challenges of access to water and the results of global warming and resultant drought. This was crucially felt between years 2016 – 2017, during the El Niño disaster in Southern Africa when many crops were lost. As a result, some farmers have adapted their produce and now grow groundnuts, pumpkin, and peppadew (similar to red chilli's) which are more resistant to drought. In addition, they find that there was an increase in alien insects and resultant diseases affecting their crops and are responding with 'natural' insecticides such as tobacco-soaked water and soap and water mixture sprays (Gender Women for Climate Justice Southern Africa, 2020).

In addition, women often have limited ability to own, sell, inherit, acquire, and use property as collateral. Such a lack of proper representation directly results in climate-influenced community actions that do not reflect the needs of women. Women in rural areas are adversely impacted by climate change vulnerability based on their role and work as primary care givers and the increased burden of water collection for household and irrigation use.

The programme will require an equitable development undertone that requires an approach that recognises that characteristics such as gender, race, class, and physical ability create overlapping and interdependent vulnerabilities in the face of climate change. The process of incorporating these aspects into the WRP will cement the systematic consideration of the differences between the conditions, situations, and needs of women and men, and the integration of gender equality concerns into the programme design.

4.1.2 GENDER IN Rural Areas

Water is necessary not only for drinking, but also for food production and preparation, care of domestic animals, personal hygiene, care of the sick, cleaning, washing and waste disposal. The provision of hygiene and sanitation are often considered women's tasks – especially so in rural contexts. Women and girls are most often the primary users, providers, and managers of water in their households and tend to have the main responsibility for household hygiene. If a water system falls into disrepair, women and girls are usually the ones forced to travel long distances over many hours to meet their families' water needs. Women and children often bear the brunt of the lack of toilets and other sanitation facilities. If there is insufficient water such as during drought, they may be punished for returning home empty-handed or for returning home late after waiting in line for hours. When school-age girls are required to spend long hours collecting water, they are at a higher risk of missing and/or not attending school resulting in negative impacts on their potential for economic and social advancement. This, in turn, may place them at a higher risk of sexual or gender-based violence in the future. Women, more than men, suffer the indignity of being forced to defecate and urinate in the open, where they may have to walk to remote locations outside the village often waiting until dark rendering them vulnerable to assault and potential rape. This can result in them choosing to drink less which can result in all kinds of health problems (SSWM, n.d.).

4.1.3 GENDER NORMS, ATTITUDES AND STEREOTYPES

To the sanitation value chain, there are entrenched social and gender norms present that will influence the design of the programme. Historically, from high to low skilled women, women are less likely to seek employment in the utilities sector (for instance, energy, water, gas, electricity etc.). Notably in 2016 and as illustrated in Table 4-1, high-skilled women accounted for 0.9% of jobs in the utility sector. Similarly in the construction sector, low-skilled women comprise 3.2% of jobs.

Table 4-1 Distribution of Employed Women across Industry by Skills and Category, 2016

Industry	High-skilled	Semi-skilled	Low-skilled	Total
Agriculture	0.4	1.3	9.2	3.9
Mining	0.5	1.3	0.6	0.9
Manufacturing	6.1	11.9	5.5	8.2
Utilities	0.9	0.6	0.1	0.5
Construction	1.4	2.1	3.2	2.3
W&R trade	9.7	31.7	19.6	22.0
Transport	2.9	3.6	0.9	2.5
Finance & business services	21.1	15.5	6.9	13.8
CSP services	57.0	31.3	14.5	31.6
Private households	0.0	0.7	39.5	14.3
Total	**100.0**	**100.0**	**100.0**	**100.0**
Employment ('000s)	***1 686***	***2 750***	***2 439***	***6 874***

Source: StatsSA, Labour Market Dynamics (2016).

One of the reasons for this bias is due to entrenched social norms alongside the water and sanitation utilities especially as the sector has been historically dominated by men and marketed as a predominantly masculine domain. Moreover, the study of STEM fields, required to work in this sector, are rife with gendered stereotypes. Gendered stereotypes give rise to roles and responsibilities that often determine women's access to rights, opportunities, resources, and decision-making access points.

For low skilled workers, technical and vocational education, and training in areas such as plumbing or meter reading are also traditionally considered male domains (World Bank Group, 2019). Women are likely to be deterred from entering water utilities because such social norms prescribe that it is an area of work that is not suitable for them or that they are incapable of performing well.

A gender assessment carried out under the World Bank's Dushanbe Water Supply and Wastewater Project in Tajikistan, for instance, revealed that among the local community, there is a widespread perception that work in the water sector is more appropriate for men. Correspondingly, social norms dictate those jobs are more suited for certain genders, for example, caregivers and nursing for women whereas construction and engineering are geared towards males. **Pervasive stereotypes and norms lead women to internalise and**

doubt certain positions creating a lacuna of mentors and leaders in the field to guide and encourage women to navigate through the real and perceived gendered barriers.

4.1.4 WOMEN IN STEM CAREERS

According to a 2018 report of the UNESCO Institute for Statistics (UIS), women in STEM represent less than 30% of researchers globally (UNESCO, 2019). This shows that there is a need for urgent attention and significant investments in women to pursue studies in and contribute to STEM fields. Women are also in the minority in STEM research occupations. In the UIS, in 2015, South Africa recorded 45% of total researchers in STEM were women however, leadership and power positions were predominantly held by men. (Univeristy of Stellenbosch, 2020). A way to encourage more women to not only research STEM related fields but also to:

- increase involvement from a leadership perspective for example, executing targeted scholarship programmes at secondary and tertiary levels;
- applying performance-based conditional cash transfer programmes that incentivize girls and women to graduate from STEM programs;
- curricula reform;
- training for vocational, college and university programs that focus on enhancing leadership skills, design, operational management and maintenance requirements for utility infrastructures coupled with incentive programmes.

Moreover, government and utility departments must play a key role in the recruitment of women and graduates to enrol in their programmes as well as linking students to potential internship and graduate programmes (Deloitte, 2017).

Figure 4-1 textbox Scholarship Program for Women in the Water Sector in the Lao

Textbox: Scholarship Program for Women in the Water Sector in the Lao People's Democratic Republic (Lao PDR)

In Lao PDR's Department of Water Supply: only 11.7 percent of its staff were women, and out of these, most were employed in administrative or financial positions. The Department of Water Supply faced challenges in finding women to fill technical and managerial positions. To address this, the ADB developed a grant project with the objective of strengthening the talent pipeline of future women engineers and leaders in the water sector (ADB 2014). The grant funded a range of comprehensive activities that supported women (1) throughout all stages of STEM employment, (2) from identifying and recruiting high school graduates at the province level, (3) to offering them a four-year scholarship to pursue an undergraduate degree in a related engineering field, (4) to providing them with a two-month internship programs in water utilities and mentorship, and (5) to offering them professional development workshops once hired. The project emphasized that targets for women's employment must be complemented with efforts to increase the supply of STEM-qualified women. The pipeline of women engineers was developed through sustained monetary and HR commitment, such as regular counselling and mentoring. Source: ADB 2014; World Bank, forthcoming.

4.1.5 HISTORICAL LEGACIES

Before 1994, the Department of Water Affairs (now DWS) consisted of technical staff who were mostly white males (Schreiner, 2013). After 1994, the drive to transform the public sector resulted in an employment equity approach that saw large numbers of black and female appointments into the department. This change brought tension alongside resentment towards the new staff as the old staff members were encouraged to shift towards transformative change. Despite the change in power in 1994, the economy remains in the hands of the white elite who hold significant bargaining power and skills as compared to poor, black women who have limited access to this power and thus limited influence in affecting change in access to water (Schreiner, 2013).

4.1.6 HYGIENE AND SANITATION

Norms of taboo and silence around menstrual hygiene, which present menstruation and menstruating adolescent girls and women as contaminated or impure, have serious impacts that include lack of access to WASH facilities that accommodate the specific needs of menstruating women. The situation is exacerbated by the existence of male-dominated WASH decision making on WASH issues resulting in a lack of sensitivity to the menstrual needs of women and girls.

Menstrual hygiene continues to receive limited attention in policies, research priorities, programmes, and resource allocation. Most sanitation programmes fail to consider women's need to manage menstruation. Furthermore, latrine design often overlooks the specific needs of women and girls. In cases where hygiene promotion programmes exist, many fail to include the issue of menstrual hygiene, focusing instead principally on hand washing practices. Lack of access to private, safe, and hygienic facilities for managing their menstruation in the workplace can also lead to loss of earnings and present barriers to economic participation.

Similarly, women experiencing perimenopause have WASH needs, particularly in terms of menstrual hygiene management as well as the need for access to safe drinking water, washing and bathing. During perimenopause, hormonal changes can result in irregular and/or heavy bleeding and sweating. There is emerging research focusing on the WASH needs of perimenopausal women in low-income countries that is worth consulting to understand their specific needs. More generally, insecurity for women using toilets includes the proximity to male lavatories, structural designs of cabins, water supply, sanitary disposal units, among other factors. Therefore, men and women must have equitable and representative say in sanitation projects to ensure their individual needs are met.

Figure 4-2 Burkina Faso - Installing Ecological Sanitation

Textbox: Burkina Faso - Installing Ecological Sanitation

In Burkina Faso ecological sanitation (in the form of urine-diverting dry toilets, with reuse) were installed. Women were reluctant to use the installed latrines when they were menstruating. They cited feelings of shame, fear of leaving traces of menstrual blood on the slab, and (misplaced) fear of contaminating fertilizer products produced from the excreta. The women were caught in the dilemma of wanting to wash away traces of blood but having been told that they must not put additional water in the faecal vault. Others worried that cloths used for menstrual protection would be visible when the faecal vault was emptied. Ensuring that women's concerns are heard and translated into action was essential to gain their buy-in for using the latrines. It was also pertinent to communicate that the fertilizers produced form sanitation waste were in no way harmed by the menstruation. Thorough research was conducted to help adapt the toilet design to cater to women's needs including adding a designated waste bin for menstrual products to be discarded (United Nations and Stockholm Environment Institute, 2020).

4.1.7 SOCIO-ECONOMIC DIMENSION AND UNEQUAL ACCESS TO RESOURCES

South Africa measures poverty using three data points: (1) food poverty line measured at R547.00; (2) lower bound poverty line measured at R785.00 (3) and the upper bound poverty line is at R1183.00, as of 2018 (Africa Check, 2019). According to the Statistics South Africa (StatsSA), 49.2% of the population over the age of 18 falls below the upper-bound poverty line. Women are more vulnerable to poverty as 52.2% of women fall below the upper bound poverty line when compared to 46.1% of men. Correspondingly, 74.8% of households are women-led and are below the upper-bound poverty line (Borgen Project, 2020). The global pandemic exacerbated the current statistics particular hunger and food security.

South Africa has the highest inequality rates in the world, with a consumption expenditure Gini coefficient of 0.63 in 2015. While inequality seems to have improved over the past 20 years when measured per capita, consumption inequality has increased since the end of apartheid. Similarly, even though black South Africans are reporting the largest increase in the average number of assets owned, within-group asset inequality among black South Africans has continued to grow. This trend seems to indicate that many of the problems from decades of apartheid have not disappeared, but rather have become a normal part of South African society (Borgen Project, 2020).

Furthermore, intergenerational mobility is low, meaning inequalities are passed down from generation to generation with little change over time. Thus, most of the population are trapped in the poverty cycle. The Government has implemented solutions to some of these wicked issues. The introduction of higher social spending, affirmative action programs and targeted government transfers contribute to alleviating the current adverse trends and obstacles.

4.1.8 UNEQUAL ACCESS TO LAND

In 2018, the World Bank reported that less than 13% of African women between the ages of 20 and 49 have sole ownership of land compared with 36% of African men (World Bank, 2018). In fact, in some African countries, fewer than 10% of women have the privilege. This points to skewed land distribution, which fails to seriously consider the critical role of ownership and its contribution to the economic empowerment of women.

A gender lens to effective land management must be applied to assist with the deepening of land inequality. This approach can assist with capacity building, engendered land unproductivity and aggravated poverty in households. As reported in the Land Audit Report of 2017, women struggle regarding ownership and access to land (Rural Development and Land Reform, 2017). The promulgation of the Extension of Security of Tenure Act, 1997 offered women an opportunity the right to land for the first time in history by including women in its definition of an occupier. In 2017, individual land ownership recorded that only 34% of individual landowners are female's ad that males own the largest size of farms and agricultural landholdings. Based on Figure 4-3 Gender owned by males own 71% of hectares followed by females with 13%.

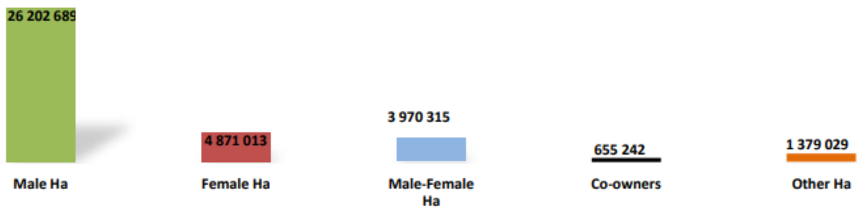

Figure 4-3 Gender owned by hectares – Source: Land Audit Report DRDLR 2017

Customary law practices in South Africa have implications for gender equity in decision making, ownership and rights of women. One such challenge is a woman's right to own land from a tribal context. Ownership of land is useful collateral to access capital and other forms of financing.

4.1.9 GENDER-BASED VIOLENCE (GBV)

Violence against women and girls which is connected to women and girls' unequal status can limit their access to water and toilets. When designing any WASH intervention, it is therefore, crucial to conduct a thorough analysis of the differing rights, needs and roles of those at risk of GBV related to WASH. It is critical to engage women, girls and other at-risk groups in the design and delivery of WASH programming. Engagement of this sort not only assists in ensuring an effective response to basic needs. It also contributes to long-term gains in gender equality and the reduction of GBV. A key example is that when there is a lack of lighting, locks, privacy and/or sex-segregated sanitation facilities, there is an increase in the risk of harassment or assault against women and girls. Inadequate building materials and poor design can also increase this risk.

In 2019, the National Government committed to create a safer and violent free South Africa. The Department of Social Department in collaboration with other Departments, introduced programmes that aim to engage with communities, traditional leaders, student organisations, youth groups and officials working in the criminal justice system. Efforts were also made to enhance the current legal and policy frameworks to be more responsive to the needs of the survivors of gender-based violence (Department of Social Development , 2019). This includes the introduction of three new bills, in 2020. The first bill intends to amend the Criminal Law (sexual offences and related matters) Act by recognising sexual intimidation as an official offence. The Criminal and Related Matters Amendment Bill ensures those accused of GBV could only be granted bail under exceptional circumstances. Finally, the Domestic Violence Amendment Bill will extend the definition of domestic violence to include victims of assault from marriage to dating as well as those in perceived romantic, intimate, or sexual relationships of any duration. It will also no longer be a requirement to apply for a protection order in person but instead can be conducted online. Correspondingly, the Department established the Gender Based Violence Command Centre (GBVCC) to provide professional psychosocial support and trauma counselling to victims of gender-based violence (domestic violence).

4.1.10 DECISION-MAKING AND PLANNING

An evaluation conducted by the World Bank found that water projects that factored women were six to seven times more effective than projects that did not (Deloitte, 2017). It was also noted that women account for 17 per cent of the water, sanitation and hygiene labour force and a mere fraction of the policymakers, regulators, management, and technical experts across the globe (Deloitte, 2017). **Correspondingly, women's involvement in decision making is crucial to avoid poor planning decisions, failed project implementation and the exclusion of women from the design and operation of programmes aimed to benefit them.** Concerns related to gender by both men and women will be addressed at design and execution phases particularly when appointing staff to operate the various facilities. A gender lens will be applied that acknowledges the inherent biases that men must dominate technological domains but avoids to internalizes these biases in practice. **The importance of involving both women and men in the management of water is not only intrinsically desirable but also practically beneficial – involving women in water projects increases project effectiveness and long-term sustainability.**

4.1.11 LACK OF SEX-DISAGGREGATED DATA

Based on the research conducted there is a lack of comprehensive gender-disaggregated data on women in South Africa in water management positions, leading water and waste utility departments and in management positions. Having this essential data will aid in measuring progress and determine whether certain interventions and initiatives are working and also resulting in an improvement in sanitation outcomes. The availability of data will also contribute to future policy and investments.

4.1.12 ROLES, RIGHTS AND DIFFERENTIAL NEEDS OF WOMEN AND MEN

As primary providers, managers, and users of water, women are often in an ideal position to assist productive change in the design and maintenance of water systems, water distribution, and policymaking. The more time women and girls spend accessing clean water for their families or caring for relatives inflicted with water-related illnesses, the less time they spend learning in school or working in the productive economy.

Gender roles in household water management are a key aspect to consider in the WRP programme as women are the primary users of water within the household. They are generally responsible for household water management: for drinking, preparing food, as well as personal and household hygiene (washing and cleaning). For women, the timing of water availability is important, as different schedules can have an impact on their workloads. **The lack of clean and safe water also leads to illness in the household and increased health care burdens for households. It is thus crucial that the programme understand the differential role and responsibilities of women in South Africa, particularly between rural and urban women and how the interventions designed in the programme will impact each class.**

Textbox: South Africa's Mabule Sanitation Project

The Sanitation Project was a project developed by the Department of Water Affairs and Forestry (DWAF). DWAF has provided funding for sanitation projects in communities where there was imbalance in gendered decision-making processes. The initiative launched established a brick-making project for latrine construction that employs mainly women, generates cash and provides the community with affordable bricks. Despite the benefits of the programme, the team faced a few bottlenecks. The community did not initially support the idea of women leading the development project. The municipality did not want to let women open bank accounts, because there was doubt that the women and the required skill to manage the funds. Husbands of the wives also opposed women participating in the project because of the taboo around sanitation issues. Thanks to this drive, the village has safe and attractive toilets as well as improved health and hygiene. There was also an increased acceptance and acknowledgment of women's leadership roles by community members as well as an increased collaboration between women and men. (Mvula Trust, n.d.)

Figure 4-4 Mabule Sanitation Project

4.2. NATIONAL AND LOCAL-LEVEL CHALLENGES TO UNLOCK

At a national level, gender equality is addressed through redress of the results of past racial and gender discrimination. This includes both representation in decision-making committees and access to water. Gender equality will therefore be considered when appointing members for catchment agencies, local-level advisory committees, water boards, and the presiding officers of tribunals. Gender equality will also consider when licenses are issued. The National Gender Policy states that 30% of all the members of village water committees must and should be women.

However, there is also the economic need for better utilization and development of human capital, especially women and girl children in the participation of the productive use of water.

The WRP will be developed in phases whereby the first phase targets projects with the highest probability of being implemented. There may be only a few projects within the first 3 years to prove the concept and build confidence in the WRP as well as the technology. Clearly the initial list of projects is likely to come from those Municipalities that have already started planning for reuse and have initiated feasibility studies. Table 4-2 provides a brief description of each municipality, indicates the ratio of males and females per municipality and highlights the population growth per annum as well as the percentage of household services per municipality. The table represents available data for the 10 selected municipalities dating to 2016. However, the most recent national Stats-SA census were conducted in early 2022 of which the results have not been released as yet.

Table 4-2: Brief description of the identified municipalities

Province	Municipality	Description	Male per 100 females (2016)	Population growth per annum	Household services (2016)
KwaZulu-Natal	eThekwini Metropolitan Municipality	• Category A (Metropolitan) municipality • Largest city in the province • Third-largest city in the country.	96.2	1.43%	a) 69.3% b) 60.8%
	Indicative Local Municipality A	• Category B (Local) municipality • Situated within the King Cetshwayo District and the largest among the five municipalities that make up the district. • Comprised of 34 wards, having the largest number of wards in the district • Third economic hub in the province after eThekwini and Msunduzi Municipalities	93.3	2.81%	45.7% 43.0%
Eastern Cape	Nelson Mandela Metropolitan Municipality	• Category A municipality • First city in South Africa to establish a fully integrated democratic local authority.	96.0	2.09%	a) 90.55 b) 77.3%
Gauteng	City of Ekurhuleni Metropolitan Municipality	• Category A municipality • One of the most densely populated areas in the province, and the country.	105.7	1.39%	a) 85.4% b) 56.55
	City of Johannesburg Metropolitan Municipality	• Category A municipality. • Johannesburg is the most advanced commercial city in Africa and the engine room of the South African economy.	100.3	2.49%	a) 88.6% b) 60.3%

	City of Tshwane Metropolitan Municipality	• Category A municipality • Single-largest metropolitan municipality in the country, comprising seven regions, 105 wards and 210 councillors.	98.5	2.60%	a) 77.2% b) 62.1%
Western Cape	City of Cape Town Metropolitan Municipality	• Category A municipality • South Africa's second-largest economic centre and second most populous city after Johannesburg. • Provincial capital and primate city of the Western Cape, as well as the legislative capital of South Africa, where the National Parliament and many government offices are located.	96.9	1.56%	a) 91.0% b) 76.7%
	Drakenstein Local Municipality.	• Category B municipality • Part of the Cape Winelands District in the Western Cape Province.	97.7	2.48%	a) 96.9% b) 84.5%
Free State	Mangaung Local Municipality	• Category A municipality.	94.2	0.37%	a) 66.7% b) 38.2%
Northern Cape	Indicative Municipality B	• Category B municipality • Located in the Frances Baard District, one of four municipalities that make up the district, accounting for a quarter of its geographical area.	98.3	0.63%	a) 87.7% b) 60.2%

*Household Services: a) Flush toilet connected to sewerage; b) Piped water inside dwelling

Of these, 6 are Metropolitan Municipalities and the most significant growth nodes of the country, all of which face significant water security challenges. Projects in these municipalities are estimated at 90% to face climate induced vulnerability. Key items to address at this local level, to achieve the national objectives of redress, access, and participation, is to:

- Establish whether municipalities Integrated Development Plan's are engendered;
- Strengthen (or establish) the enabling policy environment at municipality level to embrace gender equality, women empowerment interventions;
- Promote the use of gender mainstreaming approach in planning, budgeting, implementation, monitoring and evaluation and audit of public services at local government level;
- Facilitate a gender responsive identification, analysis and formulation of policies and programmes at local government level;
- Track, monitor and report on gender targets periodically.

4.2.1 NATIONAL

The disparities in unemployment levels between women and men in South Africa from 2008 to 2019 is shown in Figure 4-5. There is generally an increasing trend in the unemployment rate where women and youth are the most affected.

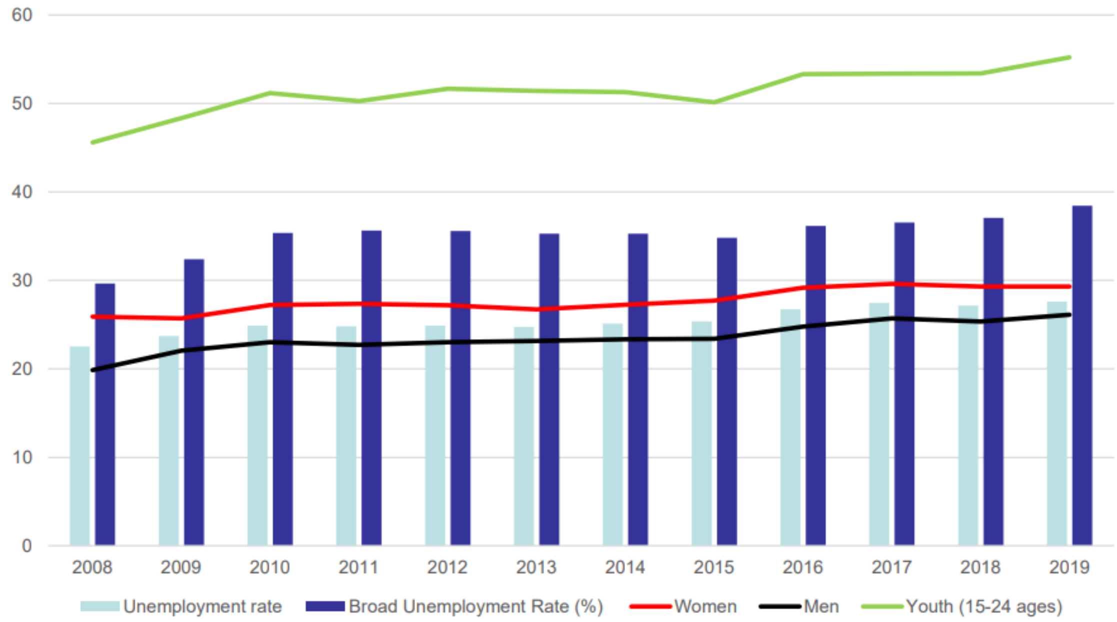

Figure 4-5: SA Unemployment levels 2008-2019

Source: (Commission for Gender Equality, 2021)

South Africa is committed to addressing gender inequality and this is evident in it being party to several international treaties including; the Beijing Declaration and Platform for Action (1995), the United Nations Women Strategic Plan (2014 – 2017), the Convention for Biological Diversity Gender Action Plan (2008), the Convention on Biological Diversity (CBD), the United Nations Convention to Combat Desertification (UNCCD), and the United Nations Framework Convention on Climate Change (UNFCCC).

The South African **Presidential Climate Commission** just transition framework sets out the actions that the government and its social partners must take to achieve a just transition and pursue long term value creation while considering the needs of all their stakeholders (Presidential Climate Commission, 2022). This includes:

- revisiting whether and how public resources have been effective in supporting improved service provision and in closing the inequality gap
- using corporate social investment to stimulate local enterprises and support skills development, pursuing the principles of Broad-Based Black Economic Empowerment including women's empowerment

- embedding environment, social and governance (ESG) principles across all operations
- tracking environmental, social, governance and climate impacts, and disclosing these impacts through best-practice reporting, including through the CDP and Johannesburg Stock Exchange.

The **NDC's** emphasise the need to raise further awareness of the financial and technical support available for promoting the strengthening of gender integration into climate policies, including good practices to facilitate access to climate finance for grassroots women's organisations and indigenous peoples and local communities. In order to sustain long and deep transformation, there also needs to be support for the implementation of transparency and building of transparency-related capacity should be provided on a continuous basis, pursuant to Article 13.14 and 13.15 of the Paris Agreement (Republic of South Africa, 2021).

The mechanisms for transforming gender relations in South Africa are collectively known within government and civil society as the "**National Gender Machinery**". Together, these mechanisms aim to promote and protect gender equality, both by mainstreaming it and by dealing with it separately. The National Gender Machinery has structures at different levels in national government, legislature, and statutory bodies. Figure 4-6 reflects how legislation and policies are cascaded and implemented through various structures and components of the national machinery for the advancement of gender equality in South Africa from government to civil society (The Office on the Status of Women, undated).

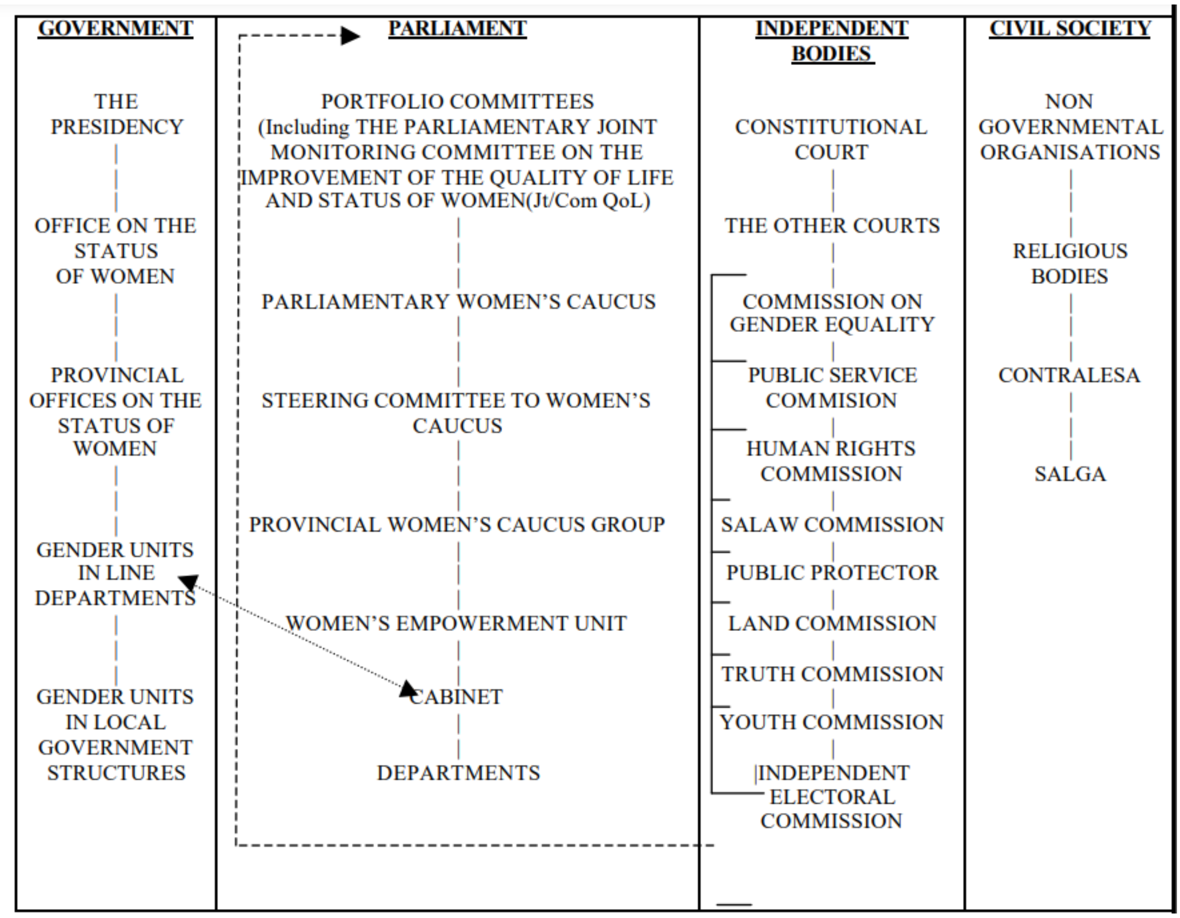

Figure 4-6: Structure of the national machinery for the advancement of gender equality in South Africa

The **Department of Justice and Constitutional Development** published a gender mainstreaming assessment report which presents a Gender Mainstreaming and Gender Budgeting Package. The report presents a Gender Mainstreaming Assessment Instrument in Table 4-3:

Table 4-3: Gender Mainstreaming Assessment Instrument

Key Results Area	Indicators
Awareness	• Reference to Gender Issues and Policy • Reference to International Instruments • Reflection of Compliance Obligations
Commitment	• Policy and Finances, Structure and Workforce Profile • Advocacy
Planning	• Strategy/Plan(s), Gender Analysis, Consistency
Policy and decision-making	• Women's Equal Participation in Decision-making • Policy and Legal Compliance • Responsiveness to Women • Organisational Priorities
Implementation	• Gender in Programmes and Projects • Integrated and Holistic Approach • Special Programmes • Mainstreaming Equality and Other Human Rights
Resources	• Engendering Budget • Human and Other Resources • Funding For Special Programmes
Capacity Building	• Education and Training • Learning Resources and Aids • Technical Support and Expertise • Specialist Skills and Professional Development
Communication/ Information	• Internal and External Communication Processes • Knowledge Generation and Information Management • Readily Available Reliable Indicators
Employment	• Gender Profile and Trends • Legal and Policy Compliance and Planning • Special Measures and Skills Development • Monitoring and Evaluation
Procurement	• Legal and Policy Compliance

	• Special Programmes
Monitoring and Evaluation	• Disaggregated Data • Accountability • Risk Management and Benchmarking
Reporting	• Gender Disaggregated Records and Reports • Standardised Reporting on Gender

The **Commission for Gender Equality** (CGE) is an independent statutory body established in terms of Section 181 of the Constitution of the Republic of South Africa. The CGE is mandated to promote respect for, protect, develop, and attain gender equality, and to make recommendations on any legislation affecting the status of women. The Commission for Gender Equality as part of its strategic objectives for 2012/2017, to date had a focus on gender transformation at local government level. The main objective of this process was to assess progress made by local government in achieving gender equality through gender transformation and gender mainstreaming approaches such as:

- Establish whether municipalities' IDP's are engendered,
- To further establish if there is an enabling policy environment at municipality level to embrace gender equality, women empowerment interventions,
- To promote the use of gender mainstreaming approach in planning, budgeting, implementation, monitoring and evaluation and audit of public services at local government level.
- To facilitate a gender responsive identification, analysis and formulation of policies and programmes at local government level

The **South African Local Government Association Women's Commission** (SWC) was established in 2010 to coordinate, promote and advocate for gender appropriate strategies and practices within member municipalities and feed into regional and continental processes.

A number of legislations and policies specifically commit local government to protecting and advancing the rights of women. These include:

- The Employment Equity Act (1999)
- The Promotion of Equality and Prevention of Unfair Discrimination Act (2000)
- National Framework for Women's Empowerment and Gender Equality (2002)
- Traditional Leadership and Governance Framework Act (2003)
- Communal Land Rights Act (2004).

4.2.2 PROVINCIAL AND LOCAL

Provincial Growth and Development Strategies (PSDS) and Integrated Development Plans (IDPs) are compiled in alignment with national objectives that promote gender equality in South Africa. Within the local government sphere the Gender Policy Framework for Local Government was developed in 2011 by the former Department of Provincial and Local Government (DPLG), currently the Department of Cooperative Governance and Traditional Affairs.

Kwa-Zulu Natal

The Kwa-Zulu Natal Provincial Growth and Development Strategy (PGDS) acknowledges the need for women and other vulnerable groups to be factored in as cross-cutting areas that need to be mainstreamed into all elements of provincial development, across all policies, plans and programmes (KwaZulu Natal Provincial Planning Commission, 2020). The Provincial Strategic Priorities relating specifically to gender are shown in Table 4-4.

Table 4-4: Outcomes, indicators, baseline, and targets relating specifically to gender in the Kwa-Zulu Natal PGDS

Priority	Outcomes	Indicator	Baseline	Target
Priority 1: Building a capable, ethical, and developmental state	Mainstreaming of gender, youth and persons with disabilities, empowerment and development institutionalised	Level of mainstreaming across public service and through the services delivered by sex, gender, age, and disability	25-year review	100% compliance to mainstreaming
	Gender, youth disability-responsive planning, budgeting, monitoring, evaluation and auditing institutionalise across government	Level of institutionalisation of Gender, Youth and disability-responsive planning, budgeting, monitoring, evaluation and auditing	New indicator	100% compliance with the frameworks
Priority 2: Economic transformation and job creation	More decent jobs created and sustained, with youth, women and persons with disabilities prioritised.	• Unemployment rate (%) • No. of jobs created	27.6%	• 20 – 24 % • 2 million
	Increased economic participation, ownership, access to resources, opportunities and wage equality for women, youth and persons with disabilities	Level of participation, ownership, employment, equity by sex/gender, age, disability, sector/industry, occupational	QLFS, LMD, EE report	50% increase for women, youth, and persons with disabilities by 2024 in each indicator category
Priority 6: Social Cohesion 2024 Impact **A diverse socially cohesive society with a common national identity**	Equal opportunity, inclusion and redress	• Inequality adjusted Human Development Index • Gender Inequality Index • Gender Pay gap (28%)	• 0.629 • 0.389 • 28%	• 10% improvement • 10% improvement • 50 % decline in the gender pay gap by 2024

eThekwini Municipality

Characterisation

The eThekwini Municipality had the fifth lowest HDI in 2018 (0.67) when compared with the other major cities in South Africa. It has also been recorded as having the highest number of people living below the food poverty line compared to other metros (eThekwini Municipality, 2020).

42.1 % of households in the metro are headed by women, having an average annual household income of R29400. 40 % of the population is employed, 80 % within the formal sector. 53.6 % of the population has completed matric and obtained a higher qualification. 44.8% of child-headed households have women as their head. The population pyramid for the municipality shows that females have a longer life expectancy. Teenage pregnancy has been identified as one of the many social development challenges.

Gender responsive initiatives

The following gender-responsive programmes were identified in the eThekwini Municipality IDP 2020/21:

Programme 2.9: Facilitating Industry Skills and Economic Inclusion acknowledges that inclusion is important in ensuring that women are integrated effectively into the workplace. The programme emphasises the need for key interventions that include the implementation of empowerment initiatives and the creation of partnerships and investing in strategic skills development.

Programme 7.3: Create mechanisms, processes and procedures for citizen participation which aims to ensure that different sectors of the communities, particularly the vulnerable groups (i.e., youth, women, disabled) participate in council processes.

Indicative Municipality A

Characterisation

The number of female-headed households increased from 36.29% in 2001 to 40.70% in 2011. About 5% of households are child and adolescent headed Households. The majority of employed within the district are males. The unemployment rate for females is higher (27.5%) than males (24.6 %). The following challenges have been identified in the municipality:

- Low skills levels and limited skills development
- High rate of unemployment
- High levels of poverty and inequalities

Gender responsive and water reuse initiatives

Indicative Municipality A is water insecure, and an adequate water supply is needed to underpin growth. Therefore, the municipality is considering alternative water sources such as desalination, water reuse, rainwater harvesting as a way to improve the water supply mix and undertaking a comprehensive feasibility study to identify the most viable solution for dealing with wastewater and associated by-products reuse generated within the City. Phase 2, the Procurement is being initiated.

As part of aligning with SDG 5, gender equality, the Indicative Municipality A has identified the following interventions:

- Established a functional Women's Forum
- Campaigns in support LGBTI,
- Preferential Procurement Women (40%), Youth (40%) and people living with disabilities (20%)
- Internship prioritising young girls
- Targeted skills development programs

The municipality has a responsibility of developing municipal gender plans as well as municipal strategies to implement them. The process of formulating a policy on Women Empowerment and Gender Equality has begun. The municipality has also successfully concluded the following programmes successfully:

- **Women Business Workshop**: where women were workshopped on the new developments in the Municipal Supply Chain Policy which allows 40% of municipal Supply Chain to women business.
- **Dialogues on Gender Based Violence**: conducted in the form of izimbizo with various sectors of the community.
- **Women in local government leaderships conference**: attended by 86 women leaders from level 11 upwards and women councillors.

Eastern Cape

- The Eastern Cape Provincial Development Plan identified the following challenges (Eastern Cape Planning Commission, 2014) Too few people work: unemployment statistics at 27.8% and an expanded rate of 43.5%
- The standard of education is poor: over the period 2000 to 2011, about 22 % of learners who entered Grade 1 progressed to Grade 12 within the 12-year period, with only 14% successfully completing matric.

Gender responsive initiatives

Proposed interventions for addressing gender-based violence are outlined as Strategic Action 4.4.2 and include making places safer through community participation, inclusive approaches and infrastructure and service provision.

Nelson Mandela Bay

Characterisation

21.6% of households listed grants as their main source of income. Nelson Mandela has the highest household access to improved sanitation of all metros in the country at 95.8% followed by Buffalo City and Tshwane at 95.1% and 83.1% respectively.

Western Cape

Although the Province has the lowest Gini co-efficient in the country, inequality has continued to increase. Poverty remains gendered and female households more likely to be unemployed and poor (Western Cape Government, 2020). Gender-based crimes and femicide have continued to impact on both the economic prospects and socio-economic development of the province.

The female labour force participation rate in the Western Cape for quarter 3 of 2019 was 60.7% compared to 76.1% for men. Women also lack access to quality skills development and further education and are generally confined to the lower paid and vulnerable sectors of the economy. The rates of gender-based violence continues to be a significant social problem, in the Western Cape, 370 women were murdered, and 7 043 sexual offenses were reported in 2018/19. Young African women are the worst affected by poverty and unemployment (Western Cape Government, 2020).

Gender is a cross cutting theme in the Provincial Strategic Plan which acknowledges that the mainstreaming of gender planning, budgeting, and monitoring and evaluation systems is critical. The following interventions have been identified:

- **Internally**, provincial staff will be supported through training in sexual harassment policies and gender mainstreaming.
- **Externally**, compliance with Domestic Violence Act by SAPS will be monitored and GBV will be a key theme in Policing Needs and Priorities reports.
    - Provide shelters for victims of violence and human trafficking, trauma support for victims of violence and sexual offences.
    - Chrysalis Academy training programme will include gender sensitivity training and the Safe Schools unit will continue their GBV interventions
    - Community Development Worker programme provide information sessions and dialogues

City of Cape Town

Characterisation

The Human Development Index is 0.75, indicating an improvement in human development, however the Gini coefficient (0.63) points to a growing gap in income distribution in the City. The pandemic has also disproportionately affected women, the poor and vulnerable. Women are (City of Cape Town, 2022):

- More likely to lose their jobs

- Less likely to gain employment
- Less likely to benefit from income support measures
- More likely to be doing extra childcare

Gender responsive and water reuse initiatives

- Objective 2: Improving access to quality and reliable services
  Project 2.1. A: Informal settlements water and sanitation project – The City will continually test and expand innovative technologies and approaches to improve the quality and sustainability of water and sanitation services in informal settlements. This includes safe access to shared facilities, specifically at night for women and children.
- Objective 4: Well-managed and modernised infrastructure to support economic growth
  Project 4.4.D: Water reuse project – The City has partnered with the Water Research Commission (WRC) to investigate and invest in wastewater reuse to meet current and future water demand.
- Objective 14: Resilient city
  Programme 14.2: Disaster risk reduction and response programme – The project aims to enhance community-based disaster risk assessment, emergency preparedness reduce overall vulnerability. The project targets at at-risk populations including women and girls. It partners with NGOs, other spheres of government, academia, commerce, and industry

The City of Cape Town has a draft Gender Policy in place (City of Cape Town, 2004).

Drakenstein Local Municipality

Characterisation

There has been a general increase in the HDI in Drakenstein, from 0.647 in 2008 to 0.723 in 2017. The unemployment rate in was recorded as 21 % in 2021, however a more realistic unemployment figure is closer to 27%. The Gini coefficient is 0.601 and the rising income equality can be attributed to an increased working age population in low skilled employment who earn low salaries The categories of people vulnerable to poverty remained largely African females, children 17 years and younger, people from rural areas, and those with no education. (Drakenstein Municipality, 2022).

Gender responsive initiatives

The Social Development predetermined objective (PDO) contains the dedication to ensure that the community is supported in respects of social issues facing vulnerable groups. The PDO addresses gender by ensuring that the municipality has a functioning Gender Forum and conducts gender-specific programs to create awareness around gender issues such as gender-based violence.

Free State

The Free State Provincial Growth and Development Strategy (PGDS) Free Sate Vision 2030 outlines the rising unemployment challenge that the province is facing due to the structure of the provincial economy which limits job creation. The declining quality of education leads to skills shortages in a context where the economy is shifting towards finance and business services which leaves most of the province's youth behind. 51 % of the Free State's population is living in poverty and income inequality in the province has worsened from 0.59 to 0.64 from 1995 to 2010 (Department of the Premier, 2012).

The unemployment rate in the Free State increased from 25.5% to 32 % from 2011 to 2012. Women and youth are the mostly affected by unemployment. The mainstreaming of vulnerable groups such as women, youth, children, and people with disabilities during the implementation of long-term programmes has been identi1fied by the municipality as a strategy to address the social development needs of the province (Free State Provincial Government, 2013)

Mangaung Municipality

Characterisation

The Mangaung Metropolitan Municipality is relatively young and mostly female. There is still a large imbalance in the society with black women still at the bottom of the beneficiation chain, black males are second to women at just above 25% unemployment rate. Young people and children between the ages 0 -14 are in majority in Mangaung, thus the municipality strives to enhance its efforts on early childhood development, youth programmes and projects aimed at supporting women development.

The Mangaung Metropolitan Municipality represents approximately 28% of the provincial population. During the period 2011 to 2019 an estimated population of the Mangaung increased from 775,028 to 878,834 – an increment of about 103 806 (1.6%) people, this is due to immigration into the city from other tows, cities and rural areas. This large influx represents both challenges and opportunities for the municipality, such as increase in demand for basic services and human settlement and opportunities are amongst others revenue income for the municipality. The life expectancy in the region (between 2016-2020) is approximately 61 years in females and 55 years for males (South African Cities Network , 2022). In 2018 the healthcare in the region was 59.1% for public healthcare usage and 22.9% for private medical aid (South African Cities Network , 2022)

Gender responsive initiatives

The 2022-2027 IDP for Mangaung Metropolitan Municipality showed its commits gender mainstreaming and employment equity by adopting plans that heed the plight of women and persons living with disabilities. This commitment is planned to be actioned by ensuring that skilled, technical, professional, and managerial posts better reflect the makeup of the economically active populace of the region in respect of race, gender, and disability. Furthermore, affirmative action measures to achieve gender transformation in the municipality are planned wherein the senior management will have key performance indicators reflecting female and persons living with disabilities' appointments.

Northern Cape

The Northern Cape has a population of 1 193 780 people of which approximately half are female, and half are male. Households with women as their head account for 38.9% of the population in the Northern Cape of which

41.6% are child-headed households with women as their head (Community Survey , 2016). Water service delivery in the province is water is sourced from a regional or local service provider is 92.2% of which 45% is piped water infrastructure in the house (Community Survey , 2016). 71.4% of the population has access to flushing toilets and 67.9% have access to refuse disposal services. Only 38% of the population are employed (Statistics South Africa, 2011).

Income disparity in the province is apparent as only 46.92% of the populations sits above the poverty line. The limited employment opportunities in rural regions contribute to higher unemployment rates. Sustainable job creation is constrained by the structural mismatch between labour and demand supply. There is an absence of low skilled jobs in manufacturing due to the structural shift towards an economy that is dependent on high skills (Office of the Premier, 2020).

The Integrated Monitoring Framework provides for the monitoring of women, people with disabilities and youth development sectors, to ensure mainstreaming and tracking implementation thereof. Economic transformation is continuously promoted through preferential procurement in order to promote meaningful participation of black people, including women, youth, people with disabilities and people living in rural areas, in the province.

The Northern Cape annual report for 2016/2017 highlights limited gendered information for the province. However, the report indicates higher municipal performance rewards for women during the period. Gender specific targeted interventions for all genders of health programmes were yet to be identified. More women across various occupation levels were provided with skills development training during the period of 2016/2017 (Northern Cape Office of the Premier , 2016). The National Women's Charter is aware of the absence of approved gender policies and municipal gender-mainstreaming strategies across municipalities in the Northern Cape which forms part of the reason for the lack of gender budgeting and mainstreaming in the province (Mokoena, 2020)

Indicative Local Municipality B

Characterisation

This Local Municipality experienced the highest population growth of 2.56% between 2009 and 2019. It is the most populated municipality in the district and the Northern Cape Province. The adult literacy rate is 78.79%. Access to sanitation services is 85.1% and access to waste removal services is 85% and an employment rate at 26.7% in 2020. 41.7% of the economically active age grouping is not participating in the economy.

The representation of women in decision making positions is reflected in the gender breakdown of council which showed that 45% of women were elected as councillors in the 2016 Elections. The proportion of men in council is slightly above the proportion of men in society, it is assumed that male leaders would have a gender sensitive approach and all councillors will work to improving the status of women in society. Much more work is required in the municipality to achieve gender equity. In terms of employment equity, the spread of women workers in the technical environment appears to be low.

Gauteng

There has been a consistent improvement in the quality of life in Gauteng. The Human Development Index has shown an improvement from 0.65 in 2008 to 0.72 in 2018, while the percentage of people living below the poverty line has declined from 32% in 2004 to 16% in 2016 However, the province still faces spatial, economic, and social inequalities, areas of economic decline. The unemployment rate in the province is 31% and income inequality, as measured by the Gini index, is 70% (Gauteng Provincial Government, n.d.).

Gauteng has shown commitment to youth and women by (Gauteng Provincial Government, n.d.):

- Offering support for women-led SMMEs and entrepreneurs and the setting aside of 40 % of procurement for women
- Mandating the State to purchase 75% of goods and services from local producers, especially women- and youth-led producers
- Providing resourced shelters and relief support for women and girls, including the provision of programmes designed to heal victims of trafficking and rehabilitate them into society
- Providing security of tenure through the issuance of title deeds, including the title deeds for women, youth and persons with disabilities
- A total of 100 000 young women have been supported by the welfare-to-work programme, and EPWP programmes targeting poorer households – strengthening them with skills to access better economic opporutnites
- 2 000 women have benefited from community food initiatives
- Prioritised women, youth and people with disabilities as beneficiaries in all government programmes

The provincial focus is on improving the policing and safety efforts, with particular emphasis on GBV and supporting the rights of women, youth, senior people, people with disabilities, military veterans and LGBTIQ+ community. As well as strengthening effective partnerships with non-governmental organisation and all relevant agencies to promote an integrated and holistic approach to the elimination of violence against women and girls.

City of Ekurhuleni Metropolitan Municipality

Characterisation

The City of Ekurhuleni Metropolitan Municipality represents a population of 3 888 873 (in 2019). The life expectancy in the region (between 2016-2020) is approximately 69 years in females and 64 years for males. In 2018 the healthcare in the region was 64.5% for public healthcare usage and 22.8% for private medical aid (South African Cities Network , 2022).

The people of Ekurhuleni experience poverty and inequality at high levels. There is a culture of dependency on social grants and an endemic skills-gap exists. The high levels of crime result in vulnerable groups such as women, children, older persons and people with disabilities, living in fear and feeling unsafe (City of Ekurhuleni, 2018).

Gender responsive initiatives

The City of Ekurhuleni developed a gender policy in 2003. There is a gender and development department, but it is not well capacitated which results in gender and climate change issues not being clearly articulated and addressed and being managed in silos. The City of Ekurhuleni aims to prioritise youth, women, disabled and households in the indigent register in job creation initiatives (City of Ekurhuleni, 2018).

City of Johannesburg Metropolitan Municipality

Characterisation

The City of Johannesburg Metropolitan Municipality represents a population of 5 738 536 (in 2019). The life expectancy in the region (between 2016-2020) is approximately 69 years in females and 64 years for males. In 2018 the healthcare in the region was 67.3% for public healthcare usage and 21.2% for private medical aid (South African Cities Network , 2022).

The city struggles with high levels of poverty, unemployment, inequality, social exclusion, and sub-standard levels of human development. These issues are further exacerbated by unequal spatial development, long and costly commutes and inadequate basic services (City of Johannesburg, 2020).

Gender responsive initiatives

The City of Johannesburg Metropolitan Municipality has a gender policy 2020 report prepared by the Integrated Social Development Policy, Planning and Research Unit. This policy aims to ensure gender mainstreaming in the plans, programmes, service delivery, institutional and management practices of the City of Johannesburg Metropolitan Municipality (City of Johannesburg, 2020). The policy aims to achieve the following objectives:

- Ensure gender responsive and gender aware delivery of services by all Core Departments and MEs to the residents of the City of Johannesburg
- Enable the creation of a gender sensitive planning environment and development of governance structures
- Challenge the direct and indirect barriers in enterprise development which prevent women from having equal access to and control over economic resources
- Increase women's easy access to finance by assessing existing programmes that provide access to finance for women and suggesting improvements to address existing gaps in a sustainable manner
- Promote access by women to key resources (e.g. employment opportunities, decision-making and business), services and facilities
- Respond to the 4 strategic outcomes set out by the Growth and Development strategy (2040), especially Outcome 1: Improved quality of life and development driven resilience for all

Gender based violence is one of the City's strategic priorities and the number of interventions implemented to respond to gender-based violence has been placed as an on the IDP scorecard which indicates that 6 000 people have been reached by interventions to respond to gender-based violence

City of Tshwane Metropolitan Municipality

Characterisation

The City of Tshwane Metropolitan Municipality represents a population of 3 649 053 (in 2019). The life expectancy in the region (between 2016-2020) is approximately 69 years in females and 64 years for males. In 2018 the healthcare in the region was 53.1% for public healthcare usage and 29.3% for private medical aid (South African Cities Network , 2022).

Gender responsive initiatives

The City of Tshwane Metropolitan Municipality takes gender equality into account in the 2022-2026 IDP. The municipality has a sustainability department and various policies in place that also feed into gender mainstreaming such as:

- Training and development policy
- Sexual harassment policy
- Revised employment equity policy
- Occupational Health policy

The city has identified the need to (City of Tshwane, 2021):

- Act as an enabler for job creation, with a specific focus on women, youth, and people with disabilities
- Have a community safety forum to deal with issues of violence against women and children
- Prioritise the needs of women, youth and people with disabilities when designing new human settlements.

4.2.3 SUMMARY

The equity-related regulatory framework, gender machinery and the statutory bodies outlined in Chapter 4.2.1 and 4.2.2 above indicate the strategic initiatives for driving the empowerment of women at national, provincial, and local levels. The strategic initiatives are intended to increase women's representation and participation, especially in decision-making platforms, as well as eliminate barriers and to protect women's rights and recourse.

The provincial and local municipality have integrated gender considerations into their strategies and plans at varying levels. KwaZulu Natal has specific outcomes, indicators and targets related to gender, whereas the other municipalities do not show that they are monitoring gender related outcomes. There are key interventions and projects in place aimed at addressing gender inequality and include the implementation of empowerment initiatives, the creation of partnerships and skills development support

The gender related challenges that many of the municipalities' face are not always in alignment with the solutions proposed in the IDPs. An example is the Indicative Municipality A which is characterised by poor access to education and low skills, whereas the strategic women empowerment initiatives are aimed at preferential procurement for women. The vulnerable people in this municipality would be unlikely to be able to meet supply the municipality's technical and procurement requirements. This is a similar case in the Sol Plaatje local municipality. It is important to strengthen the capacity of vulnerable groups to take advantage of empowerment opportunities.

Gender responsiveness at a municipal level is focused on gender-based violence, which is to deal with the high incidences that have been reported. This represents a limited response to addressing gender inequality and the broader challenges impacting women such as the lack of access to basic services, education and skills. There is a need to promote awareness of the opportunities available to women and grassroots organisations and improve their capacity to participate in project implementation and decision-making. By engaging diverse perspectives on experiences this ensures that the product, services, and systems are designed to address gender differences and barriers.

Limitations:

- Obtaining disaggregated data on gender for the public sector is particularly challenging, since data is not always readily available in a completely disaggregated manner.
- Lack of understanding of gender mainstreaming as an approach
- Limited human and financial resources
- There is a disconnect between national and provincial and municipal realities

4.3. KEY GENDER-RELATED CHALLENGES FOR THE PROGRAMME TO ADDRESS

South Africa is a water-scarce country, and the sustainable provision of water is one of its most significant challenges. Basic water-related services including the provision of potable water and access to sanitation and stormwater drainage are not readily available to a significant proportion of the population.

The implementation of a national Water Reuse Programme ('WRP') through the scale-up of water reuse approaches and water reuse infrastructure in municipalities would significantly enhance water security in South Africa. A successful WRP should be able to demonstrably indicate that climate change resilience objectives will be achieved **(by strengthening the country's adaptive capacity against water stress and scarcity)** and should be able to measurably maximize climate change adaptation and mitigation in a manner that meets the criteria and requirements of the GCF, thus enhancing the probability of successful GCF funding being achieved.

When designing the programme, consideration was made to how the programme will address gender-related challenges at a programme level. The contribution of women to the design, operation and maintenance of the incumbent water systems will assist to reflect their needs and preferences in the overall

WRP. This user-centred approach will ensure that immediate and anticipated concerns regarding cultural contexts will improve public perception, relating to the reuse of effluent, particularly regarding acceptability on religious or cultural grounds and will remain sensitive to the responsiveness of the consumers. Similarly, WRP will aim to attract, retain, and promote women into the operational aspects of the programme by encouraging and enabling the appointment of women as engineers, technicians, mechanics, operators, system architects and utility manager. Their contribution will help to shape the design, construction of the chosen water reuse project archetype. Globally there is a need to promote women into the utility sector so they can contribute to the initiatives and apply a gendered lens. This gap, however, is a global challenge as shown in Figure 4-7 Average Share of Employees in a Water Utility that are Women (2018-2019)Figure 4-7 below.

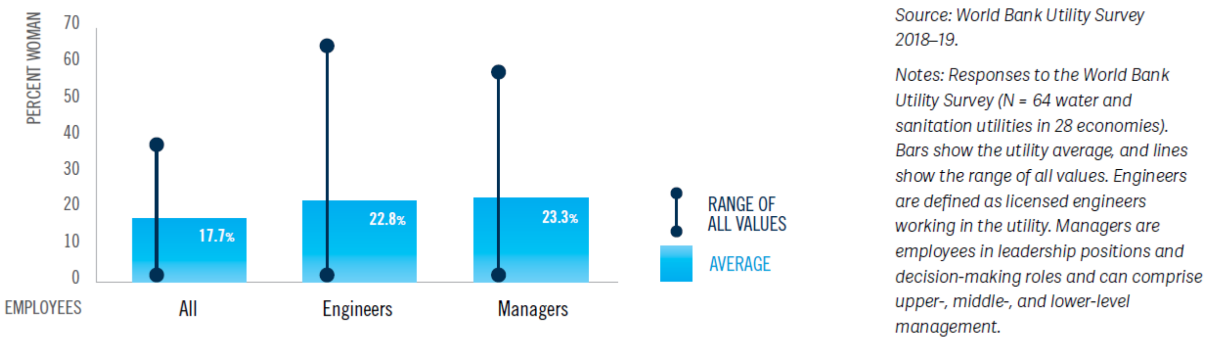

Figure 4-7 Average Share of Employees in a Water Utility that are Women (2018-2019)

The programme will thus consider the following programmatic challenges that will be addressed in the activities and incorporated into the final design and implementation of the WRP. The challenges were discussed with key gender specialists as well as supplemented by extensive data collection and research as noted below in Table 4-5 below.

Table 4-5 Programme Challenges

#	Programmatic Challenge	Description
1.	**Modernisation**	When designing the required technology (for example the advanced technology required to treat wastewater effluent), there is a possibility that as it will be capital and technologically intensive it can lead to a reduction in employment opportunities for low-skilled women who would previously have been appointed to assist with some aspects of the operational process.
2.	**Working Conditions**	Working conditions are not gendered and are not designed to reflect their needs and preferences, e.g., personal protective gear, infrastructure, sanitary facilities.
3.	**Gender Mainstreaming**	Key stakeholders are unaware of gender terminology, mainstreaming solutions.
4.	**Configuration of waste management systems**	Existing solid waste management systems have not been configured to collect and process menstrual waste, diapers etc. causing significant issues. Also, the advanced treatment

		technology is not equipped to process ill-discarded menstrual products and diapers as it was not considered or due to feasibility constraints.
5.	**Equipment Guidelines**	Equipment is designed by and for men often lack appropriate training for women
6.	**Gender-responsive budget**	Gender responsive budgeting to fund gender interventions is not considered
7.	**Procurement Process**	Failure to promote involvement of women in the procurement process e.g., provision of goods, services, labour
8.	**Health Impacts**	Health impacts due to water quality can impact men and women differently for biological reasons influencing their options.
9.	**Target Market**	The main beneficiaries of water are women as they conduct domestic duties and are thus the key target market.
10.	**Poor Perception**	Lack of understanding/poor perception on the benefits and needs for wastewater reuse

It should be noted that a Gender Specialist will be appointed to ensure that the implementation of the projects align with the broader Programme objectives on Gender, as well as monitor and track the diversity and gender of the implementation teams as well as the broader impact on women and girl children. The recruitment timing of this will be dependent on the Water Partners Office establishment and having a strong portfolio of projects underway.

5. ACTION PLAN INTERVENTIONS AND ACTIVITIES

The WRP is committed to applying a gender lens to its project design and execution. The team is also acutely aware of the possible failures that may occur when limited consideration is applied to understanding the gender inequalities in a country and at a programme level. These inequalities will hamper the effective implementation of the programme and reap unintended consequences for those most vulnerable in society if they are not adequately addressed.

Based on the Findings above, Gender integration needs to happen at two levels;

I. **Programme Level** - The Programme is to provide the broader overarching: framework and mechanisms for gender integration.
II. **Project Level** – This level should address the specific genders gaps in the localities identified.

The actions that are needed to be addressed in these levels are presented below.

5.1. PROGRAMME-LEVEL INTERVENTIONS

A. Planning and Governance

A.1. Embed gender considerations in the establishment and operationalisation of the Water Partnership Office (WPO): such as targets on the percentage of women in decision-making positions at varying levels and points of programme implementation. There will also be human resource targets set to ensure that modernisation of technology does not adversely impact the economic empowerment of women. Contract documents will include gender responsive clauses outlining the contractor's responsibility for the health and safety of all workers. Enforceable accountability measures assist in the effective promotion of integration of gender considerations at the personal, unit and institutional levels.

A.2 Engendered procurement: This includes ensuring that procurement and contracting processes are governed by labour laws and mandates equal wages for an equal value of work. Additional considerations include:

- The programme will endeavour to procure equipment from women-led small and medium enterprises during the public procurement process and infusing gender equality throughout the supply chain.
- The programme will attempt to support corporate supply chain diversification in line with environmental, social and governance considerations.
- The programme will endeavour to procure technology that is tested with a gendered approach.
- The programme will apply gender provisions as required in BBBEE Acts and labour provisions will be adhered to and improved on where possible.

- These targets/guidelines will be included in the framework/guidelines of project specific gender action plans and the WRP will require project sponsors to look at these elements when they prepare GAPs for their projects.

A.3. Budget for gender activities: A ringfenced budget is implemented at the programme level that enables the listed initiatives to be realised. Without a firm financial commitment, there is a possibility that the activities will be side-lined until, or if, a budget is made available. A ringfenced budget ensures that financial parameters are set, and the activities are manageable and implemented without financial limitations. This budget needs to link to staffing, planning and operationalisation of activities.

B. Capacity Building and Awareness

B.1. Engender WRP brand development and communications strategies: The public awareness campaigns during the implementation phase of the programme will consider the low uptake and buy-in from the community to water reuse. The campaigns will highlight points that will ease the various gender-related issues that may be perceived from the use of the water, e.g., menstruation, the use of chemicals at the advanced technological treatment plant etc., and how it will not adversely affect women physically, biologically, and physiologically. Throughout the design and implementation phases, the project managers from the programme management office will safeguard that GESI implications are consistently discussed and brought up as a standing point to all the meetings, both internally and with stakeholders. This will ensure legitimacy that GESI is not brought in as a 'tick box' exercise for the programme but as a key component to being analysed and discussed. This will also include the stakeholder engagement plan.

B.2. Develop gender-focussed knowledge products: The interventions aimed at addressing the gender specific needs of women and girls in the anticipated project areas will be identified during the initial project conceptualisation phase. Progress in the form of specific project outputs, outcomes will be measured consistently throughout all the projects falling under this programme. The team will thus develop knowledge-products that address the impacts of water insecurity on women and girls and adapt the challenges to solutions to drive gender-responsive interventions. The knowledge products will also aid in the effectiveness of different types of water projects in addressing gender inequalities related to water insecurity.

B.3. Strengthen institutional capacity and awareness: The programme will ensure that there is institutional awareness in the WPO and municipalities that promote gender equality. Capacity building and training programmes will aim to promote institutional awareness of the gender implications of water infrastructure. During the implementation stage, gender specialists and environmental and social experts will be secured to assist with community development and addressing gender equality issues that the project team may confront. The experts will also drive training on participation, leadership and management of sanitation, health and hygiene, microplanning, participatory learning methods, self-help group strengthening. Managerial and executive leadership training will be offered internally to support women to participate in key strategic and decision-making roles the programme.

C. Monitoring, Evaluation and Reporting

C.1. Design of gender framework and monitoring and reporting strategies: The gender monitoring framework for the programme needs to be flexible enough to incorporate the varying project activities and robust enough to capture important baseline data for tracking the gender considerations at a programme level. The collection of data on both women and men enables the tracking of gender impacts and assessing the programme benefits for women and men.

C.2. Appoint a gender specialist: A gender specialist is required to ensure that the objectives of the programme are being translated into the project and local-level interventions. The recruitment timing of this will be dependent on the WPO establishment, but is recommended for the first year of establishment.

C.3. Measure gender disaggregated outcomes: This will include collating data water security, time poverty and household allocation of tasks to assist with measuring progress against meeting the gender-specific data requirements and targets.

C.4. Undertake continuous monitoring, evaluation, and learning: Once the programme is in its implementation phase, there will be an opportunity to refine or add activities and interventions that are more relevant and gender sensitive. Knowing the systemic issues will allow the implementation team to design more effective and sustainable outcomes, mirroring the diverse and inclusive needs with a targeted and inclusive approach and response.

5.2. PROJECT-LEVEL INTERVENTIONS

Based on the assessment, the following actions are required at a Project Level. Many of these will be informed by the programme-level framing and requirements. These project-level activities are not budgeted in the Gender Action Plan, as much of these costs will be transferred to project-level budgets once they are initialised.

A. Planning and Governance

A.1. Alignment to programme-level gender monitoring framework: At programme-level, guidance and tools are provided on how project-level gender mainstreaming needs to be done and what targets and indicators need to feed into the programme level gender monitoring framework.

A.2. Map institutional/governance gender elements at project-level: The roll out of gender related project interventions will be dependent on the institutional context which varies across the different municipalities. An institutional map of each project will help to better understand the gender elements at the specific municipal level, identify gaps and seek opportunities to align with broader programme interventions.

A.3. Budget for gender activities: Gender activities are appropriately resourced at project level in terms of human resources and budgets. This should be outlined in project-level Terms of Reference (ToR).

A.4. Align to Programme Procurement: The programme will provide strict guidance on gender-specific procurement requirements. It is important that at project-level, these are adhered to.

B. Capacity Building and Awareness

B.1. Share Repository of resources, templates, and tools: The consistent application of the same methods, templates and tools will be used to monitor, track the data that has been collected at a project level and allow for better integration with the programme level gender monitoring framework.

B.2. Create Awareness: The projects will combine social protection and water projects with initiatives to sensitise communities about unequal gender relations, encourage gender sensitive allocation of project benefits and empower women in household and community decision-making. If well designed and implemented, projects can provide an opportunity to shift gender norms and encourage behaviour changes. Women can contribute to and lead community decision-making processes, provided that their time, mobility, and social constraints are considered and addressed.

B.3. Implement training activities: The programme will actively recognise and address the existing and growing interactions between municipal service projects, job creation, environmental and social protection, water insecurity and gender inequalities. There are already clear areas of overlap between these issues, and the links will only become further entrenched with the increased impacts of climate change and the social exclusion of women from decision-making positions. Failure to acknowledge these links and coordinate action across sectors can reduce or reverse the intended impacts of individual programmes on poverty, vulnerability, water security or female empowerment. Accordingly, the programme will consider training programmes that address these key themes and explain them in an easy to digest and simple fashion. The training programmes will expose municipal officials and key stakeholders to gender terminology and broader consideration so that the burden for reporting and measuring progress to achieving gender equality is not strictly on women but is a requirement for all involved in the project.

C. Monitoring, Evaluation and Reporting

C.1. Determine the Baseline*:* Information on gender and sex-disaggregated data is limited at a local level and relies heavily on work already conducted by a range of different stakeholders. Ensuring that sex-disaggregated data is collected and shared with the initiation of each project is important. This will support and feed into the Programme-Level M&E Framework. Project level ToR's should include this as an update of the gender assessment activity to be undertaken. Budgeting for this will then fall under the project implementation, and not the Programme budget.

C.2. Report on M&E activities (with disaggregated data, where available)

Key activities include:

- Collect sex disaggregated data to track gender equality results and assess gender impacts.

- Monitor access, participation, and benefits among women and men and incorporate remedial action that redresses any gender inequalities in project implementation.
- Ensure women and men can participate in monitoring and / or evaluation processes.
- Integrate gender evaluation questions and components in the Evaluation TORs
- Identify good practices and lessons learned on project outcomes / outputs or activities that promote gender equality and / or women's empowerment.
- Consider and integrate lessons learned from previous projects with gender dimensions into project formulation where relevant.

Reflections on the following is:

- Compliance
- Employment Gender Profiles
- Training and Capacity Building
- Site-level benefits and beneficiaries
- Sexual exploitation, abuse and sexual harassment
- Gender resources utilisation (human and finance)

5.3. RISK AND MITIGATION: INDEPENDENT GRIEVANCE MECHANISM

An Independent Grievance Redress Mechanism process is established to resolve all complaints that arise from the programme. The mechanisms will be aligned with the GCF's independent redress mechanism and experiences. An example of such a mechanism will be enable members of the public to submit complaints about the programme through various mechanisms such as public participation community meetings, at the Municipal office and through their Ward Councillors. In addition, members could opt to use the Municipal Help Desk and Suggestion Box as located at central offices. Municipal Employees can approach their Equity Officer and Labour Relations units through Employee Assistance Programme (under Human Resources), and their relevant trade or professional union.

An outline of the Grievance Redress Mechanism is outline in the Environmental and Social Management Framework.

6. CONCLUSION

Gender is not one-dimensional or simplistic. It is complex and multi-layered, and practices and processes should reflect them

From our assessment, there are sound legal, regulatory and policy frameworks that aim to achieve gender equality in the region, from an international, regional, and local perspective. However, there are still lingering patriarchal norms, practices, and customs; poverty; historical legacies, gender-based violence and lack of employment opportunities in senior leadership position that hamper women from achieving economic parity.

Despite these broader challenges and progressive achievements, how will the project contribute toward gender equality and women's empowerment? Through the various activities as outlined in the Gender Action Plan the programme will aim to address the current social and gender inequalities that will impact upon the success of the programme. Some of the initiatives include:

- Applying quotas in female participation in sanitation policies for water committee members, Boards and agencies.
- Integrating a monitoring system with gender monitoring indicators.
- Ensuring a 30-40% target for the beneficiaries of the program to be women.
- Raising Public awareness campaigns aimed at creating an appreciation and understanding of the benefits of water reuse.
- Enabling all water stakeholders—from the implementing agencies to the beneficiaries to build requisite skills and knowledge for gender-sensitive services and management.

Ultimately, gender disparity is not only a challenge in South Africa but across the globe. Through programmes like the WRP, some of these issues can be intricately addressed while avoiding exacerbating existing social inequalities. As the WRP scales and evolves, greater in-depth research will need to be conducted to complement this current gender assessment report in the future.

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WATER REUSE PROGRAMME - PROGRAMME GENDER ACTION PLAN

The purpose of the Gender Action Plan is to provide gender specific and time-bound framework within which to operationalize the relevant Gender Policies and the findings from this gender assessment. Implementation of the Gender Action Plan will require ensuring that the WRP and the Water Partnerships Office has the established competencies, tools, and processes to achieve the outlined results. This gender action plan focuses upon the programme level where separate gender action plans will be prepared for individual projects that speak to the current action plan below. It is envisaged that support from the Gender Forum in DBSA will help support various elements of this plan, in the absence of a dedicated Gender specialist.

IMPACT STATEMENT: The improved economic empowerment of women through the establishment of a national water reuse programme that integrates gender-responsive initiatives and interventions into the design and implementation of projects as well as in the procurement, management and operational functions. This will be done by (1) enhancing the capacity of women in senior leadership in utilities, (2) continuous learning and adapting the programme to gender sensitive practices, (3) improving the working conditions, procurement processes, creating gender-specific contracting provisions and appointment of staff, (4) improving the protective gear, equipment and sanitary facilities on site (5) selecting projects that are gender-responsive and that do not exacerbate existing social inequalities and (6) ensure that that there are appropriate budget allocations for gender activities. OUTCOME STATEMENT: Women are empowered to actively participate and benefit from the WRP and the improved water supply value chain that is developed.			
Programme Level			
Intervention A: Planning and Governance			
Action	**Indicator and Target**	**Timeframe/Budget**	**Responsible**
A.1 Embed gender considerations in the establishment and operationalisation of the Water Partners Office (WPO) a) Develop appropriate gender mainstreaming policy for the WPO b) Apply gender specific quotas to staff complement c) Apply gender specific quotas in senior decision-making positions d) Engendered participation in project management processes	a) Gender responsive and inclusive HR Policies developed and approved by WRP Oversight Committee ○ 100% of gender issues are considered in the GM Policy (including procurement policy, SEAH and grievance mechanisms)	Y1 of establishment of the WPO 10 000 USD	DBSA Gender Specialist and Gender Forum WRP oversight Committee Review by Programme Gender Specialist once appointed.

e) Gender-sensitive working conditions Contracts with gender-responsive provisions such as: maternity and paternity leave, childcare and pension provisions; offering flexible hours and remote working for employees; Creating separate sanitation facilities for men, women, and gender-neutral parties; establishing breakaway, prayer, and breastfeeding rooms	o Communication to 100% of employees, once approved b) % of women in WPO o 40% target (national target is a minimum of 30%[1]) c) % of women in senior, decision-making positions o 40% target d) % of female participation in sanitation policies for water committee members, Boards, and agencies o 40% Target e) Contracts in place outlining the gender-sensitive work conditions o 100% of appointment contracts ae gender sensitive o 100% of operational office modalities consider gender in their protocols		
A.2 Engendered procurement policy a) Ensuring procurement processes (women in the value chain) target historically disadvantaged run companies and/or women-led companies b) Ensure all projects have Terms of Reference and Requests for Proposals are engendered	a) Contracts to reflect quota for historically disadvantaged companies or women-led companies o 40% of contracts to be contracted to b) Engendered ToRs and RfPs are produced o 100% of ToRs and RfPs are engendered	Y1 of establishment 10 000USD	DBSA Gender Specialist and Gender Forum Review by Programme Gender Specialist once appointed.
A.3 Budget for gender activities **a)** Ringfencing of a gender-specific budget that links to programmatic activities (staffing,	a) Secured budget allocations for gender-activities	Throughout programme	DBSA Gender Specialist and Gender Forum

[1] https://www.gov.za/sites/default/files/gcis_document/201904/42391gon567.pdf

planning and operationalisation of activities)	○ Annual budget for gender-responsive activities is secured		Review by Programme Gender Specialist once appointed.
Intervention B: Capacity Building and Awareness			
Action	**Indicator/Target**	**Timeframe/Budget**	**Responsible**
B.1 Engender WRP brand development and communications strategies a) Develop specific strategy for communicating the gender activities b) Ensure that communication materials are gendered	a) Gender communication strategy completed ○ Approved by senior WPO management b) Publication of an annual report that reflects gendered activities ○ At least 2-3 gender-activities are implemented each year	Y1 of establishment Throughout the programme	DBSA Gender Specialist and Gender Forum Programme Gender Specialist once appointed
B.2 Publish gender-focused knowledge products a) Develop gender-focused knowledge products at programme level to inform project-level outcomes • Identify barriers to women being appointed to senior leadership positions within utility departments • Research initiatives to attract women to STEM careers • Unpack current social inequalities in the wastewater sector and publishing findings • Identify the main skills women workers require for capacity building including communication skills, literacy, computer, management • Research the gender, water insecurity and climate change links	a) Engendered knowledge product developed and published ○ At least 2 knowledge products published per year	Y2 onwards 50 000USD	Programme Gender Specialist once appointed
B.3 Strengthen institutional capacity and awareness	a) Senior management capacity built	Annual	Programme Gender Specialist

a) Develop specific training interventions for all staff (including diversity and inclusion workshops, and training on specific gender policy requirements, such as SEAH, and monitoring of gender activities, procurement) b) Develop capacity of senior management regarding gender mainstreaming and its processes (Oversight and grievance mechanisms)	○ 2 training sessions held annually ○ 100% orientation and training on guidance on labour standards and gender terminology ○ b) Senior management capacity built ○ 1 training sessions held annually	15 000USD	
Intervention C: Monitoring, Evaluation and Reporting			
Action	**Indicator/Target**	**Timeframe/Budget**	**Responsible**
C.1 Design of gender framework and monitoring and reporting strategies a) Develop a gender M&E Framework and Strategy, including a tracking tool	a) Gender M&E framework and Strategy in place ○ Gender M&E Framework and strategy signed off by senior management	Y2 30 000USD	Programme gender specialist and WPO Management
C.2 Appoint a gender specialist a) Undertake recruitment and appointment of a gender specialist for the WRP Programme	a) Gender specialist appointed ○ Programme Gender Specialist approved and appointed by senior management	Y2 2000 USD (recruitment)	WPO Management
C.3 Measure gender disaggregated outcomes a) Implement M&E Framework and Tracking tool that collects and collates sex-disaggregated data	a) Quarterly reports reflect sex-disaggregated outcomes ○ Quarterly reports submitted and approved	Quarterly, once the projects are initiated	Tracked and implemented by the Programme Gender Specialist
C. 4 Undertake continuous monitoring, evaluation, and learning b) Implement M&E Framework and Tracking tool that ensures the following: a) Compliance b) Employment Gender Profiles c) Training and Capacity Building d) Site-level benefits and beneficiaries	a) Quarterly reports developed ○ Quarterly reports submitted and approved	Annually once the projects are initiated	Tracked and implemented by the Programme Gender Specialist

e) Sexual exploitation, abuse and sexual harassment f) Gender resources utilisation (human and finance)			

Project Level			
Intervention A: Planning and Governance			
Action	**Indicator/Target**	**Timeframe/Budget**	**Responsible**
A.1 Alignment to programme level gender framework	a) Project level gender mainstreaming plans developed ○ 100% of all projects have a gender mainstreaming plan ○ 100% of all project gender mainstreaming plans are aligned to programme policies and guidance	Y2 onwards 100 000 USD	Project owners and supporting service providers to develop project level plans WPO Oversight (Gender specialist)
A.2 Map institutional/governance gender elements at project-level	a) Project level gender mainstreaming assessment undertaken to clarify gaps and roles and responsibilities to address at municipal levels ○ 100% of all project gender mainstreaming plans link to municipal gaps and opportunities	Y2 onwards 100 000 USD	Project owners and supporting service providers to develop project level plans WPO Oversight (Gender specialist)
A.3 Budget for gender activities	a) Budget for gender mainstreaming in each project has been determined and is allocated ○ 100% of all project gender mainstreaming plans are costed ○ Budget allocations for gender mainstreaming in all projects is secutred	Y2 onwards 50 000 USD	Project owners and supporting service providers to develop project level plans WPO Oversight (Gender specialist)
A.4 Align to Programme Procurement	a) Procure equipment from women led SMME during public procurement process ○ 30% of all equipment is purchased from women -led SMMEs	Y2 onwards 500 000 USD	Project owners and supporting service providers to develop project level plans

	b) Procure technology that is tested with a gendered approach ○ 75% of technology procured is sensitive to gender c) % Of machinery and equipment procured from companies that are owned by historically disadvantaged individuals ○ 50% of machinery and equipment purchased from companies owned by historically disadvantaged individuals (in accordance with broad based black economic empowerment) d) Capacitation programmes are developed at project level to up- skill workers to ensure that the modernisation of technology does not adversely impact the economic empowerment of women ○ 100% of all projects have a gendered capacitation and training programme		WPO Oversight (Gender specialist)
Intervention B: Capacity building and awareness			
Action	**Indicator/Target**	**Timeframe/Budget**	**Responsible**
B.1 Shared Repository of resources, templates, and tools	a) A database of templates and tools has been developed and operationalised ○ 100% of all gender policies, guidance material and tools are available at project level through a shared portal ○ Training session held on the use of these policies, guidance materials and tools help at the inception phase of all project.	Y1 150 000 USD	WPO Oversight (Gender specialist)
B.2 Create awareness	b) Municipal officials and key stakeholders are capacitated regarding gender and its project level implications ○ 100% of projects have an awareness creation session held at the beginning		Project owners and supporting service providers to develop project level plans

	○ of each project regarding gender mainstreaming and its importance ○ 100% of all projects have a mid-term project awareness session on gender mainstreaming and its progress		WPO Oversight (Gender specialist)
B.3 Implement training activities	a) Project level training interventions are gendered and available to all women at project level ○ 100% of training interventions consider gender aspects ○ 100% of project training are attended by at least 50% women ○ All projects host at least 2 gender training sessons per year.	Y2 onwards 200 000 USD	Project owners and supporting service providers to develop project level plans WPO Oversight (Gender specialist)
Intervention C: Monitoring, evaluation, and reporting			
Action	**Indicator/Target**	**Timeframe/Budget**	**Responsible**
C.1 Determine the Baseline	a) Project level baseline assessments are undertaken for each project ○ 100% projects included a baseline gender assessment b) Collating the findings from the assessment to support the current gender assessment and action plan ○ Finding from the assessments are integrated into project level gender mainstreaming plans and monitoring of project impacts	Y2 onwards 75 000USD	Project owners and supporting service providers to develop project level plans WPO Oversight (Gender specialist)
C.2 Report on M&E activities (with disaggregated data, where available)	a) Compliance at the project level is assessed ○ 90% compliance with policies, guides and tools in ensured in all projects b) Employment Gender Profiles ○ Gender employment targets for all projects achieved c) Training and Capacity Building	Y2 onwards 200 000USD	Project owners and supporting service providers to develop project level plans WPO Oversight (Gender specialist)

	○ All training and capacity building interventions are monitored and reported upon in terms of gender d) Site-level benefits and beneficiaries ○ 100% of projects are monitored for impact on beneficiaries, using gender disaggregated data collection e) Sexual exploitation, abuse and sexual harassment ○ 100% of all cases of SEAH are addressed in accordance with policies and guidelines ○ 100% of all SEAH cases are sensitively documented in alignment with policies and protocols ○ In 100% of SEAH cases, appropriate actions and interventions are taken and reported. f) Gender resources utilisation (human and finance) ○ 100% of projects monitor and document the utilisation of gender resources, both in terms of time and budget ○ Corrective action to address utilisation of under 50% (on both time and budget) is documented and reported		